3. **How many latitudes are there on the globe drawn at 1 degree interval?**

(a) 180 (b) 178

(c) 179 (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (c)

If parallels of latitude are drawn at an interval of one degree, there will be 89 parallels in the northern and the southern hemisphere each. The total number of parallels is drawn including the equator will be 179.

iii. Standard Time

1. **How much is the difference between Indian Standard Time (I.S.T.) and Greenwich Mean Time (G.M.T.)?**

(a) $+4\frac{1}{2}$ hours (b) $+5\frac{1}{2}$ hours

(c) – 5 hours (d) $-4\frac{1}{2}$ hours

45th B.P.S.C. (Pre) 2001

Ans. (b)

Indian Standard Time is calculated on the basis of 82°30' E longitude. This Meridian passes through Mirzapur district (Near Prayagraj) in the state of Uttar Pradesh. It also passes through Jagdalpur of Chhattisgarh. The Indian Standard Time is ahead of Greenwich Mean Time by 5 hours and 30 minutes.

2. **On which plateau, the Tropic of Cancer and the Indian Standard Time Line intersect each other?**

(a) Bundelkhand (b) Baghelkhand

(c) Malwa (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (b)

Tropic of Cancer is 23.5^0 N and Indian Standard Time Line is 82.5^0 E. Significantly Tropic of Cancer passes through eight states lies Rajasthan, Gujarat, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram while Indian Standard Time line passes through five states–Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha and Andhra Pradesh. They both intersect at Korea district in Chhattisgarh. Korea district lies in Baghelkhand Plateau.

3. **What is the local time of Thimphu (Bhutan) located at 90° East longitude when the time at Greenwich (0°) is 12:00 noon?**

(a) 6:00 p.m. (b) 4:00 p.m.

(c) 7:00 p.m. (d) 6:00 a.m.

69th B.P.S.C. (Pre) 2023

Ans. (a)

Difference between Greenwich and Thimpu (Bhutan) = 90^0 of longitudes
Total time difference = 90 × 4 = 360 minute
= 360/60 hour = 6 hour
= 6 hours/Local time of Thimpu is 6 hours more than that at Greenwich i.e. 6.00 p.m.

2. PHYSICAL DIVISIONS

i. Natural Regions of India

1. **Arrange the following ranges from North to South in sequence :**

1. Ladakh **2. Karakoram**

3. Pir Panjal **4. Zanskar**

Select the correct answer using the codes given below.

(a) 1, 3, 2, 4 (b) 2, 1, 4, 3

(c) 3, 4, 1, 2 (d) More than one of the above

(e) None of the above

B.P.S.C. School Teacher (Class 11-12) 15-12-2023

Ans. (b)

The correct sequence of ranges from North to South is Karakoram Range → Ladakh range → Zaskar range → Pir Panjal. Hence the correct answer is option (b).

2. **As per geographical area, arrange the following physiographic units of India in ascending order**

1. Central highlands **2. Great plains**

3. Coastal plains **4. Northern Mountains**

Select the correct answer using the codes given below.

(a) 3, 1, 2, 4 (b) 1, 2, 3, 4

(c) 2, 4, 1, 3 (d) More than one of the above

(e) None of the above

B.P.S.C. School Teacher (Class 11-12) 15-12-2023

Ans. (a)

As per geographical area, the correct ascending order of graphic units of India is Coastal plains → Central Highlands → Great Plains → Northern Mountains.

3. **Consider the following statements regarding formation of landforms in India :**

I. Structurally, the Meghalaya Plateau is an extended part of the Deccan Plateau.

II. **The Valley of Kashmir was formed in a synclinorium.**

III. **The Gangetic plain was formed in a fore deep.**

IV. **The Himalayas originated as a result of triangular convergence of the Indian Plate, the European Plate and the Chineses Plate.**

Which of these statements are correct?

(a) I, II and III (b) I, III and IV

(c) I and III (d) II and IV

47th B.P.S.C. (Pre) 2005

Ans. (a)

Meghalaya Plateau is an extension of the Peninsular Plateau. It is believed that due to the force exerted by the north eastward movement of the Indian plate at the time of the Himalayan origin, a huge fault was created between the Rajmahal hills and the Meghalaya plateau. Later, this depression got filled up by the deposition activity of numerous rivers. Today, the Meghalaya and Karbi Anglong plateau stand detached from the main Peninsular Block. Thus statement (1) is correct. The Valley of Kashmir was formed in a synclinorium. The Indo-Gangetic basin is an active foreland basin having east-west elongated shape. The basin formed in response to the uplift of Himalaya after the collision of India and China plates (Dewey and Bird, 1970). Suess (1893-1909) was the first geologist to suggest that the Indo-Gangetic depression is a 'fore-deep' and was formed in front of the high crust-waves of the Himalayas as their southward migration was resisted by the rigid landmass of the Peninsula.

Thus statement (3) is also correct. The Himalayan mountain range and Tibetan Plateau have formed as a result of the collision between the Indian plate and the Eurasian plate. Chinese plate is not mentioned in the theory of plate tectonics. Thus statement (4) is wrong.

4. **Which of the following is *not* a peculiarity of middle Gangetic plain agroclimatic region?**

(a) Located in Ganga and its tributaries' drainage area

(b) Extended in more than two States

(c) Receives more than 100 cm rain annually

(d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (e)

Middle Gangetic Plain agroclimatic region is a fertile plain drained by Ganga and its tributaries. It is extended in more than two states and receives more than 100 cm rain annually. Hence option (e) is the correct answer

ii. Northern Mountainous Region

1. **The foothills region of Himalayas is –**

(a) Trans-Himalayas (b) Shiwalik

(c) Great Himalayas (d) Aravali

43rd B.P.S.C. (Pre) 1999

Ans. (b)

The foothills region of the Himalaya is called Shiwalik. These are the outermost ranges of the Himalayas . They extend over a width of 10-50 Km and have an altitude varying between 900 to 1100 meters. These ranges are composed of unconsolidated sediments brought down by rivers from the main Himalayan ranges located farther north.

2. **Which of the following areas is not landslide-prone area?**

(a) Western Ghats

(b) Eastern Ghats

(c) Himalayan Regions

(d) More than one of the above

(e) None of the above

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Ans. (b)

Landslides are the gravitational movement of rocks, masses of earth, or debris through the mountain slopes. Hilly slopes become unstable due to groundwater pressure, earthquakes, erosions and volcanic eruptions, which causes landslides. Western Ghats and Himalayan Regions are landslide prone area while Eastern Ghats areas is not landslide prone area. Hence option (b) is the correct answer.

3. **The highest peak in the Eastern Ghats of India is :**

(a) Anai Mudi (b) Kanchenjunga

(c) Mahendragiri (d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (c)

Mahendragiri is the highest peak in the Eastern Ghats of India. It is located in Odisha, in the Rayagada block of the Eastern Ghats.

4. **Shiwalik series was formed in –**

(a) Aozoic (b) Paleozoic

(c) Mesozoic (d) Cenozoic

42nd B.P.S.C. (Pre) 1997

Ans. (d)

Shiwalik or outer Himalaya has formed approximately 2-5 million years ago in Pliocene Epoch means in Cenozoic Era.

5. **Someshwar Range is a part of which of the following mountain ranges?**
   (a) Satpura Range (b) Vindhyan Range
   (c) Eastern Ghats (d) Shiwalik Range
   (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (d)

Someshwar Ranges is an extension of Shiwalik Ranges in Bihar. They are situated in the northeastern part of the state, near the border with Nepal. The hills are named after the famous Someshwar Temple, which is located in the region. It is the Northernmost part of Bihar. Which is also the highest point in Bihar. It extends from Triveni Canal to Bhikhnathori.

6. **Kargil is located on the bank of–**
   (a) Suru river
   (b) Jhelum river
   (c) Indus river
   (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (a)

Kargil is located on the banks of the Suru River, a tributary of the Indus River, in the Kargil district of Ladakh, India. Kargil District is situated at a distance of 205 Kms from Srinagar and 230 kms from Leh. It is connected to Srinagar and Leh through National Highway. The District remains cut off with rest of the world during the winter season for more than seven months. But Leh – Kargil road remains open throughout the year.

7. **Atal Tunnel is across which one of the following Himalayan ranges?**
   (a) Zanskar (b) Western Pir Panjal
   (c) Ladakh (d) Eastern Pir Panjal
   (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

Atal tunnel is across Eastern Pir Panjal range. It connects Manali and Lahaul-Spiti Valley. It is 9.02 km long and was built by Border Roads Organisation (BRO). The tunnel reduces the distance by 46 km between Manali and Leh and the travel time by about 4 to 5 hours. Strategically, it provides better connectivity to the armed forces in reaching Ladakh.

8. **Atal Tunnel/Rohtang Tunnel opened recently, connects**
   (a) Sonmarg-Drass (b) Jammu-Srinagar
   (c) Leh-Kargil (d) Leh-Manali
   (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (d)

Atal Tunnel (also known as Rohtang Tunnel) is a highway tunnel built under the Rohtang pass in the eastern Pir Panjal Range of Himalaya, which connects Leh to Manali in Himachal Pradesh.

9. **Jawahar Tunnel is located near**
   (a) Bara Lacha (b) Banihal Pass
   (c) Chang La (d) Khardung La
   (e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (b)

Banihal has always been known as the gateway connecting Jammu and Srinagar which was made famous by the Jawahar tunnel and now the Banihal - Qazigund rail tunnel. Jawahar Tunnel also called Banihal Tunnel or Banihal pass.

10. **The part of the Himalayas lying between Satluj and Kali rivers is known as which of the following?**
   (a) Kumaon Himalayas
   (b) Assam Himalayas
   (c) Nepal Himalayas
   (d) More than one of the above
   (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Kumaon Himalayas lying between Satluj and Kali rivers. he part of the Himalayas lying between the rivers Teesta and Dihang is known as Assam Himalayas and part of Himalayas lying between the Kali and Teesta is known as Nepal Himalayas.

iii. Mountain Ranges and Hills of South and Central India

1. **At which of the following hills the Eastern Ghats join the Western Ghats?**
   (a) Palni Hills (b) Anaimudi Hills
   (c) Nilgiri Hills (d) Shevaroy Hills
   (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Nallamalai Hills are situated between the Krishna River and the Pennar River, stretched from north to south, parallel to the Coromandel Coast on the Bay of Bengal. Javadi Hills are located on the Eastern Ghats spread across parts of Vellore (Now in Tirupathardis) and Tiruvannamalai district of Tamil Nadu State. Renowned as 'Queen of Hills'. Shevaroy Hills of Eastern Ghats in Salem district of Tamil Nadu. Nilgiri Hills are located at the junction of Western Ghats and Eastern Ghats. Doddabetta is the highest mountain in the Nilgiri Hills at 2,637 metres. There are around 15 tribal groups in the Nilgiris. Among them the Badagas, Kotas, and Todas are the main tribal groups of the region. The Anaimalai Hills form the southern portion of the Western Ghats and span upto the border of Kerala and Tamil Nadu in southern India. Anaimudi is the highest peak in the Anaimalai Hills at 2695 metres Palni hills located in Tamil Nadu. These four hills are located in Andhra Pradesh, Tamil Nadu and Kerala state in India from north to south. Hence, the correct option is (c).

2. Indian Oceanists discovered a high mountain with a height of 1505 meter at the bottom of Arabian sea about 455 km west South West from Bombay, the mountain is called-

(a) Kailash II

(b) Raman Sagar mountain

(c) Kanya Sagar Parvat

(d) Bombay Parvat

40th B.P.S.C. (Pre) 1995

Ans. (b)

Indian Oceanists have discovered three marine mountain ranges in which one is located in Indian Ocean basin; Second is Sagar Kanya in Eastern Arabian Sea and third one is a high mountain with height of 1505 meter at the bottom of Arabian Sea about 455 km west south west from Mumbai. This mountain is named after famous scientist C.V.Raman as Raman Sagar mountain.

3. The major water divides between Arabian Sea and Bay of Bengal drainage are

(a) Delhi ridge, Malwa plateau and Bundelkhand

(b) Delhi ridge, Aravalli range, Sahyadri and Amarkantak hills

(c) Delhi ridge, Malwa plateau and Chota Nagpur plateau

(d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

Indian drainage system may be divided on various bases. On the basis of discharge of water (orientations to the sea), it may be grouped into (i) Arabian sea drainages and (ii) Bay of Bengal drainage. They are separated from each other through the Delhi Ridge, the Aravallis, Sahyadris and Amarkantak hills. Nearly about 77% of the drainage area consisting of the Ganga the Brahmaputra, the Mahanadi, the Krishna etc. is oriented towards the Bay of Bengal which is comprising the Indus, the Narmada, the Tapi, the Mahi and the Periyar systems discharge their waters in the Arabian sea.

4. Western Ghats in Maharashtra and Karnataka is known as –

(a) Nilgiri mountain (b) Sahyadri

(c) Deccan plateau (d) None of these

44th B.P.S.C. (Pre) 2000

Ans. (b)

Western Ghats is known as Sahyadri in Maharashtra, Goa, Karnataka. The Western Ghats has a high altitudinal variation and the average elevation is 1200 metres.

5. The Hills situated closer to Kanyakumari are :

(a) Anaimalai Hills

(b) Nilgiri Hills

(c) Cardamom Hills

(d) Shevaroy Hills

(e) None of the above/more than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Cardamom Hills are closest to Kanyakumari which is approximately 365 km from Cardamom Hills. Cardamom Hills are part of the Western Ghats located in southeast Kerala and southwest Tamil Nadu.

iv. Mountain Peaks

1. Highest mountain peak in India is–

(a) K2 Godwin Austin

(b) Kanchenjungha

(c) Nanda Devi

(d) Mount Everest

42nd B.P.S.C. (Pre) 1997

Ans. (a)

Mount K2 also known as Godwin Austen is the highest peak of India and the second highest peak of the world. It is located in Ladakh in Karakoram range. Its height is approx 8611m.

v. Passes

1. The Pass, which is situated at the highest elevation, is

(a) Zoji La (b) Rohtang

(c) Nathu La (d) Khyber

(e) None of the above/more than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

According to first four options Nathula is situated at the highest elevation. Nathula is in Sikkim at an elevation of approximately 4404 mts followed by Rohtang (Himachal Pradesh) 3978 mts (approx), Zojila (Ladakh) 3528 mts (approx) and Khyber (Pakistan) 1070 m (approx). But Yangzi Diwan is the India's highest Pass, since none of the above is also given as option. So, option (e) will be the right answer.

2. The Khyber Pass connects which one of the following pairs of cities?

(a) Kandahar-Quetta

(b) Kabul-Islamabad

(c) Ghazni-Bannu

(d) Herat-Tehran

(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (*)

The Khyber pass is a mountain pass that links Afghanistan (Kabul) and Pakistan (Peshawar). Throughout its history it has been an important trade route between Central Asia and South Asia and is a strategic military location. It is the route used by Alexander the Great, Cyrus the Great, Genghis Khan, Babur, Chandragupta Maurya, Darius I and countless others

3. The snow-line in Himalayas lies between

(a) 4300 to 6000 meters in East

(b) 4000 to 5800 meters in West

(c) 4500 to 6000 meters in West

(d) None of the above

39th B.P.S.C. (Pre) 1994

Ans. (e)

The altitude in a particular place above which some snow remains on the ground throughout the year is called snow line. The snowline in the Himalayas has different heights in different parts. On an Average it has height of 5500 – 6000m in Northern Part and 4500- 6000m in Southern Part of the Himalaya . In this way snowline in Himalayas lies between 4300 m to 6000 m.

vi. Plateaus

1. Which one of the following is not a part of the Meghalaya Plateau?

(a) Bhuban Hills (b) Garo Hills

(c) Khasi Hills (d) Jaintia Hills

(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

Garo, Khasi, Jaintia are parts of Meghalaya Plateau whereas Bhuban Hills is located in Barak Valley in Assam, which is part of Himalaya.

2. Extra peninsular mountains of India were formed during

(a) Aozoic Era (b) Palaeozoic Era

(c) Mesozoic Era (d) Cenozoic Era

41st B.P.S.C. (Pre) 1996

Ans. (d)

Extra peninsular mountains of India were formed in Cenozoic Era. Himalayan Ranges are example of extra peninsular mountains of India.

3. Chota Nagpur Plateau –

(a) is a front sloping (b) is a pitfall

(c) is a foothill (d) is a plain subland

40th B.P.S.C. (Pre) 1995

Ans. (a)

Chota Nagpur plateau covers much of Jharkhand as well as adjacent parts of Odisha, West Bengal, Bihar and Chhattisgarh.

The Chotanagpur plateau extends over an area of approximately 87239 sq. km. Chotanagpur consists of a series of plateaus at different levels of elevation. On Central Western portion the 'Patland' is extended whose height is about 1100 mt. In this the sharp break in slope is marked by steep scarps. Hence it is a front sloping.

4. Which one of the following districts does not have Dharwar geological formations?

(a) Munger (b) Rohtas

(c) Jamui (d) Nawada

(e) None of the above/more than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Munger, Jamui and Nawada in south-eastern Bihar are part of Dharwar geological formations. Rohtas is a part of Vindhyan rocks geological formations.

5. **Coastal line of India is –**
(a) 6,200 km. long (b) 6,100 km. long
(c) 5,985 km. long (d) 6,175 km. long

39th B.P.S.C. (Pre) 1994

Ans. (*)

India is surrounded by sea on three sides. The length of it's total coastline is 7516.6 km. India's coastline consists of Bay of Bengal in the east, Indian Ocean in south and Arabian Sea in the west. India's coastline extends to nine States and four Union Territories. These are –

Gujarat	– 1214.70km
Maharashtra	– 652.60 km
Goa	– 101 km
Karnataka	– 280 km
Kerala	– 569.70 km
Tamil Nadu	– 906.90 km
Andhra Pradesh	– 973.70 km
Odisha	– 476.40 km
West Bengal	– 157.50 km
Dadra and Nagar Haveli and Daman and Diu	– 42.50 km
Lakshadweep	– 132 km
Puducherry	– 47.60 km
Andman Nicobar Island	– 1962 km

The length of mainland India's coastline is 5422.6 km and offshore is 2094 km.

6. **Compare the Himalayan river with the peninsular river based of the following comparison -**
   1. **Most of the Himalayan rivers are perennial, whereas most of the Peninsular rivers are rain-fed.**
   2. **The gradient of the Himalayan river is steeper than the Peninsular river.**
   3. **The peninsular river causes more erosion on its way in comparison to the Himalayan river.**

Choose the correct answer from the options given below.
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
(c) none of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

Most of the Himalayan rivers are perennial and receive water from Himalayan glaciers, while most of the peninsular rivers are dependent on rainfall. The slope of the Himalayan river is more steep than that of the peninsular river. The Himalayan river is an example of the Yuva river and causes more erosion than the peninsular river.

7. **Which district of Uttarakhand is not situated along the Tibbet boundary?**
(a) Uttarkashi (b) Chamoli
(c) Almora (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (c)

Almora district of Uttarakhand is not situated along the Tibbet boundary. Three border with Tibbet are Chamoli, Pithoragarh and Uttarkashi.

8. **Which of the following comprises of Purvanchal?**
(a) The Jaintia Hills
(b) The Khasi Hills
(c) The Garo Hills
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (e)

The Purvanchal Range, or Eastern Mountains, is a sub-mountain range of the Himalayas in North-East India. The Purvanchal comprises :
- The Patkai Hills
- The Naga Hills
- Manipur Hills
- The Mizo Hills

The Patkai Hills are the hills on India's north-eastern border with Burma or Myanmar. The Naga Hills lies on the border of India and Burma.Mizo Hills is a mountain range in southeastern Mizoram state.Mizo Hills form part of the north ArakanYoma system.

9. **Arabica variety of coffee was initially introduced for cultivation in which hills?**
(a) Anamalai Hills
(b) Baba Budan Hills
(c) Nilgiri Hills
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (b)

The two varieties of coffee grown in India are Arabica and Robusta. It was initially introduced in the 17th century in the Baba BudanGiri hill ranges of Karnataka.The most commonly used coffee beans are Arabica and Robusta grown in the hills of Karnataka (Kodagu, Chikmagalur and Hassan), Kerala (Malabar region) and Tamil Nadu (Nilgiris District, Yercaud and Kodaikanal).

vii. Island groups of Arabian Sea

1. **The second largest island of the World is–**
   (a) Borneo (b) Madagascar
   (c) New Guinea (d) None of the above

BPSC Agriculture Department-2024

Ans. (c)

New Guinea is the second-largest island in the world, after Greenland.

1. **Lakshadweep island is situated –**
   (a) In South West India
   (b) In South India
   (c) In South East India
   (d) In East India near West Bengal

38th B.P.S.C. (Pre) 1992

Ans. (a)

Lakshadweep is a coral island located in the Arabian sea and its capital is Kavaratti. It is located in South West of India.

2. **Islands group Lakshadweep is -**
   (a) Accumulation of coral reef
   (b) Accumulation of volcano substances
   (c) Soil sedimentation
   (d) None of the above-mentioned is true

39th B.P.S.C. (Pre) 1994

Ans. (a)

See the explanation of the above question.

3. **Lakshadweep consists of how many Islands?**
   (a) 17 (b) 27
   (c) 36 (d) 47

45th B.P.S.C. (Pre) 2001

Ans. (c)

The Lakshadweep group of islands comprises 36 islands group.

3. STATES UNION TERRITORIES OF INDIA

i. States

1. **Vidarbha is a regional name in India and it is a part of –**
   (a) Gujarat (b) Maharashtra
   (c) Madhya Pradesh (d) Orissa (Odisha)

41st B.P.S.C. (Pre) 1996

Ans. (b)

Vidarbha is the north-eastern region of the State of Maharashtra. Presently, there are two divisions under this region – Nagpur and Amravati.

2. **The Pat region is located in –**
   (a) Bihar (b) Jharkhand
   (c) Madhya Pradesh (d) Meghalaya

44th B.P.S.C. (Pre) 2000

Ans. (b)

'Pat Region is located in the Chota Nagpur region. This region has generally flat topped Plateau.. The Maximum part of Chota Nagpur plateau lies in the State of Jharkhand. So, the correct answer is (b).

3. **Jhumri Telaiya (famous for request of songs on radio) is located in which state ?**
   (a) Bihar (b) Jharkhand
   (c) Odisha (d) West Bengal

47th B.P.S.C. (Pre) 2005

Ans. (b)

Jhumri Telaiya is a city in the Koderma district of Jharkhand State of India.

4. **As per the area, what is the correct descending order of the following States?**
   (1) Andhra Pradesh (2) Bihar
   (3) Madhya Pradesh (4) Uttar Pradesh
   Code :
   (a) 3, 2, 4, 1 (b) 1, 2, 3, 4
   (c) 4, 3, 2, 1 (d) 3, 4, 1, 2

40th B.P.S.C. (Pre) 1995

Ans. (d)

According to Census 2011, descending order of the given States in terms of area is as follows:

Madhya Pradesh	–	3,08,252 sq km
Uttar Pradesh	–	2,40,928 sq km
Andhra Pradesh (after bifurcation)	–	1,60,205 sq km
Bihar	–	94,163 sq km

5. **The three largest States of India in order of area are**
   (a) Rajasthan, Madhya Pradesh, Maharashtra
   (b) Madhya Pradesh, Rajasthan, Maharashtra
   (c) Maharashtra, Rajasthan, Madhya Pradesh
   (d) Madhya Pradesh, Maharashtra, Rajasthan

52nd B.P.S.C. (Pre) 2008
53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

The total area of India is 3,287, 263 sq. km. which is 2.4% of the total area of the world.
India ranks 7th in terms of area in the world and its population is 17.5 % of the total population of the world.
The largest States of India in order of area are:
1. Rajasthan (3,42, 239 sq. km)
2. Madhya Pradesh (3,08,252 sq. km)
3. Maharashtra (3,07,713 sq. km)
Which account for 30% of the total area of India.

6. In India the third largest state according to population and twelfth largest state in area is –

(a) Maharashtra (b) Madhya Pradesh
(c) Karnataka (d) Bihar

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

Bihar is the 12th largest State of India by area and 3rd largest State of India by population. Its population is 104,099,452 (2011) and area is 94,163 km^2.

7. How many States and Union Territories surround the state of Assam?

(a) 6 (b) 7
(c) 8 (d) 9

45th B.P.S.C. (Pre) 2001

Ans. (b)

Assam is surrounded by 7 States namely West Bengal, Nagaland, Mizoram, Meghalaya, Arunachal Pradesh, Manipur and Tripura. It does not share boundary with any Union Territories.

8. Which one of the following districts of India is the largest in terms of geographical area?

(a) Leh (b) Kutch (Kachchh)
(c) Jaisalmer (d) Barmer
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

According to State Website Kutch district in Gujarat is the largest areawise district (45,674 km^2) in India. Kutch (45,674 km^2) > Leh (45,100 km^2) > Jaisalmer (38,401 km^2) > Barmer (28,387 km^2).

9. Capital of Gujarat is –

(a) Godhra (b) Baroda
(c) Gandhinagar (d) Ahmedabad

45th B.P.S.C. (Pre) 2001

Ans. (c)

Gandhinagar is the capital of the state of Gujarat.

10. Capital of Rajasthan is-

(a) Jaipur (b) Udaipur
(c) Jodhpur (d) Ajmer

45th B.P.S.C. (Pre) 2001

Ans. (a)

The capital city of Rajasthan is Jaipur. Jaipur is also known as Pink City. Hawa Mahal is situated in Jaipur.

11. Which States of India have common border with Myanmar?

(a) Manipur, Mizoram, Nagaland, Tripura
(b) Arunachal Pradesh, Nagaland, Manipur, Mizoram
(c) Arunachal Pradesh, Assam, Manipur, Mizoram
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

The states that share their border with Myanmar are, Arunachal Pradesh, Nagaland, Manipur and Mizoram. India–Myanmar border has the Free Movement Regine which allows the tribes living along the border to travel across the boundary without visa restrictions.

12. How many States of India have common border with Pakistan and Bangladesh?

(a) 6 (b) 5
(c) 7 (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (e)

The India and Pakistan border runs along Jammu and Kashmir, Ladakh, Punjab, Rajasthan and Gujarat. India & Bangladesh border passes through riverine and low-lying areas in West Bengal, Assam and hilly terrains in Tripura, Mizoram and Meghalaya.

13. Which one of the following statements is not correct?

(a) Gujarat has the longest coastline in India.
(b) Jog Waterfall is the highest waterfall in India.
(c) Kovalam is the longest sea beach in India.
(d) The Great Boundary Fault lies in the State of Gujarat and Rajasthan.
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (e)

The given correct statement–

- Gujarat has the longest coastline in India.
- Jog waterfall is the second highest plunge waterfall in India which is situated on Sharavathi River in Karnataka.
- Marina Beach in Chennai along the Bay of Bengal is India's longest and world's second longest beach. Hence option (c) is not correct.
- The Great Boundary Fault lies in the State of Gujarat and Rajasthan. Hence option (e) is current but PBSC Official Answer Key if (c).

14. The Pitti an uninhabited island is known for–

(a) Bird sanctuary (b) Asiatic lion
(c) Mangrove Forest (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (a)

The Pitti Island is located in the Union Territory of Lakshadweep in India. It is also named as Pakshipitti which when separated makes it 'pakshi' and 'pitti' and pakshi means 'bird'. It is an uninhabited coral islet which has been declared a Bird Sanctuary under the Wildlife Protection Act, 1972. It is surrounded by the Kavaratti Island in the north, Agatti Island in the east and Amini Island in the south west.

15. When the Laccadive, Minicoy and Amindive were named as Lakshadweep?

(a) 1975 (b) 1973
(c) 1990 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (b)

Formerly the Union Territory of Lakshadweep was known as Laccadive, Minicoy, and Amindivi Islands, a name that was changed to Lakshadweep by an act of Parliament in 1973. Lakshadweep, a gathering of coral islands comprises 12 atolls, three reefs, and lowered shoals. Of the twenty seven islands, just eleven are occupied. They lie distributed in the Arabian Sea (off Kerala coast).

4. SPECIES/TRIBES

1. Which one of the north-eastern states of India has the highest concentration of tribal population?

(a) Nagaland (b) Mizoram
(c) Meghalaya (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (b)

The northeastern state of India with the highest proportion of tribal people as a percentage of its total population, according to the 2011 Census, is Mizoram (94.4%). Tribal people make up about 8.9% of India's overall population. The Bhil tribe is the largest in India, and Madhya Pradesh has the highest number of tribes, followed by Odisha.

2. Which of the following trees is worshipped by Mundas and Santhals of Chota Nagpur region?

(a) Tamarind (b) Mango
(c) Kadamba (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Many tribal communities believe that all creations of nature are scared and must be reversed and preserved. The Mundas and the Santhals of Chota Nagpur region worship Mahua and Kadamba tree.

3. The Dravidian races are mainly confirmed at which of the following parts of India?

(a) South India (b) North-Western India
(c) North-Eastern India (d) North India
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

The Dravidian races are mainly confined to South India. This race is related to Dravidian language family that includes Tamil, Kannada, Malyalam, Telugu etc. They are also found in central and southeast India.

4. Which amongst the following states has not identified tribal community?

(a) Maharashtra (b) Chhattisgarh
(c) Haryana (d) Karnataka

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

According to the 2011 census Punjab, Haryana, Chandigarh, Delhi and Puducherry have not identified tribal communities.

5. Gaddis are inhabitants of –

(a) Madhya Pradesh
(b) Himachal Pradesh
(c) Arunachal Pradesh
(d) Meghalaya

44th B.P.S.C. (Pre) 2000

Ans. (b)

Gaddi tribe lives in the Dhauladhar Range of Western Himalaya which extends into Kangra and Chamba districts of Himachal Pradesh. Gaddis Tribes also live in UT Jammu and Kashmir and Ladakh. Gaddis Tribe relate themselves to the dynasty of Garhwal Rulers of Uttarakhand. This is one of the main tribes of Dhauladhar Range which constitute a population of around 1.5 lakh. The lifestyle of Gaddis tribe is different from other tribes. The Main tribes of Dhauldhaar Range are Gaddis, Laddakhi Gujjar, Bakarwal, Lahauli, Bari etc.

6. Bodos are inhabitants of –

(a) Garo hills (b) Santhal Pargana

(c) Amazon Basin (d) Madhya Pradesh

43rd B.P.S.C. (Pre) 1994

Ans. (a)

Bodo tribe is an ethnic and linguistic group that mainly resides in the Garo hill of Meghalaya. Santhal Pargana is a Santhal majority division of Jharkhand.

7. Garo Tribes are of-

(a) Assam (b) Manipur

(c) Mizoram (d) Meghalaya

42nd B.P.S.C. (Pre) 1997

Ans. (*)

According to the 2011 Census, the Garo tribe is found in Meghalaya, Assam and Mizoram. Thus it is not possible to select any one option as correct answer.

8. Where is Bhil Tribe founded?

(a) Assam (b) Jharkhand

(c) West Bengal (d) Maharashtra

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

The name 'Bhil' is derived from the word villu or billu, which according to the Dravidian language is known as 'Bow and Arrow'. They belong to Proto Austroloid group of tribes. They are found in States like Madhya Pradesh, Maharashtra, Gujarat, Chhattisgarh and Rajasthan etc. Bhils are also found in the northeastern part of Tripura. The famous dance among the Bhils is Ghoomar.

9. Which one of the above following pairs of Tribe and State is not matched ?

(a) Bhils-Gujarat (b) Gaddis-Himachal Pradesh

(c) Kotas -Tamil Nadu (d) Todas-Kerala

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

Todas tribe lives in the Nilgiri mountains of Tamil Nadu and also live in Karnataka. While Bhils are found in Gujarat, Gaddis in Himachal Pradesh and Kotas in Tamil Nadu. So, (a), (b) and (c) are correctly matched. The correct answer would be option (d).

10. Bharmour tribal region is located in

(a) Sikkim

(b) Himachal Pradesh

(c) Uttarakhand

(d) Ladakh

(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (b)

Bharmour Tribal region is located in the state of Himachal Pradesh. Bharmour was the ancient capital of Chamba district.

11. Non-scheduled population concentrated in central valley of Manipur is called

(a) Meitei (b) Mishmi

(c) Kuki (d) Apatanis

(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (a)

Meitei people are ethnic group native to the state of Manipur. Meitei people settled mainly Imphal Valley region of Manipur. They settled in Nagaland, Meghalaya, Tripura and Assam etc. They are also present in the neighboring countries of Myanmar and Bangladesh.

i. Ganga Drainage System

1. The confluence of the rivers Son and Ganga is located in which district of Bihar?

(a) Buxar

(b) Patna

(c) Bhojpur

(d) Nalanda

(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Ganga is a Snow-fed and major river of the Indian subcontinent. It originates in Uttarakhand and crosses the Uttar Pradesh and enters the boundary of Bihar at Chausa a village near Buxar river Son meets the Ganga river about 16 km. upstream of Dinapur (Danapur) in the Patna district of Bihar.

2. **The depth of Gangetic alluvial soil below the land surface is about –**

(a) 6000 meters (b) 600 meters
(c) 800 meters (d) 100 meters

39th B.P.S.C. (Pre) 1994

Ans. (a)

According to Oldham the depth of Gangetic alluvial soil below the land surface is about 4000-6000 meters and according to Glany, it is about 2000 meters.

3. **At which of the following towns the Alaknanda and the Bhagirathi combine to form River Ganga?**

(a) Haridwar
(b) Rishikesh
(c) Rudraprayag
(d) Devprayag
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

The Ganga is the most important river of India. It rises in Gangotri glacier near Gaumukh in Uttarkashi district of Uttarakhand. Here it is known as Bhagirathi. Bhagirathi meet Alaknanda at Devprayag, hereafter it is known as Ganga.

4. **The nomenclature of 'Diara' lands of Ganga River is related to**

(a) Diya (b) Dahar
(c) Devara (d) Daira
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (a)

Diara has been derived from the word Diya meaning earthen lamp. Keeping in conformity with the shape of the Diya, the bowl like system on the surface (depression) situated between the natural levees on either side of the river appear like small Diyas when rain water gets accumulated in term during the rainy session. These pieces of lands also contribute partially or fully towards the diara land. Although the diara lands of eastern India, namely Brahmaputra Diaras, Ganga Diaras, Saryu Diaras and Mahanadi Diaras.

5. **The longest flowing river in India is-**

(a) Mahanadi (b) Godavari
(c) Ganga (d) Narmada

40th B.P.S.C. (Pre) 1995

Ans. (c)

The length of flowing rivers of India (mentioned in the question) is given below:

Ganga river	–	2525 km
Godavari river	–	1465 km
Narmada river	–	1312 km
Mahanadi river	–	851 km

6. **Ganga Plain has been described as a**

(a) Pediplane (b) peniplane
(c) Geosyncline (d) Karst plane
(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

The Gangetic plain is described as a geosyncline or geomorphic trough. It was mainly formed in the third phase of the Himalayan range formation process.

7. **Among the following tributaries, which one is the part of the Ganga river basin?**

(a) Sankh
(b) North Koel
(c) South Koel
(d) Barakar
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (e)

North Koel is a river flowing from Jharkhand, a tributary of the Son River, it joins the right bank of the Son River. The Son River is the second largest (first Yamuna) tributary of the Ganges River from the southern part of the Ganges near Patna.
The Southern Koel River flows into Jharkhand and Odisha. The Southern Koel River flows in association with the Northern Karo River as the Koel River, which joins the Sankh River near Rourkela in Odisha as the Brahmani River. The Barakar River is the main tributary of the Damodar River, which originates from the Koderma Plateau in Jharkhand and Joins Damodar via Hazaribagh. Damodar joins the Hooghly River in the lower Ganges basin. The Damodar River Basin is a sub-basin of the Ganges River Basin. Thus, Barakar and North Koel are sub-rivers of the Ganges river valley. Hence the correct answer to this question is option (e).

8. **Which one of the following rivers joins the Ganga at Fatuha?**

(a) Son (b) Punpun

(c) Sakri (d) Balan
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

The Punpun river originates from the plateau state of Jharkhand and enters the Aurangabad district of Bihar and Joins the Ganges near Fatuha in Patna district. Morhar, Butani and Madar are tributaries of Punpun.

9. Which of the following mountain ranges divides Ganga Plain and Deccan Plateau?

(a) Vindhyan Range
(b) Himalayan Range
(c) Aravalli Range
(d) Eastern Ghats
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

The Vindhya Range vary may be a sophisticated and discontinuous chain of ridges, ridges, highlands, and upland escarpments in western India. The term principally refers to the escarpment and its hilly extensions that runs north of and roughly parallel to the Narmada River in Madhya Pradesh. The term "Vindhyas" was utilized in an exceedingly broader sense and embedded a series of mountain ranges between the Indo-Gangetic Plain and thus the Deccan plateau. The western end of the Vindhya range is located in the state of Gujarat, near the state's border with Rajasthan and Madhya Pradesh, at the eastern side of the Kathiawar peninsula. A series of hills connects the Vindhya extension to the Aravalli Range near Champaner.

ii. Brahmaputra Drainage System

1. In India, 'Yarlung Zangbo River' is known as:

(a) Ganga (b) Indus
(c) Brahmaputra (d) Mahanadi

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Yarlung Zangbo is known as Brahmaputra river in India. It originates at Angsi glacier in western Tibet.

2. Near Mansarowar lake in Tibet, there is the source of the river –

(a) Brahmaputra (b) Satluj
(c) Indus (d) All the above

39th B.P.S.C. (Pre) 1994

Ans. (d)

The source of river Brahmaputra, Indus, and Sutlej, is near Mansarovar Lake in Tibet.

3. Manas is the tributary of river :

(a) Godavari (b) Mahanadi
(c) Krishna (d) Brahmaputra

44th B.P.S.C. (Pre) 2000

Ans. (d)

Manas is the tributary of Brahmaputra river. Other major tributary of Brahmaputra are – Tista, Kameng, Subansiri, Dhansari, Burhi Dihing, Dibang, and Kopili. Brahmaputra is known as Tsangpo or Sanpu in Tibet, Dihang in Arunachal Pradesh, Brahmaputra in Assam and Jamuna in Bangladesh. Brahmaputra river flows through Tibet region of China, India, and Bangladesh.

iii. The South Indian Rivers

1. Which of the following rivers has largest catchment area?

(a) Kalinadi (b) Ponnani
(c) Periyar (d) More than one of the above
(e) None of the above

B.P.S.C. Headmaster 07-12-2023

Ans. (b)

Bharathapuzha is known as Ponnani River in Kerala. Total drainage area of the basin is 6186 sq.km. out of which nearly 71% lies in Kerala State. The state-wise distribution of the drainage are Tamil Nadu 1786 sq. km. and Kerala 4400 sq. km.

2. The river which flows through a fault trough, is?

(a) Narmada (b) Brahmaputra
(c) Ganga (d) Krishna

44th B.P.S.C. (Pre) 2000

Ans. (a)

Originating at the Amarkantak Hill, the Narmada river flows westwards over a length of 1,312 km before falling through the Gulf of Cambay into the Arabian Sea. It passes through rift valley between Vindhyan and Satpura ranges.

Damodar River: Damodar River begins near Khamar Pat hills on the Chota Nagpur Plateau in Jharkhand.

Mahanadi River: This river originates near Sihawa in Dhamtari district, Chhattisgarh. Hirakud XDam, one of the longest dams of the world, is located on this river near Sambalpur, Odisha.

Tapti River - It originates in Betul district of Madhya Pradesh. Surat is situated on the bank of this river.

3. **Westward flowing rivers are –**
(i) Narmada (ii) Tapti (iii) Rapti
Code :
(a) (i) and (ii) (b) (ii) and (iii)
(c) (i) and (iii) (d) (i), (ii) and (iii)

43rd B.P.S.C. (Pre) 1999

Ans. (a)

Narmada, Tapti, and Mahi are three major westward-flowing rivers in India. The Rapti River is the tributary of the Ghaghara River which flows North- West to South -East direction. So, option (a) is the correct answer.

4. **Which one of the following mountain peaks is not correctly matched?**
(a) Abor-Jharkhand (b) Dunagiri-Uttarakhand
(c) Mikir-Assam (d) Nimgiri-Odisha
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (a)

Correctly matched mountain peaks and its respective state–

(a)	Abor	—	Arunachal Pradesh
(b)	Dunagiri	—	Uttarakhand
(c)	Mikir	—	Assam
(d)	Nimgiri	—	Odisha

5. **River Cauvery flows through –**
(a) Gujarat, Madhya Pradesh, Tamil Nadu
(b) Karnataka, Kerala, Tamil Nadu
(c) Karnataka, Kerala, Andhra Pradesh
(d) Madhya Pradesh, Maharashtra, Tamil Nadu

42nd B.P.S.C. (Pre) 1997

Ans. (b)

Cauvery River rises in Brahmagiri hills of Kodagu district in Karnataka and flows about 800 km. in the drainage area of 81,155 sq. km. It empties into the Bay of Bengal. Cauvery basin extends over Karnataka, Kerala & Tamil Nadu and U.T. of Puducherry. This river forms islands of Srirangpatnam, Shivasamudram, and Srirangam in the Deccan Plateau.

6. **Consider the following river and identify the tributaries of Kaveri river :**
1. Amaravati 2. Bhavani
3. Dudhganga 4. Hemavati
Select the correct answer the codes given below
(a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

The Kaveri River (Dakshin Bharat Ki Ganga) rises from Brahmagiri range in Karnataka. Tributaries of the Cauvery River–
Left Bank–The Harangi, The Hemavati, The Shimsha and the Arkavati.
Right Bank–Lakshmantirtha, The Kabbani, The Suvarnavati, The Bhavani, The Noyyal and The Amaravati.
Dudhganga is the tributary of Jhelum.

7. **Son, Narmada, and Mahanadi originate from –**
(a) Pulamu hills (b) Amarkantak
(c) Eastern Ghats (d) Aravali

44th B.P.S.C. (Pre) 2000

Ans. (b)

Son, Narmada and Mahanadi originate from Amarkantak Plateau. The source of the Mahanadi river is in Dhamtari district of Chhattisgarh.

8. **Chandra and Bhaga river in flow through the region:**
(a) Spiti (b) Ladakh
(c) Lahaul (d) Kargil
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (e)

The Chenab River is formed by the confluence of two rivers, Chandra and Bhaga, at Tandi. The Chandra and Bhaga originate from the South-West and North-West faces of Barelacha pass respectively in Himalayan canton of Lahul and spiti valley in Himachal Pradesh.

iv. Other Rivers

1. **Which of the following rivers is not flowing eastward?**
(a) Gangavalli (b) Narmada
(c) Godavari (d) Krishna
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (e)

The two major west flowing rivers are the Narmada and the Tapi. The Sabarmati, Mahi and Luni are other rivers of the Peninsular India which flow westwards. Gangavalli River (also called Bedthi River) originates from the Western Ghats the south of Dharwad (Near Someshwara temple) as Shalmala and flows in the west direction to meet the Arabian sea just after the Ganga temple. Here the River embraces the name Gangavalli from the Goddess Ganga; the village in this area carries the same name Gangavalli. The Narmada River is also known as the Rewa River. The River is originated from Maikala range near Amarkantak. It is a West flowing river. The River flows through Gujarat, Chhattisgarh Madhya Pradesh, and Maharashtra.

2. The river Subarnarekha originates near which of the following villages?

(a) Ormanjhi (b) Mandar

(c) Hehal (d) Nagri

69th B.P.S.C. (Pre) 2023

Ans. (d)

The Subarnarekha originates near Nagri village in Ranchi district of Jharkhand. The principal tributaries of the river are Kanchi, Kharkai and Karkari. It is bounded on the North-West by the Chhotanagpur Plateau, in the South-West by Brahmani basin, in the South by Burhabalang basin and in the South-East by the Bay of Bengal.

3. The smallest river/tributary in length (in km) is

(a) Burhi Gandak (b) Kosi

(c) Punpun (d) Kamala

(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (c)

The river, Punpun, a small tributary of Ganga, originate in hills of the Palamau district which falls in the state of Jharkhand known as Chhotanagpur region. It enters in Bihar at Aurangabad.

4. The Pattiseema Project is associated with the integration of which of the following rivers?

(a) Krishna and Kaveri

(b) Krishna and Godavari

(c) Godavari and Mahanadi

(d) Ganga and Brahmaputra

(e) None of the above | More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Pattiseema Lift Irrigation Project is related to the interlinking of Krishna and Godavari rivers. This project has been implemented by the Government of Andhra Pradesh.

5. Which one of the following river basins has the smallest basin area (in sq.km)?

(a) Brahmani (b) Cauvery

(c) Mahanadi (d) Tapi

(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (a)

Here, are the basin area of the following River -

Brahmani River	approx 39033 sq.km
Cauvery River	approx 81155 sq.km
Mahanadi River	approx 1,41589 sq.km
Tapi River	approx 65145 sq.km

Therefore, in the given option Brahmani River basin is smallest in area.

6. Which of the following drainage systems fall into the Bay of Bengal?

(a) Ganga, Brahmaputra, and Godavari

(b) Mahanadi, Krishna, and Cauvery

(c) Luni, Narmada, and Tapti

(d) Both (A) and (B)

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

Ganga, Brahmaputra, Godavari, Mahanadi, Krishna, Cauvery, Pennar, Subarnarekha and Brahmani are drainage systems (rivers) which flow into the Bay of Bengal while Narmada, Tapti, Sabarmati, Mahi, except Luni are important rivers that fall into the Arabian Sea.

7. Indravati-Sabari Rivers in Dandakaranya are the tributaries of

(a) Mahanadi River (b) Godavari River

(c) Krishna River (d) Damodar River

(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (b)

Indravati-Sabari Rivers in Dandakaranya are the tributanies of Godavari River. Godavari River is the largest river in peninsular India and known as the Dakshina Ganga. The Godavari Basin is the second largest basin after the Ganga basin.

8. River Tel is a tributary of which of the following rivers?

(a) Bagmati (b) Ghaghara

(c) Gandak (d) Kamla

(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

Tel is a tributary of Mahanadi river not Krishna river. Other tributaries of Mahanadi are Seonath, Hasdeo, Mand, Ib, Ong and Jonk.

9. **River Damodar emerges from –**
(a) Tibbet
(b) Chotanagpur
(c) Near Nainital
(d) Western slope of Somesar hills

39th B.P.S.C. (Pre) 1994

Ans. (b)

Damodar river also known as the Sorrow of Bengal, originates in the Kharmarpat Hill of Chota Nagpur Plateau.

10. **Which of the following is a land-bounded river ?**
(a) Tapti (b) Krishna
(c) Luni (d) Narmada

40th B.P.S.C. (Pre) 1995

Ans. (c)

Luni river originates in Aravali range, south-west to Ajmer. It disappears in the Rann of Kutch traversing a distance of 511 km. Thus, it does not come in contact with the sea. Rest other rivers given in the option flow into the sea.

11. **Which of the following rivers, the maximum shifting, of course, has taken place in –**
(a) Son (b) Gandak
(c) Kosi (d) Ganga

44th B.P.S.C. (Pre) 2000

Ans. (c)

The Kosi River has seven tributaries of which Arun is most important. It originates from the north of Himalayas. The Kosi River is known for its dangerous floods and especially for maximum shifting of its course. Thus, it can be said that the Kosi River changes its course for maximum times among all rivers of India.

12. **The Triveni Canal has been constructed on which of the following rivers?**
(a) Kosi (b) Sone
(c) Gandak (d) Mayurakshi

69th B.P.S.C. (Pre) 2023

Ans. (c)

Triveni Canal has been constructed on the river Gandak. Significantly, it is mainly used for irrigation in the North-Western part of Bihar. Gandak is the tributary of river Ganga and joins the Ganga at Sonpur near Patna.

13. **Triveni Canal receives water from river-**
(a) Son (b) Kosi
(c) Gandak (d) Mayurakshi

45th B.P.S.C. (Pre) 2001

Ans. (c)

The Triveni canal receives water from river Gandak in Bihar.

14. **The other name of river Gandak is**
(a) Burhi Gandak (b) Mahananda
(c) Narayani (d) Punpun

69th B.P.S.C. (Pre) 2023

Ans. (c)

In Nepal, the Gandak River is also referred to as the Gandaki and Narayani River. Significantly, the Gandak comprises two streams, namely Kaligandak and Trishulganga. It enters the Ganga plain in Champaran district of Bihar and joins the Ganga at Sonpur near Patna.

v. Cities Located on the Bank of Rivers

1. **The largest city located on the bank of river Ganga is –**
(a) Varanasi (b) Patna
(c) Kanpur (d) Allahabad

44th B.P.S.C. (Pre) 2000

Ans. (c)

According to the data of Census- 2011, the population of urban areas mentioned in the question is given below

Kanpur (U A)	–	2,920,067
Patna (U A)	–	2,046,652
Varanasi (U A)	–	1,435,113
Allahabad (Now Prayagraj) (U A)	–	1,216,719

2. **Which one of the following cities is not located on the bank of river Ganga?**
(a) Fatehpur (b) Bhagalpur
(c) Uttarkashi (d) Kanpur

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

The prime tributary of Ganga river is Bhagirathi which originates from Gangotri glacier at Gomukh, Uttarkashi district in Uttarakhand. The confluence of Alaknanda and Bhagirathi at Devprayag forms river Ganga. Thus, Uttarkashi is related to Bhagirathi not Ganga. Fatehpur, Bhagalpur, and Kanpur are located on the bank of river Ganga.

3. **Pahalgam in Kashmir is located on the bank of river:**
(a) Shyok (b) Jhelum

(c) Lidder (d) Chenab
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (c)

Pahalgam is located on the bank of river Lidder. It is situated at an altitude of 2,130 m. It is also the starting point of the annual Amarnath Yatra.

4. Which of the following statements about Lake Superior of North America continent is true?

(a) It is world's greatest saltwater lakes group.
(b) It forms a geographical boundary between Canada and USA.
(c) It makes approximately 1700 km boundary between the two countries.
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The Lake Superior of North American continent forms a geographical boundary between Canada and USA. It is a world's largest freshwater lake by area.

vi. Waterfalls and Lakes

1. In India, Vembanad Lake is located in–

(a) Kerala (b) Odisha
(c) Rajasthan (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (a)

Vembanad is the longest lake in India, as well as the largest and longest in the state of Kerala.The lake has its source in four rivers, Meenachil, Achankovil, Pampa and Manimala. It is separated from the Arabian Sea by a narrow barrier island and is a popular backwater stretch in Kerala.

2. Hundru falls is formed on –

(a) Indrawati (b) Kaveri
(c) Subarnarekha (d) None of the above

39th B.P.S.C. (Pre) 1994

Ans. (c)

Hundru fall is located on the course of the Subarnarekha river. It is 45km away from main city Ranchi. Its height is 98 mt.

3. The Sivasamudram falls is located on the river –

(a) Cauvery (b) Krishna
(c) Godavari (d) Mahanadi

44th B.P.S.C. (Pre) 2000

Ans. (a)

Sivasamudram falls is located on the River Cauvery.

4. Niagara Falls in North America is located between:

(a) Lake Superior and Lake Michigan
(b) Lake Michigan and Lake Huron
(c) Lake Huron and Lake Ontario
(d) Lake Ontario and Lake Erie
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (d)

Niagara falls is located between Lake Ontario and Lake Erie. It is comparised of three waterfalls - American Falls, Horeshoe Falls and Bridal Veil Falls.

5. Which one of the following is a freshwater lake?

(a) Chilka (b) Sambar
(c) Wular (d) Loktak
(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (e)

Wular (J&K) and Loktak (Manipur) are among the fresh water lake from the given alternatives. While Chilka (Odisha) and Sambhar (Rajasthan) are salt water lakes.

6. Lake Taupo is located in which country?

(a) Fiji (b) New Zealand
(c) Japan (d) None of the above

BPSC Agriculture Department-2024

Ans. (b)

Lake Taupo is located in New Zealand, in the center of the North Island. It is rich in volcanic and heritage, and is within a short drive of many sightseeing, adventure, and leisure activities.

CLIMATE

i. Monsoon

1. Which of the following atmospheric features is associated with the performance of Indian summer monsoon?

(a) Tropical Easterly Jet Stream (b) Shift of ITCZ
(c) Tibetan High (d) La Nina
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (e)

Factors influencing performance of the Indian Summer monsoon are,
- Tropical Easterly Jet Stream
- Tibetan High
- Formation of La Nina
- Shift of ITCZ

2. Which type of climate is in eastern Bihar according to Thornthwaite's scheme?

(a) C2 (b) C1
(c) A (d) More than one of the above
(e) None of the above

B.P.S.C. Headmaster 07-12-2023

Ans. (a)

The 'C2' Type (moist sub-humid) of climate is experienced in large parts of Odisha, Jharkhand, eastern Bihar and Chhattisgarh region. A small zone of moist sub humid climate is also experienced along Nilgiri hills, Western Ghat Mountains and along Coastal Maharastra.

3. During the monsoon season in India, most of the cyclones have their origin

(a) between 8° N and 13° N latitude
(b) between 10° N and 15° N latitude
(c) between 16° N and 21° N latitude
(d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 11-12) 15-12-2023

Ans. (b)

During the monsoon season in India, most of the cyclones have their origin between 10° N and 15° N latitude.

4. The Indian monsoon is indicated by seasonal displacement because of –

(a) Differential temperature of land and sea
(b) Cold wind of middle Asia
(c) Excess similarity of temperature
(d) None of the above

39th B.P.S.C. (Pre) 1994

Ans. (a)

The Indian monsoon is indicated by seasonal displacement because of the differential temperature of land and Sea. Some of the important concepts about the origin of Monsoon are:
(1) Thermal concept: It propounds that the primary arrival of the annual cycle of the Indian monsoon circulation is the differential heating effect of the land and the Sea.
(2) Dynamic Concept: Put forward by Flohn, according to this concept, monsoon is the result of seasonal migration of planetary winds and air pressure belts.

5. Amritsar and Shimla are almost on the same latitude, but their climate difference is due to-

(a) The difference in their altitudes
(b) Their distance from sea
(c) Snowfall in Shimla
(d) Pollution in Amritsar

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Amritsar ($31^0 29'$–$32^0 03'$ N) and Shimla ($30^0 45'$–$31^0 44'$ N) are almost on the same latitude but their climate difference is due to the difference in their altitudes. Amritsar is located at 756 feet and Shimla at 7866.10 feet above the mean sea level.

6. Declining of monsoon is indicated by:

(i) Clear Sky
(ii) Pressure condition in Bay of Bengal
(iii) Rising temperature on land

Select your answer using the following code –

(a) Only i (b) Both i and ii
(c) i, ii and iii (d) Both ii and iii

40th B.P.S.C. (Pre) 1995

Ans. (c)

When monsoon returns from the north, there is a slight increase in temperature on Land, after that there is a gradual decrease in the temperature on land. The low pressure covers the Bay of Bengal and the sky is clear. Thus all statements are correct.

7. According to Koppen's climatic classification, the climate of North Bihar may be explained as:

(a) Cwg (b) Aw
(c) CA'w (d) CB'w
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

According to Koppen's Climatic classification, the climate of North Bihar may be explained as Cwg.

8. Rajasthan receives very little rain because –

(a) The monsoon fails to reach this area
(b) It is too hot
(c) There is no water available and thus the winds remain dry
(d) The winds do not come across any barriers to cause the necessary uplift to cool the air
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

Rajasthan receives very little rainfall, as the winds do not cross any barriers, due to which they do not attain the required height to cool down. The Aravalli mountain ranges do not become an obstacle in the way because they are lined with the direction of the South-West monsoon winds. Hence, most of Rajasthan (Western part), becomes a rain shadow (dry) area.

9. According to G. T. Trewartha, the area of Am type climate in India is

(a) Meghalaya plateau
(b) Western Ghats
(c) Coromandel coastal region
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

As per G.T. Trewartha, the area of Am type Climate in India in Western Ghats. This region receives annual rainfall about 250 cm and its annual temperature is about 27°C.

ii. Rainfall

1. India gets maximum rainfall mainly from –

(a) North-East Monsoon (b) Retreating Monsoon
(c) South-West Monsoon (d) Convectional rainfall

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (c)

During April and May when the Sun Shines between Equator and the Tropic of Cancer, the large land mass in the north of India gets intensely heated. This causes the formation of intense low pressure in the northwestern part of the subcontinent followed by the northward shift in the position of Inter Tropical Convergence Zone (ITCZ). The south-west monsoon may thus, be seen as the continuation of the southeast trades deflected towards the Indian subcontinent after crossing the Equator. Southwest monsoon enters India in two rain-bearing systems. First from the Bay of Bengal causing rainfall over the plains of north India. Second from the Arabian Sea current of the south-west monsoon which brings rain to the west coast of India. According to data provided by Indian Meteorological Department for the year 2022. South west monsoon (June to September) actual rainfall of India is 106.5% of Long Period Average (L.P.A.) of Annual Rainfall.

2. Mango shower is –

(a) Shower of mangoes
(b) Dropping of mangoes
(c) Rainfall in March-April in Bihar and Bengal
(d) Crop of mango

43rd B.P.S.C. (Pre) 1994

Ans. (*)

Mango shower is the name of pre-monsoon showers in Karnataka and Kerala that help in the ripening of mangoes. It is also known as April rains or summer showers. The reason for the Mango showers is the thunderstorms over the Bay of Bengal. The shower prevents mangoes from dropping prematurely. So Mango shower is related to crop of mango. So answer will be considered as option (d).

3. Which of the following blows from Mediterranean Sea to the Northwest part of India?

(a) Western Disturbance (b) Norwester
(c) Mango Shower (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 09-12-2023

Ans. (a)

A western disturbance is an extra tropical storm originating in the mediterranean region that brings sudden winter rain to the north western parts of the Indian subcontinent.

4. Cherrapunji is located in –

(a) Assam (b) Manipur
(c) Meghalaya (d) Mizoram

41st B.P.S.C. (Pre) 1996

Ans. (c)

Cherrapunji (Now Sohra) is located in Meghalaya, one of the north-eastern States of India. It has held the record for highest rainfall after Mawsynram.

5. Which one of the following is the driest place?

(a) Mumbai (b) Delhi
(c) Leh (d) Bangalore

B.P.S.C. 56th to 59th (Pre) 2015

Ans. (c)

Leh is the region of lowest rainfall in India. It is the driest place in India.

6. The driest place in India is

(a) Leh
(b) Bikaner
(c) Jaisalmer
(d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (c)

The driest place in India is Jaisalmer. It is located in Rajasthan. The state of Rajasthan is also the home to the hottest and driest desert in India, the Thar Desert.

7. Which of the following receives heavy rainfall in the month of October and November?

(a) Hills of Garo, Khasi and Jaintia
(b) Coromandel Coasts
(c) Plateau of Chota Nagpur
(d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 11-12) 15-12-2023

Ans. (b)

Most parts of the country receive rainfall from June to September. But some parts like the Coromandel coasts coast gets a large portion of its rain during October and November. Mawsynram in the southern ranges of the Khasi Hills receives the highest average rainfall in the world.

iii. Winter Rainfall

1. Which of the following, the state receives rainfall in winter season–

(a) Kerala (b) Tamil Nadu
(c) West Bengal (d) Odisha

44th B.P.S.C. (Pre) 2000

Ans. (b)

Coastal areas of Tamil Nadu receive winter rainfall due to retreating North-East Monsoon. Additionally, Punjab also receives winter rainfall due to Western Disturbance.

2. The place of origin of Western Disturbances active in North- Western India in winter season is

(a) Asia Minor
(b) Western Asia
(c) Mediterranean coastal area of Western Asia
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

According to the India Meteorological Department (IMD), Western disturbances are storms that originate in the Caspian or Mediterranean Sea, and bring non-monsoonal rainfall to the North Western India. They are labelled as an extra-tropical storm originating in the Mediterranean, is an area of low pressure that brings sudden showers, snow and fog in North Western India.

Natural Disasters

1. Which of the following coasts of India is most affected by violent tropical cyclones?

(a) Malabar (b) Coromandel
(c) Konkan (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 11-12) 15-12-2023

Ans. (b)

Tropical cyclones are violent storms that originate over oceans in tropical areas and move over to the coastal areas bringing about large scale destruction caused by violent winds, very heavy rainfall and storm surges. This is one of the most devastating natural calamities. They are known as Cyclones in the Indian Ocean, Coromandel coasts of India are most affected by violent tropical cyclones.

2. Which of the following statements is true regarding earthquakes?

(a) P-waves are transverse waves.
(b) The place on the surface above focus is called epicentre.
(c) Studying animal behaviour is one of the methods of prediction of earthquakes adopted by local people.
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

The Focus is the place inside Earth's crust where an earthquake originates. The point on the Earth's surface directly above the focus is the epicenter. When energy is released at the focus, seismic waves travel outward from that point in all direction. An earthquake generates both transverse wave(s) and longitudinal (e) sound waves in the earth. There are many methods of prediction of earthquakes but studying animal behaviour is one of the methods of prediction of earthquakes adopted by local people.

3. Match List-I (Natural Hazards) with List-II (Regions) and select the correct answer using the codes given below the lists:

List-I (Natural Hazards)	List-II (Regions)
A. Floods	**1. Himalayan foothill Zone**
B. Earthquakes	**2. Jharkhand and Northern Odisha**

C. Droughts **3. Plains of Uttar Pradesh and Bihar**

D. Cyclones **4. Mid-Eastern India.**

Code :

	A	B	C	D
(a)	3	1	2	4
(b)	3	1	4	2
(c)	2	3	1	4
(d)	4	2	3	1

47th B.P.S.C. (Pre) 2005

Ans. (b)

"Flood Forecast & Warning Organisation" was set up by Central Water Commission in 1969 to establish forecasting sites on Inter-State rivers at various flood-prone places in the country. The "National Flood Forecasting and Warning Network" of Central Water Commission, which comprised of 338 flood forecasting sites. The most drought-affected area is middle east India. Generally, the Cyclones affect-eastern coast of India. Central eastern India is most affected by cyclones originating in the Bay of Bengal. Cyclones are generated more frequently in Odisha and West Bengal.

4. Richter scale is used to measure

(a) volcano eruption (b) flood intensity

(c) earthquake (d) More than one of the above

(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (c)

An earthquake is measured with a machine called seismograph. The magnitude of the earth is measured on Richter scale. An earthquake of 2.0 or less can be felt only a little. An earthquake over 5.0 can cause damage from things falling. A 6.0 or higher magnitude is considered very strong and 7.0 is classified as a major earthquake.

SOILS

i. Black Soil

1. Which of the following terms is also used for black soil?

(a) Regur

(b) Bangar

(c) Black cotton soil

(d) Khadar

(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (e)

Black soil is also called as Regur soil or Black cotton soil, since cotton has been the most common traditional crop in areas where they are found. It is the third major group of soil in India and forms the top part of the earth's surface, that includes disintegrated rock, humus, inorganic, and organic materials. Black soils are derivatives of trap lava and are spread mostly across interior Gujarat, Maharashtra, Karnataka, and Madhya Pradesh on the Deccan lava plateau and the Malwa Plateau, where there is both moderate rainfall and underlying basaltic rock.

2. Regur is the name of –

(a) Red soil (b) Alluvial soil

(c) Black soil (d) Lateritic soil

42nd B.P.S.C. (Pre) 1998

44th B.P.S.C. (Pre) 2000

Ans. (c)

The black soil is also called 'Regur', cotton soils and tropical Chernozem, etc. This soil is mainly found in part of Maharashtra, Madhya Pradesh part of Karnataka, Andhra Pradesh, Gujarat, and Tamil Nadu. Geographically, Black soil is spread over 5.46 lakh sq km.

3. Regur soil is most widespread in –

(a) Maharashtra (b) Tamil Nadu

(c) Andhra Pradesh (d) Jharkhand

44th B.P.S.C. (Pre) 2000

Ans. (a)

See the explanation of the above question.

ii. Alluvial Soil

1. Which one of the following soils is deposited by rivers?

(a) Red soil

(b) Black soil

(c) Alluvial soil

(d) Lateritic soil

(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

Alluvial soil is formed by the sediments released from the rivers. Alluvial soil is mixture of sand, silt and clay. Alluvial soils are divided into two subclasses -

1. New Alluvium or Khadar - It is deposited by floods annually, which enriches the soil by depositing fine silts.

2. Older Alluvium or Bhangar - It represents a system of older alluvium, deposited away from the flood plains.

2. **Which soil is predominantly found in the districts of Muzaffarpur, Darbhanga and Champaran?**
(a) Black soil (b) Newer alluvium
(c) Older alluvium (d) Red soil
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

Newer alluvium soil is found in the districts of Muzaffarpur, Darbhanga and Champaran. Newer alluvium soil (khadar) is created by the decomposition of sediments from the flood waters brought by the rivers Gandak, Burhi Gandak, Bagmati etc., every year. This soil helps in the production of crops like paddy, wheat, Jute and Sugarcane.

3. **Old Kachhari clay of Gangetic plain is called**
(a) Bhabar (b) Bhangar
(c) Khadar (d) Khondolyte

41st B.P.S.C. (Pre) 1996

Ans. (b)

Geologically, the alluvium of the Great Plain of India is divided into newer or younger Khadar and older Bhangar soils. Bhangar is found on the higher level beyond the reach of floods. Bhangar soil contains calcareous concretions and is generally pale reddish brown in colour.

4. **Which soil particles are present in loamy soils?**
(a) Sand particles (b) Clay particles
(c) Silt particles (d) All types of particles

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

Generally, Loam soil has 40% sand particles, 40% clay particles and 20% silt particles. Thus, option (d) is correct.

Soils: Miscellaneous

1. **Karewas soils, which are useful for the cultivation of Zafran (a local variety of saffron), are found**
(a) Kashmir Himalaya
(b) Garhwal Himalaya
(c) Nepal Himalaya
(d) Eastern Himalaya
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

Karewa soil is found in the Kashmir Valley and is used for growing local saffron called Zafran.

2. **For which cultivation Karewas are famous?**
(a) Banana (b) Saffron
(c) Mango (d) Grapes
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre)-2021

Ans. (b)

Karewa glacier, clay and other substances is deposited as a thick layer on the glacier Karewa, is famous for the cultivation of Zayran (native saffron)

3. **Soil water available to plants is maximum in :**
(a) clay soil (b) silty soil
(c) sandy soil (d) loamy soil

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Clay soil has the highest water retention capacity. Therefore soil water available to plants is maximum in it. Clay soil has less than 50% silt, 50% clay and some amount of sand. This soil slows air ventilation, due to which it sustains water.

Soil Erosion and Conservation

1. **When running water cuts through clayey soils and makes deep channels, they lead to**
(a) gully erosion (b) sheet erosion
(c) deforestation (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 11-12) 15-12-2023

Ans. (a)

The denudation of the soil cover and subsequent washing down is described as soil erosion. The running water cuts through the clayey soils and makes deep channels as gullies. This leads to gully erosion. Sometimes water flows as a sheet over large areas down a slope. In such cases, the topsoil is washed away. This is known as sheet erosion.

2. **Soil erosion can be checked by**
(a) Excess grazing
(b) Removal of plants
(c) Afforestation
(d) Increasing number of birds

40th B.P.S.C. (Pre) 1995

Ans. (c)

Soil conservation includes all the methods to check soil erosion and maintain soil fertility. Some of the effective ways to prevent soil erosion are – afforestation, contour tillage, banning shifting cultivation, etc.

Natural Vegetation

1. **In India, the State with the largest area under dense deciduous forest cover is :**
   (a) Odisha (b) Maharashtra
   (c) Madhya Pradesh (d) Chhattisgarh
   65th B.P.S.C. (Pre) 2019

Ans. (c)

Madhya Pradesh has the largest area under dense deciduous forest cover.

2. **Which one of the following States has the highest percentage of area under forests?**
   (a) Himachal Pradesh (b) Assam
   (c) Andhra Pradesh (d) Arunachal Pradesh
   (e) None of the above / More than one of the above
   63rd B.P.S.C. (Pre) 2017

Ans. (e)

According to the India State of Forest Report, 2021, in the state Mizoram (84.53%) has the highest percentage of area under forest.

3. **The maximum irrigation potential of India is created through:**
   (a) Major Projects
   (b) Minor projects & Major Projects
   (c) Minor Projects
   (d) Medium Projects
   52nd B.P.S.C. (Pre) 2008

Ans. (c)

Irrigation Projects in India are classified into three categories:

1. **Minor Irrigation Projects:** All groundwater and surface water schemes that have a Culturable Command Area (CCA) up to 2,000 hectares individually are classified as Minor Irrigation Schemes. The minor irrigation projects comprise all groundwater development schemes such as dug wells, private shallow tubewells, deep public tubewells, and boring and deepening of dugwells, and small surface water development works such as storage tanks, lift irrigation projects, etc. more than 62 percent irrigation potential of India is created through minor irrigation projects.
2. **Medium Irrigation Projects:** Those having a CCA between 2,000 hectares and 10,000 hectares fall under the category of medium irrigation projects.
3. **Major Irrigation Projects:** Irrigation projects having Culturable Command Area (CCA) of more than 10,000 hectares each are classified as major projects. 38 percent of the irrigation potential of India is created through major and medium irrigation projects.

4. **The Saran irrigation canal is drawn from the river**
   (a) Son (b) Ganga
   (c) Kosi (d) Gandak
   44th B.P.S.C. (Pre) 2000

Ans. (d)

The Saran irrigation canal has been built across the Gandak River in Balmiki Nagar from where few canals have been drawn. Saran Irrigation Canal is one of them in Bihar.

5. **Which of the following canal systems irrigate areas of Bihar?**
   (a) Upper Ganga Canal (b) Triveni Canal
   (c) Sharda Canal (d) Eastern Yamuna Canal
   (e) None of the above/More than one of the above
   60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Triveni Canal is made for irrigation in northwestern part of Bihar. This canal is related to Gandak irrigation project.

6. **Indira Gandhi Canal, irrigating Thar Desert, originates from**
   (a) Bhakra Dam
   (b) Hathni Kund Barrage
   (c) Harike Barrage
   (d) Ranjit Sagar Dam
   (e) None of the above/More than one of the above
   B.P.S.C. CDPO 2018

Ans. (c)

Indira Gandhi Canal is the longest canal in India, which starts from Harike Barrage near Harike and end at the irrigation facilities in the Thar desert in Rajasthan. The Indira Gandhi Canal was earlier known as the Rajasthan canal.

7. **For what purpose is water used the most in India?**
   (a) Domestic (b) Industries
   (c) Irrigation (d) More than one of the above
   (e) None of the above
   BPSC (Tre-3) {Class 6-8}–2024

Ans. (c)

In India, the primary use of water is for irrigation purposes. Agriculture accounts for the largest share of water usage, with over 87% of the total water consumption being used for irrigation of crops and farmlands. The other major uses of water in India include domestic/household purposes, industrial activities, and power generation. However, irrigation remains the single largest consumer of water resources in the country.

8. **Which of the following states is known for bamboo drip irrigation system?**
   (a) Meghalaya
   (b) Jharkhand
   (c) Assam
   (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (a)

The bamboo drip irrigation system is generally based over the gravity and the steep topography of the Meghalayan hills that helps to facilitate the flow of water through the network. Bamboo culms are basically cut longitudinally in half along with their length and these portions are considered as the channels as well as diversions with the help of which the water flows.
Bamboo drip irrigation system is a 200-year-old system of tapping stream and spring water by using bamboo pipes. This irrigation system is prevalent in Meghalaya.

Multi-Purpose River Valley Projects

i. Bhakra Nangal Dam

1. **Bhakra Nangal is a joint project of –**
   (a) Haryana, Punjab and Rajasthan
   (b) Haryana, Punjab and Delhi
   (c) Himachal Pradesh, Haryana, Punjab
   (d) Punjab, Delhi, Rajasthan

42^{nd} B.P.S.C. (Pre) 1997

Ans. (a)

Bhakra Nangal Project is a joint venture of the successor states of the erstwhile, state of Punjab and state of Rajasthan and Haryana States designed to harness the precious water of the Satluj for the benefit of the concerned States. The project has been named after the two dams built at Bhakra and Nangal on the Satluj River. It is notable that the command area of this project is expanded in Himanchal Pradesh, Punjab, Haryana and Rajasthan.

2. **On which river is the Bhakhra-Nangal dam built?**
   (a) Ravi (b) Indus
   (c) Chenab (d) Satluj

45^{th} B.P.S.C. (Pre) 2001

Ans. (d)

See the explanation of the above question.

ii. Dam at the River Cauvery

1. **Which of the following is the oldest hydropower station in India?**
   (a) Mayurakshi (b) Machkund
   (c) Pallivasar (d) Shivasamudram

43^{rd} B.P.S.C. (Pre) 1999

Ans. (d)

The oldest hydropower plant is in the hills of Darjeeling in West Bengal. It was commissioned in the year 1897. The hydroelectric power station near Shivasamudram was set up on the Cauvery river in Karnataka in 1902. It is the second oldest power station in India. Thus, option (d) is the correct answer.

iii. Nagarjuna Sagar Dam

1. **'Nagarjuna Sagar Multipurpose Project' is on which river?**
   (a) Tapti (b) Kosi
   (c) Godavari (d) Krishna

56^{th} to 59^{th} B.P.S.C. (Pre) 2015

Ans. (d)

Nagarjuna Sagar Project is a multipurpose scheme on River Krishna. The project was completed in 1967. The dam is located in Nalgonda district of Telangana and Guntur district of Andhra Pradesh. The project provides irrigation mainly to Palnadu Nalgonda, Khammam, Prakasam, Krishna and Guntur districts.

iv. Hirakud Dam

1. **Which of the following dams is located on the Mahanadi River?**
   (a) Hirakud Dam (b) Dudhawa Dam
   (c) Ghumarapadar Dam (d) More than one of the above
   (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (d)

Hirakund, Dudhawa and Ghumarapadar dams are located on the Mahanadi River. Dudhawa and Ghumarapadar dams are built in Chattisgarh while Hirakund dam is built in Odisha. It is the highest dam in India, while Bhakra Nargal Dam is the largest dam in India.

2. **On which river is the Hirakud Dam constructed?**
   (a) Shivnath (b) Narmada

(c) Mahanadi (d) Son

44th B.P.S.C. (Pre) 2000

Ans. (c)

Hirakud dam is built across river Mahanadi in the State of Odisha. Length of the main dam is about 4801 km. The Hirakud dam has been built 15 km away from Sambalpur town.

v. Damodar Valley Project

1. **Maithon, Belpahari and Tilaya dams are constructed on the river-**
   (a) Damodar (b) Barakar
   (c) Konar (d) Bokaro

45th B.P.S.C. (Pre) 2001

Ans. (b)

Maithon, Belpahari and Tilaya Dams were constructed on the Barakar River which is the main tributary of Damodar River in Eastern India. These dams were constructed in the first phase of the Damodar River Valley project.

Agriculture

1. **Towards the end of summer, there are pre-monsoon showers which are a common phenomenon in Kerala and coastal areas of Karnataka. Locally, they are known as**
   (a) Mango showers (b) Blossom showers
   (c) Norwesters (d) More than one of the above
   (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Towards the end of summer, there are pre-monsoon showers which are a common phenomena in Kerala and coastal areas of Karnataka. Locally, they are known as mango showers since they help in the early ripening of mangoes. Blossom showers–with this shower, coffee flowers blossom in Kerala and nearby area. Norwesters–These are dreaded evening thunderstorms in Bengal and Assam their notorious nature can be understood from the local nomenclature of "kal baisakh". A calamity of the month of Baisakh. These showers are useful for tea, jute and rice cultivation. In Assam, these storms are known as "Bardoisila".

2. **Dryland agriculture in India is practised in areas having annual rainfall less than**
   (a) 150 cm (b) 125 cm
   (c) 100 cm (d) 75 cm
   (e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (d)

Dryland Agriculture refers to growing of crops entirely under rainfed condition. Based on the amount of rainfall received, dryland farming is confined to regions having annual rainfall less than 75 cm.

3. **Double cropping in agriculture means raising of**
   (a) Two crops at different times
   (b) Two crops simultaneously
   (c) One crop along with another crop
   (d) None of these

43rd B.P.S.C. (Pre) 1999

Ans. (a)

The practice of consecutively producing two crops of either like or unlike commodities on the same land within the same year or same growing season at different times is called double cropping. An example of double cropping might be to harvest a wheat crop by early summer and then plant corn or soybeans on that acreage for harvest in the fall. This practice is only possible in regions with long growing seasons. While the cultivation of two or more crops simultaneously on the same field is called Intercropping.

4. **Which one of the following is the pathway to increase productivity in agriculture?**
   (a) Efficient irrigation (b) Quality seeds
   (c) Use of pesticides (d) Use of fertilizers
   (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

All the statements given above will increase productivity in agriculture.

5. **Which is not the source of agricultural finance in India?**
   (a) Co-operative Societies
   (b) commercial bank
   (d) none of these
   (c) Regional Rural Banks

44th BPSC (Pre) 2000

Ans. (c)

The major sources of agricultural finance in India are 1. Commercial banks 2. Co-operative banks 3. Regional rural banks. Hence it is clear that the correct answer is option (c).

6. **Which of the following is the local name of 'Jhumming'?**
(a) Bewar (b) Dahiya
(c) Podu (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (d)

The Shifting cultivation is a form of agricultural practice or a cultivation system in which an area of ground is cleared of vegetation and cultivated for a few years and then abandoned for a new area until its fertility has been naturally restored. Its practice is known as Jhum in North-Eastern India Vivar and Dahiya in Bundelkhand region, Podu in Andhra Pradesh region etc.

7. **Which of the following explains the term 'mulching'?**
(a) Covering the bare ground between crops with a layer of organic matter like straw
(b) Covering the contours with stones, grass, etc.
(c) Cropping along terraces made on steep slopes
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

The term 'mulching' is defined as a layer of material applied to the surface of soil. Reasons for applying mulch include conservation of soil moisture, improving fertility and health of the soil, reducing weed growth and enhancing the visual appeal of the area.

8. **The correct pair of local name of shifting agriculture and related country is**
(a) Chena—Bangladesh
(b) Milpa—Rhodesia (Zimbabwe)
(c) Ladang—Malaysia
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

The correct pair of local name of shifting agriculture and related country is–

Local Name of shifting Agriculture	Related Country/ Regions
Chena	Sri Lanka
Milpa	Central America
Ladang	Malaysia

9. **Millets are called miracle grains and crops of the future. Why?**
(a) Drought-tolerant crops
(b) Require few external inputs
(c) Nutritionally comparable to rice and wheat
(d) More than one of the above
(e) None of the above

B.P.S.C. Headmaster 07-12-2023

Ans. (d)

Millets can grow in Saline soil. They can thus be grown as an important solution for farmers grappling with climate changes sea level rise which leads to an increase in soil salinity, heatwaves, droughts, floods etc. Millet is nutritionally comparable to rice and wheat. Due to the peculiar nature, they are termed as the "miracle grains" or the "crops of the future."

Green Revolution

1. **Which one of the following most appropriately describes the nature of the Green Revolution of the late sixties of 20th century?**
(a) Intensive cultivation of green vegetable
(b) Intensive agriculture district programme
(c) High-yielding varieties programme
(d) Seed-Fertilizer-Water technology
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)

Green Revolution means using high-yielding varieties of seeds, modifying farm equipment, and substantially increasing chemical fertilizers. In, India Green Revolution began in 1966-67. High yield varieties programme started in India with the help of Rockefeller Foundation based in the U.S.A.

2. **After Independence India progressed maximum-**
(a) In the production of Rice
(b) In the production of Pulses
(c) In the production of Jute
(d) In the production of Wheat

40th B.P.S.C. (Pre) 1995

Ans. (d)

India achieved maximum growth in the production of wheat after Independence. This growth was propelled by the Green Revolution introduced in 1966-67.

3. What is true about the second green revolution in India?

1. It aims at further increasing the production of wheat and rice in areas already benefited from the green revolution.

2. It aims at extending seed-water-fertilizer technology to areas which could not benefit from green revolution.

3. It aims at increasing yields of crops other than those used for the green revolution in the beginning.

4. It aims at integrating cropping with animal husbandry, social forestry and fishing

Select the correct answer from the code given below :

(a) 1 and 2 (b) 2 and 3
(c) 2 and 4 (d) 1 and 4

47th B.P.S.C. (Pre) 2005

Ans. (c)

The Second Green Revolution in India aims at extending seed, water, fertilizer and technology to areas that could not be benefited from the Green Revolution and integrating agriculture with animal husbandry, social forestry and fishing. It does not aim to yield the crops other than those used in the Green Revolution in the beginning. Thus, statements 2 and 4 are correct.

4. Select the component of the Green Revolution by using the given code :

1. High-yielding varieties of seeds

2. Irrigation

3. Rural Electrification

4. Rural roads and marketing

Code :

(a) Only (1) and (2) (b) Only (1),(2) and (3)
(c) Only (1),(2) and (4) (d) All four

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

In 1965, the introduction of high-yielding varieties of seeds (hybrid seeds), increased use of chemical fertilizers, irrigation, rural electrification and connectivity to the market places through roads led to the increase in production needed to make the country self-sufficient in food grains which were called as Green Revolution. Thus all the four components were part of the Green Revolution.

5. The local names of rice crops produced in Northern India in winter are

(a) Aman, Aghani and Aus (b) Aman, Sali and Aghani
(c) Aman, Aus and Sali (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

The main rice growing season in the country is the "Kharif" It is known as winter rice as per the harvesting time. the sowing time of winter rice is June-July and it is harvested in Non-December. Winter rice is known as 'Aman' in West Bengal, 'Sali' in Assam 'Sarrad' in Orissa, 'Agahoni' in Bihar and Uttar Pradesh etc. Whereas Autumn rice crop is known as 'Aus' in West Bengal.

6. Rainbow Revolution is related with –

(a) Green-revolution (b) White-revolution
(c) Blue-revolution (d) All the above

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

On 28 July 2000, the new National Agricultural Policy was announced by the Central government. In this policy the concept of Rainbow Revolution was introduced. The various colours of the Rainbow Revolution indicate various farm practices such as Green Revolution (Foodgrains), White Revolution (Milk), Yellow Revolution (Oilseeds), Blue Revolution (Fisheries), Red Revolution (Tomatoes and meat),Golden Revolution (Fruits), Grey Revolution (Fertilizers) and so on. Thus, the concept of the Rainbow Revolution is an integrated development of crop cultivation, horticulture, forestry, fishery, poultry, animal husbandry and food processing industry.

7. The Black Revolution is related to the

(a) Fish production (b) Coal production
(c) Crude oil production (d) Mustard production
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

Black Revolution is associated with Crude Oil Production.
Other important Revolutions are as follows –
Pink Revolution - Onion.
Red Revolution - Tomato and Meat
Blue Revolution - Fisheries
Yellow Revolution - Oilseed

FOOD CROPS

i. Rabi Crops

1. Cash Crop does not consist –

(a) Sugarcane (b) Cotton
(c) Jute (d) Wheat

39th B.P.S.C. (Pre) 1994

Ans. (d)

Cash crops include Sugarcane, Cotton, Jute, Tobacco, Oilseeds etc and Wheat, Rice etc. are food crops.

2. **The highest wheat-producing state of India is-**
(a) Haryana (b) Punjab
(c) Bihar (d) Uttar Pradesh
53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

Three largest wheat producing states, when this question was asked, were – Uttar Pradesh, Punjab and Haryana. According to 2022-23 Economic Survey data, in 2021-22 (4th AE) three largest producers of wheat are –
(i) Uttar Pradesh (ii) Madhya Pradesh (iii) Punjab.

3. **Which is the maximum wheat producing state of India?**
(a) Uttar Pradesh (b) Punjab
(c) Madhya Pradesh (d) Haryana
45th B.P.S.C.(Pre) 2001

Ans. (a)

See the explanation of the above question.

4. **Which State of India is the highest producer of Millet?**
(a) Rajasthan (b) Maharashtra
(c) Tamil Nadu (d) More than one of the above
(e) None of the above
B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (a)

The major millets producing states in India are Rajasthan, Karnataka, Maharashtra, Uttar Pradesh, Haryana, Gujarat, Madhya Pradesh, Tamil Nadu, Andhra Pradesh and Uttarakhand. As per Agricultural and Processed Food Products export Development Authority (APEDA), with the contribution of 28.61 percent of the total millet production, Rajasthan is the highest producing state in India.

5. **Which of the following countries was the leading producer of wheat in 2020?**
(a) India (b) Russia
(c) United States of America (d) China
(e) None of the above / More than one of the above
B.P.S.C. CDPO 2022

Ans. (d)

Wheat is the third most-produced cereal-after rice and maize– and the second most produced for human consumption. China is the world's largest wheat producer and has yielded more than 2.4 billion tonnes of wheat in the last 20 years, around 17% of total production. Russia is the largest global wheat exporter, exporting volumes worth more than $7.3 billion in 2021. However the Russia-Ukraine war has caused massive disruptions to the global wheat market and adjacent industries.

ii. Kharif Crops

1. **Which of the following is not a cash crop?**
(a) Jute
(b) Groundnut
(c) Jowar
(d) Sugarcane
(e) None of the above/More than one of the above
60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

(A) Food crops
(i) Rice, Wheat, Maize, Millets.
(ii) Pulses
(B) Non-food crops (cash crops)
(i) Groundnut, Rapeseed, mustard
(ii) Fiber - Cotton, Jute
(iii) Plantation crops - Tea, coffee, Rubber
(iv) Others- Sugarcane, Tobacco etc.

2. **Chief food crop of India is –**
(a) Wheat (b) Rice
(c) Maize (d) Pulses
41st B.P.S.C. (Pre) 1996

Ans. (b)

Rice is the chief food crop of India, Wheat is the second most important food crop of India. Thus, option (b) is the correct answer.

3. **Which of the following is the most important food crop in terms of cropped area?**
(a) Wheat (b) Maize
(c) Barley (d) Rice
40th B.P.S.C. (Pre) 2000
41st B.P.S.C. (Pre) 1996

Ans. (d)

According to Economic Survey 2022-23, Rice occupies largest cropping area (46.4 million hectares). Wheat is grown on 30.5 million hectare area during 2021-22 (4th AE).

4. **The region known as the Rice Bowl of India is –**
(a) Kerala and Tamil Nadu
(b) Delta region of Krishna-Godavari
(c) North East region
(d) Indus Gangetic Plain
40th B.P.S.C. (Pre) 1995

Ans. (b)

According to the given options, the delta region of Krishna and Godavari is known as the **'Rice Bowl of India**. This region comes under Andhra Pradesh. Rice is cultivated on 2.25 million hectare area (2021-22 4th A.E.) of the total cropped area of Andhra Pradesh. Almost two third of total rice production is cultivated during the Kharif crop. Here, most important area for the production of rice is Rayalseema which is extended on Krishna-Godavari delta region. Thus, (b) is the correct answer.

5. In 2020, which of the following crops had the highest acreage in India?

(a) Pulses (b) Coarse cereals
(c) Wheat (d) Rice (Paddy)
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (d)

In 2020, about 321.79 lakh ha area coverage under rice as compared to 274.19 lakh ha. during the corresponding period of last year in 2019-20. Thus, 47.60 lakh ha more area has been covered compared to last year.

6. Choose the correct sequence of the States of India, according to ascending order of rice production in the year 2018-19.

(a) Punjab, Uttar Pradesh, Rajasthan, Haryana, Madhya Pradesh
(b) Uttar Pradesh, Punjab, Haryana, Madhya Pradesh, Rajasthan
(c) Rajasthan, Haryana, Madhya Pradesh, Punjab, Uttar Pradesh
(d) Punjab, Rajasthan, Haryana, Madhya Pradesh, Uttar Pradesh
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (e)

According to figures of Agricultural statistics at a glance, 2022 (data 2021-22 4th A.E.) top 5 Rice producing states in India are West Bengal, Uttar Pradesh, Punjab, Telangana and Odisha.

iii. Zaid Crop

1. Which of the following is a Zaid crop?

(a) Moong (b) Urad
(c) Watermelon (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (c)

In between the rabi and the kharif seasons, there is a short season during the summer months known as the Zaid season. Some of the crops produced during 'Zaid' are watermelon, muskmelon, cucumber, vegetables and fodder crops while moong, urad are Kharif crops.

2. Which of the following is grown during the Zaid season?

(a) Muskmelon (b) Cucumber
(c) Watermelon (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

A short cropping season between Rabi and Kharif is called Zaid season. Cultivation of Watermelon, Muskmelon, Cucumber is done mostly during this season. Hence option (d) is the correct answer.

Cash Crops

1. The largest producer of cotton in India is –

(a) Maharashtra (b) Gujarat
(c) Punjab (d) Haryana

44th B.P.S.C. (Pre) 2000

Ans. (b)

According to the data of 2021-22 (Agricultural statistics at a glance, 2022) (4th AE), the three largest cotton producing states in India are 1. Gujarat, 2.Maharashtra, 3. Telangana.

2. Which Indian state has the largest number of Cotton Textile Mills?

(a) Madhya Pradesh (b) Maharashtra
(c) Gujarat (d) West Bengal
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

Maharashtra is the state having the largest number of cotton Textile mills among the given options. Tamil Nadu has the largest number of Cotton textile mills in India.

3. The correct sequence in decreasing order of the four sugarcane producing states in India is:

(a) Maharashtra, U.P., Tamil Nadu, Andhra Pradesh
(b) U.P., Maharashtra, Tamil Nadu, Andhra Pradesh
(c) Maharashtra, U.P.,Andhra Pradesh, Tamil Nadu
(d) U.P., Maharashtra, Andhra Pradesh,Tamil Nadu.

44th B.P.S.C. (Pre) 2000

Ans. (b)

According to the data available at the time when this question was asked, the four sugarcane producing states in descending order of their sugarcane production were as follows –

(1) Uttar Pradesh (2) Maharashtra
(3) Tamil Nadu (4) Andhra Pradesh

According to Agricultural statistics at a glance, 2022 (data 2021-22 4th A.E.) data, the five largest sugarcane producing states are –

(1) Uttar Pradesh (2) Maharashtra
(3) Karnataka (4) Gujarat
(5) Tamil Nadu

4. Which States in India are the largest producers of sugarcane?

(a) Bihar and Uttar Pradesh
(b) Uttar Pradesh and Rajasthan
(c) Andhra Pradesh and Jammu and Kashmir
(d) Punjab and Himachal Pradesh

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

When the question is asked answer will be an option (a). According to data of the year 2021-22 (4th A.E.), 8 largest sugarcane producing states are – Uttar Pradesh, Maharashtra, Karnataka, Gujarat, Tamil Nadu, Bihar, Haryana and Punjab.

5. Which one among the following States of India is called 'Sugar Bowl'?

(a) Uttar Pradesh (b) Maharashtra
(c) Bihar (d) Haryana
(e) None of the above/More than one of the above.

64th B.P.S.C. (Pre) 2018

Ans. (a)

Uttar Pradesh is called the 'Sugar Bowl of India'. It is the largest producer of sugarcane in India.

Oilseeds

1. Which is the most suitable crop for dryland farming?

(a) Sugarcane (b) Jute
(c) Wheat (d) Groundnut

45th B.P.S.C.(Pre) 2001

Ans. (d)

Dryland agriculture refers to growing crops entirely under rainfed condition in given option. Groundnut is the most suitable crop for dryland farming. It is a tropical plant that requires a long and warm growing season. It grows well in area receiving 50 to 125 cm. of well distributed rainfall during growing season. Dryland farming is a type of dryland agriculture which is suitable for the cultivation of crops in areas receiving rainfall up to 100 cm.

'Silk'

1. The country, which is the largest silk producer in the world, is

(a) India (b) China
(c) Brazil (d) Japan
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

China is the largest producer of silk in the world followed by India.

PLANTATION CROPS

(i) Coffee

1. Which one of the following coffee-growing areas is not in Karnataka?

(a) Chikmagalur (b) Coorg
(c) Baba Budangin (d) Pulneys
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

The Pulneys hill range is situated adjacent to the popular Kodaikanal hill resort in Tamil Nadu.

(ii) Tea and Rubber

1. India is the best producer and consumer of

(a) Rice (b) Tea
(c) Oilseeds (d) Pulses

39th B.P.S.C. (Pre) 1994

Ans. (*)

During the time when this question was asked, India was leading in production and consumption of pulses and tea. According to the data released by Tea Board India, in 2022, India came down to the 2nd position in production of Tealeaves but in terms of production and consumption of pulses from the time of the question to present, India is still maintaining its first position. It is notable that the consumption of pulses in India is so high that even though it is the largest producer of pulses in the world, India still imports pulses from other countries.

2. India produces more than its need –

(a) Tea (b) Foodgrains
(c) Petroleum (d) Petro-chemicals

43rd B.P.S.C. (Pre) 1999

Ans. (a)

In the options given in the question hour. India used to import cereals, petroleum and petro-chemicals and export tea, while. The percentage share of tea in India exports in the year 2021-22 is about 0.18 percent. Ranks fourth among tea exporting countries.

3. India's largest rubber producer state is

(a) Andhra Pradesh (b) Karnataka
(c) Kerala (d) Tamil Nadu

43rd BPSC (Pre) 1999

Ans. (c)

Kerala is the largest rubber producing state in India. About 75.68% (2019-20) of the total rubber production in the country is produced here. Ernakulam, Kottayam, Kozhikode and Kollam are the major rubber producing districts of Kerala. According to data for the year 2022 (FAO), India placed 5th in the world in the terms of natural rubber in Primary Forms Production.

4. Which one of the following hills do not have tea plantations?

(a) Kanan Devan (b) Nilgiri
(c) Darjeeling (d) Girnar
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

Tea producing states are Assam, West Bengal, Tamil Nadu Kerala and Karnataka. Kanan Devan, Nilgiri and Darjeeling are in Kerala, Tamil Nadu and West Bengal respectively. But, Girnar is in Gujarat.

5. Which one of the following States is the leading producer of rubber in India?

(a) Tamil Nadu (b) Kerala
(c) Karnataka (d) Andhra Pradesh
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

See the explanation of the above question.

Agriculture : Miscellaneous

1. In which one of the following agricultural products, India is the highest producer in the world?

(a) Cotton (b) Oilseeds
(c) Tea (d) Tobacco
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (*)

India's rank in oil seeds production in the world–
Castor oil seeds — First
Groundnut (excluding shelled) — Second
Rapeseed — Fourth
(Source – FAOSTAT)

- India ranked First Position in Cotton production which is about 6.16 Million Metric Tonnes in the year 2021-22.
- In Tea production India ranked second after China.
- In Tobacco (unmanufactured) production India ranked second.

2. The largest Jute producing state in India is –

(a) Andhra Pradesh (b) Bihar
(c) Tamil Nadu (d) West Bengal

41st B.P.S.C. (Pre) 1996

Ans. (d)

According Directorate of Jute Development to the figures released in 2021-22 (4th A.E.), the largest Raw Jute producing state in India is West-Bengal followed by Bihar and Assam.

3. The only state which produces saffron in India is

(a) Himachal Pradesh (b) Assam
(c) Jammu-Kashmir (d) Meghalaya
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

The largest amount of saffron in India is produced in the Kashmir valley. Iran has ranked first in production of saffron globally.

4. The major source of methane in India is

(a) wheat field (b) sugarcane field
(c) rice field (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The major source of methane in India is rice field. The common sources of methane are oil and natural gas agricultural activities, coal mining and wastes. It is a powerful greenhouse gas. It is flammable and used as a fuel worldwide.

Animal Husbandry

1. India's share in meat and meat preparation exports in the year 2017 was

(a) 5% (b) 6%
(c) 2% (d) 3%
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (e)

India's share in meat and meat preparation export was 2.5 percent in 2018, in World exports.

2. **Which of the following is not a breed of Rajasthan?**
   (a) Tharparkar (b) Rathi
   (c) Hallikar (d) Mewati
   (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (e)

Rajasthan state has three native cattle breeds viz Rathi, Tharparker and Nagori, having great deal of endurance. Hallikar is a breed of cattle native to the state of Karnataka. Mewati, also known as Kesi, is a breed of Haryana.

Mineral Resources

(A) Rock System

1. **Bihar's geographical structure consists of rock systems. Match the rock systems with their descriptions :**

A. Dharwar rock system	**1. Created by the rapid deposition of alluvium by Himalayan and Peninsular rivers**
B. Vindhyan rock system	**2. Can be found in the West Champaran district**
C. Quaternary rock system	**3. Part of the oldest Archaean rock system**
D. Tertiary rock system	**4. Sandstone, limestone, dolomite, quartzite and shale are the main constituents of this rock system**

Select the correct answer using the codes given below.

	A	B	C	D
(a)	3	4	1	2
(b)	1	3	2	4
(c)	2	3	4	1
(d)	4	2	3	1

69th B.P.S.C. (Pre) 2023

Ans. (a)

The correct match is following–

Dharwar rock system	Part of the oldest Archaean rock system
Vindhyan rock system	Sandstone, limestone, dolomite, quartzite and shale are the main constituents of this rock system
Quaternary rock system	Created by the rapid deposition of alluvium by Himalayan and Peninsular rivers
Tertiary rock system	Can be found in the West Champaran district

2. **Which of the following rocks is not a metamorphic rock?**
   (a) Quartzite
   (b) Gneiss
   (c) Dolomite
   (d) None of the above

BPSC Agriculture Department-2024

Ans. (c)

Dolomite is not a metamorphic rock. Dolomite is a sedimentary carbonate rock, while metamorphic rocks are formed when existing rocks are transformed by intense heat or pressure.

3. **Most rich state in minerals in India is –**
   (a) Rajasthan
   (b) Madhya Pradesh
   (c) Bihar
   (d) Odisha

39th B.P.S.C. (Pre) 1994

Ans. (c)

During the time, when this question was asked, the state of Bihar was rich in mineral production. According to Indian Bureau of Mines report. Present (2021-22) situation in terms of mineral production on value Odisha > Chhattisgarh > Rajasthan > Karnataka > Jharkhand.

B. Metallic Minerals

i. Iron-Ore

1. **What is the most famous feature of Chiriya located in West Singhbhum?**
   (a) Iron-ore mining
   (b) Dam
   (c) Bird Sanctuary
   (d) National Park
   (e) None of the above More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

The most famous feature of the Chiriya mine located in West Singhbhum is the iron-ore mining.

2. The two states of India, most richly endowed with iron ore, are

(a) Bihar and West Bengal
(b) Madhya Pradesh and Odisha
(c) Bihar and Odisha
(d) Madhya Pradesh and West Bengal

56th to 59th B.P.S.C. (Pre) 2015

Ans. (*)

According to Indian Minerals Yearbook, 2021. The States that are rich in Iron Ore in thousand tonnes (Haematite) are as follows : Odisha (39%), Jharkhand (20%), Chhattisgarh (19%). While Magnetite is found in Karnataka (69.50%), Andhra Pradesh (13.10%), Rajasthan (7.10%) and Tamil Nadu (4.70%).

3. Iron Ore is not available in which of the following Indian state-

(a) Bihar (b) Madhya Pradesh
(c) Odisha (d) Punjab

38th B.P.S.C. (Pre) 1992

Ans. (d)

India has four major iron-ore producing regions –

East Region	Jharkhand, Odisha
Central India and West Region	Madhya Pradesh, Chhattisgarh, Maharashtra
Peninsular Region	Karnataka, Goa
Other areas	Rajasthan, Gujarat, Haryana, Andhra Pradesh, Kerala, West Bengal

This is notable that at the time when this question was asked, Jharkhand was part of Bihar. Hence, the correct answer would be (d).

4. Which of the following Iron ores is mined at Bailadila?

(a) Haematite (b) Siderite
(c) Limonite (d) Magnetite
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

Haematite iron ore is mined at Bailadila which is located in Dantewada district of Chhattisgarh. Haematite is a high quality iron ore. It is mainly found in the states of Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa, Maharashtra and Andhra Pradesh etc. Magnetite which is comparatively High grade in terms of iron content is found in the states of Karnataka, Andhra Pradesh, Rajasthan, Tamil Nadu etc.

ii. Silver

1. In which of the following states of India, silver is not found?

(a) Odisha (b) Andhra Pradesh
(c) Gujarat (d) Jharkhand

43rd B.P.S.C. (Pre) 1999

Ans. (c)

According to **'Indian Mineral Yearbook 2021'**, silver (in terms of ore) reserves are found in Rajasthan, Karnataka, Jharkhand, Andhra Pradesh, Madhya Pradesh, Uttarakhand, Odisha, Meghalaya, Sikkim, Tamil Nadu, Maharashtra, . Thus, reserves of silver are not found in Gujarat.

iii. Copper

1. Ghatshila of Jharkhand is famous for which mineral production?

(a) Mica (b) Bauxite
(c) Copper (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (c)

Ghatshila is best known for copper mines because they are Asia's first copper mines and the world's second deepest mines. Indian Copper Corporation Ltd. was established by a British company in 1930 at Ghatshila consisting of a cluster of underground copper mines, concentrator plants and smelters.

2. At which of the following places the Copper Industry is located?

(a) Tarapur (b) Titagarh
(c) Ranchi (d) Khetri
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

The correct match is as follows :

Copper Fields		**State**
Tarapur	–	Maharashtra
Titagarh	–	West Bengal
Khamman	–	Telangana
Khetri	–	Rajasthan
Ranchi	–	Jharkhand

In given option copper industry located in Khetri (Rajasthan).

3. **Khetri Belt of Rajasthan State is famous for:**
   (a) Copper mining (b) Gold mining
   (c) Mica mining (d) Iron ore mining
   (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

The Khetri belt of Rajasthan State is famous for copper mining. It is located in the Neem Ka Thana district of Rajasthan State.

4. **Which State of India is the largest producer of copper?**
   (a) Madhya Pradesh (b) Rajasthan
   (c) Jharkhand (d) Chhattisgarh
   (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (a)

According to Indian Minerals year book, 2021 Madhya Pradesh was the leading producer state of copper concentrates accounting for about 60% of the production during 2020-21.

5. **Which of the following States has the largest deposits of copper ore?**
   (a) Rajasthan (b) Jharkhand
   (c) Madhya Pradesh (d) Karnataka
   (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

The total metal content out of the total copper resources is 12.16 million tonnes of which 2.73 million tonnes constitute reserves. According to 58th edition of Indian Minerals Yearbook, 2019, largest reserves/resources of copper ore to the tune of 813 million tonnes (53.81%) are in the State of Rajasthan, followed by Jharkhand with 295 million tonnes (19.54%) and Madhya Pradesh with 283 million tonnes (18.75%).

C. Non-Metallic Minerals-Mica

1. **Which of the following Indian states is the biggest producer of Mica?**
   (a) Andhra Pradesh (b) Bihar
   (c) Jharkhand (d) Rajasthan

47th B.P.S.C. (Pre) 2005

Ans. (d)

According to Indian Mineral Yearbook 2021, Rajasthan was the Biggest producer of Mica.

D. Energy Mineral-Coal

1. **Which of the following river valleys is rich in coal reserves in India?**
   (a) Mahanadi River Valley
   (b) Damodar River Valley
   (c) Son River Valley
   (d) More than one of the above
   (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (d)

In India over 97 percent of coal reserves occur in the valley of Damodar, Son, Mahanadi and Godavari. About 80 percent of coal deposits in India are of a bituminous type and are of non-cooking grade. The most important Gondwara coal field in India is located in Damodar Valley.

2. **Where was first coal mine in India mined?**
   (a) Jharia (b) Raniganj
   (c) Dhanbad (d) Asansol
   (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (b)

In India, the first coal mine was dug on the banks of Damodar River at Raniganj. It was first mined in 1774 AD.

3. **Which of the following two States are the largest producers of coal in India?**
   (a) Bihar and West Bengal
   (b) Jharkhand and Chhattisgarh
   (c) Andhra Pradesh and Kerala
   (d) Madhya Pradesh and Odisha

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (b)

The largest coal producing state in India is Odisha followed by Chattisgarh and Jharkhand. On the basis of Ministry of Coal Statistics data in January 2024, but as per the given option (b) is the correct answer.

4. **The descending order of the state in coal production is-**
   (a) Bihar, Madhya Pradesh, West Bengal
   (b) Madhya Pradesh, West Bengal, Bihar
   (c) West Bengal, Madhya Pradesh, Bihar
   (d) Bihar, West Bengal, Madhya Pradesh

43rd B.P.S.C. (Pre) 1999

Ans. (a)

In the year 1999 (when the question was asked) the production of coal in undivided Bihar (including Jharkhand) was more than that of undivided Madhya Pradesh (including Chhattisgarh). Thus, the correct answer was option (a). According to Indian Minerals Yearbook 2021 (2020-21 (P)), the three largest coal-producing states in India are : 1. Chhattisgarh, 2. Odisha, 3. Madhya Pradesh.

5. Which State in India is the leading producer of Thorium?

(a) Kerala (b) Bihar
(c) Jharkhand (d) Madhya Pradesh
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

In given option Kerala is the leader in thorium Production, Monozite is the main source of thorium. According to Indian Minerals Yearbook 2021 Maximum Resources of Monazite in State Andhra Pradesh. Indian Rare Earths Limited produces thorium by processing Monazite obtained from Manavalakurichi (Tamil Nadu) at its plant located at Udyagmandal (Aluva) in Ernakulam district.

6. The Korba coalfield is located in –

(a) Odisha (b) Chhattisgarh
(c) West Bengal (d) Assam

44th B.P.S.C. (Pre) 2000

Ans. (b)

Korba coalfield is located in Korba district of Chhattisgarh in the basin of the Hasdeo River.

7. Coal mines in Jharkhand are located at :

(a) Jharia (b) Jamshedpur
(c) Ranchi (d) Lohardaga

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Jharia Coalfield is located in Dhanbad district of Jharkhand. Most of the Coking Coal Reserve is found here. Jamshedpur is famous for Iron & Steel industry; Lohardaga is famous for the production of Bauxite and Ranchi is famous for heavy machines.

8. What is chiefly found at Jharia in Jharkhand?

(a) Thorium (b) Silk
(c) Gold (d) Coal

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

See the explanation of the above question.

9. Consider the following problems being faced by the Indian coal industry-

I. Poor quality of coal and bottlenecks in the coal movement.

II. Low utilisation capacity of washeries.

III. Growing dependence on the import of coking coal.

IV. Administered prices.

Which of the above are correct?

(a) II, III and IV (b) I, II, III and IV
(c) I, III and IV (d) I, II and III

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Currently, coal industry in India is facing a lot of challenges which are low-quality coal, deficiency in coal washing establishment, high import of coking coal and administered prices etc.

10. Combustion of underground coal occurs in the state of -

(a) West Bengal (b) Bihar
(c) Jharkhand (d) Odisha
(e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

Jharia coalfield of Jharkhand has suffered a coal bed fire since 1910.

ii. Petroleum and Natural Gas

1. The oldest oil field in India is –

(a) Bombay High, Maharashtra
(b) Ankleshwar, Gujarat
(c) Navgam, Gujarat
(d) Digboi, Assam

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

Digboi, a town area in Tinsukia district in the north-eastern part of Assam, is the birthplace of oil Industry in India. Digboi refinery, commissioned on 11th December, 1901 is India's oldest operating refinery and one of the oldest operating refineries in the world. The historic Digboi Refinery has been termed as 'Gangotri of the Indian Hydrocarbon Sector' Digboi Refinery field is part of Brahmaputra Valley Oilfield. Other Refineries in Brahmaputra Valley are :
Naharkatiya field, Moran Hugrijan field, Rudra Sagar, Lakwa and Surma Valley.

2. **When did the first oil crisis/energy crisis occur in India?**

(a) During 1950's and 1960's (b) During 1930's and 1940's

(c) During 1990's and 2000's (d) During 1970's and 1980's

56th to 55th B.P.S.C. (Pre) 2011

Ans. (d)

First oil/energy crisis occurred during the year 1973 when oil exporting countries (OPEC) suddenly increased the oil price about four times. In India, industry and transportation was developed in such a way that demands for petroleum was increasing rapidly. Hence, due to the rise in petroleum prices, there was a huge oil crisis during 1970-80.

3. **The correct pair of Indian petroleum producing area and its location is**

(a) Surma Valley—Assam

(b) Lunej Area—Gujarat

(c) KG Basin—Bay of Bengal

(d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

The correct pair of Indian petroleum producing area and its location is–

Area	Location
Surma Valley	Assam
Lunej Area	Gujarat
KG basin	Andhra Pradesh
Kg – D6	Andhra Pradesh and Bay of Bengal

E. Miscellaneous : Minerals

1. **In which one of the following minerals, India leads in production in the world?**

(a) Sheet mica (b) Copper

(c) Gypsum (d) Iron ore

(e) None of the above/More than one of the above.

64th B.P.S.C. (Pre) 2018

Ans. (a)

According to Indian Minerals Yearbook 2021 and World Mineral Production 2017-2021

Minerals	Country (Leading in production)
Sheet mica	India
Copper (Mine)	Chile
Gypsum	USA
Iron Ore	Australia

2. **What are the main mineral constituents of the oceanic crust?**

(a) Silica and alumina (b) Silica and nickel

(c) Silica and magnesium (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

The uppermost surface of earth is called crust. It is the thinnest of all the layers. It is about 35 km on the continental masses and only about 5 km on the ocean floor. The main mineral constituents of the continental mass are silica and alumina. It is thus called. Sial (si – silica and al – aluminium). The oceanic crust mainly consists of silica and magnesium, it is therefore called sima (si – cilica and ma – magnesium).

3. **Which of the following statements is correct?**

(a) Natural gas is found in Dharwar rock formation.

(b) Mica is found in Kodarma.

(c) Cuddapah series is famous for diamonds.

(d) Petroleum reserves are found in Aravalli hills.

69th B.P.S.C. (Pre) 2023

Ans. (b)

Natural gas is found in sedimentary basin, generally in the areas of marine sediments. Hence statement (a) is incorrect. Significantly, Kodarma district of Jharkhand State has been famous world wide for mica mining, specially for ruby mica. The main reserve of mica is found under the forest of wild life Sanctuary of Kodarma. Cuddapah series has been found in Copper, Iron, Lead, Diamond, Uranium etc. Petroleum is found in Krishna, Godavari, Cauvery basin, Assam, Gujarat and Rajasthan.

4. **Ilmenite, which is widely distributed along the Indian coastline, is a mineral of?**

(a) Tungsten (b) Titanium

(c) Gallium (d) Tin

(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Ilmenite ($FeO.TiO_2$) and Rutile (TiO_2) are the two major minerals of titanium. This mineral is found in a wide area in the coastal areas of India from Saurashtra Coast (Gujarat) to Digha Coast (West Bengal).

5. **The most important ore of aluminum is -**

(a) Bauxite (b) Calamine

(c) Calcite (d) Galena

(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

Among the advanced alternatives, the most important ore of aluminum is bauxite. It is the main source of aluminum and gallium in the world.

6. Radium was obtained from -

(a) limestone (b) pitchblende

(c) Rutile (d) hematite

39th B.P.S.C. (Pre) 1994

Ans. (b)

Radium was discovered by Marie Curie and Pierre Curie from pitchblende a material that contains uranium.

7. Which of the following is a centre of textiles?

(a) Bokaro (b) Bhilwara

(c) Sanand (d) Tarapur

(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (b)

The industrial town of Bhilwara, situated in the Mewar region of Rajasthan, is a famous hub for textiles in India. Bhilwara is also known as the textile city of India. The city is known for its production of high-quality cotton, silk, and woolen fabrics, which are exported to various parts of the world. The textile industries of the Bhilwara district display an annual growth rate of 8 to 10 percent and are widely popular for exporting textile products like synthetic yarn, woolen commodities, cotton yarn, and fabrics. Apart from textiles, the city is also known for its mining and agriculture industries.

Electrical Energy

i. Hydroelectricity

1. Rana Pratap Sagar Hydro-electricity Project on Chambal River is located in the State of:

(a) Madhya Pradesh (b) Rajasthan

(c) Uttar Pradesh (d) Chhattisgarh

(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (b)

The Rana Pratap Sagar Project location on the Chambal River at Rawatbhata (Chittorgarh district) in Rajasthan.

2. In which hill station the electricity supply in India commenced in 1897?

(a) Darjeeling (b) Mussoorie

(c) Shimla (d) Mahabaleshwar

(e) None of the above / More than one of the above

B.P.S.C. Project Manager Prelims 2021

Ans. (a)

The first small hydro project (Sidrapong) of commissioned in the hills of Darjeeling in 1897 marked the development of hydropower in India.

3. What is the share of hydroelectric power in the total electricity produced in India?

(a) 10 percent (b) 12 percent

(c) 20 percent (d) 22 percent

(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

Hydro-electric power contributed to 10.6% of the total electricity produced during question hour (2016-17). According to figures of Ministry of Power total installed capacity in India as on June, 2024 was 446190 MW.

ii. Energy : Miscellaneous

1. Which of the following is not a conventional source of energy?

(a) Coal (b) Natural gas

(c) Wind energy (d) More than one of the above

(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (c)

Conventional Source of energy is also known as non-renewable sources of energy and are available in limited quantities. Coal, Petroleum, wood, natural gas are the examples of conventional sources of energy whereas Wind energy, is an example of non-conventional source of energy. Hence, option (c) is the correct answer.

2. Which one of the following statements about energy production and consumption in India is not correct?

(a) During the last decade, energy production in India has shown a declining trend.

(b) The per capita energy consumption in the world is the lowest in India

(c) The non-conventional sources of energy contribute less than one percent of the total commercial energy produced in India.

(d) The industry is the major energy consuming sector in India.

47th B.P.S.C. (Pre) 2005

Ans. (c)

The non-conventional sources of energy contribute approx more than 1% of the total commercial energy produced in India. Thus, statement (c) is wrong.

3. In India, per capita consumption of energy in 1994 was –

(a) 300 kg. of oil equivalent

(b) 360 kg. of oil equivalent

(c) 243 kg. of oil equivalent

(d) 343 kg. of oil equivalent

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

In India, per capita consumption of energy in 1994 was oil equivalent to 244 kg Oil which was much less than world standards for per capita consumption of energy of 1471 kg.

4. Which one of the following States is a leading producer of solar energy in India?

(a) Telangana (b) Karnataka

(c) Andhra Pradesh (d) Rajasthan

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

Rajasthan (Dec-2023) top the list of States is a leading Producer of Solar Power in the country.

5. Energy sources that do not increase carbon emissions include

(a) solar cells (b) windmills

(c) nuclear power plants (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

Renewable energy is energy that comes from a source that won't run out. They are natural and self-replenishing, and usually have a low- or zero-carbon footprint. Examples of renewable energy sources include wind mills, solar cells, nuclear energy and hydroelectric, including tidal energy.

Industry

Iron and Steel Industry

1. World's famous iron ore deposits named Krivoy Rog is located in–

(a) Ukraine (b) Sweden

(c) Belarus (d) None of the above

BPSC Agriculture Department-2024

Ans. (a)

The Krivoy Rog iron ore deposits are in the Kryvyi Rih Iron Basin (Kriv bass) in central Ukraine, about 330 km southeast of Kyiv and 150 km north of the Black Sea.

2. Which of the following States is the largest producer of iron in India?

(a) Jharkhand (b) Bihar

(c) Odisha (d) West Bengal

(e) None of the above / More than one of the above

B.P.S.C. Project Manager Prelims 2021

Ans. (c)

The production of Iron Ore at about 253.97 million tonnes in 2021-22 registered an increase of 23.86% over the previous year. Almost the entire production of Iron Ore (98.62%) accrued from Odisha (53.82%), Chhattisgarh (16.27%), Karnataka (15.88%), Jharkhand (9.74%) and Madhya Pradesh (2.91%). ***(Source : Ministry of Mines)***

3. Stainless steel is an alloy of –

(a) Iron and Copper (b) Iron and Zinc

(c) Iron and Chromium (d) Iron and Graphite

38th B.P.S.C. (Pre) 1992

Ans. (c)

Stainless steel is a metal alloy, made up of steel mixed with elements such as chromium, nickel, molybdenum, silicon, aluminium and carbon. Iron mixed with carbon to produce steel is the main component of stainless steel. Hence, option (c) is correct.

4. Some iron and steel plants have been planned along the western coast of India. What is the major reason for this locational shift in this industry?

(a) Increased nuclear power generation in the Western Coastal region

(b) The occurrence of high-grade iron ore deposits in Goa and parts of Madhya Pradesh and the comparative ease of exporting steel from here.

(c) The decline in international demand for Indian iron ore from the Western coastal region.

(d) Adoption of sponge from technology.

47th B.P.S.C. (Pre) 2005

Ans. (b)

Some iron and steel plants have been planned along the western coast of India. The major reasons for the locational shift of this industry are occurrence of high-grade iron ore deposits in Goa & parts of Madhya Pradesh as well as sea-ports for comparative case of exporting steel and consumption of iron ore in the concerned field for manufacturing.

5. **TISCO plant is located near –**
   (a) Patna (b) Darbhanga
   (c) Dhanbad (d) Tatanagar

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

TISCO (Tata Iron & Steel Company Limited) is prime steel company of India. It was established in 1907.

6. **In India dry point settlement is found in–**
   (a) Flood-prone area (b) Bikaner
   (c) Aravali region (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (d)

Both Bikaner and certain parts of the Aravali region are suitable locations for dry point settlements in India.

7. **Badampahar iron ore mine is located in–**
   (a) Durg and Bastar districts
   (b) Gua and Noamundi districts
   (c) Mayurbhanj and Kendujhar districts
   (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (c)

Badampahar mines are located in the Mayurbhanj and Kendujhar districts in Odisha, where high-grade hematite ore is found. Hematite has the best physical qualities required to make steel, jewellery, or precious gems. Hematite iron ore is mined in the Jharkhand district of Singbhum, near the towns of Gua and Noamundi.

Miscellaneous

1. **'Rat Hole' mining is done in which state of India?**
   (a) Arunachal Pradesh (b) Nagaland
   (c) Assam (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (e)

Rat-hole mining, aptly named for its resemblance to rodent burrows, is an illegal and highly hazardous method of extracting coal prevalent in certain pockets of India, particularly the state of Meghalaya.

2. **Khetri mines in Rajasthan are famous for**
   (a) Copper (b) Iron-ore
   (c) mica (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (a)

Khetri is situated at the foothills of the Aravalli Range, which hosts copper mineralization, giving rise to 80 km long metallogenetic province from Singhana in the north to Raghunathgarh in the south, popularly known as Khetri Copper Belt.
Khetri Nagar is a town in the Jhunjhunu district of Rajasthan in India. Khetri Nagar is also very well-known with the name of 'Copper'.

3. **Which one of the following pairs of region and State is not correctly matched?**
   (a) Wayanad-Kerala
   (b) Dandakaranya-Madhya Pradesh
   (c) Malnad-Karnatak
   (d) Vidarbha-Maharashtra
   (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

Correctly matched region and states -

Wayanad	Kerala
Dandakaranya	Chhattisgarh
Malnad	Karnataka
Vidarbha	Maharashtra

The Dandakaranya region is a vast forested area in Central India covering parts of five states including Chhattisgarh, Telangana, Maharashtra, Odisha and Andhra Pradesh.

4. **At which of the following places the newsprint paper industry is located?**
   (a) Satana (b) Durgapur
   (c) Nepanagar (d) Kanpur
   (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

Nepanagar is an industrial town located in Burhanpur district of Madhya Pradesh state, which is famous for newsprint industry.

5. **Dalmianagar of Bihar is famous for**
   (a) Silk (b) Cement
   (c) Leather (d) Jute

56th to 59th B.P.S.C. (Pre) 2015

Ans. (b)

Dalmianagar is one of the oldest and biggest industrial towns of India.

6. **The central stretch of the western coast is known as–**
   (a) Kannad plain (b) Malabar coast
   (c) Konkan coast (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (a)

The central stretch of the west coast is known as the Kannad plain. This plain is primarily situated in Maharashtra. The Kannad plain is the central part of the three parts of the western coastal plain.The Kannad plain is further followed by the southern stretch of the west coast, and it runs between Karnataka and Kerala.

7. **Where was the first fertilizer plant of India set up?**
   (a) Nangal (b) Sindri
   (c) Aluva (d) Trombay

39th B.P.S.C. (Pre) 1994

Ans. (*)

The first fertilizer plant of India was commissioned in Sindri (then Bihar now Jharkhand) in 1951 as public sector undertaking. First Fertilizer plant in India was established at Ranipet, Tamilnadu in 1906.

8. **Match the following places with industries and select the correct answer using the codes given below:**

List-I (Place)	**List-II (Industry)**
A. Bengaluru	**1. Iron & Steel**
B. Korba	**2. Copper**
C. Jamshedpur	**3. Aircraft**
D. Malanjkhand	**4. Aluminium**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	1	4	3
(c)	4	3	2	1
(d)	3	4	1	2

(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

The correct match is as follows –

(Place)		**(Industry)**
Bengaluru	-	Aircraft
Korba	-	Aluminium
Jamshedpur	-	Iron & Steel
Malanjkhand	-	Copper

9. **The largest number of cotton mills in Tamil Nadu are found in :**
   (a) Chennai (b) Coimbatore
   (c) Madurai (d) Salem
   (e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

The largest number of cotton mills are found in Coimbatore, Tamil Nadu. There are around 919 cotton mills in Coimbatore.

Transport

i. Road Transport

1. **In total there are how many National Highways in India and approximately what is their total length?**
   (a) 34 and 16,000 kms. (b) 44 and 24, 000 kms.
   (c) 54 and 32,000 kms. (d) 64 and 40,000 kms

45th B.P.S.C. (Pre) 2004

Ans. (d)

When this question was asked the number of National Highways in India was 64 and length was 40,000 kms. According to recent data on 31 December 2022, the number of national highways is around 599 and the total length is 144955 kilometres. (According to Annual Report 2022-23)

2. **What is the Golden Quadrilateral?**
   (a) Rail lines joining metros
   (b) Major Air Routes
   (c) National Highway Project
   (d) Cold Trade Routes
   (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

Golden Quadrilateral is a National Highway project which was started in 2001 connecting India's four top metropolitan cities, Delhi, Mumbai, Chennai and Kolkata. Its total length is 5846 km.

3. **........ carries 40 percent of road traffic of India –**
   (a) National Highways (b) State roads
   (c) District roads (d) Village roads

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (a)

According to Annual Report 2022-23 the Indian road network of 63.32 lakh km, is the second largest in the world. Which is constructed and maintained by the Public Work Department (PWD) and National Highway Authority of India (NHAI). National Highway Network consist of approx 1,44,955 km. length.

ii. Rail Transport

1. **The railways in India was first opened in –**
   (a) 1853 (b) 1854
   (c) 1855 (d) 1859

44th B.P.S.C. (Pre) 2000

Ans. (a)

During the tenure of Lord Dalhousie, country's first railway was built by the Great Indian Peninsula Railway (GIPR), which opened on 16 April, 1853 between Bombay to Thane.

2. **The width between two rails of Broad gauge is –**
   (a) 6 1/2 ft (b) 5 1/2 ft
   (c) 5 ft (d) 4 1/2 ft

38th B.P.S.C. (Pre) 2008

Ans. (b)

Indian Railway uses three types of gauges :
(1) **Meter gauge -** The distance between two bars is 1 metre or 1000 mm.
(2) **Broad gauge -** The distance between two bars is 1.676 metres or 1676 mm.
(3) **Narrow gauge -** The distance between two bars is 762 mm. and 610 mm.

3. **The Railway Zone Headquarter Hajipur is located in –**
   (a) Chhattisgarh (b) Uttar Pradesh
   (c) Jharkhand (d) Bihar

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (d)

Currently, Indian Railway has been divided among 17 zones (Headquarters). Among these headquarters of Indian Railway in Bihar, only the East Central Railway is located at Hajipur. Hajipur Headquarters started functioning since 1 October, 2002.

4. **Which one of the following numbered Rajdhani trains covers the longest distance?**
   (a) 12429 Bangalore City Junction
   (b) 12431 Trivandrum Central
   (c) 12433 Chennai Central
   (d) 12435 Dibrugarh Town

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

The Thiruvananthapuram Rajdhani train (12431/32) covers the longest distance. It runs from Thiruvananthapuram Central to Hazrat Nizamuddin, which covers about 3,149 km. Dibrugarh Town travels 2459 kilometres, the Bengaluru Rajdhani Express covers 2365 kilometres and the Chennai Rajdhani Express covers 2175 kilometres.

iii. Water / Air Transport

1. **Paradip is located in the state of –**
   (a) Kerala (b) Maharashtra
   (c) Odisha (d) Andhra Pradesh

44th B.P.S.C. (Pre) 2000

Ans. (c)

Paradip port is situated in Odisha, 210 nautical miles south of Kolkata and 260 nautical miles North of Vishakhapatnam. It can accommodate vessels up to 2 lakh DWT. The port started as a mono commodity port intended mainly to cater to the export of iron ore from Odisha to Japan. Paradip port is developed for reducing the traffic of Kolkata & Vishakhapatnam ports.

2. **Marmugao seaport is situated in –**
   (a) Odisha (b) Tamil Nadu
   (c) Goa (d) Kerala

40th B.P.S.C. (Pre) 1995

Ans. (c)

Marmugao seaport is an important port of Goa, located at the entrance of Zuari estuary and occupies the prestigious position in handling the traffic. This is a major export port from where about 39% of the total iron is exported from the country.

3. **India's 13th major port is going to be set up in which State?**
   (a) Kerala (b) Gujarat
   (c) Maharashtra (d) Tamil Nadu
   (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

India's 13th major port will be set up at Vadhavan in Maharashtra. It will be developed on "landlord model" (where infrastructure is leased to private firms or industries and chemical plants).

4. **Jawaharlal Nehru Port is located in the State of:**
   (a) Goa (b) Gujarat
   (c) Andhra Pradesh (d) Maharashtra
   (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (d)

Jawaharlal Nehru Port (JNP), also known as Nhava Sheva Port is situated in the state of Maharashtra.

5. **Out of total number of shipyards in the country, what is the number of private shipyards?**
   (a) 18 (b) 19
   (c) 20 (d) 21
   (e) None of the above / More than one of the above

B.P.S.C. Project Manager Prelims 2021

Ans. (*)

As per information available, there are 36 shipyards/ship building companies/ship repairing companies, out of which 7 ship building companies are in the public sector and rest are in the private sector. (***Source***–Statistics of India's ship building and ship repairing industries 2020-21)

6. **Which of the following is a landlocked port?**
   (a) Paradwip (b) Vishakhapatnam
   (c) Tuticorin (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (b)

The port of Visakhapatnam is a landlocked harbour as it is surrounded by land and the water passage is towards the ocean or sea. Therefore, this is the correct option. Visakhapatnam port is one of the 13 major ports in India and the only major port in Andhra Pradesh. It is located between the Kolkata and the Chennai ports. It handles a large volume of cargo and is the third largest state owned port in that way.

Miscellaneous

1. **Topographical map of India is prepared by which organization?**
   (a) The Geological Survey of India
   (b) The Survey of India
   (c) The Zoological Survey of India
   (d) More than one of the above
   (e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (b)

The topographical maps are prepared by "The Survey of India". Survey of India, The National Survey and Mapping Organization of the country under the Department of Science & Technology, is the oldest Scientific Department of the Govt. of India.

2. **When was 'Speed Post Service' launched by the Indian Postal Department in competition to the 'Courier service'?**
   (a) 1988 (b) 1987
   (c) 1989 (d) 1986

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The Indian Postal Department started the Speed Post Service in August 1986. This Service provides time-bound and express delivery of letters and parcels between specified stations in India. If the speed post is not delivered within the time then the postal department refunds postal charges to the customers.

3. **Coral reefs are not found in which one of the following regions?**
   (a) Gulf of Cambay
   (b) Gulf of Mannar
   (c) Gulf of Kachchh
   (d) Lakshadweep and Minicoy Island
   (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

Coral reefs in India are found at Gulf of Mannar, Gulf of Kachchh, Lakshadweep and Minicoy Islands, Andaman and Nicobar Islands, Netrani island in Karnataka and Malvan in Maharashtra. But not found in Gulf of Cambay.

4. **Where are more favorable conditions found in India for tidal and wave energy production?**
   (a) Ganga river (b) Gulf of Kutch
   (c) Khambhat bay (d) Mannar bay
   (e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (c)

In India, the Gulf of Khambhat, the Gulf of Kutch in Gujarat on the western coast, and the Gangetic delta in the Sundarban regions of West Bengal provide ideal conditions for utilising tidal energy.

5. **Which one of the following programmes was initiated during the Sixth Five -Year Plan?**
   (a) Integrated Rural Development
   (b) Rural Literacy Development
   (c) Rural Railways
   (d) Advanced Communication Links for Rural People
   (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

During the Sixth Five-Year Plan (1980-85) Integrated Rural Development programme was implemented, and also National Rural Employment Programme (NREP) and Rural Landless Employment Guarantee Programme (RLEGP) etc.

6. **Telephone in India was introduced in –**

(a) 1951 (b) 1981
(c) 1851 (d) 1861

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (*)

In 1850, experimental Electric Telegraph started for the first time in India between Calcutta (Kolkata) and Diamond Harbor. In early 1881, Oriental Telephone Company Limited of England opened telephone exchange at Calcutta (Kolkata), Bombay (Mumbai), Madras (Chennai) and Ahmedabad. On 28th January 1882, the first formal telephone service was established. So, none of the options seems to be correct.

7. **The first hydro power station in India started in -**

(a) Paikra (b) Koyna
(c) Bhakra Nangal (d) Shivasamudram

43rd BPSC (Pre) 1999

Ans. (d)

The first hydroelectric plant in India was established in 1897 AD. It took place at Sidrapong/Sidrbagh near Darjeeling in West Bengal. The second hydro power plant in India came from Shivasamudram (in 1902 AD) in Mysore district of Karnataka. The plant was built to supply electricity to the Kolar gold mines.

8. **Dandakaranya region lies in the State of:**

(a) Odisha (b) Chhattisgarh
(c) Andhra Pradesh (d) Telangana
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (e)

Dandakaranya covers about 89078 square kilometers of land, including regions of Telangana, Maharashtra, Andhra Pradesh, Chhattisgarh and Odisha States.

9. **The historical Dakshinapath region is situated in between**

(a) Vindhyachal range and Krishna river
(b) Vindhyachal range and Cauvery river
(c) Satpura range and Nilgiri
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (d)

Dakshinapatha is traditionally regarded as the land south of Vindhyas. It includes the region between Vindhyachal range and Krishna River, Vindhyanchal range and Cauvery River, etc. Hence, option (d) is the correct answer.

10. **With reference to the population characteristics of India, consider the following statements -**

1. **Growth rate of population in the decade 1991-2001 was about 21%**
2. **The percentage of male and female literacy in 2001 as compared to 1991 was increased.**
3. **According to the 2001 census, there are 35 such big cities in the country, in which about 48% of the total urban population resides.**
4. **According to the census of 1991, it shows a decrease in the number of unproductive consumers.**

Which of these statements are correct?

Code

(a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 2, 3 and 4

47th BPSC (Pre) 2005

Ans. (c)

The growth rate of population in the decade 1991-2001 was 21.54%. The gap between male and female literacy was 24.84% in the year 1991, which decreased to 21.69% in the year 2001. According to the 2001 census, 48% of the urban population does not reside in 35 major cities. The growth rate of population in the decade 2001-2011 was 17.7%.

Literacy (Male and Female)				
Year	Total literacy	Male Literacy	Female Literacy	Difference
1991	52.21%	64.13%	39.29%	24.84%
2001	65.38%	75.85%	54.16%	21.69%
2011	73%	80.9%	64.6%	16.3%

11. **According to the 2011 Census, The percentage of the urban population of India to total population was approximately -**

(a) 21 (b) 31
(c) 36 (d) 40
(e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

According to the data of Census 2011, the urban population is 31.1 percent and the rural population is 68.9 percent of the total population of India.

12. Name the third research center of India in Antarctica -

(a) Bharti (b) Swagatam

(c) Hindustan (d) Maitri

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

'Dakshin Gangotri' is India's first research station in Antarctica. It was established in the year 1983 established its second research center "Maitri" has been operational since 1988. India's third research center is 'Bharati' in Antarctica. The newly constructed Bharati station is operational since March 2012.

13. In which of the following areas coral reef is not found?

(a) Gulf of Cambay

(b) Gulf of Mannar

(c) Gulf of Kutch

(d) Lakshadweep and Minicoy Islands

(e) None of the above/more than one the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

According to 'ESSO–Indian National Centre for Ocean Information Services', the coral reef areas of India are

1. Gulf of Kutch
2. Malvan area
3. Lakshadweep Islands
4. Gulf of Mannar
5. Andaman and Nicobar Islands

Thus option (a) is the correct answer.

14. Which state is the first in India who has made roof top rainwater harvesting structure legally compulsory to all houses?

(a) Haryana (b) Rajasthan

(c) Tamil Nadu (d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (c)

- Rooftop rainwater harvesting structure is the part of the technique through which rainwater is captured from the roof catchments and stored in reservoirs.
- The act of collecting, diverting and storing water from rain events for later reuse is called rainwater harvesting.
- Tamil Nadu is the first and the only state in India which has made roof top rainwater harvesting structure compulsory to all the houses across the state.

WORLD GEOGRAPHY

The Universe

i. General Concept

1. Time taken by the Sun to revolve around the centre of our galaxy is –

(a) 2.5 crores years (b) 10 crores years

(c) 25 crores years (d) 50 crores years

40th B.P.S.C. (Pre) 1995

Ans. (c)

Sun is one of the star of our galaxy (Milky Way). Every star of the galaxy revolves around the galactic centre and the time taken by a star to revolve around this centre depends upon the distance between the galactic centre and the star. Sun is relatively far from the centre, so it takes more time to revolve around. Sun is at 7.94 KPC or 25.896 thousand light years away from the centre and with a speed of 720 thousand Km./hr., it takes 22.5 to 25 crore years, to revolve around the centre of the Milky Way galaxy. This period is called 'Cosmic Year or Galactic Year'.

2. How many constellations are in our space?

(a) 87 (b) 88

(c) 89 (d) 90

44th B.P.S.C. (Pre) 2000

Ans. (b)

According to the International Astronomical Union, there are 88 constellations in the space. Most of these imaginary patterns can be seen from the southern hemisphere. It will take a full year to get a glimpse of all the constellations.

3. The group of stars that indicates the direction of polaris, is

(a) Saptarishi (b) Mrig

(c) Scorpio (d) Taurus

41st B.P.S.C. (Pre) 1996

Ans. (a)

Saptarishi is a group of stars that indicate the direction of the Pole. Hence, option (a) is the correct answer.

4. Which elements are abundant in the formation of interior layer of the earth?

(a) Silica and aluminium (b) Silica and magnesium

(c) Basalt and silica (d) Nickel and ferrum

(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

The elements nickel and ferrum are abundant in the formation of Nife, the inner most (core) layer of the Earth. Nife is abbreviation of Nickel and Ferrum (Iron).

5. **Who among the following postulated the concept of geographical cycle of erosion?**
(a) A. Holmes (b) W. M. Davis
(c) S. W. Wooldridge (d) Kober
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre)-2021

Ans. (b)

W.M. Davis (William Morris Davis) has presented the concept of geological erosion cycle. On the other hand, Walther Penck Rejected the theory of Davis and introduced the concept of Morphological Analysis.

ii. The Solar System

1. **Which of the following does not belong to the solar system?**
(a) Asteroids (b) Comets
(c) Planets (d) Nebula

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The solar system was formed around 4.6 billion years ago. The solar system consists sun (star), cosmic dust or masses attached with each other by gravitational force, planets, dwarf Planets, natural satellite, asteroids, comets, meteoroids and cosmic dust while Nebulae is not a part of Solar System. A nebula is a giant cloud of dust and gases in space.

2. **The scientist who first discovered that the earth revolves around the sun was –**
(a) Newton (b) Dalton
(c) Copernicus (d) Einstein

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Nicolaus Copernicus was an astronomer, scientist, mathematician of Poland. In 1514, Copernicus propounded the Heliocentric Theory of the solar system in his work 'Commentariolus' (Little Commentary). Notably, Indian astronomers Varahamihira propounded the same theory around a thousand years before Copernicus in the sixth century. He mentioned that the moon revolves around the earth and the earth revolves around the sun.

3. **What is the distance between Sun and Earth?**
(a) 93 million miles (b) 103 million miles
(c) 83 million miles (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (a)

As per NASA, the distance from Earth to the Sun is 93 million miles (149 million kilometers) and as per NCERT, the solar output received at the top of the atmosphere varies slightly in a year due to the variations in the distance between the earth and the sun. During its revolution around the sun, the earth is farthest from the sun (152 million km). This position of the earth is called aphelion and when the earth is nearest to the Sun (147 million km), this position is called perihelion.

4. **The distance between the Earth and the sun is greater during**
(a) Aphelion (b) Perihelion
(c) Summer solstice (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (a)

The solar output received at the top of the atmosphere varies slightly in a year due to the variations in the distance between the Earth and the sun. During its revolution around the sun, the Earth is farthest from the sun (Approx. 152 million km) on 4th July. This position of the Earth is called aphelion. On 3rd January, the Earth is the nearest to the sun (Approx. 147 million km). This position is called perihelion. Therefore, the annual insolation received by the Earth on 3rd January is slightly more than the amount received on 4th July.

iii. The Sun

1. **When does solar eclipse occur?**
(a) When the sun comes between the earth and the moon.
(b) When the Earth comes between the Sun and the Moon.
(c) When the Moon comes between the Earth and the Sun
(d) None of the above

47th B.P.S.C. (Pre) 2005

Ans. (c)

A solar eclipse occurs when the Moon comes between Earth and the Sun, and the Moon casts a shadow over the Earth. A solar eclipse can only take place at the phase of New Moon (Amavasya), when it passes directly between the Sun and the Earth and its shadow falls upon the Earth surface.

2. **Which is the chief heavenly body of solar system?**
(a) Earth (b) Jupiter
(c) Saturn (d) Sun
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre)-2021

Ans. (d)

The main celestial body of the solar system is sun. Sun is located at the centre of solar system.

iv. The Mercury

1. **Which of the following planets does not have a satellite?**
(a) Mars (b) Mercury
(c) Neptune (d) Pluto

44th B.P.S.C. (Pre) 2000
42nd B.P.S.C. (Pre) 1997

Ans. (b)

There are only two planets in our solar system having no natural satellite which are - (i) Mercury (ii) Venus. The planet having satellite are as follows –
Earth-1 Mars-2
Jupiter-95 Saturn-146
Uranus-28 Neptune-16
In addition to these, Pluto (Dwarf Planet) has five (5) known satellites.

v. The Venus

1. **The hottest planet of the solar system is –**
(a) Mercury (b) Venus
(c) Mars (d) Earth

41st B.P.S.C. (Pre) 1996

Ans. (b)

Average temperature of all Planet in our Solar System is as follows –

Mercury	=	$167^{o}C$
Venus	=	$464^{o}C$
Earth	=	$15^{o}C$
Mars	=	$-65°C$
Jupiter	=	$-110^{o}C$
Saturn	=	$-140^{o}C$
Uranus	=	$-195^{o}C$
Neptune	=	$-200^{o}C$

Thus, Venus is the hottest planet in our Solar System and Neptune is the coldest planet.

2. **Which planet is considered as the Earth's twin planet due to similarity in shape and size?**
(a) Mars (b) Jupiter
(c) Venus (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Planet Venus is considered as the Earth's twin planet due to the similarity in shape and size and having very similar composition.

3. **Which planet is known as the 'Evening Star'?**
(a) Mars (b) Jupiter
(c) Venus (d) Saturn

44th B.P.S.C. (Pre) 2000

Ans. (c)

Venus is known as the Evening Star and also as the Morning Star while Mars is called the Red Planet and Saturn has lowest density among all planets (lower than water also).

vi. The Earth

1. **Time taken by light emitted from sun to reach earth is –**
(a) 2 minutes (b) 1 minutes
(c) 8 minutes (d) 16 minutes

38th B.P.S.C. (Pre) 1992

Ans. (c)

The radiation from the sun which is popularly known as sunlight is a mixture of electromagnetic waves. Sunlight takes an average of 8 minutes and 19 seconds to travel from the Sun to the Earth. These electromagnetic energy waves travel with a speed of 3.0×10^{18} m/sec. Thus, the closest answer given in option (c) is true.

2. **In completing one revolution of the sun, Earth takes approximately –**
(a) 365 days (b) 365.25 days
(c) 365.5 days (d) 365.75 days

41st B.P.S.C. (Pre) 1996

Ans. (b)

The Earth takes 365.25636 days or 365 days, 5 hours 59 minutes and 9.51 second to complete orbit around the Sun. This period is known as 'Solar Year'. The solar year is also known as The tropical year. In the year 2000, the increase of 5 Minute had been noted in comparison to other years. In this period the time was calculated to 365 days, 6 hours 13 minutes (in place of 9 minutes), 53.26 second (in place of 9.51 second). Such year is known as 'Anomalistic Year'.

3. **Approximately how much distance per minute does earth cover while revolving?**

(a) 49 km (b) 59 km

(c) 69 km (d) 79 km

44th B.P.S.C. (Pre) 2001

Ans. (*)

Earth's equatorial perimeter is about 40,090 Km. This distance is covered by Earth in 24 hours and this indicates that cruising speed of Earth on its axis per hour is 40,090.28 Km/ 24 hour = 1670 Km/ hour (approx). According to this calculation the cruising speed per minute is = 1670 Km/ 60 min = 27.8 Km/ min. Thus, none of the given answers is correct.

Note : In question rotating should be on the place of revolving.

4. **Who among the following was the first to explain that the rotation of the earth on its own axis accounts for the daily rising and setting of the Sun?**

(a) Aryabhata (b) Bhaskara

(c) Brahmagupta (d) Varahamihira

(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

Aryabhata was the first to explain that the rotation of the earth on its own axis accounts for the daily rising and setting of the sun. His works are compiled in Aryasiddhanta. Bhaskara-I was a proponent of Bhedabheda School of Vedanta Philosophy. Varahamihira's notable works include Brihat Sanhita, Pancha Sididhantika.

5. **The possibility of a desert on earth is more**

(a) Nearby 0° latitude (b) Nearby 23° latitude

(c) Nearby 50° latitude (d) Nearby 70° latitude

41st B.P.S.C. (Pre) 1996

Ans. (b)

Desert climate is also Known as arid climate. The perfect climate for the desert is Tropical climate which is spread mostly between 15°-30° between North and South of the Equator. There are four main geographical situations for the formation of desert –

(1) The presence of high air pressure near 30^0 latitude. Sahara and Australia desert fall under this climate.

(2) The western coastal side of continent found between 20º-30º latitude. Here cold oceanic current is reason for desert formation. Arizona desert, Mexican desert, Atacama desert, Namibian desert come under this climate.

(3) Death Valley (North America), Patagonian desert (Argentina) and Peruvian desert are the deserts found in mountains rain shadow region.

(4) The desert situated in the mid part of the Continents, far from the humid air. Gobi desert (China and Mongolia), Australian desert and Great basin desert (USA) come under this group.

6. **The animals found in desert are called**

(a) Deserticolous

(b) Desertimalia

(c) Desertimals

(d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (a)

The term 'deserticolous' refers to organisms that inhabit desert regions. Animals in deserts are exposed to limited quantities of food, low and unpredictable water supplies, high and widely varying temperatures, loose substrates, high incident radiation, and lack of concealment because of scant vegetation.

7. **Geodesy is the science which deals with –**

(a) The dating of terrestrial rock.

(b) Measurement of Earth's dimensions

(c) Measurement of elevation and depression of the earth

(d) Recording of changes made by the crust

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Geodesy is the science of accurately measuring and understanding the three fundamental properties of the Earth. These properties are the geometric shape of the earth, the orientation of the earth in space and the gravitational field of the earth; along with it also studies the changes in the above properties with time. Thus geodesy is the science which deals with the measurement of the dimensions of the earth.

8. **When a bar magnet is placed in the earth's magnetic field, with its north pole pointing towards the geographical north, the number of neutral points around the magnet is**

(a) 2 (b) 4

(c) 6 (d) 5

(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

If the magnet's north pole points to geographical north, the magnet's and earth's fields will point in opposing directions along the equatorial line of the magnet, yielding two neutral spots on the equatorial line that are equidistant from the magnet's axis. Hence option (a) is correct.

vii. The Jupiter

1. Which is the largest planet in the solar system?

(a) Jupiter (b) Neptune
(c) Uranus (d) Saturn

41st B.P.S.C. (Pre) 1996

Ans. (a)

According to size planets of our solar system (in Diameter) are as follows:
Jupiter – (142,984 Km.)
Saturn − (120,536 Km.)
Uranus − (51,118 Km.)
Neptune −(49,528 Km.)
Earth − (12,756 Km.)
Venus − (12,104 Km.)
Mars − (6,792 Km.)
Mercury − (4,879 Km.)

viii. The Saturn

1. Time Saturn takes to complete one revolution around the Sun-

(a) 18.5 years (b) 36 years
(c) 29.5 years (d) 84 years

44th B.P.S.C. (Pre) 2000

Ans. (c)

Saturn takes near about 29.456 Earth's years to complete one revolution around the Sun. Hence, option (c) is the correct answer.

2. Which of the following planets has the least density?

(a) Earth (b) Mars
(c) Venus (d) Saturn
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

Planet	**Density (kg/m^3)**
Earth	5514
Mars	3934
Venus	5243
Saturn	687

Hence Saturn is the correct answer.

ix. The Uranus, Neptune and Pluto

1. For one revolution around the Sun, Uranus takes –

(a) 84 years (b) 36 years
(c) 18 years (d) 48 years

44th B.P.S.C. (Pre) 2000

Ans. (a)

Uranus's period of revolution around the Sun is nearabout 84 Earth's years. A day on Uranus is approximately 17 hours and 14 minutes.

2. The year is largest on –

(a) Pluto (b) Jupiter
(c) Neptune (d) Earth

39th B.P.S.C. (Pre) 1994

Ans. (a)

The year is largest on Pluto because it is farthest among the other celestial objects from the Sun. Pluto takes the period of 247.9 earth year to complete one revolution around the Sun.

3. The theory of continental drift was developed by?

(a) J.J. Wilson (b) A. Wegener
(c) Du Toit (d) H. Hess
(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Continental drift theory was developed by Alfred Wegener. According to Wegener, the division of Pangea began about 225–200 million years ago. Pangea first split into two large continental bodies, Laurasia and Gondwanaland, as the northern and southern part, respectively. After this, Laurasia and Gondwanaland gradually split into many smaller parts, which form today's continent.

The Earth

i. Latitudes

1. The country which has the longest north-south (latitudinal) extension of its territory is :

(a) Russia (b) Chile
(c) China (d) Brazil
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Chile has the longest north-south (latitudinal) extension of its territory between approx. 17^{o} South and 56^{o} south.

2. **The Tropic of Cancer covers the maximum area of**
   (a) Ranchi Plateau (b) Malwa Plateau
   (c) Chota Nagpur Plateau (d) More than one of the above
   (e) None of the above

B.P.S.C. School Teacher (Class 6-8) 09-12-2023

Ans. (b)

The maximum length of the tropic of cancer is in Madhya Pradesh. Malwa plateau comprises maximum area of Madhya Pradesh that why Tropic of Cancer covers maximum area of Malwa Plateau.

ii. The Longitude

1. **What would be the time at 30° East of Greenwich, if it is 12 noon at Greenwich?**
   (a) 2 p.m. (b) 1 p.m.
   (c) 10 a.m. (d) More than one of the above
   (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

$\boxed{24\,\text{hour} = 360^0}$

1 hr = 15^0

$\boxed{2\,\text{hour} = 30^0}$

Given that
The place is located at 30^0 East of Greenwich i.e. 2 hour or 120 minutes ahead of GMT.

So, $\boxed{\text{The time at } 30^0 \text{ east of greenwich}}$, if it is 12 noon at greenwich, then is 2 p.m.

2. **The time at Cairo is 2 hours ahead of Greenwich. Hence, it is located at-**
   (a) 30° W longitude (b) 30° E longitude
   (c) 28° E longitude (d) 28° W longitude

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Cairo (Egypt) is located at $30^0\,3'$ N latitudes and $31^0\,14'$ E longitude. Time of Cairo is 2 hours ahead of Greenwich. Therefore its position approximately falls near the 30^0 East longitude. The Earth being spherical in shape rotates on its axis by 360^0 in 24 hours. Therefore, its movement for 1^0 of longitude takes the time of 4 minutes. So, it can be easily calculated that in 120 minutes (2 hours) angle for longitude will be 30^0 , Therefore, option (b) should be the correct answer.

3. **The basis of deciding the standard time of any place is–**
   (a) Longitude (b) Latitude
   (c) International Date Line (d) Prime Meridian

43rd B.P.S.C. (Pre) 1999

Ans. (d)

Prime Meridian line is also known as International Meridian or Greenwich meridian. This line passes through England's Royal Greenwich Observatory. Line situated at Zero degree longitude determines the International Standard Time. 180^0 longitude is called as International Date Line. Prime Meridian (0^0 longitude) passes through the following countries –
(1) United Kingdom, (2) France, (3) Spain, (4) Algeria, (5) Mali, (6) Burkina-Faso, (7) Togo, (8) Ghana.

4. **International Date Line passes through –**
   (a) Africa (b) Asia
   (c) Pacific ocean (d) Atlantic ocean

44th B.P.S.C. (Pre) 2000

Ans. (c)

International date line is an imaginary line on the Earth which separates two consecutive days on the calendar. This line passes through 180^0 opposite to Greenwich (England) from the Pacific Ocean. This line turns when any land area falls in the way of this line.

iii. Tropic of Cancer

1. **Which of the following cities is located in the North of Tropic of Cancer?**
   (a) Bhopal (b) Aizawl
   (c) Ranchi (d) More than one of the above
   (e) None of the above

B.P.S.C. Headmaster 07-12-2023

Ans. (b)

Tropic of Cancer is a latitude located at 23.50 degrees. Tropic of Cancer passes through 8 Indian states including–Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram (Aizawl).

iv. The Tropic of Capricorn

Day & Night

1. **The shortest day of the year in Northern hemisphere is on –**
   (a) 21 December (b) 22 December
   (c) 21 June (d) 22 June

47th B.P.S.C. (Pre) 2005

Ans. (b)

The shortest day of the year in Northern hemisphere occurs during Winter solstice on 21/22 December. Contrary to it on 20/21 June longest day occurs in Northern hemisphere during the Summer solstice. Opposite to it, in Southern hemisphere, the shortest day occurs on 20/21 June and longest day on 21/22 December.

2. The Longest day in southern Hemisphere is

(a) 22 June (b) 22 December
(c) 21 March (d) 22 September

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (b)

See the explanation of the above question.

3. In which hemisphere, roaring forties, furious fifties and shrieking sixties are blowing?

(a) Northern Hemisphere (b) Southern Hemisphere
(c) Eastern Hemisphere (d) Western Hemisphere
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre)-2021

Ans. (b)

Roaring forties, furious fifties and shrieking sixties winds flow in the Southern hemisphere of the earth. They were given these names on the basis of the extensity of the Westerly winds nearby these latitude in the Southern hemisphere.

v. The Geological History

1. The southern continent broken from Pangaea is called

(a) Pacific Ocean (b) Laurasia
(c) Gondwanaland (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

Alfred Wegener a German Meteorologist was from "the continental drift theory" in 1912. This was regarding the distribution of the oceans and the continents. According to wegener, all the continents formed a single continental mass and mega ocean surrounded the same. The supercontinent was named PANGAEA, which meant all earth. The Mega Ocean was called PANTHALASSA, meaning all water. He argued that, around 200 million year ago, the super continent, Pangaea, began to split, Pangaea first broken into two large continents masses as Laurasia and Gondwanaland forming the northern and Southern components respectively. Laurasia and Gondwanaland continued to break into various smaller continents that exist today.

2. Continents have drifted apart because of-

(a) Volcanic eruptions
(b) Tectonic activities
(c) Folding and faulting of rocks
(d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Continents rest on massive slabs of rocks called tectonic plates, these plates are always moving and interacting in a process called Plate Tectonic Activity. Through these tectonic activities, continents shift position on Earth's Lithosphere.

3. Cocos Plate the between:

(a) Central America and Pacific Plate
(b) South America and Pacific Plate
(c) Red Sea and Persian Gulf
(d) Asiatic Plate and Pacific Plate
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (a)

In Carboniferous period, the whole land mass on the Earth was unified and was named as Pangea. This Pangea Supercontinent was surrounded by a superocean which was named 'Panthalassa' by Wegener. Due to the process of continental drift, the Pangea split into a northern most and southern most supercontinent. The northernmost landmass was known as Laurasia. The southernmost supercontinent was known as Gondwana land which split into South-America, Australia, Africa, Madagascar, Antarctica and Peninsular India. Cocos plate is a small lithospheric plate which lies between Central America and Pacific plate.

4. One astronomical unit is the average distance between

(a) the Earth and the Sun
(b) the Earth and the Moon
(c) the Jupiter and the Sun
(d) the Pluto and the Sun
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (a)

An Astronomical unit (AU) is the average distance between Earth and the Sun, which is about 93 million miles or 150 million kilometer. Astronomical units are usually used to measure distance within our Solar System.

5. 'Rust Bowl' of the USA is associated with which one of the following regions?

(a) Great Lakes region (b) Alabama region

(c) California region (d) Pittsburg region
(e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (e)

Rust Bowl or Rust belt, a geographic region that was formerly a manufacturing or industrial powerhouse but now in deep decline phase as manufacturing hubs were moved overseas, the Great lake region and Pittsburg region comes under it.

6. Folding is the result of-

(a) Epeirogenic force (b) Coriolis force
(c) Orogenic force (d) Exogenic force

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

The Earth's surface experiences different types of forces. The orogenic force takes millions of years to build a mountain from plain and sea bed. These forces due to the interaction between tectonic plates crumpled and pushed upward to form mountain range. Thus folding is the result of orogenic force.

7. An effective Coriolis force results in

(a) Solar System
(b) Earth rotation
(c) Interior of the earth
(d) Colorado and Gulf Stream
(e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

The direction of the winds blowing on the surface is determined by the air pressure and the rotation of the earth. The deflection in the direction of the winds is due to the repulsive force generated by the axial motion of the earth. Due to the discovery of this force by G.G. Coriolis, it was named Coriolis force. Thus one of the given options is the cause of the effective Coriolis force due to the rotation of the earth.

8. Which of the following statements is true about troposphere?

(a) Its average height is 13 km.
(b) It is the topmost layer of the atmosphere.
(c) The temperature at this layer increases with the height.
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

The height of the troposphere varies from place to place and from season to season the height of the trophosphere at the pole is about 6–8 km. while at the equator it is about 16–18 km. its average height is approximetly 13 km.

9. Ocean thermal energy is produced due to

(a) pressure difference at different levels in the ocean
(b) temperature difference at different levels in the ocean
(c) energy stored in waves of the ocean
(d) tides rising out of the ocean
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

Ocean thermal energy is produced due to temperature difference at different levels is the ocean. Energy from the sun heats the surface water of the ocean. In tropical regions, surface water can be much warmer than deep water. This temperature difference can be used to produce electricity and to desalinate ocean water. This whole process is known as Ocean Thermal Energy Conversion (OTEC).

10. The theory that states "pieces of the Earth's crust are in constant, slow motion driven by movement in the mantle" is called

(a) the theory of continental drift
(b) the theory of Pangaea
(c) the theory of plate tectonics
(d) the theory of plate boundaries

69th B.P.S.C. (Pre) 2023

Ans. (c)

The theory of plate tectonics states-pieces of the Earth's crust are in constant, slow motion driven by movement in the mantle. According to this theory, the earth surface is composed of several plates. There are seven major plates– Eurasian, African, Indo-Australian, Pacific, North American, South American and Antarctic plates. Significantly plates constitute the entire surface of the earth.

11. The process that continually adds new crust is

(a) subduction
(b) earthquake
(c) seafloor spreading
(d) convection

69th B.P.S.C. (Pre) 2023

Ans. (c)

The process that continually add new crust is seafloor spreading. Significantly, new crust is created by seafloor spreading. Old crust is destroyed by subduction. Seafloors spreading are only an aspect of plate tectonic, another is subduction.

Rocks

1. **The formation of 'mushroom rock' in desert region is an example of**
   (a) abrasion (b) erosion
   (c) deflation (d) attrition
   (e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (a)

The rocks having broad upper part and narrow base resembling an umbrella or mushroom are called mushroom rock or pedestal rock. These undercut, mushroom-shaped rock are formed due to abrasion works of wind.

2. **Regarding a sedimentary rocks which of the following statements is true?**
   (a) These are such rocks whose structure depend on temperature and pressure
   (b) These rocks are crystal
   (c) These rocks are deposited in layers
   (d) These rocks cannot be formed in water

39th B.P.S.C. (Pre) 1994

Ans. (c)

Sedimentary rocks are formed by the accumulation of sediments. This sediment or debris accumulates in low lying areas like-lakes, Oceans. These sedimentary materials may be formed from eroded fragments of pre-existing rocks or even from the remains of plants or animals and fossils, most frequently found in sedimentary rocks, which comes in layers, called Strata. Thus all of the given statements are true related to sedimentary rocks.

3. **Which statement of the following is true for igneous rocks?**
   (a) These have little fossils
   (b) They have porous for water
   (c) They are both crystal and non-crystal
   (d) These rocks have no silicas

40th B.P.S.C. (Pre) 1995

Ans. (c)

The origin of igneous rocks is mainly related to the process of a volcanic eruption. These types of rocks are formed by the cooling and solidification of material called Magma or Lava. Igneous rocks have both, crystalline and non-crystalline structure and thus Statement given in option (c) is true but all other given statements are wrong because these rocks are composed of silicate minerals, fossils and layers are not found in these types of rocks and also water supply through igneous rocks is very low.

4. **Minerals are deposited and accumulated in the strata of which of the following rocks?**
   (a) Sedimentary rocks (b) Metamorphic rocks
   (c) Igneous rocks (d) More than one of the above
   (e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (a)

Minerals are deposited and accumulated in the strata of sedimentary rocks. In sedimentary rocks a number of minerals occur in beds or layers. They have been formed as a result of deposition, accumulation and concentration in horizontal strata. Coal and some forms of iron ore have been concentrated as a result of long periods under great heat and pressure. Another group of sedimentary minerals include gypsum, potash salt and sodium salt.

5. **What are the last signs of the activities of a volcano?**
   (a) Geysers (b) Craters
   (c) Famaroles (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (c)

Fumaroles are the last sign of the activities of a volcano. These are openings in the Earth's surface that emit steam and volcanic gases, such as sulfur dioxide and carbon dioxide. They can occur as holes, cracks, or fissures near active volcanoes or in areas where magma has risen into the earth's crust without erupting.
Fumaroles can vent for centuries or quickly go extinct, depending on the longevity of its heat source.

The Volcano

1. **Which of the following features is the product of vulcanicity?**
   (a) Geosyncline (b) Escarpment
   (c) Atoll (d) Fold mountain
   (e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (e)

Heightened volcanic activity encourage volcano tourist to explore volcanic feature such as Strombolian eruption, lava flows, geysers and hot spring, lava lakes, crater lakes, fumaroles, boiling mud ponds, hot river and travertine terraces created by volcanic hot Springs and their mineral deposits.

2. **Mt. Etna is –**

(a) A mountain (b) A mountain peak
(c) A volcano (d) A plateau

43rd B.P.S.C. (Pre) 1994

Ans. (c)

Mount Etna is an active volcano on the East-coast of Sicily, Italy. Its maximum height is about 3357 meter.

3. **Mauna Loa is an example of**

(a) Active volcano
(b) Dormant volcano
(c) Dead volcano
(d) Plateau in the volcano region

39th B.P.S.C. (Pre) 1994

Ans. (a)

In the Hawaiian language, Mauna Loa means the 'Long Mountain'. It is one of the active volcanoes. Mauna Loa had its first well documented eruption in 1843. It is spread over half of the Hawaii Island (USA).

4. **Kilimanjaro is a –**

(a) Volcano (b) Island
(c) Peak (d) River

44th B.P.S.C. (Pre) 2000

Ans. (a)

Kilimanjaro is the highest mountain in Africa. It is situated in the North-Eastern part of Tanzania. It has three main volcanic cone–Kibo, Mawenzi and Shira. Highest peak of Mt. Kibo is Uhuru whose name was Kaiser-Wilhelm-Spitze before it is an igneous mountain which forms igneous rocks through erupted lava.

The Earthquake

1. **Which one of the following statements about the 2004 Indian Ocean earthquake and the resulting Tsunami is not correct?**

(a) The earthquake originated due to slipping of about 1,200 km of fault line by 15 m along the subduction zone where the India Plate subducts the Burma plate at the Sunda Trench.
(b) The resulting Tsunami devastated the shores ranging from the coast of Indonesia to the east coast of Africa, some 8,500 km away from the epicentre.
(c) As per the current estimates, the quake-generated Tsunami killed more than 50 lakh people, in addition to unaccounted dead bodies swept out to sea.
(d) Bangladesh had very few casualties because the quake-affected fault line was in a nearly north-south orientation, the greatest strength of the tsunami waves was in an east-west direction.

47th B.P.S.C. (Pre) 2005

Ans. (c)

Under sea earthquake in the Indian Ocean on 26 December, 2004 produced a Tsunami that caused huge natural disaster and more than 2-3 lakh people lost their lives and missing. The earthquake took place in the Indian Ocean off the western coast of Sumatra island of Indonesia. In regards to the options the details of Tsunami is as follows :
Statement (a) – An estimated 1300 Km. of faultline slipped about 15 metre. The earthquake occurred on the interface of the India and Burma tectonic plates. Thus statement (a) is almost right.
Statement (b) – The waves devasted on the shores of Indonesia, Sri Lanka, India as far as Somalia on the east coast of Africa. Hence This statement is also nearly true.
Statement (c) – Approx 2-3 Lakhs people were estimated to have lost their lives or missing people. Thus statement (c) is wrong.
Statement (d) – Bangladesh, which lies at the northern end of the Bay of Bengal had very few casualties. So, statement (d) is true.

2. **The Dilatancy theory explains**

(a) prediction of earthquakes (b) intensity of fire
(c) gathering of áir pressure (d) origin of earthquakes

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (a)

Dilatancy is the property of soil material that refers to a change in its volume in response to shearing under a normal or confining stress. The Dilatancy theory explains the prediction of earthquakes because when the stress on the edge overcomes the friction, there is an earthquake that releases energy in waves that travel through the earth's crust and cause the shaking that we feel.

3. **Which one of the following earthquakes is very severe and disastrous?**

(a) Volcanic earthquake
(b) Tectonic earthquake
(c) Plutonic earthquake
(d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (b)

Tectonic earthquakes are indeed the most significant in terms of their potential for widespread damage. They occur due to the movement of the Earth's tectonic plates, which can cause intense shaking over large areas. This can lead to severe consequences for communities, infrastructure, and the environment.

In contrast, plutonic and volcanic earthquakes are generally more localized. Plutonic earthquakes are associated with the movement of magma beneath the Earth's surface, while volcanic earthquakes are linked to volcanic activity. These tend to impact smaller regions, although they can still be dangerous, especially if they are tied to volcanic eruptions or other geological hazards.

The Continents

1. The Arctic region and the Antarctica continent are situated near

(a) the Amazon Basin

(b) the Sahara Desert

(c) the North and South Poles

(d) More than one of the above

(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

The Arctic is a polar region located in the northernmost part of Earth. The Antarctica is Earth Southernmost continent it contains the geographic South Pole and is situated in the Antarctic region of the Southern Hemisphere.

2. Among the following continents, which one has the highest number of countries?

(a) Europe (b) Asia

(c) Africa (d) North America

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

The number of countries in the following continents is as follows :

Africa54

Europe	—	49
Asia	—	48
North America	—	23
Oceania	—	14
South America	—	12

3. As per area which of the following is the largest continent?

(a) Europe (b) Africa

(c) North America (d) South America

39th B.P.S.C. (Pre) 1994

Ans. (b)

There are total of 7 continents in the world. Asia, Africa, North America, South America, Antarctica, Europe and Australia. Asia is the largest continent having an area of 44,568,500 Sq.km.

The area of other continents is as follows–

Africa	-	30,065,000 sq. Km.
North America	-	24,473,000 sq. Km.
South America	-	17,819,000 sq. Km.
Antarctica	-	14,200,000 sq. Km
Europe	-	9,948,000 sq. Km.
Australia	-	7,741,220 sq. Km.

The World Mountain

1. The Kota Kinabalu is the highest peak in

(a) Myanmar (b) Indonesia

(c) Thailand (d) Malaysia

(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (d)

The Kota Kinabalu, also known as Mount Kinabalu is the highest peak in Southeast Asia. It is located in the Malaysian state of Sabah on the Island of Borneo.

2. Highest peaks of the world are mostly found in which type of Mountains?

(a) Old folded mountains

(b) Young folded mountains

(c) Residual mountains

(d) Block mountains

45th B.P.S.C. (Pre) 2001

Ans. (b)

World's many highest peaks including the highest peak Mount Everest are situated in the Himalayas. Himalaya is a young folded mountain. Thus, it is clear that the world's highest peaks are situated in the young folded mountains.

3. The largest mountain series of the World is –

(a) Himalaya (b) Andes

(c) Rockies (d) Alps

41st B.P.S.C. (Pre) 1996

Ans. (b)

The Andes is the longest mountain range in the world, located along the entire western coast of South America. The Andes mountain range is 7000 km. to 8900 km. long. Rocky mountain is the major mountain system of North America extending over 4800 km. The Himalaya is a prominent mountain range in South Asia situated at the northern end of India and also located in other neighbour South Asian nation having the length of approximately 2500 km.

4. A mountain range of Europe is –

(a) Alps (b) Himalaya
(c) Andes (d) Rocky

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (a)

The Alps is a mountain range located in Europe immediately North of the Mediterranean Sea. This range is extended in mainly France, Austria, Germany and Switzerland. The highest peak of Alps Mountain Range is Mont Blanc (4,807 meter high). The Alps mountain is an example of young folded mountain in the world. The Alps Mountain range is not a part of England.

5. Which of the following areas has the lower snow line?

(a) Equatorial region (b) Himalayan region
(c) Alps region (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 09-12-2023

Ans. (c)

Snow line is an Umbrella term for different interpretations of the boundary between snow covered surface and snow free surface. Beyond the tropics, the snow line becomes progressively lower as the latitude increases, to just below 3000 meters in the Alps and falling all the way to sea level itself at the ice caps near the poles.

6. The Pennines (Europe), the Appalachians (America) and the Aravalli (India) are examples of

(a) Young mountains (b) Old mountains
(c) Block mountains (d) Fold mountains

56th to 59th B.P.S.C. (Pre) 2015

Ans. (b)

Pennines (Europe), the Appalachians (North America) and the Aravalli (India) are examples of an old mountain. Old fold mountains are characterised by having stopped growing higher due to the cessation of upward thrust caused by stopping of movement of tectonic plates in the Earth crust.

7. Pyrenees Mountain Range in separate the countries

(a) Spain and France
(b) Spain and Portugal
(c) France and Germany
(d) Germany and Switzerland
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (a)

The Pyrenees is a chain of mountains that forms a natural boundary between France and Spain. They extend from the Bay of Biscay to the Mediterranean Sea.

The Plateaus

1. It is known as 'Roof of the World'?

(a) Aravali (b) Satpura
(c) Pamir (d) Myanmar

44th B.P.S.C. (Pre) 2000

Ans. (c)

The Pamir Plateau or the "Pamir" is called the roof of the world because of its high altitude. The southern border of Central Asia is determined by Pamir Plateau. It is a convergence of many mountains.

The Arid Regions/ Deserts

1. Match List-I with List-11 and select the correct answer using the codes given below the Lists:

List-I	List-II
A. Dasht-e Lut	**1. USA**
B. Nubian Desert	**2. Iran**
C. Mojave Desert	**3. Sudan**
D. Rub-al-Khali	**4. Saudi Arabia**

Codes :

	A	B	C	D
(a)	2	3	4	1
(b)	4	3	1	2
(c)	2	3	1	4
(c)	4	1	3	2

(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (c)

Correctly matched Deserts and their respective countries–

(a)	Dasht-e Lut	—	2. Iran
(b)	Nubian Desert	—	3. Sudan
(c)	Mojave Desert	—	1. USA
(d)	Rub-al-khali	—	4. Saudi Arabia

Hence, option (c) will be the right answer.

2. The Desert 'Takla Makan' is located in the –

(a) Kazakhstan (b) Turkmenistan
(c) Uzbekistan (d) China

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

Takla Makan Desert is situated in Xinjiang province of China.

3. On which coast is the St. Petersburg city in Russia located?

(a) Black Sea (b) Caspian Sea
(c) Baltic Sea (d) North Sea
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (c)

St. Petersburg, city and port, extreme north - western Russia. It is Situated on the River Neva, at the head of the Gulf of Finland on the Baltic Sea.

4. In which country is the Gobi desert located?

(a) Mexico (b) Somalia
(c) Mongolia (d) Egypt

38th B.P.S.C. (Pre) 1992

Ans. (c)

Gobi Desert stretches into northern China and southern Mongolia. This desert is surrounded by Alashan Tienshan and Altai mountain.

5. The Great Victorian Desert is located at –

(a) Australia (b) India
(c) Egypt (d) North Africa
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

The Great Victorian desert is located in South Australia and Western Australia. It is the largest desert in Australia. It is spread over 348,750 sq. km of area.

6. Which of the following features has been produced by wind deflation?

(a) Lake Chad (b) Qattara Depression
(c) River Nile (d) Lake Tube

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (b)

Qattara Depression is a depression in northwestern Egypt. It lies below sea level, and its bottom is covered with salt pans, sand dunes and salt marshes. It is produced by wind deflation and by the interplay of salt weathering and wind erosion.

The Grasslands

1. The Sargasso Sea is a part of the

(a) Indian Ocean (b) Arctic Ocean
(c) North Atlantic Ocean (d) South Atlantic Ocean
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

The Sargasso Sea is a water region in the North Atlantic Ocean. In this region Sargassum (seaweed/algae) is found in abundance, due to which it was named Sargasso.

2. Grasslands of Brazilian Plateau are known as

(a) Pampas (b) Campos
(c) Velds (d) Downs
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (b)

Grassland of Brazilian plateau are known as Campos.
Other -
Argentina - Pampas
South Africa - Veld
Australia - Down

3. What is the name of mid-latitude grassland in South America?

(a) Prairie (b) Pampas
(c) Veld (d) Steppes
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Mid-latitudes or temperate belt grasslands of South America are known as Pampas. It is spread over mostly in Argentina. Uruguay and southern parts of Brazil. Other mid-latitude grasslands are Prairie (North America), downs (Australia), etc.

Countries and their Borders

1. Which country has the largest coastline?

(a) U.S.A.
(b) Australia
(c) Canada
(d) India

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

The longest coastline in the world is of Canada with (202,080 km) followed by Indonesia (54,716 km), Greenland (Denmark) (44,087 km), Russia (37,653 km) and Philippines (36,289 km).

2. **Which one of the following countries is the largest country without borders in terms of geographical area?**
   (a) New Zealand (b) Philippines
   (c) Japan (d) Cuba
   (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Country		**Area**
New Zealand	-	2,68,838 sq. km
Philippines	-	3,00,000 sq. km
Japan	-	377915 sq. km
Cuba	-	110,860 sq. km

3. **Israel has common borders with:**
   (a) Lebanon, Syria, Jordan and Egypt
   (b) Lebanon, Syria, Turkey and Jordan
   (c) Cyprus, Turkey, Jordan and Egypt
   (d) Turkey, Syria, Iraq and Yemen

47th B.P.S.C. (Pre) 2005

Ans. (a)

Israel is a Mediterranean coastal country in West Asia of Middle East. It shares its western border with Mediterranean Sea, the northern border with Lebanon, the north-eastern border with Syria, eastern border with Jordan and the south-western border with Egypt.

4. **Which of the following countries does not have a land border with the Dead Sea?**
   (a) Lebanon (b) Jordan
   (c) Israel (d) Palestine
   (e) None of the above / More than one of the above

67 B.P.S.C. (Pre) 2022

Ans. (a)

Among the given options, Lebanon does not have a land border with the Dead Sea. The Dead Sea extends into Jordan, Israel and the West Bank region. According to the official map of the Government of Palestine, the West Bank region comprises most of the territory of Palestine. According to official website of Dead Sea. The Dead Sea is a salt lake located in the Judean desert of Southern Israel, bordered by Jordan to East.

The Land-Locked Countries

1. **Hyderabad city is located in which of the following countries?**
   (a) Afghanistan (b) India
   (c) Pakistan (d) Turkmenistan
   (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (e)

Hyderabad is the name of the city exists in both the countries– India and Pakistan. Hyderabad (in India)is the capital and largest city of the Indian state of Telangana. It lies on the Deccan Plateau along the banks of the Musi River, in the northern part of Southern India. This city is known as the City of Pearls, as it had once flourished as a global center for trade of rare diamonds, emeralds as well as natural pearls. Hyderabad (in Pakistan), city, south-central Sind province, southeastern Pakistan. It lies on the most northerly hill of the GanjoTakkar ridge, just east of the Indus River. It is 2nd biggest City of Sindh and 5th largest in Pakistan. In olden days Hyderabad City name was "NeroonKot" by Ghulam Shah Kalhora.

2. **Afghanistan shares its longest international boundary with**
   (a) Iran (b) Pakistan
   (c) China (d) Turkmenistan
   (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

Afghanistan is a landlocked country situated at the hub of South Asia and Central Asia. Afghanistan shares its land boundary with seven countries in which it shares its maximum border with Pakistan .

3. **Which one of the following is a landlocked country?**
   (a) Belgium (b) Hungary
   (c) Romania (d) Ukraine
   (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

Land-locked Countries are those which do not have a coastline. There are total 45 land-locked countries in the world which are as follows:

Afghanistan	Andorra	Armenia
Austria	Azerbaijan	Belarus
Bhutan	Bolivia	Botswana
Burkina Faso	Burundi	Central African Republic

Chad	Check Republic	Ethiopia
Hungary	Kazakhstan	Kirgizstan
Kosovo	Laos	Lesotho
Lichtenstein	Luxembourg	North Macedonia
Malawi	Mali	Moldova
Mongolia	Nepal	Niger
Paraguay	Rwanda	San Marino
South Sudan	Serbia	Slovakia
Eswatini	Switzerland	Tajikistan
Turkmenistan	Uganda	Uzbekistan
Vatican City	Zambia	Zimbabwe

Thus, Liberia is not a land-locked country.

Capital Cities of the World

1. Which of the following is not a capital city of a country?

(a) Canberra (b) Sydney
(c) Wellington (d) Riyadh

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

In the above options, Sydney is not a capital city of any country rather it is a port city of Australia. Canberra is the capital of Australia, Wellington is the capital of New Zealand and Riyadh is the capital of Saudi Arabia.

2. Which one of the following cities experiences an early sunset?

(a) Sydney (b) Hong Kong
(c) Tokyo (d) Singapore
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (a)

The Timing of sunset is influenced by several factors, including geographical location, time of year and local time zone. Sydney, Australia, being located in the Southern Hemisphere, experience earlier sunsets compared to cities located farther east .

3. Which one of the following statements is not correct?

(a) The capital of Myanmar is Naypyidaw.
(b) Maximum people of Fulani tribe are in Nigeria.
(c) 'Korosi' volcano is situated in USA.
(d) Jarvis, Baker are the examples of coral islands.
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (c)

The correct given statements–

- The capital of Myanmar is Naypyidaw.
- The Fulani, also known as Fula or Fulbe, are widely distributed across several West African Countries. Nigeria has the longest Fulani population.
- Jarvis, Baker are the examples of coral Islands.
- Korosi Volcano, also known as Mount Korosi, is situated in Kenya.

Hence, option (c) will be the right answer.

Geographical Sobriquets

1. Which of the following is known as 'Mistress of Eastern Sea'?

(a) Sri Lanka (b) Pakistan
(c) Burma (d) India

38th B.P.S.C. (Pre) 1994

Ans. (a)

'The Mistress of Eastern Sea' title is given to Sri Lanka. Sri Lanka is an island situated in the Indian Ocean, at the base of the Indian Sub-Continent, 880 km. North to the Equator. It is also known as 'The Pearl of Indian Ocean'.

2. Name the city of Russia that is known as 'Venice of North'.

(a) Vladivostok (b) St. Petersburg
(c) Novosibirsk (d) Moscow
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (b)

Saint Petersburg is a city in Russia which is located on the Neva River. St. Petersburg is one of a group of 7 cities which are often called, "The Venice of the North"; it shares this name with Amsterdam, Bruges, Copenhagen, Hamburg, Manchester and Stockholm.

3. On the basis of fossils which is the origin place of man?

(a) Rift Valley of Africa
(b) Central Asia
(c) Jerusalem
(d) More than one of the above
(e) None of the above

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Ans. (a)

Based on fossil evidence, the origin of humans is Africa. Fossils of early humans who lived between 6 and 2 million years ago are all from Africa.

The Hydrosphere

1. **The surface of the earth covered with water is approximately?**
   (a) One-fourth (b) Half
   (c) Two-thirds (d) Three-fifth (3/5)

41st B.P.S.C. (Pre) 1996

Ans. (c)

Around 29% (29.1%) of the total surface Area of the Earth is land (about 148172000 sq. km) and 71% part (70.9%) is covered by Ocean. Earth's surface covers with water in both liquid and frozen form. Hence, the surface of earth covered with water is approximately more than two-thirds of total surface area.

2. **On the sea level, nearest place to the center of earth is?**
   (a) North pole (b) Tropic of Capricorn
   (c) Tropic of Cancer (d) Tropic of Equator

38th B.P.S.C. (Pre) 1992

Ans. (a)

The nearest point to the centre of the Earth lie near Arctic Ocean (at the depth of 4 km from the surface) which is at the 6353 km far from the centre of the Earth.

3. **A ridge of 64000 km. length and 2000 km to 2400 km width passing through the north and south Atlantic oceanic basin enter into the south pacific oceanic basin through Indian oceanic and then from the middle of Australia and Antarctic. This ridge is –**
   (a) Socotra-Lakshadweep - Chagos-Ridge
   (b) Pacific - Antarctica Ridge
   (c) Dolfin - Challenger Ridge
   (d) Mid - Oceanic Ridge

47th B.P.S.C. (Pre) 2005

Ans. (d)

Above qualities are related to Mid Oceanic Ridge. Mid Oceanic Ridge encircles the globe in the length of approximately 40390 miles or 65000 kilometer, covering Arctic, Atlantic, Indian and the Pacific Oceans. This is the single largest geological phenomena upon the surface of Earth.

4. **Where is 'Ninety East Ridge' situated?**
   (a) Pacific Ocean (b) Indian Ocean
   (c) Atlantic Ocean (d) Arctic Ocean

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Ninety East Ridge (90° E Ridge) is the name of a ridge located in the Indian Ocean. Located near 90° east longitude (meridian), this ridge is 5000 km long parallel to it and stretched across 34° south latitude to 10° northern latitude with an average width of approx 200 km.

The Ocean Currents

1. **Gulf Stream is –**
   (a) a river in the Gulf
   (b) an oceanic current
   (c) another name of Jet Stream
   (d) a surface wind

44th B.P.S.C. (Pre) 2000
42nd B.P.S.C. (Pre) 1997

Ans. (b)

Gulf stream is a powerful warm current in the North Atlantic Ocean. It originates at the Gulf of Mexico as the Florida current which merges in Antilles current and flows through the Cape Hatteras and then flows along the east coast of United States of America. Spanish explorer Juan Ponce de Leon first discovered the Gulf Stream in 1513.

2. **Which one of the following is not an area of Mediterranean type of climate?**
   (a) Central California (b) Northern New Zealand
   (c) Central Chile (d) Southern tip of Africa
   (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

Northern New Zealand does not have a mediterranean type of climate because it experience a different climate pattern known as a temperate maritime climate. This climate is influenced by the surrounding ocean, which moderates temperatures and leads to relatively mild winters and summers without the distinct dry summers characteristic of Mediterranean Climates.

3. **Cold water current flowing along Namibian Coast is known as**
   (a) Agulhas Current
   (b) Benguela Current
   (c) Humboldt Current
   (d) Kuroshio Current
   (e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (b)

Flowing along the coast of South Africa, Namibia and Angola, the Benguela current is the eastern boundary of a large gyre in the South Atlantic Ocean. The current mixes water from the Atlantic and Indian Oceans as they meet off the capes of South Africa.

4. **Where does the El Niño current flow?**
   (a) The Pacific Ocean
   (b) The Indian Ocean
   (c) The Bay of Bengal
   (d) More than one of the above

69th B.P.S.C. (Pre) 2023

Ans. (a)

According the observatory of NASA, the EI Nino is evident in the east ward flowing wind in the tropical western and central pacific EI Nino means little boy in Spanish. South American fisherman first noticed periods of unusually warm water in the Pacific Ocean.

5. **Which of the following currents is not a warm current?**
   (a) Alaska Current (b) Mozambique Current
   (c) Antilles Current (d) None of the above

BPSC Agriculture Department-2024

Ans. (d)

Warm currents bring warm water into cold water areas and are usually observed on the east coast of continents in the low and middle latitudes (true in both hemispheres). In the northern hemisphere they are found on the west coasts of continents in high latitudes. Alaska, Mozambique and Antilles currents are warm currents. Hence option (d) is the correct answer.

Salinity of the Ocean

1. **Main Source of Salinity of the Sea is –**
   (a) Rivers
   (b) Land
   (c) Wind
   (d) Ash ejected from the Volcano

53rd to55th B.P.S.C. (Pre) 2011

Ans. (b)

Main source of the salinity of sea water is the land. While rivers bring various types of salt contents towards the sea, other factors which regulate the salinity of the sea are vaporisation, wind, river water, rain, oceanic currents and the volcano etc. Average salinity of the sea is about 35‰.

2. **Which of the following seas has the highest average salinity?**
   (a) Black Sea (b) Yellow Sea
   (c) Mediterranean Sea (d) Dead Sea
   (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

Among the given options Dead Sea has the highest average salinity (238%). Assal Lake has the highest average salinity in the world i.e. 348%.

3. **The average salinity of water of Arabian Sea is-**
   (a) 25 ppt (b) 35 ppt
   (c) 45 ppt (d) 55 ppt

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Between 0°-10° latitudes in the Indian Ocean, the salinity is 35.14‰ but in the Bay of Bengal, it decreases to 30‰. At the same time salinity of Arabian Sea is found to be 36‰ because of comparatively dry weather, more vaporization takes place and water brought by rivers is also less. Therefore, option (b) is correct.

High Tide, Low Tide

1. **The high tide in the Ocean is caused by**
   (a) Earthquake (b) Sun
   (c) Stars (d) Moon

47th B.P.S.C. (Pre) 2005

Ans. (d)

Spring-tide occurs in the ocean on the day of Full Moon or New Moon while at 7th or 8th day after the Full Moon and the New Moon neap-tide occurs. Since, the Moon is nearer to the Earth in comparison to other stars, so its gravitational force has a greater effect upon the Earth. Due to this effect, high sea waves are produced in oceans. Tide producing power of the moon is 2.17 times more than the Sun.

Rivers of the World

1. **Which of the following are rivers?**
   (a) Victoria and Chad (b) Murray and Darling
   (c) Suez and Panama (d) Erie and Huron
   (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (b)

The Murray–Darling Basin is a large geographical area in the interior of southeastern Australia, encompassing the drainage basin of the tributaries of the Murray River, Australia's longest river, and the Darling River, a right tributary of the Murray and Australia's third-longest river. The Basin is home to 16 internationally significant wetlands, 35 endangered species and 120 species of native and migratory birds. The Suez Canal has an extension of 193 kilometers, and connects the Mediterranean Sea with the Red Sea. The Panama Canal is 80 kilometers long, and connects the Pacific and Atlantic Oceans. Others are lakes of the world–

Lake Name	Surface Area	Type	Countries on shoreline
Victoria	(68,870km^2)	Fresh water	Uganda, Kenya, Tanzania
Huron	(59,600km^2)	Fresh water	Canada, U.S.
Erie	(25,700km^2)	Fresh water	Canada, U.S.

2. Consider the following statements :

1. Lake Victoria is the third largest freshwater lake in the world by surface area.

2. It is one of the great lakes of Africa.

3. It is bordered by four countries—Tanzania, Uganda, Rwanda and Kenya.

4. The only outflow from Lake Victoria is the Nile River, which exits the lake near Jinja, Uganda.

Which of the above statements are incorrect?

(a) 1 and 2 (b) 2 and 4

(c) 3 and 4 (d) 1 and 3

69th B.P.S.C. (Pre) 2023

Ans. (d)

Lake Victoria is one of the greatest lake in African continent. Lake Victoria is the second largest freshwater lake in the world by surface area. Lake Victoria is bordered by Tanzania, Uganda and Kenya. Significantly Lake Victoria is included in Great Lake Region. Hence, statement 1 and 3 are incorrect.

3. Which one of the following river in Russia is flowing towards Pacific Ocean?

(a) Ob (b) Lena

(c) Amur (d) Yenisey

(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (c)

The Amur River is the only river listed that flows into the Pacific Ocean. The other rivers including the Ob, Lena and Yenisey, all flow into the Arctic Ocean. The Amur River is one of the longest rivers in Asia and forms part of the border between Russia and China.

4. The longest river of South America –

(a) Nile (b) Amazon

(c) Mississipi (d) Ganga

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (b)

Amazon is the longest river in South America. This river is about 6437 km long. Ships can sail up to 1,600 km distance. Apart from this, Nile (6,695 km) of Africa, Mississippi-Missouri (3779 km) of U.S.A., and the river Ganga (2,525 km) of India are the prominent rivers. The Nile River flows through eleven countries like Burundi, DRC Congo, Egypt, Ethiopia, Eritrea, Kenya, Rwanda, South Sudan, Sudan, Tanzania and Uganda. Nile River falls into the Mediterranean Sea.

5. Which of the following countries forms a part of Amazon River basin?

(a) Argentina (b) Chile

(c) Venezuela (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

The Amazon River Basin covers more than approx 6,100,000 km^2 of the land area of the South American Continent extending into Bolivia, Brazil, Columbia, Ecuador, Guyana, Peru, Suriname and Venezuela.

Towns and Cities along the Banks of Rivers

1. 'River Seine' flows through the town –

(a) London (b) Paris

(c) Rome (d) Frankfurt

38th B.P.S.C. (Pre) 1992

Ans. (b)

Spree River: It is a tributary of the Havel River and its length is 403 kilometer. The Spree flows through Saxony, Brandenburg and Berlin cities of Germany.

Potomac River: This river flows through south-west Maryland and Washington DC and falls into the Chesapeake Bay.

Seine River: Seine is the second largest river of France after Loire River. It flows in northern France through Paris.

Manzanares river : The Manzanares is one of the main rivers in Central Spain which passes through Madrid.

2. **Match the rivers with the cities through which they are flowing and select the correct answer using the codes given below:**

City	River
A. Rotterdam	**(1) Seine**
B. Paris	**(2) Potomac**
C. Budapest	**(3) Rhine**
D. Washington	**(4) Danube**

Code :

	A	B	C	D
(a)	2	3	1	4
(b)	1	3	4	2
(c)	3	1	4	2
(d)	4	3	2	1

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

The correct match of given cities with their related rivers is as follows :

(City)		**(River)**
Rotterdam	—	Rhine
Paris	—	Seine
Budapest	—	Danube
Washington D.C.	—	Potomac

Landforms by River

1. **The greatest delta of the world is formed by –**

(a) Ganga and Brahmaputra (b) Mississippi- Missouri
(c) Yangtze-Kyang (d) Huang-Hoe

41st B.P.S.C. (Pre) 1996

Ans. (a)

The greatest delta in the world is the Ganges Delta. The Ganges Delta is a river delta in Bangladesh and West Bengal, India. It is known as Ganges–Brahmaputra-Meghna or Sunderban Delta also. The Ganges Delta has the shape of a triangle and is considered to be an "arcuate" delta (arc-shaped). It covers approx. 105641 sq. km.

2. **Grand Canyon is –**

(a) a gorge (b) a large cannon
(c) a river (d) an old cannon

43rd B.P.S.C. (Pre) 1994

Ans. (a)

The Grand Canyon is a gorge carved by the Colorado River in the State of Arizona, United States. The Grand Canyon is 446/447 km long, average width 6.4 to 29 km. and attains a depth of over 1,600 meters.

3. **Tipaimukh Dam is an issue between which of the following countries?**

(a) India and Myanmar (b) India and China
(c) India and Bangladesh (d) India and Pakistan

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

Tipaimukh Dam is an issue between India and Bangladesh. It is a proposed embankment dam on the river Barak in Manipur. The main aim of this project is flood control and hydel power generation for the benefit of the surrounding location. Bangladesh argue that this may affect the seasonal rhythm of the river and have an adverse effect on downstream agriculture and fisheries of Bangladesh.

The Islands

1. **Which one among the following is the largest island in area?**

(a) Borneo (b) Great Britain
(c) Madagascar (d) Sumatra
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

In the given option, Borneo's area is the highest (751929 sq. km.), Madagascar (587041 sq. km.), Sumatra (472784 sq. km.) and Great Britain area (209331 sq. km.).

2. **Which one of the following countries has the highest number of islands?**

(a) Philippines (b) Indonesia
(c) Maldives (d) Cuba
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Indonesia with 17,508 islands has the highest number of islands in the given option, but in the world, Sweden has the highest number of islands.

3. **Greenland is a part of which one of the following countries?**

(a) Denmark (b) Finland
(c) Canada (d) United Kingdom
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

Greenland is the largest island in the world. Greenland belongs to Denmark. Nuuk is the capital and largest city of Greenland. The elevation of Nuuk above the sea level is 1-3 meter.

4. **Fiji Island is located in –**
(a) Atlantic ocean (b) Pacific ocean
(c) Indian ocean (d) Arabian sea

44th B.P.S.C. (Pre) 2000

Ans. (b)

Fiji is an island country in the South Pacific Ocean. Its closest neighbours are Vanuatu to the west, Tonga to the east and Tuvalu to the north. Fiji is an archipelago of 333 islands of which the two major islands, Viti Levu and Vanua Levu, are inhabited by 87% of the total population.

5. **The largest island of Japan in terms of the geographical area is.**
(a) Hokkaido (b) Honshu
(c) Shikoku (d) Kyushu
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

Japan is a group of four main islands Hokkaido, Shikoku, Kyushu & Honshu. Honshu is the largest island where capital Tokyo is situated. In these islands, Hokkaido (83,453.57 km^2) is in the north and mainland Honshu (approx. 228000 km^2) and Shikoku & Kyushu are situated at south-west of the archipelago.

6. **Sunda Trench lies parallel to the island of**
(a) Java (b) Maldives
(c) Sumatra (d) Bhaurikas
(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (e)

The Sunda Trench is located in the Indian Ocean. Its expansion is mainly parallel to the islands of Sumatra and Java of the Indonesian archipelago. Earlier this trench was also known as Java trench.

The Straits

1. **Malacca Strait facilitates movement from –**
(a) Indian Ocean to China Sea
(b) Red Sea to Mediterranean Sea
(c) Atlantic Ocean to Pacific Ocean
(d) Mediterranean Sea to Black Sea

44th B.P.S.C. (Pre) 2000

Ans. (a)

Strait of Malacca runs between Indonesia and Malaysia. The Strait of Malacca connects Indian ocean (Andaman sea) to Pacific Ocean (South China Sea). Thus, option (4) is correct.

2. **Which strait connects the Red Sea and the Indian Ocean?**
(a) Bab-el-Mandeb (b) Hormuz
(c) Bosphorus (d) Malacca

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Bab-el-Mandeb Strait connects the Red Sea to the Indian Ocean (Gulf of Aden). The Strait of Hormuz links the Persian Gulf with the Gulf of Oman. The Strait of Malacca connects the Indian Ocean (Andaman Sea) to the Pacific Ocean (South China Sea). The Bosphorus Strait (or Bosphorus) connects the Black Sea to the Sea of Marmara.

The Canals

1. **Suez Canal Connects –**
(a) Black sea with Red sea
(b) Mediterranean sea with Caspian sea
(c) Red sea with Mediterranean Sea
(d) None of these

38th B.P.S.C. (Pre) 1993

Ans. (c)

The Suez Canal is an artificial sea-level waterway located in Isthmus of Suez of Egypt. It was 193.30 km. long. This canal connects the Mediterranean Sea and the Red Sea. It was made operational in 1869. Tewfik port (Suez port) is located to the south of Suez Canal and Port Said is located to its north. It is the largest artificial and navigational canal in the world. Suez Canal reduced the maritime distance between Europe and India by 7,000 km.

2. **Panama Canal Connects –**
(a) North America and South America
(b) Pacific Ocean and Atlantic Ocean
(c) Red Sea and Mediterranean Sea
(d) Indian Ocean and Pacific Ocean

43rd B.P.S.C. (Pre) 1999

Ans. (b)

The Panama Canal is a major ship canal of Central America. It was constructed by cutting across the Isthmus of Panama. It connects the Atlantic Ocean to the Pacific Ocean. This canal has significantly reduced the distance between San Francisco (on the western coast) and New York (on the eastern coast).

3. **Which of the following is known as the longest canal in India?**
   (a) Setu samudram Shipping Canal
   (b) Sharda Canal
   (c) Indira Gandhi Canal
   (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (c)

The Indira Gandhi Canal is the longest canal in India. Origin of this canal is from Harike Barrage situated in Punjab. From Harike, 204 Km. long Indira Gandhi Feeder off-takes, which has 170 Km. length in Punjab & Haryana and balance 34 Km. in Rajasthan.
This canal enters Rajasthan at Hanumangarh. From tail of Indira Gandhi Feeder 445 Km. long Indira Gandhi Main Canal starts which passes through Sri Ganganagar and Bikaner districts and ends at Mohangarh in Jaisalmer.
Total length of the canal is 649 km.

The Atmosphere

1. **The layer of atmosphere having lowest temperature is**
   (a) stratosphere (b) troposphere
   (c) mesosphere (d) More than one of the above
   (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The atmosphere consists of different layers with varying density and temperature. The column of atmosphere is divided into five different layers depending upon the temperature condition. They are troposphere, stratosphere, mesosphere, thermosphere and exosphere. The top of the mesosphere is the coldest area of the Earth's atmosphere.

2. **Ozone layer refers to –**
   (a) The atmospheric condition of Antarctica
   (b) Modern invention done on the planet Saturn
   (c) The layer about 10-20 km. below the surface of Earth
   (d) The layer of atmosphere of 15-20 km. above the surface of Earth

38th B.P.S.C. (Pre) 1992

Ans. (d)

The ozone layer protects us from the harmful ultra-violet radiations of the Sun. In atmosphere, it is mainly found in the lower portion of the Stratosphere from about 15–35 km above Earth. Ozone is a molecule composed of three oxygen atoms. Its colour is blue.

3. **The height of the ozone layer above the surface of the earth is-**
   (a) 15-20 km (b) 40-50 km
   (c) 70-80 km (d) 110-120 km

41st B.P.S.C. (Pre) 1996

Ans. (a)

See the explanation of the above question.

4. **Which of the following options is the correct sequence of layers of atmosphere starting from the Earth's surface?**
   (a) Troposphere → Stratosphere → Mesosphere → Thermosphere → Exosphere
   (b) Troposphere → Mesosphere → Stratosphere → Thermosphere → Exosphere
   (c) Troposphere → Mesosphere → Thermosphere → Stratosphere → Exosphere
   (d) More than one of the above
   (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

Earth's atmosphere has a series of layers, each with its unique characteristics. Going away from the ground level, those layers are as :–
Troposphere→Stratosphere→Mesophere→Thermosphere →Exosphere.

The Insolation

1. **Clear sky nights are cooler than cloudy sky nights due to**
   (a) condensation (b) radiation
   (c) induction (d) conduction

39th B.P.S.C. (Pre) 1994

Ans. (b)

The cloudy night contains more water vapour than a clear night. The heat emitted from the Earth's surface is trapped by the clouds and emitted back towards the earth. As a result cloudy night seems warmer than clear night. If the sky is clear, heat emitted from the Earth's surface freely escapes into space, resulting in colder temperatures.

The Forests

1. **In India, tropical rain forests are found in**
   (a) Karnataka (b) Himachal Pradesh
   (c) Kerala (d) More than one of the above
   (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (d)

India is a land of great variety of natural vegetation. Himalayan neights are marked with temperate vegetation. The western ghats (Karnataka, Kerala, etc.) and the Andaman Nicobor Islands have tropical forests.

2. **Dense-forests on the earth are mostly found –**
(a) Nearby Equator
(b) Nearby tropic of Cancer
(c) Nearby tropic of Capricorn
(d) Nearby Poles

41st B.P.S.C. (Pre) 1996

Ans. (a)

Tropical evergreen forests grow in the regions near the Equator (10^{0}N-10^{0}S). These regions are hot and receive heavy rainfall throughout the year. As there is no particular dry season, the trees do not shed their leaves altogether. This is the reason they are called evergreen. Three major regions of evergreen forest in the world are-

Amazon River Basin- Brazil, Bolivia, Peru, Ecuador, Colombia, Venezuela, French Guyana, Guyana, and Suriname.

Congo basin- Democratic Republic of Congo, Cameroon, Central African Republic, Gabon, Equatorial Guinea, Republic of the Congo.

Borneo- Mekong Southeast Asia basin–The region is composed of Two Sub region the Island of Borneo and the Mekong.

3. **Which country of the world is the largest producer and exporter of soft timber and wood pulp?**
(a) U.S.A. (b) Norway
(c) Sweden (d) Canada

45th B.P.S.C. (Pre) 2001

Ans. (d)

During the period this question was asked, Canada was the leading country for production and export of softwood and wood pulp. According to the latest World Food and Agriculture Statistical yearbook 2023 (Data-2021), the United States is the top wood pulp producing country with 49685 million tonnes in 2021. Brazil, China, Canada, Sweden and Finland were at second, third, fourth and fifth place respectively.

The World Climate

1. **Which one of the following forest fires/bushfires is not a Mediterranean climate type?**
(a) California (b) Turkey
(c) Victoria (d) Siberia
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (d)

Siberia is not a Mediterranean Climate type. While California, Turkey and Victoria experience Mediterranean Climates characterized by hot, dry summers and mild, wet winters conducive to forest fires or bushfires, Siberia generally has a continental, climate with long, cold winters and short warm summers, which differs from the Mediterranean climate pattern.

2. **In which of the following areas, Mediterranean climate does not prevail?**
(a) Central Chile (b) Cape Town
(c) Adelaide (d) Pampas
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (d)

Western part of Central Chile, Adelaide and Cape Town region have Mediterranean climate. The Pampas are fertile South American lowlands that cover more than 750,000 km^2 and include the Argentine province of Buenos Aries, La Pampa, Santa Fe, Entre Rios and Cordoba etc. all of Uruguay and the southernmost Brazillian state, Rio Grande do Sul. The climate is temperate with precipitation of 600 to 1200 mm. The climate is generally temperate, gradually giving away to more subtropical climate in the north and to a semi-arid climate on the western fringes.

3. **Which of the following pair matching's is correct?**
(a) Mediterranean Sea Zone-Summer rain
(b) Equatorial Zone-rain with thunder in the noon
(c) Monsoon Zone-Heavy rain throughout the year
(d) Desert zone- winter rain

39th B.P.S.C. (Pre) 1994

Ans. (b)

The Mediterranean sea climatic region receives rainfall in winter.
In equatorial regions, convectional rain occurs with thunder at noon almost throughout the year. In Monsoon regions, the major rainfall is received by Monsoon winds. In Desert regions, rainfall is either negligible or indefinite. Hence, the correctly matched pair is option (b).

4. **Match correctly :**

A. The hottest			**1. Chile**
B. The coldest			**2. Cherrapunji**
C. The wettest			**3. Antarctica**
D. The driest			**4. Sahara**

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	2	3	1	4
(d)	3	2	4	1

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (b)

The correct match is as follows :

The hottest	-	Sahara Desert
The coldest	-	Antarctica
The wettest	-	Cherrapunji (Now Sohra)
The driest	-	Chile

According to NASA's satellite data (2003-09), Dasht-e-Lut desert in Iran (highest temperature 70.7^{0}C recorded in 2005) is the hottest place. According to Guinness World Record, the wettest place on earth is Mawsynram (11872 mm annual rain).

5. **Which of the following process is called soil mulching?**
   (a) Covering bare ground with a layer of organic matters
   (b) To build barriers through stones
   (c) Rocks are piled up to slow down the flow of water
   (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 6-8}–2024

Ans. (a)

Mulching is the process of covering the topsoil with plant material such as leaves, grass, twigs, crop residues, straw etc. it helps in soil conservation.

The Soil

1. **Which of the following statements are true regarding Esker and Drumlin?**
   1. **Eskers are ridges of crude bedded gravel and sand.**
   2. **Drumlins are mostly made of boulders and clay.**
   3. **Egg basket topography is characteristic of escarpous areas.**
   4. **While eskers are formed by the currents of glaciers, drumlin are produced by the action of glaciers. Select the correct answer from the options given below.**

   (a) 1, 2 and 3 (b) 1, 2 and 4
   (c) 3 and 4 only (d) 1 and 2 only
   (e) none of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Eskers are long, narrow, ridges with steep slopes, formed by the deposition of debris by glaciers or ice melt streams. Esker is made up of gravel, sand and gravel. The landforms formed by glacial deposits are a type of mound formed by drumlin boulder clay, shaped like an upside down boat or a cut inverted egg. Egg basket topography is characteristic of drumlined areas. Drumlins are produced by the action of glaciers.

2. **Capillaries are most effective in –**
   (a) Clayey soil (b) Silt soil
   (c) Sandy soil (d) Loamy soil

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Smaller the pore of the soil, more effective is the capillary activity and bigger the pore of any soil, less effective will be its capillary activity. In clay soil, the cell activity is most effective and it is least in sandy soil. The effectiveness of the capillary activity in these four soils are as follows :
Clayey > silt soil > loamy soil > sandy soil

3. **Conservation of soil is the process in which –**
   (a) Barren land is converted into fertile
   (b) Soil is aerated
   (c) Soil is eroded
   (d) Soil is conserved from harm

39th B.P.S.C. (Pre) 1994

Ans. (d)

In a broad sense, the meaning of soil conservation is not only controlling erosion, rather it aims at maintaining high-level fertility of the land. The improvement in land use is called soil conservation. Hence, correct answer is (d).

4. **The formal development of Terrarossa takes place in that part of land which consists of –**
   (a) Limestone (b) Cynite
   (c) Granite (d) Sandstone

39th B.P.S.C. (Pre) 1994

Ans. (a)

Terrarossa is a type of red clay soil produced by the weathering of limestone rocks.

The Major Tribes of the World

1. **The native African tribe 'Pygmies' is found in which of the following river basins –**
   (a) Niger (b) Congo
   (c) Nile (d) Zambezi

45th B.P.S.C. (Pre) 2001

Ans. (b)

Pygmy tribes live in equatorial rain forests of Congo river basin in Africa. It is hunter- gatherer tribe. The Pygmy tribe traditionally live in single-family huts called *mongulus,* made of branches and leaves on the tree.

2. Pygmies are inhabitants of –

(a) Africa (b) Asia

(c) Australia (d) South America

42nd B.P.S.C. (Pre) 1997

44th B.P.S.C. (Pre) 2000

Ans. (a)

See the explanation of the above question.

3. Eskimos are inhabitants of –

(a) Canada (b) Mongolia

(c) Malaya (d) Sri Lanka

44th B.P.S.C. (Pre) 2000

Ans. (a)

Eskimo tribe lives in Tundra region of Canada & Greenland, the Maasai live in Kenya, the Bedouin tribe lives in Saudi Arabia and the Bushmen tribe lives in Kalahari desert (mainly in Botswana and some parts of South Africa and Namibia).

4. One of the following pairs is a mismatch. Find it?

(a) Masai - Central Eastern Africa

(b) Sakai - Malaysia

(c) Bedouin - Arabian Peninsula

(d) Kirghiz - Central Asia

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re-Exam) 2020

Ans. (e)

Masai - Central Eastern Africa
Sakai - Indonesia and Malaysia
Bedouin - Arabian Peninsula
Kirghiz - Central Asia

The Languages

1. What is Esperanto?

(a) Highest mountain of Latin America

(b) Seaport city of Spain

(c) The name of a game

(d) An artificial language to serve as world language

38th B.P.S.C. (Pre) 1992

Ans. (d)

Esperanto is an International auxiliary language devised by Dr. Ludwik Lejzer Zamenhof.

Economic Geography

Agriculture & Livestock

1. Among the following countries, which one is the largest producer of saffron in the world?

(a) Spain (b) Greece

(c) New Zealand (d) Iran

(e) None of the above/ More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

Iran is the largest producer of saffron in the world. About 90 to 95% of the world's total saffron production in Iran.

2. Most attractive, warmest and lightest wool of the world 'Shahtoosh' is produced in–

(a) Nepal (b) Uzbekistan

(c) China (d) Bangladesh

39th B.P.S.C. (Pre) 1994

Ans. (c)

Shahtoosh is the name given to a specific kind of shawl, which is woven with hair of the Tibetan antelope (Chiru) by master craftsman. They are found almost entirely in China, Tibet, Southern Xinjiang, and Western Qinghai; a few are also found across the border in Ladakh, India. Thus, the correct option is (c).

3. Which of the following is not a wine-producing province of France?

(a) Burgundy (b) Sachsen

(c) Alsace (d) Champagne

(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (b)

Sachsen is Germany easternmost and one of the smallest wine - growing region. There are many wine producing region in France. Alsace, Bordeaux, Loire, Champagne, Burgundy and Rhone valley etc are dominated French wine regions.

The Minerals

(i) Coal

1. Which of the following countries has the largest reserve of coal deposits?

(a) United States of America (b) China

(c) South Africa (d) India

(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

The world's largest coal reserves are concentrated mainly in five countries, which have been able to exploit this natural asset for their own industrial development. According to figures published in the BP Statistical Review of World Energy 2020, there are more than one trillion tonnes of proven coal resources worldwide. Global production in 2019 topped 7.9 billion tonnes, with China, India and the US among the world's top coal-producing countries. As of January 2020, the United States has the largest recoverable coal reserves with an estimated 249 billion short tons of coal (23% share of the global total) remaining, according to the U.S. Energy Information Administration.

2. **Which of the following countries is the leading country in the reserve of Hard Coal?**
   (a) Nepal (b) China
   (c) New Zealand (d) India

40th B.P.S.C. (Pre) 1995

Ans. (b)

China was at the top in terms of the reserve of hard coal when this question was asked followed by the USA and India. According to the BP Statistical Review of World Energy, 2021 data the USA ranked first in the proven reserve of Anthracite and bituminous coal at the end of 2022 followed by China & India respectively.

3. **The highest coal-producing country in the world is-**
   (a) India (b) USA
   (c) China (d) Russia

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

According to World Mineral Production 2017-21, China ranks first in highest coal production in the year 2021.

4. **Common type of coal is –**
   (a) Bituminous (b) Sub-bituminous
   (c) Anthracite (d) Coke

39th B.P.S.C. (Pre) 1994

Ans. (a)

Coal is mainly a free carbon mixture of a compound of carbon. Coalification is the transformation of plant matter into coal by chemical process. On the basis of the carbon content, there are generally four types of coal- Anthracite, Bituminous, Lignite and Peat. According to USGS the four type (ranks) of coal are Anthracite, Bituminous, Sub-bituminous and Lignite. The Precarsor to Coal is Peat. Among the above four types of coal, bituminous is considered as common type of coal, anthracite is of very good quality and peat is the lowest grade of coal. More than 90 percent of the total coal obtained in India is found in the Gondwana rocks of ancient times. The Gondwana era coal is mainly of bituminous type.

ii. The Gold

1. **Which of the following activities is performed at Coolgardie?**
   (a) Coal mining (b) Copper mining
   (c) Gold mining (d) Forestry
   (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

World's maximum Gold reserved is found in Australia. About 68 percent of its gold is derived from Western Australia. Two of the important gold mines Kalgoorlie and Coolgardie are situated in Western Australia.

2. **Johannesburg is famous for –**
   (a) Gold mining (b) Tin mining
   (c) Mica mining (d) Iron-ore mining

42nd B.P.S.C. (Pre) 1997

Ans. (a)

Johannesburg is the biggest city of South Africa and capital of Gauteng Province. It is famous for its Gold and Diamond Mining.

3. **Which of the following countries produced the highest annual gold production (in tonnes) in 2019?**
   (a) Russia (b) Australia
   (c) China (d) United States of America
   (e) None of the above / More than one of the above

66th BPSC (Pre) 2020

Ans. (c)

According to the available data of World Minerals Production 2017-2021 till the year 2021, China has the highest gold production in the world. Russia, Australia and USA are second, third and fourth Gold Production Countries in the world.

iii. The Aluminium

1. **The largest producer of aluminium in the world is –**
   (a) France (b) India
   (c) U.S.A. (d) Italy

56th to 59th B.P.S.C. (Pre) 2015

Ans. (b)

According to World Mineral Production 2017-2021. The largest producers of aluminium in the world (in the year 2021) are China > Russia > India > Canada > UAE.

iv. The Tin

1. **Tin is found-**
   (a) In Placer deposits
   (b) In Metamorphic rocks
   (c) In little silica Igneous rocks
   (d) In all these

39th B.P.S.C. (Pre) 1994

Ans. (a)

About 80% of World's Tin is found in placer deposits. These deposits are a natural concentration of heavy minerals caused by the effect of gravity on moving particles. Placer deposits are usually situated near river beds and valleys or on sea floor and are spread in South East Asia.

v. The Petroleum

1. **Which one among the following countries of the world, except the United States of America, is the largest crude oil producer?**
   (a) Russia (b) China
   (c) Saudi Arabia (d) Canada
   (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

According to World Mineral Production 2017-2021. In the year 2021 USA was the largest crude Petroleum producer followed by Russia.

2. **Which country has the largest reserves of oil?**
   (a) United States (b) China
   (c) Russia (d) Venezuela
   (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

In the year 2016, when this question was asked, option (c) was the correct answer but in present Venezuela is First place. The five countries with the largest Petroleum Reserve, according to Indian Mineral Year Book - 2021 are as follows:

(Country)		**Reserve deposit (billion ton)**
Venezuela	-	48.0
Saudi Arabia	-	40.9
Canada	-	27.1
Iran	-	21.7
Iraq	-	19.6

vi. The Uranium

1. **Which one of the following countries is the largest producer of uranium in the world?**
   (a) Kazakhstan (b) Canada
   (c) Australia (d) France
   (e) None of the above/ More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

According to the World Nuclear Association (Data-2022) Kazakhstan is the largest producer of Uranium.

2. **Radium is extracted from the mines of –**
   (a) Limestone (b) Pitchblende
   (c) Rutile (d) Hematite

39th B.P.S.C. (Pre) 1994

Ans. (b)

Radium is a radioactive element with symbol Ra. It is extracted from Uranium ore Pitchblende.

Mineral: Miscellaneous

1. **Commercial sources of energy purely consist of-**
   (a) Power, coal, oil, gas, hydro-electricity and uranium
   (b) Coal, oil, firewood, vegetable waste and agricultural waste
   (c) Power, coal, animal dung and firewood
   (d) Coal, gas, oil and firewood

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Commercial sources of energy purely consist of power, coal, oil, gas, hydro-electricity and uranium.

2. **Which of the following countries is the leading producer of electronics?**
   (a) Japan (b) South Korea
   (c) China (d) Germany
   (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

According to Electronics Export Data, in 2023-24 without a doubt, China remains the major electronics exporter. China is the Titan of Electronics Manufacturing. China's strong technology infrastructure, skilled labor, and huge production facilities cement its position as the epicenter of electronic manufacturing. Also, China has gained the title of "world's factory" due to its huge production capabilities. The United States is the biggest electronics exporter in terms of electronics innovation and exports. Home to some of the world's IT behemoths, the United States not only leads in cutting-edge technology but also has a prominent presence in worldwide electronics exports.

Industries

1. **Which of the following is the largest metal trading centre?**
   (a) Johannesburg (b) New York
   (c) London (d) Singapore

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

London Metal Exchange (LME) is the world's largest industrial-trading metal and cost-risk management center. The exchange also offers contracts on ferrous metals and precious metals.

The Transport

1. **Match List- I with List- II and select the correct answer using the codes given below:**

List- I	**List II**
(A) European trans-Continental railway	**(1) Paris to Istanbul**
(B) Trans-Andean railway	**(2) Leningrad to Vladivostok**
(C) Trans-Siberian railway	**(3) Leningrad to Volgograd**
(D) Orient Express	**(4) Buenos Aires to Valparaiso**
	(5) Paris to Warsaw

Code :

	A	B	C	D
(a)	5	4	2	1
(b)	1	4	3	2
(c)	5	1	2	3
(d)	1	2	3	4

47th B.P.S.C. (Pre) 2005

Ans. (a)

European trans-continental railway travel from Paris to Warsaw. Trans-Andean railway is the most important rail route of South America. It travels from Valparaiso to Buenos Aires of Argentina. The Trans-Siberian railway is the longest (about 9300 km.) railway in the world. It has connected Moscow with Vladivostok.

Note: The main track of Trans-Siberi and Railway extends from Vladivostok to Moscow. By using a connecting service it can be travelled to Saint Petersburg in West. Orient Express is most important rail route of Europe. It runs across eight countries of Europe. It connects Paris & Le Harare to Istanbul.

The Ports and Harbours

1. **Which one of the following is not a sea-port city?**
   (a) Tokyo (b) Canberra
   (c) New York (d) London

40th B.P.S.C. (Pre) 1995

Ans. (b)

Canberra, the capital of Australia is not surrounded by Sea.
Tokyo - Capital of Japan and prominent seaport situated on Honshu island.
New York - Sea-port city of the USA located on the banks of Hudson River in New York City.
London - Port situated on the banks of Thames.
Hence Canberra is not a seaport city.
Thus, option (b) is the correct answer.

2. **Who wrote Periplus of the Erythraean Sea?**
   (a) Ctesias (b) Pliny
   (c) Ptolemy (d) Strabo
   (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

The Periplus of the Erythraean Sea or Periplus of the Red Sea is a Greco-Roman periplus written in Koine Greek that describes navigation and trading opportunities from Roman Egyptian ports like Berenice Troglodytica along the Coast of Red Sea and others along the Horn of Africa, the Sindh region of Pakistan, along with southwestern regions of India. The text has been ascribed to different dates between the first and the third century. A mid-first century date is now the most commonly accepted.
While the author is unknown, it is clearly a first-hand description by someone familiar with the area and is nearly unique in providing accurate insights into what the ancient European world knew about the lands around the Indian Ocean.

3. **Which one of the following is known as the 'Coffee Port' of the world?**
   (a) Sao Paulo (b) Santos
   (c) Rio de Janeiro (d) Buenos Aires
   (e) None of the above / More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

The Port of Santos is located in Brazil. It was established in 1546 for exporting coffee at large scale. Now it is known as Coffee Port due to highest coffee export in the world.

General Mental Ability

1. The value of

$$\frac{\mathbf{18}}{\left(\mathbf{5}+\sqrt{\mathbf{7}}\right)}+\frac{\mathbf{4}}{\left(\sqrt{\mathbf{7}}+\sqrt{\mathbf{3}}\right)}+\frac{\mathbf{1}}{\left(\sqrt{\mathbf{3}}+\sqrt{\mathbf{2}}\right)}+\frac{\mathbf{1}}{\left(\sqrt{\mathbf{2}}+\mathbf{1}\right)}$$

(a) 4 (b) 3
(c) 5 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (a)

$$=\frac{18}{\left(5+\sqrt{7}\right)}+\frac{4}{\left(\sqrt{7}+\sqrt{3}\right)}+\frac{1}{\left(\sqrt{3}+\sqrt{2}\right)}+\frac{1}{\left(\sqrt{2}+1\right)}$$

We know that–

$$=\frac{18\left(5-\sqrt{7}\right)}{\left(5+\sqrt{7}\right)\left(5-\sqrt{7}\right)}+\frac{4\left(\sqrt{7}-\sqrt{3}\right)}{\left(\sqrt{7}+\sqrt{3}\right)\left(\sqrt{7}-\sqrt{3}\right)}+$$

$$\frac{\left(\sqrt{3}-\sqrt{2}\right)}{\left(\sqrt{3}+\sqrt{2}\right)\left(\sqrt{3}-\sqrt{2}\right)}+\frac{\sqrt{2}-1}{\left(\sqrt{2}+1\right)\left(\sqrt{2}-1\right)}$$

$$=\frac{18\left(5-\sqrt{7}\right)}{\left[5^2-\left(\sqrt{7}\right)^2\right]}+\frac{4\left(\sqrt{7}-\sqrt{3}\right)}{\left[\left(\sqrt{7}\right)^2-\left(\sqrt{3}\right)^2\right]}+\frac{\sqrt{3}-\sqrt{2}}{\left(\sqrt{3}\right)^2-\left(\sqrt{2}\right)^2}$$

$$+\frac{\sqrt{2}-1}{\left[\left(\sqrt{2}\right)^2-1^2\right]}$$

$$\left[\because (a-b)(a+b)=a^2-b^2\right]$$

$$=\frac{18\left(5-\sqrt{7}\right)}{25-7}+\frac{4\left(\sqrt{7}-\sqrt{3}\right)}{7-3}+\frac{\sqrt{3}-\sqrt{2}}{3-2}+\frac{\sqrt{2}-1}{2-1}$$

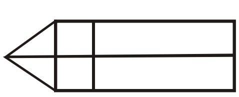
$$=\frac{18\left(5-\sqrt{7}\right)}{18}+\frac{4\left(\sqrt{7}-\sqrt{3}\right)}{4}+\frac{\left(\sqrt{3}-\sqrt{2}\right)}{1}+\frac{\sqrt{3}-1}{1}$$

$$=5-\sqrt{7}+\sqrt{7}-\sqrt{3}+\sqrt{3}-\sqrt{2}+\sqrt{2}-1$$

$= 5–1$

$= 4$

2 How many rectangles are there in the figure given below?

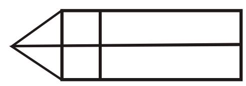


(a) 9 (b) 12
(c) 8 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (a)

The given figure–



Rectangle formed from (1) figure = 4

Rectangle formed from (2) figure = 4

Rectangle formed from (4) figure = 1

Total rectangle = 4+4+1 = 9

3 A player makes 7 complete revolutions of a circular path to complete a race of 2200 metres. The radius of the circular path is :

$$\left(\pi=\frac{\mathbf{22}}{\mathbf{7}}\right)$$

(a) 42 metres (b) 45 metres
(c) 50 metres (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

Ans. (c)

A player makes 7 complete revolutions of a circular path to complete a race of 2200 meter.

$\therefore$ 7 rounds = 2200 meters

1 round = $\frac{2200}{7}$ meters

We know that–

1 round = circumference of the path then,

$\frac{2200}{7}$ meters = $2\pi r$ ($\therefore$ where r is the radius of path)

$$\frac{2200}{7}=2\times\frac{22}{7}\times r$$

$$r=\frac{100}{2}$$

$$\boxed{r=50\,\text{meters}}$$

4 If the capacity of a cylindrical tank is 1848 m^3 and the diameter of its base is 14 m, the depth of the tank is :

$\left(\pi = \frac{22}{7}\right)$

(a) 8 m (b) 12 m
(c) 16 m (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

Ans. (b)

Given that–
Volume of the tank is 1848 m^3 and diameter of its base is 14 m also, we know that–
Volume of Cylinder = pr^2h
(∵ where is the depth of the cylinder)

$1848 = \frac{22}{7} \times 7 \times 7 \times h \quad (\because 2r = 14 \Rightarrow r = 7)$

1848 = 22 × 7 × h

$h = \frac{1848}{22 \times 7}$

$\boxed{h = 12 \text{ meters}}$

5 If $\frac{1}{8}$th of a number is 30, what will be 62% of that number?

(a) 181.3 (b) 178.24
(c) 148.8 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

Ans. (c)

Let, the number is x.
According to the question–

$\frac{1}{8} \times x = 30$

$x = 240$

$62\% \text{ of } x = 240 \times \frac{62}{100} = \frac{14880}{100}$

$= \frac{1480}{10}$

$\boxed{62\% \text{ of } x = 148.8}$

6 The salary of an officer is increased by 25%. By what percent should the new salary be decreased to restore the original salary?

(a) 25 (b) 22.5
(c) 20 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

Ans. (c)

Let officer salary was 100 rupees.
Now, after the increment of 25%

$= 100 \times \frac{125}{100}$

=125

To restore the original salary, the required percentage decrement
= 125 – 100
=25

decreased percentage = $\frac{25}{125} \times 100$

=20%

7. If $x + \frac{1}{x} = 5$, what is the value of $x^2 + \frac{1}{x^4}$?

(a) 525 (b) 527
(c) 529 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

Ans. (b)

Given that–

$x + \frac{1}{x} = 5$

Squaring on both sides–

$x^2 + \frac{1}{x^2} + 2 = 25$

$x^2 + \frac{1}{x^2} = 23$

again, squaring on both sides–

$x^4 + \frac{1}{x^4} + 2 = 529$

$\boxed{x^4 + \frac{1}{x^4} = 527}$

8 A bag contains 5-rupees, 2-rupees and 1-rupee coins in the ratio 2 : 3 : 4. The total value of all the coins is ₹2,000. How many coins of 2-rupee are there in the bag?

(a) 200 (b) 250
(c) 400 (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

A bag contains 5 rupees, 2 rupees and 1 rupee coins in a ratio 2 : 3 : 4.

Then,

Denomination	–	5	2	1
Ratio	–	2x :	3x :	4x
Total	–	10x +	6x +	4x

= 20x

According to the question–

20x = 2000

x = 100

then 2 rupees coins will be $\frac{600}{2} = 300$

Hence, option (e) will be correct.

9. The interior angle of a regular polygon exceeds its exterior angle by 108°. How many sides does the polygon have?

(a) 10 (b) 9

(c) 8 (d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

Ans. (a)

We know that–

Interior angle + Exterior angle = 180^0 ....(i)

According to the question–

Interior angle = 108^0 + exterior angle

Interior angle – exterior angle = 108^0 ....(ii)

from equation (i) & (ii) □

2 × Interior angle = $180^0 + 108^0$

2 × Interior angle = 288^0

Interior angle = 144^0

From equation (i)–

Exterior angle = $180^0 - 144^0$

Exterior angle = 36^0

also we know that–

$$\text{exterior angle} = \frac{360^0}{x} \qquad \{\because \text{where } x \text{ is side}\}$$

$$36^0 = \frac{360^0}{x}$$

$$x = \frac{360^0}{36^0}$$

$$\boxed{x = 10}$$

10. Which of the following is the value of (x+1/x)²?

(a) $x^2 + 1/x^2$ (b) $x^2 - 1/x^2$

(c) $x^2 + 1/x^2 + 1$ (d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (e)

$$\left(x+\frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 \times x \times \frac{1}{x}$$

$$= x^2 + \frac{1}{x^2} + 2$$

Hence, option (e) is the correct answer.

11. An article is at 10% more than the CP. If discount of 10% is allowed then which of the following is right?

(a) 1% gain

(b) 1% loss

(c) no gain no loss

(d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (b)

Let, the CP of an article = Rs. 100.

∴ Marked price of an article = $\text{Rs. } 100 + \frac{100 \times 10}{100}$

= Rs. 110

Discount % = 10%

∴ Selling Price of an article = 110 – 110 × 10%

= Rs. 99

∵ CP > SP

$$\therefore \text{Loss \%} = \frac{CP - SP}{CP} \times 100 = \frac{100-99}{100} \times 100 = 1\%$$

Hence, option (b) is the correct answer.

12. By what number should 81 be divided to get a perfect cube?

(a) 3

(b) 6

(c) 7

(d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (a)

81 = 3×3×3×3

$81 = 3^3 \times 3$

Hence, 81 should be divided by 3 to get a perfect cube.

13. The difference between two whole numbers is 66. The ratio of the two numbers is 2:5. The two numbers are :

(a) 60 and 6 (b) 100 and 33

(c) 110 and 44 (d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Let, the two whole numbers be a and b.

According to question–

a ~ b = 66 ....(i)

and $\frac{a}{b}=\frac{2}{5}$

$a=\frac{2b}{5}$ ....(ii)

$\frac{2b}{5}\sim b=66$

$\frac{3b}{5}=66$

$\boxed{b=110}$ and $\boxed{a=44}$

Hence, the two whole numbers are 44 and 110.

14. In a parallelogram ABCD, angle A and angle B are in the ratio 1:2. Find the angle A.

(a) 30° (b) 45°

(c) 60° (d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

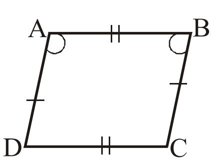
Ans. (c)

Let, the angles A and B of parallelogram ABCD are x and 2x. As we know, the sum of adjacent angles of a parallelogram is 180^0.

□A + □B = 180^0

x + 2x = 180^0

3x = 180^0



$\boxed{x=60^0}$

$\boxed{2x=120^0}$

Hence, the value of angle A is 60^0.

15. The height of a cylinder whose radius is 7 cm and the total surface area is 968 cm² is :

(a) 15 cm (b) 17 cm

(c) 19 cm (d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (a)

Given that,

Radius of a cylinder = 7 cm

Total surface area of cylinder = 968 cm^2

Let, the height of a cylinder = x cm.

According to the question–

Total surface area of cylinder = 2prh + 2pr^2

968 = 2pr(h+r)

$968=2\times\frac{22}{7}\times7(x+7)$

$22=x+7$

$\boxed{x=15\,\text{cm}}$

Hence, the height of a cylinder is 15 cm.

16. If $(-3)^{m+1}\times(-3)^5=(-3)^7$, then the value of m is :

(a) 5

(b) 7

(c) 1

(d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (c)

$(-3)^{m+1}\times(-3)^5=(-3)^7$

$\because(-3)^{(m+1)+5}=(-3)^7$

$\therefore(m+1)+5=7$

$\boxed{m=1}$

Hence, the value of m is 1.

17. If x and y are inversely proportional, then:

(a) y/x = constant

(b) xy = constant

(c) x/y = constant

(d) More than one of the above

(e) None of the above

BPSC (Tre-3) {Class 11-12}–2024

Ans. (b)

$x\propto\frac{1}{y}$

$x=\frac{c}{y}$ {where C = constant}

$\boxed{xy=\text{constant}}$

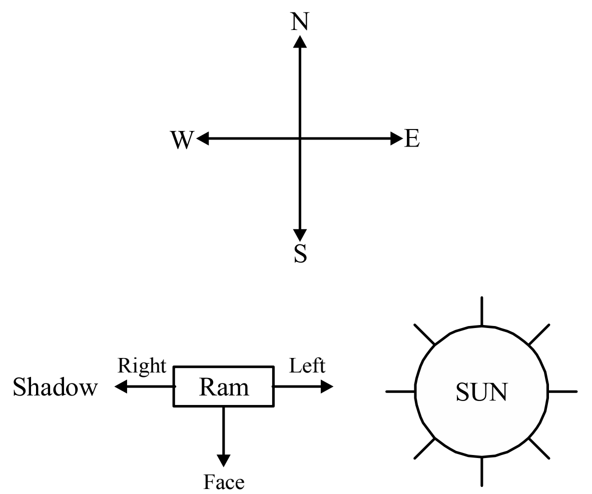
Hence, option (b) is the correct answer.

18. Early morning, Ram was standing on the ground in front of an effigy. The shadow of the effigy was appearing towards the right side of Ram. In which direction is Ram facing?

(a) East
(b) West
(c) South
(d) None of the above

BPSC Agriculture Department-2024

Ans. (c)



Hence, Ram is facing towards south direction.

19. If the 3rd day of a month is Friday, which day will be the 6th day before 27th of this month?

(a) Tuesday (b) Wednesday
(c) Friday (d) None of the above

BPSC Agriculture Department-2024

Ans. (a)

Given that,

3rd Day of a month = Friday

6th Day before 27th = 21st

∴ 21st Day of a month = 3rd + 18 {18 = 21–3}

= 3rd + 21–3

= Friday – 3

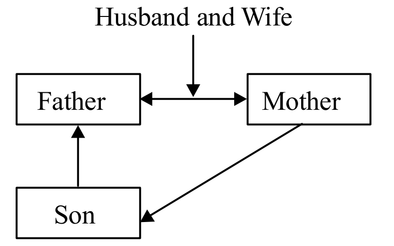
= Tuesday

20. Two persons are going. One is the father of the son of the other. What is the relationship between the two?

(a) Mother–Son (b) Father–Son
(c) Wife–Husband (d) None of the above

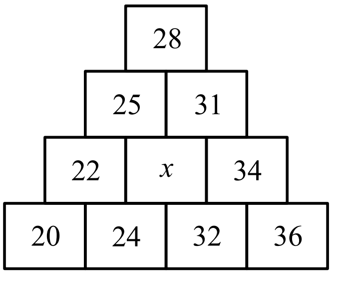
BPSC Agriculture Department-2024

Ans. (c)



Two persons are going. If one is the father of the son of the other. The relationship between the two is wife-husband.

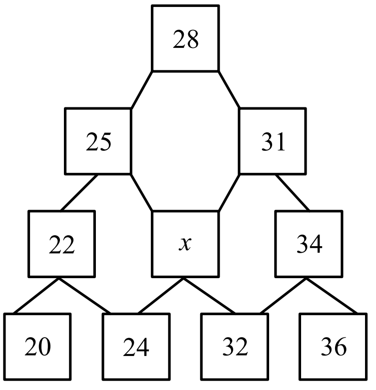
21. In the following table, obtain the missing number x :



(a) 28 (b) 30
(c) 32 (d) None of the above

BPSC Agriculture Department-2024

Ans. (a)



$$28 \Rightarrow \frac{25+31}{2} = \frac{56}{2} = 28$$

$$34 \Rightarrow \frac{32+36}{2} = \frac{68}{2} = 34$$

$$22 \Rightarrow \frac{20+24}{2} = \frac{44}{2} = 22$$

$$\therefore 25 \Rightarrow \frac{22+x}{2},\ 31 \Rightarrow \frac{x+34}{2},\ \frac{24+32}{2} = x$$

$$\boxed{x=28} \quad \boxed{x=28} \quad \boxed{x=28}$$

22. In a certain code language, 'GAME' is written as '$ ÷ * %' and 'BEAD' is written as '# % ÷ ×'. How will the word 'MADE' be written in that code language?

(a) $ ÷ × %

(b) * ÷ $ %

(c) * ÷ × %

(d) # ÷ × %

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Ans. (c)

In the given coding language, there is certain code for certain word–

Just like—		In the same way—
G — $	B — #	M — *
A — ÷	E — %	A — ÷
M — *	A — ÷	D — ×
E — %	D — ×	E — %

Hence, option (c) will be correct option.

23. Q travels towards East. M travels towards North. S and T travel in opposite directions. T travels towards right of Q. Which of the following is definitely true?

(a) M and S travel in the opposite directions

(b) S travels towards West

(c) T travels towards North

(d) M and S travel in the same direction

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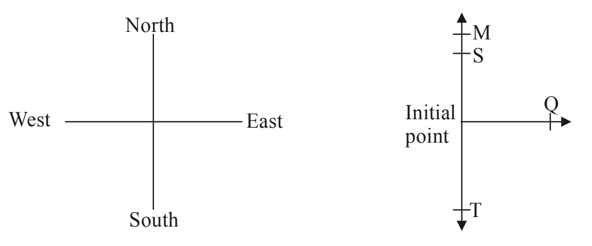
Ans. (d)



From the above arrangement, it is clear that M & S are travelling in the same direction. Hence, option (d) will be the correct option.

24. A man 'Ramesh', who owns a plot of land of 100 square yards, increases his plot of land by acquiring 10% more from his neighbor 'Suresh', who also owns 100 square yards land. After 2 years, he sells back 10% of the total plot to the neighbor. Which of the following is correct?

(a) Ramesh's land is more than Suresh

(b) Suresh's land is more than Ramesh

(c) Both are equal

(d) None of the above

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Ans. (b)

According to the question—

Ramesh $\rightarrow$ 100 sq. yard

Suresh $\rightarrow$ 100 sq. yard

After the acquisition of 10% of land from his neighbour Suresh by 10% $\rightarrow$

Ramesh $\rightarrow 100+100\times\frac{10}{100}=110$ sq. yard

Suresh $\rightarrow 100-100\times\frac{10}{100}=90$ sq. yard

After 2 years Ramesh sells backs 10% of this total land to his neighbour—

Ramesh $\rightarrow 110-110\times\frac{10}{100}=99$ sq. yard

Suresh $\rightarrow 90+110\times\frac{10}{100}=101$ sq. yard

Hence, Suresh land is more than Ramesh land, which is mention in the option (b).

25. Find the odd pair among the following options.

(a) Millinery : Hats

(b) Brewery : Alcohol

(c) Stationery : Paper

(d) Snobbery : Shoes

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Ans. (d)

In the given question the option (d) is odd among other options because–

Millinery means women's Hat and other related goods and Brewery mean a company that makes beer or a place where beer is made & the third option Stationery means the things needed for writing where as Snobbery means behaviour and opinions that are typical of a snob: that means it is not related to shoes.

26. What should come in place of question mark (?) in the following number series?

132 156 ? 210 240 272

(a) 196 (b) 182

(c) 199 (d) 204

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Ans. (b)

The given number series—

132   156   182   210   240   272

+24   +26   +28   +30   +32

Hence, option (b) will be the right answer.

27. Select the missing number from the given alternatives :

44	**49**	**37**
52	**?**	**41**
58	**35**	**53**

(a) 56 (b) 77
(c) 66 (d) 63

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Ans. (b)

44	**49**	**37**	$-R_1$
52	**? = 77**	**41**	$-R_2$
58	**35**	**53**	$-R_3$

from $R_1 \rightarrow (44 - 37) \times 7 = 49$
from $R_3 \rightarrow (58 - 53) \times 7 = 35$
Similarly $R_2 \rightarrow (52 - 41) \times 7 = 11 \times 7 = 77$
Hence, option (b) will be the right answer.

28. Read the given statement and conclusions carefully. Decide which of the given conclusions logically follow(s) from the statement.
Statement : In a one-day cricket match, the total runs made by a team were 200, of which 160 runs were made by spinners.
Conclusions :
1. 80% of the team consisted of spinners.
2. The opening batsmen were spinners.
Select the correct answer.
(a) Only 1 follows the statement
(b) Only 2 follows the statement
(c) Both 1 and 2 follow the statement
(d) Neither 1 nor 2 follows the statement

69th B.P.S.C. (Pre) 2023

Ans. (d)

Statement : Total run = 200 runs
Runs made by Spinners = 160
Conclusion : (1) 80% of team is consisted of spinners–
This conclusion is wrong, here 80% of runs made by spinners but it does not clarify that the team 80% player is spinners.
(2) The opening batsman were spinners–
This conclusion is also wrong because, there is nothing about this conclusion is mentioned in the statement.
Therefore, neither conclusion (1) nor conclusion (2) follows the statement.
Hence, option (d) will be the right answer.

29. In a code language, if GREAT is written as 718222620 and MONK is written as 13121411, then how will VIGOROUS be written in the same language?
(a) 22187121812619 (b) 21177121811619
(c) 22187131813620 (d) 21187111711620

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Ans. (a)

Here, the vowels in the word is coded with its opposite place value.
Just like—

Place value	Code	Place value	Code
G—7	7	M—13	13
R—18	18	O—15	opposite 12
E—5	opposite 22	N—14	14
A—1	opposite 26	K—11	11
T—20	20		

In the same way—

Place value	Code
V—22	22
I—9	opposite 18
G—7	7
O—15	opposite 12
R—18	18
O—15	opposite 12
U—21	opposite 6
S—19	19

Hence, the given word "Vigorous" is written as 22187121812619 which is in option 'a'.

30. Find the odd one in the following groups :
Q,W,Z,B B,H,K,M W,C,G,J M,S,V,X
(a) Q,W,Z,B (b) M,S,V,X
(c) W,C,G,J (d) B,H,K,M

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Ans. (c)

The given letter groups—

Q   W   Z   B
+6   +3   +2

M   S   V   X
+6   +3   +2

W   C   G   J
+6   +4   +3

B   H   K   M
+6   +3   +2

Hence, 'W, C, G, J' is different from the other that is given in

31. Four branches of a company are located at M, N, O and P. M is in the North of N at a distance of 4 km; P is in the South of O at a distance of 2 km; N is in the Southeast of O by 1 km. What is the distance between M and P in km?

(a) 5·34 (b) 6·74

(c) 28·5 (d) None of the above

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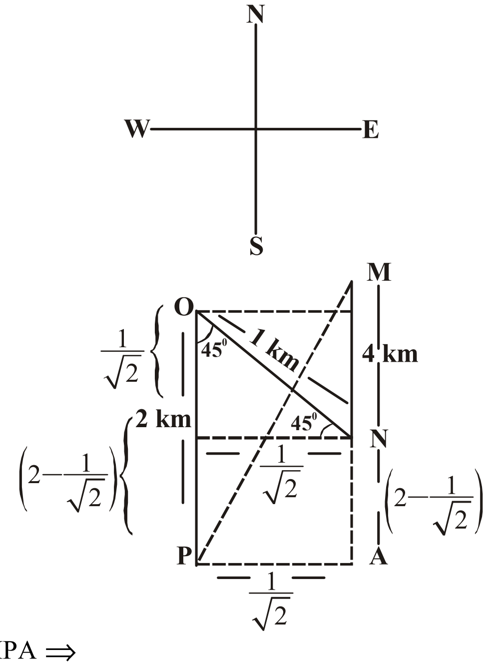
Ans. (a)

Four branches of the company– M, N, O, P



In $\triangle$ MPA $\Rightarrow$

$$MP^2=\left(\frac{1}{\sqrt{2}}\right)^2+\left[4+\left(2-\frac{1}{\sqrt{2}}\right)\right]^2$$

$$MP^2=\frac{1}{2}+16+\left(2-\frac{1}{\sqrt{2}}\right)^2+8\left(2-\frac{1}{\sqrt{2}}\right)$$

$$=\frac{1}{2}+16+4+\frac{1}{2}-2\times2\times\frac{1}{\sqrt{2}}+16-\frac{8}{\sqrt{2}}$$

$$=21-\frac{4}{\sqrt{2}}+16-\frac{8}{\sqrt{2}}$$

$$=37-\frac{12}{\sqrt{2}}$$

$$=37-6\sqrt{2}$$

$$=37-6\times1.414$$

$$=37-8.484$$

$$=28.516$$

$$\boxed{MP=\sqrt{28.516}\approx5.34\ km}$$

32. If $\sqrt{\left(1+\frac{27}{169}\right)}=\left(1+\frac{x}{13}\right)$, then the value of x is

(a) 1 (b) $-\frac{14}{13}$

(c) -27 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

$$\sqrt{\left(1+\frac{27}{169}\right)}=\left(1+\frac{x}{13}\right)$$

$$\sqrt{\left(\frac{169+27}{169}\right)}=1+\frac{x}{13}$$

$$\sqrt{\frac{196}{169}}=1+\frac{x}{13}$$

$$\frac{14}{13}=1+\frac{x}{13}\ \Big|\ \frac{-14}{13}=1+\frac{x}{13}$$

$$14=13+x\ |\ -14=13+x$$

$$\boxed{x=1}\ \Big|\ \boxed{x=27}$$

33. The radius of a solid sphere of some metal is 3 cm. How long a wire of diameter 4 mm can be drawn from it?

(a) 8·8m (b) 9m

(c) 9·1m (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Radius of a sphere = 3 cm

and we are supposed to make a wire of diameter 4 mm. So the length of the wire will be as follows—

Let the length of the wire is h, radius of sphere is 'r' & radius of cylinder is R

Volume of solid sphere = volume of cylinder

($\because$ Wire is in shape of cylinder)

r = 3 cm

R = 2 mm $\Rightarrow$ 0.2 cm

$$\frac{4}{3}\pi r^3=\pi R^2h$$

$$\frac{4}{3}\times3\times3\times3=0.2\times0.2\times h$$

$$36=0.2\times0.2\times h$$

$$h=\frac{36}{0.04}$$

h = 900 cm

$$\boxed{h=9\ meter}$$

34. On selling 30 chairs, a furniture seller gets a loss equal to selling price of 6 chairs. His loss percent is

(a) 16 (b) $16\frac{1}{3}$

(c) $16\frac{2}{3}$ (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

On selling 30 chairs, a furniture seller get a loss equal to selling price of 6 chairs.
Now loss = cost price – selling price
According to question–
6 selling price (SP) = 30 cost price (CP) – 30 selling price (SP)
6 SP + 30 SP = 30 CP

$$\frac{SP}{CP}=\frac{30}{36}$$

$$L\%=\frac{CP-SP}{CP}\times100=\frac{36-30}{36}\times100$$

$$L\%=\frac{6}{36}\times100=\frac{100}{6}$$

$$\boxed{L\%=16\frac{2}{3}\%}$$

35. The product of the digits of a two-digit number is 36. If 45 is subtracted from the number, then the digits interchange mutually. The number is

(a) 84 (b) 73

(c) 49 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (e)

Let the digit = 10x+y
According to the question
x × y = 36 ....(i)
and
(10x+y) – 45 = 10y+x
(10x+y) – (10y+x) = 45
9x – 9y = 45
x – y = 5 ....(ii)
from (i) & (ii)
x = 9 & y = 4
so the initial number is = 94

36. If a + b + c = 0, then what is the value of $\frac{a^2}{bc}+\frac{b^2}{ca}+\frac{c^2}{ab}$?

(a) 1 (b) 2

(c) 3 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Given that–
a + b + c = 0
so a3+b3+c3 = 3abc ....(i)
Now

$$\frac{a^2}{bc}+\frac{b^2}{ca}+\frac{c^2}{ab}$$

$$\frac{a^3}{abc}+\frac{b^3}{bca}+\frac{c^3}{cab}$$

$$\frac{a^3+b^3+c^3}{abc}$$

from equation (i)

$$\frac{3abc}{abc}$$

$$\boxed{\frac{a^2}{bc}+\frac{b^2}{ca}+\frac{c^2}{ab}=3}$$

37. The solution of $4^{x+1}+4^{1-x}=10$ is

(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$

(c) 0 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

$$4^{x+1}+4^{1-x}=10$$

$$4^x\times4+4^1\times4^{-x}=10$$

$$4^x\times4+4\times\frac{1}{4^x}=10$$

$$4^x+\frac{1}{4^x}=\frac{5}{2}$$

Let 4^x = P then–

$$P+\frac{1}{P}=\frac{5}{2} \quad ....(i)$$

$$\left(P-\frac{1}{P}\right)^2=\left(P+\frac{1}{P}\right)^2-4$$

$$=\frac{25}{4}-4$$

$$=\frac{9}{4}$$

$$\left(P-\frac{1}{P}\right)=\pm\frac{3}{2} \qquad \text{.................(ii)}$$

from the equation (i) & (ii)

2P = 4

P = 2

Now

$4^x = 2$

$$4^x=\sqrt{4}=4^x=4^{1/2}$$

$$so\ \boxed{x=\frac{1}{2}}$$

38. On dividing x^3-3x^2+x+2 by a polynomial $g(x)$, the quotient and remainder are (x^2-x+1) and (-2x+4) respectively. Then $g(x)$ is

(a) x-2 (b) x^2+x+1

(c) x^2-1 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

Dividend = Divisor×Quotient+Remainder

Now

$$x^3-3x^2+x+2=g(x)\times(x^2-x+1)+(-2x+4)$$

$$g(x)\times(x^2-x+1)=x^3-3x^2+x+2+2x-4$$

$$=x^3-3x^2+3x-2$$

$$g(x)\times(x^2-x+1)=x^3-3x^2+3x-2$$

$$g(x)=\frac{x^3-3x^2+3x-2}{x^2-x+1}$$

$$g(x)=\frac{(x-2)(x^2-x+1)}{(x^2-x+1)}$$

$$\boxed{g(x)=(x-2)}$$

39. A toy is in the form of a cone mounted on a hemisphere of diameter 7 cm. The total height of the toy is 14·5 cm. Then the volume of the toy is $\left(\pi=\frac{22}{7}\right)$

(a) 231 cm^3

(b) 331 cm^3

(c) 131 cm^3

(d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

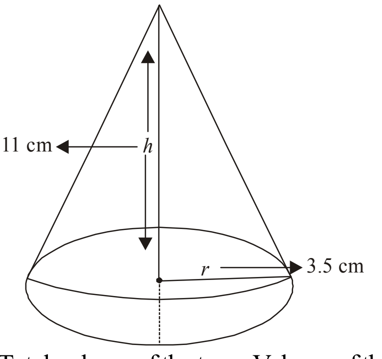
Ans. (a)

Given that–

Diameter of Hemisphere = 7 cm

Total height of the toy = 14.5 cm

Now



Total volume of the toy = Volume of the Hemisphere + Volume of the cone.

$$\text{Total volume}=\frac{2}{3}\pi r^3+\frac{1}{3}\pi r^2 h$$

$$=\frac{\pi}{3}r^2(2r+h)$$

$$=\frac{22}{7\times3}\times\frac{7}{2}\times\frac{7}{2}\left(2\times\frac{7}{2}+11\right)$$

$$=\frac{22}{3}\times\frac{7}{4}\times18$$

=11×21

$$\boxed{Total\ volume=231\ cm^2}$$

40. In a club, all the members participate either in Tambola or in Rummy. 320 participate in Rummy, 350 participate in Tambola and 220 participate in both. How many members does the club have?

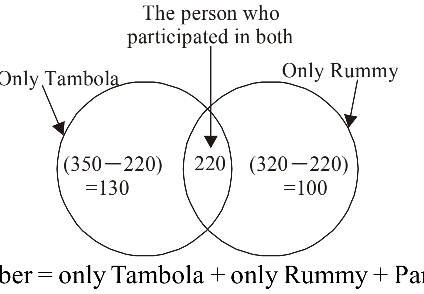
(a) 440 (b) 445

(c) 450 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)



Total member = only Tambola + only Rummy + Participent of both

=130+100+220

$$\boxed{Total\ member=450}$$

41. The sum of the reciprocals of A's ages (in years) 2 years ago and 2 years from now is 2/3. The present age of A is

(a) 5 years (b) 7 years
(c) 4 years (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Let A's present age be x.
The sum of the reciprocals of A's age 2 years ago (x-2) and 2 years from now (x+2) is given by–

$$\frac{1}{(x-2)}+\frac{1}{(x+2)}$$

According to the question–

$$\frac{1}{x-2}+\frac{1}{x+2}=\frac{2}{3}$$

$$\frac{x+2+x-2}{x^2-4}=\frac{2}{3}$$

$$\frac{2x}{x^2-4}=\frac{2}{3}$$

$3x = x^2 - 4$
$x^2 - 3x - 4 = 0$
$x^2 - 4x + x - 4 = 0$
$x(x-4) + 1(x-4) = 0$
$x = 4, -1$
Hence, the answer will be 4

42. For what value of k, the system of equations $x - ky = 2$ and $3x + 2y = -5$ has a unique solution?

(a) $-\frac{2}{3}$ (b) 1
(c) 2 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

The given equation–
$x - ky = 2$ ....(i)
$3x + 2y = -5$ ....(ii)
If the given equation have unique solution, then–

$$\frac{1}{3}\neq\frac{-k}{2}$$

$$-3k\neq 2$$

$$\boxed{k\neq\frac{-2}{3}}$$

42. The radius of a circle is 14 cm and the area of a sector is 102·7 cm². The central angle of the sector is

(a) 30^0 (b) 60^0
(c) 45^0 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

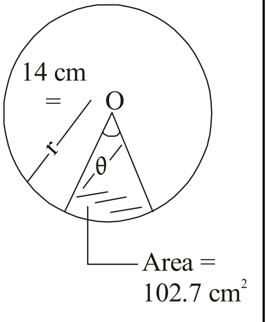
We know that–

$$\text{Area of sector} = \pi r^2\frac{\theta}{360}$$

$$102.7=\frac{22}{7}\times 14\times 14\times\frac{\theta}{360}$$

$$\theta=\frac{102.7\times 45}{11\times 7}$$

$$\boxed{\theta\approx 60^0}$$



44. What is the compound interest of ₹ 15,625 for 9 months when the rate of interest is 16% per annum and the interest is payable quarterly?

(a) ₹ 1,961 (b) ₹ 1,951
(c) ₹ 2,051 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The given sum (P) = 15625
(t) Time = 9 month
rate of interest (r) = 16%
if the interest is payable quarterly than the interest rate for a quarter–
12 m $\rightarrow$ 16%
3m $\rightarrow$ 4%
and No. of quarter in 9 month = 3
We know that–

$$A=P\left(1+\frac{r}{100}\right)^t$$

$$A=15625\times\left(1+\frac{4}{100}\right)^3$$

$$=15625\times\frac{26}{25}\times\frac{26}{25}\times\frac{26}{25}$$

A = 17576
Now total interest = A – P
= 17576 – 15625

$$\boxed{Total\ Interest = ₹\ 1951}$$

45. If the product of the roots of the equation $3kx^2$-$25kx$+ k+8 = 0 is 3, then k is

(a) 1 (b) 4
(c) 3 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

The given equation–
$3kx^2 - 25kx + (k+8) = 0$ ....(i)
if the equation is in the form of the equation given below & their roots is , then–
$ax^2 + bx + c = 0$ ....(ii)
then

$$\alpha + \beta = \frac{-b}{a}$$

$$\& \ \alpha . \beta = \frac{c}{a}$$

on comparing equation (i) & (ii)
a=3k, $b = -25k$, $c = (k+8)$
Product of roots is given i.e. 3

$$\text{so } \frac{k+8}{3k} = 3$$

$k + 8 = 9k$
$8 = 8k$

$\boxed{k = 1}$

46. The value of $\log_{\sqrt{2}}(32)$ is

(a) $\frac{5}{2}$ (b) 5
(c) 10 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Let
$\log_{\sqrt{2}}(32) = x$
We can also write this log function as–

$$32 = \left(\sqrt{2}\right)^x$$

$$32 = \left[(2)^{1/2}\right]^x$$

$$2^5 = (2)^{x/2}$$

on comparing both side–

$$\frac{x}{2} = 5$$

$\boxed{x = 10}$

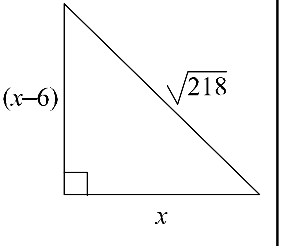
47. The altitude of a right-angled triangle is 6 cm less than its base. If the hypotenuse is $\sqrt{218}$ cm, the other two sides (in cm) are

(a) 5 and 11 (b) 9 and 15
(c) 10 and 16 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (e)

Let base of right angle triangle = x
so Altitude of the right angle triangle = (x–6)
According to the question–
by pythagoras theorem–



$$\left(\sqrt{218}\right)^2 = (x)^2 + (x-6)^2$$

$$218 = x^2 + x^2 + 36 - 12x$$

$$218 = 2x^2 + 36 - 12x$$

$$109 = x^2 + 18 - 6x$$

$$x^2 + 18 - 6x - 109 = 0$$

$$x^2 - 6x - 91 = 0$$

$$x^2 - (13-7)x - 91 = 0$$

$$x^2 - 13x + 7x - 91 = 0$$

$$x(x-13) + 7(x-13) = 0$$

$$(x+7)(x-13) = 0$$

$$x = -7, 13$$

Length cannot be in negative so we take the positive value so that–
$x = 13$

$\boxed{Base = 13}$

$\boxed{Altitude = 13 - 6 = 7}$

48. Arrange the following fractions in ascending order :

$$\mathbf{\frac{13}{18}, \frac{29}{36}, \frac{41}{45}, \frac{51}{60}}$$

(a) $\frac{13}{18} < \frac{29}{36} < \frac{51}{60} < \frac{41}{45}$ (b) $\frac{13}{18} < \frac{51}{60} < \frac{29}{36} < \frac{41}{45}$

(c) $\frac{51}{60} < \frac{41}{45} < \frac{29}{36} < \frac{13}{18}$ (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

$$\frac{13}{18}, \frac{29}{36}, \frac{41}{45}, \frac{51}{60}$$

Now,

$$\frac{13}{18}, \frac{29}{36}, \frac{41}{45}, \frac{51}{60}$$
$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$
$$\frac{130}{180}, \frac{145}{180}, \frac{164}{180}, \frac{153}{180}$$

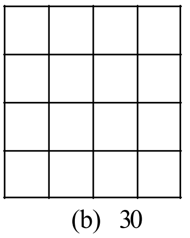
In this case, when denominator is equal than the highest valued numerator will be greater so the ascending order–

$$\frac{130}{180} < \frac{145}{180} < \frac{153}{180} < \frac{164}{180}$$

$$\frac{13}{18} < \frac{29}{36} < \frac{51}{60} < \frac{41}{45}$$

Hence, option (a) will be the right answer.

49. How many squares are there in the given figure?

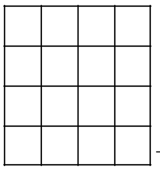


(a) 25 (b) 30
(c) 32 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

When the row and column are equal then–



$$-1^2 \to 1$$
$$-2^2 \to 4$$
$$-3^2 \to 9$$
$$-4^2 \to \underline{16}$$
$$30$$

Hence, the total number of squares in the given figure will be 30.

50. A man fills a basket with eggs in such a way that the number of eggs added on each successive day is the same as the number already present in the basket. This way the basket gets completely filled in 24 days. After how many days, the basket was one-fourth full?

(a) 6 (b) 12
(c) 22 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

If a man fills a basket with eggs in such a way that the number of eggs added on each successive day is the same as the number already present in the basket, then it goes like this

1 2 4 8 16 32 64 .... 2^{23}

if the basket gets completely filled in 24 days, it means the basket was half full on the 23rd day. Therefore, it was one-fourth full on the 22nd day.

51. The sum of two numbers is 480. If the bigger number is decreased by 2% and the smaller number is increased by 10%, then the numbers obtained are equal. The larger number is approximately

(a) $\frac{3320}{13}$

(b) $\frac{3300}{13}$

(c) $\frac{3120}{13}$

(d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Let the bigger number is x & other is y

According to the question–

$x + y = 480$ ....(i)

Now bigger is decreased by 2% and smaller number is increased by 10% then–

$$\frac{98x}{100} = \frac{110y}{100}$$

$$\frac{x}{y} = \frac{110}{98} = \frac{55}{49}$$

from equation (i)

$$x + \frac{49x}{55} = 480$$

$$104x = 480 \times 55$$

$$x = \frac{480 \times 55}{104}$$

$$x = \frac{60 \times 55}{13}$$

$$\boxed{x = \frac{3300}{13}}$$

52. The next number in the finite sequence 486, 324, 216, 144, … is

(a) 96 (b) 76
(c) 36 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

The Given Series–

$486 \xrightarrow{\times\frac{2}{3}} 324 \xrightarrow{\times\frac{2}{3}} 216 \xrightarrow{\times\frac{2}{3}} 144 \xrightarrow{\times\frac{2}{3}} \boxed{96}$

Hence, option (a) will be the right answer.

53. Which number should replace '?' in the following series?
240, 120, ? , 180, 360, 900

(a) 120 (b) 130
(c) 150 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

The given series–

$240 \xrightarrow{\times\frac{1}{2}} 120 \xrightarrow{\times 1} 120 \xrightarrow{\times\frac{3}{2}} 180 \xrightarrow{\times 2} 360 \xrightarrow{\times\frac{5}{2}} 900$

Hence, option (a) will be the right answer.

54. The term in the blank
_____ : FICTION :: GRIND : BMDIY is

(a) LOIZOUT (b) KNHYNTS
(c) IMGXMSR (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Like as–

$B \xrightarrow{+5} G$
$M \xrightarrow{+5} R$
$D \xrightarrow{+5} I$
$I \xrightarrow{+5} N$
$Y \xrightarrow{+5} D$

Similarly–

$F \xrightarrow{+5} K$
$I \xrightarrow{+5} N$
$C \xrightarrow{+5} H$
$T \xrightarrow{+5} Y$
$I \xrightarrow{+5} N$
$O \xrightarrow{+5} T$
$N \xrightarrow{+5} S$

Hence, option (b) will be the right answer.

55. There are 5 routes to go from Prayagraj to Patna and 4 routes to go from Patna to Kolkata. How many ways are possible for going from Prayagraj to Kolkata via Patna?

(a) 45 (b) 20
(c) 54 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (B)

Since there are 5 routes from Prayagraj to Patna and 4 routes from Patna to Kolkata, the total number of ways to go from Prayagraj to Kolkata via Patna is the product of the two route options.
= 5 routes from Prayagraj to Patna × 4 routes from Patna to Kolkata
$= 5 \times 4$
$= 20$
Hence, there a 20 possible ways which is given in the option (b).

56. Select the most suitable statement.

(a) Every circle is an ellipse.
(b) No circle is hyperbola.
(c) Every square is a rectangle.
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

From option (a) – Every circle is an ellipse is correct because, the formal definition of an ellipse is the set of all points such that the sum of the distance between those points and two fixed points is constant.
From option (c) – "Every square is a rectangle" is correct while every rectangle is not correct.
Hence, option (d) will be the right answer.

57. The word CODE is coded as 15161991 and STEP is coded as 18769122. Then SPOTE is coded as

(a) 1822761691 (b) 1822167691
(c) 1816227691 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The given code for the words–
CODE → 15 16 19 91
S T E P → 18 76 91 22
so the code for the word 'SPOTE' is–
= S P O T E → 18 22 16 76 91
Hence, option (b) will be the right answer of the given question.

58. The area of a circle is $40\pi^2$. The area of a square with sides equal to radius of the circle is

(a) 4π (b) $4\pi^2$
(c) 40π (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (c)

Let the radius of the circle is a then the side of squate will also be 'a'–
According to the question–
$\pi a^2 = 40\pi^2$
$\boxed{a^2 = 40\pi}$
($\because$ Area of square of sides a = a^2

ight answer.

59. Select the odd word in the group :
Book, Paper, Pencil, Pen, Eraser

(a) Book (b) Pencil
(c) Eraser (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

In the given words–'Book' is odd one because "Paper, Pencil, Pen, Eraser, is used to write while book is used to "Study". Therefore, option (a) will be the right answer.

60. If in English alphabet, A = 1, D = 2, P = 4, then which of the following denotes 3?

(a) H (b) I
(c) J (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Place value	Code
A = 1 →	$(1)^2$
D = 4 →	$(2)^2$
P = 16 →	$(4)^2$
∴ I = 9 →	$(3)^2$

Therefore, option (b) will be the right answer.

61. Two numbers are in the ratio 21:10. If their HCF is 11, then the sum of the numbers is

(a) 341 (b) 141
(c) 241 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)

Ratio of two numbers are – 21 : 10
HCF = 11
Let the numbers are – $21x$ & $10x$
so the number will be – $21x \times 11 = 231x$
— $10x \times 11 = 110x$
The sum of numbers will be
$= 231x + 110x$
$= 341x$
Therefor sum of numbers will be either 341 or multiples of 341.

62. A profit of 10% is made when a smartphone is sold at P rupees and there is 4% loss when the smartphone is sold at Q rupees. Then P :Q is

(a) 55 : 48 (b) 45 : 34
(c) 110 : 96 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (d)

Let the price of smartphone = 100 ₹
When it is sold at 10% profit

$$P = 100 \times \frac{110}{100} = 110 \text{ ₹}$$

When it is solid at 4% loss

$$Q = 100 \times \frac{96}{100} = 96 \text{ ₹}$$

Now $\boxed{\frac{P}{Q} = \frac{110}{96} = \frac{55}{48}}$

Hence, option (d) will be the right answer.

63. In how many different ways can the letters of the word EDUCATION be arranged so that the vowels come together?

(a) 7200 (b) 14400
(c) 12100 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

The given word–
E D U C A T I O N
Vowels – E, U, A, I, O
Consonants – D, C, T, N
If vowels considered as one group then total ways to arrange the consonants and group = 5! = 120
Also the group of vowels can be arranged in – 5!
ways ⇒ 5! ⇒ 120
Total Number of arrangement–
= 5! × 5!
= 120 × 120
= 14400
Hence, option (b) will be the right answer.

64. If $\left(\frac{3}{4}\right)^{-6} \times \left(\frac{4}{3}\right)^{-8} = \left(\frac{3}{4}\right)^{n}$ then the value of n is–

(a) 1 (b) 4
(c) 3 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (e)

$$\left(\frac{3}{4}\right)^{-6} \times \left(\frac{4}{3}\right)^{-8} = \left(\frac{3}{4}\right)^{n}$$

$$\left(\frac{3}{4}\right)^{-6} \times \left(\frac{3}{4}\right)^{8} = \left(\frac{3}{4}\right)^{n}$$

$$\left(\frac{3}{-}\right)^{-6+8} = \left(\frac{3}{-}\right)^{n}$$

$\left(\frac{3}{4}\right)^2 = \left(\frac{3}{4}\right)^n$

n = 2

Hence, option (e) will be the right answer.

65. The simplified form of $(4^{-1} + 8^{-1}) \div \frac{1}{\left(\frac{3}{2}\right)^{-2}}$ is–

(a) $\frac{1}{6}$ (b) $\frac{2}{3}$

(c) $\frac{1}{2}$ (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

$\left(4^{-1} + 8^{-1}\right) \div \frac{1}{\left(\frac{3}{2}\right)^{-2}}$

$= \left(\frac{1}{4} + \frac{1}{8}\right) \div \left(\frac{3}{2}\right)^2$

$= \frac{(2+1)}{8} \div \frac{9}{4}$

$= \frac{3}{8} \times \frac{4}{9} = \frac{1}{6}$

66. If $A : B = \frac{1}{2} : \frac{3}{8}$, $B : C = \frac{1}{3} : \frac{5}{9}$ and $C : D = \frac{5}{6} : \frac{3}{4}$, then the ratio A : B : C : D is–

(a) 6 : 8 : 9 : 10 (b) 8 : 6 : 10 : 9

(c) 4 : 6 : 8 : 10 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

$A : B = \frac{1}{2} : \frac{3}{8} \Rightarrow 4 : 3$

$B : C = \frac{1}{3} : \frac{5}{9} \Rightarrow 3 : 5$

$C : D = \frac{5}{6} : \frac{3}{4} \Rightarrow 10 : 9$

Now

A : B : C = 4 : 3 : 5

C : D = 10 : 9

Now

A : B : C : = 8 : 6 : 10 : 9

Hence, option (b) will be the right answer.

67. Suresh and Shashi complete a work in 11 days and 17 days respectively. They received a remuneration of ₹5,600 after completing the work jointly. Their shares in the remuneration are respectively–

(a) ₹ 3600 and ₹ 2000 (b) ₹ 3400 and ₹ 2200

(c) ₹ 3500 and ₹ 2100 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

Day — Total work — efficiency

Suresh (11) — LCM (11, 17) = 187 — 17

Shashi (17) — — 11

Total remuneration will be distributed as per the ratio of their efficiency-

$(17 + 11) \rightarrow 5600$

$1 \rightarrow 200$

Suresh = 17 × 200

= 3400

Shashi = 11 × 200 = 2200

68. *A*'s weight is 25% of *B*'s weight and 40% of *C*'s weight. What percentage of *C*'s weight is *B*'s weight?

(a) 160 (b) 180

(c) 50 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

According to the question-

A's weight = 25% of B's Weight = 40% of C's weight

Now

25% B = 40% of C

$\frac{B}{C} = \frac{40}{25}$

$\frac{B}{C} = \frac{8}{5}$

Percentage of B's weight of C's weight

$= \frac{8}{5} \times 100 = 160\%$

69. The diameter of a copper sphere is 6 cm. The sphere is melted and is drawn into a long wire of uniform circular cross-section. If the length of the wire is 36 cm, then its radius is–

(a) 0.5 cm (b) 2 cm

(c) 1.5 cm (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Let the radius of sphere = R
radius of long wire = r
length of the wire = h
According of the question-
2R = 6cm
R = 3 cm
h = 36 cm
Volume of the sphere = volume of the wire

$$\frac{4}{3}\times\pi\times3\times3\times3=\pi\times r^2\times36$$

$\boxed{r = 1\text{ cm}}$

Hence, option (e) will be the right answer.

70. Two circles touch internally. The sum of their areas is 116π cm^2 and the distance between their centres is 6 cm. The radii of the circles are respectively–

(a) 16 cm and 10 cm (b) 10 cm and 4 cm
(c) 12 cm and 6 cm (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

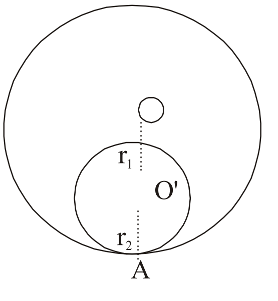
Ans. (b)

According to the question-
$(r_1 - r_2) = 6$ cm ..... (i)
$\pi r_1^2 + r_1^2 = 116\pi$
$r_1^2 + r_2^2 = 116$ ..... (ii)
from equation (i)–
$(r_1 - r_2)^2 = 6^2$
$r_1^2 + r_2^2 - 2r_1r_2 = 36$
$116 - 2r_1r_2 = 36$
$2r_1r_2 = 80$
Now
$(r_1 + r_2)^2 = r_1^2 + r_2^2 + 2r_1r_2$
$(r_1 + r_2)^2 = 116 + 80$
$(r_1 + r_2)^2 = 196$
$r_1 + r_2 = 14$ .... (iii)
from (i) and (iii) –
$2r_1 = 20$ cm

$\boxed{r_1 = 10cm}$

again
$2r_2 = 8$ cm

$\boxed{r_2 = 4cm}$



Hence, option (b) will be the right answer.

71. A fast train takes 3 hours less than a slow train for a journey of 600 km. If the speed of the slow train is 10 km/hr less than that of the fast train, then the speeds of the two trains are respectively

(a) 50 km/hr and 40 km/hr
(b) 30 km/hr and 20 km/hr
(c) 40 km/hr and 30 km/hr
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Let the speed of the high speed train = x
then slow train speed = $(x - 10)$
According to question-

$$\frac{600}{(x-10)}-\frac{600}{x}=3$$

$$\frac{x-x+10}{x(x-10)}=\frac{3}{600}=\frac{1}{200}$$

$2000 = x(x - 10)$
$x^2 - 10x - 2000 = 0$
$x^2 - (50 - 40)x - 2000 = 0$
$x^2 - 50x + 40x - 2000 = 0$
$x(x - 50) + 40(x - 50) = 0$
$(x - 50)(x + 40) = 0$
$x = 50$ km
Hence, High speed train speed is 50 km/h and slow speed train speed is = 50 – 10 = 40 km/h

72. The value of x satisfying the equations

$$\mathbf{\frac{1}{x}+\frac{1}{y}=8,\ \frac{1}{y}+\frac{1}{z}=12,\ \frac{1}{z}+\frac{1}{x}=10}\ \textbf{is}$$

(a) 3 (b) $\frac{1}{3}$
(c) $3\frac{1}{3}$ (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

$\frac{1}{x}+\frac{1}{y}=8$ ....(i)

$\frac{1}{y}+\frac{1}{z}=12$ ....(ii)

$\frac{1}{z}+\frac{1}{x}=10$ ....(iii)

From equation (ii) – (i) –

$\frac{1}{y}+\frac{1}{z}-\frac{1}{x}-\frac{1}{y}=12-8$

$\frac{1}{z}-\frac{1}{x}=4$ .... (iv)

from equation (iii) and (iv) –

$\frac{1}{}+\frac{1}{}+\frac{1}{}-\frac{1}{}=4+10$

$$\frac{2}{z}=14$$

$$\boxed{z=\frac{1}{7}}$$

from equation (iii)

$$\frac{1}{z}+\frac{1}{x}=10$$

$$7+\frac{1}{x}=10$$

$$\frac{1}{x}=3$$

$$\Rightarrow \boxed{x=\frac{1}{3}}$$

Hence, option (b) will be the right answer.

73. There is 50% increase in an amount in 5 years at simple interest. What will be the compound interest of ₹12,000 after 3 years at the same rate?

(a) ₹3972 (b) ₹6240
(c) ₹3120 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Let the principle amount (P) = 100

after 5 year at simple interest the amount will be $=100\times\frac{150}{100}$

$=150$

Now

simple interest = 150 - 100 = 50

$$50=\frac{100\times5\times r}{100}$$

r = 10%

now, at the same rate compound interest of ₹12000

so-

$$A=12000\times\frac{110}{100}\times\frac{110}{100}\times\frac{110}{100}$$

A = 12 × 11 × 11 × 11

A = 12 × 1331

Now compound Interest = A - P

= 15972 – 12000

Compound Interest = 3972

74. $37\frac{1}{2}\%$ of the candidates in an examination were girls. 75% of the boys and $62\frac{1}{2}\%$ of the girls passed, and 342 girls failed. The number of boys failed was

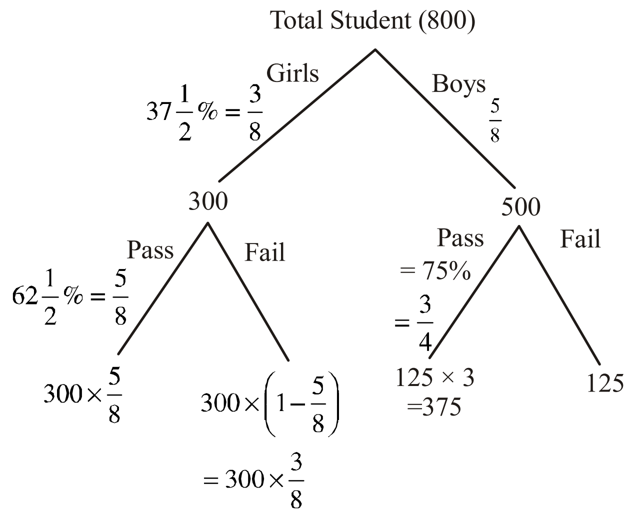
(a) 370 (b) 380
(c) 360 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

$$37\frac{1}{2}\%=\frac{3}{8}$$

Let Total student in the class = 800



According to the question

$$300\times\frac{3}{8}\rightarrow342$$

Now failed boys-

$$125\rightarrow125\times342\times\frac{8}{3\times300}$$

$$\rightarrow5\times342\times\frac{2}{9}$$

$$\rightarrow5\times38\times2$$

Number of boys failed = 380 boys

75. Two possible rational numbers between $-\frac{2}{3}$ and $\frac{1}{2}$ are

(a) $\frac{2}{6},\frac{3}{5}$ (b) $-\frac{2}{6},\frac{2}{6}$
(c) $-\frac{1}{6},\frac{4}{6}$ (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

In order to determine two possible number

b/w $\frac{-2}{3}$ and $\frac{1}{2}$ have to draw number line-

$$\frac{-2}{}\rightarrow\frac{-4}{}$$

$\frac{1}{2} \rightarrow \frac{3}{6}$

$-\frac{4}{6}$, $-\frac{3}{6}$, $-\frac{2}{6}$, $-\frac{1}{6}$, 0 , $\frac{1}{6}$, $\frac{2}{6}$, $\frac{3}{6}$

There are the possible rational numbers between, $\frac{-2}{3}$ and $\frac{1}{2}$

Hence option (b) will be the right answer.

76. A father is five times as old as his son, but after 15 years, he will remain only twice as old as his son. What is the age of the father?

(a) 20 years (b) 25 years
(c) 30 years (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

Let the age of son = x
then, the age of father = 5x
According to the question-
$(5x + 15) = 2 \times (x + 15)$
$5x + 15 = 2x + 30$
$3x = 30 - 15$
$3x = 15$
$x = 5$
Now the age of Father $= 5 \times 5 = 25$ years

77. Neelu buys lemons at the rate of 2 lemons in one rupee and sells them at the rate of 5 lemons in three rupees. Her profit is

(a) 15% (b) 18%
(c) 20% (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

Cost price of 2 lemons (cp) = ₹1
Selling price of 5 lemons (sp) = ₹3
Now
2cp = ₹1 ....(i)
5sp = ₹3 ....(ii)
multiply by 3 on the (i) equation-
6cp = ₹3 ....(iii)
from equation (iii) and (ii)-
6 cp = 5 sp

$$\frac{sp}{cp} = \frac{6}{5}$$

$$P\% = \frac{sp - cp}{cp} \times 100$$

$$P\% = \frac{6 - 5}{5} \times 100$$

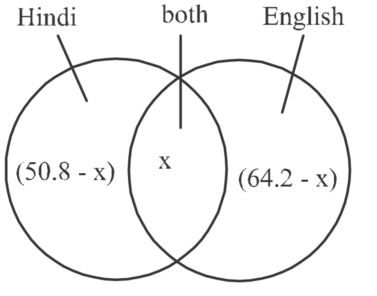
P% = 20%

78. In a school, every student offers either Hindi or English or both. 50·8% offer Hindi and 64·2% offer English. If the total number of students is 500, how many students offer both Hindi and English?

(a) 60 (b) 65
(c) 75 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)



According to question
$(50.8\% - x) + x + (64.2\% - x) = 100\%$
$50.8\% - x + x + 64.2\% - x = 100\%$
$115.0\% - x = 100\%$
$x = 15\%$
Now the number of students who offer both Hindi and English

$$= 500 \times \frac{15}{100} = 75 \text{ Student}$$

79. What is the value of the following?

$$x^{a(b-c)}.x^{b(c-a)}.x^{c(a-b)}$$

(a) 0 (b) 1
(c) x (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

$x^{a(b-c)}.x^{b(c-a)}.x^{c(a-b)}$
$= x^{ab-ac+bc-ba+ca-cb}$
$= x^0$
$= 1$

80. How many 4-digit numbers can be formed using the digits 0, 1, 2, 3, 4 without repetition?

(a) 120
(b) 5^4
(c) 4^5
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

We have 5-digit
0, 1, 2, 3, 4
we are supposed to form 4-digit number using there digits with repetition
(i) If we chose First digit we have 5 options
(ii) If we chose second digit we have 4 options
(iii) If we chose Third digit we have 3 options
(iv) If we chose fourth digit we have 2 options
Now the total number of four digit no.
$= 5 \times 4 \times 3 \times 2$
$=120$

81. If the word HEIGHT is coded as 96108921 and LOOSE is coded as 131616206, then the code for MOBILE, in the same language, is

(a) 1416310136 (b) 1416411136
(c) 1416312136 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

The given coded word-
like as-

	Place value		Code
H	8	$\xrightarrow{+1}$	9
E	5	$\xrightarrow{+1}$	6
I	9	$\xrightarrow{+1}$	10
G	7	$\xrightarrow{+1}$	8
H	8	$\xrightarrow{+1}$	9
T	20	$\xrightarrow{+1}$	21

	Place value		Code
L	12	$\xrightarrow{+1}$	13
O	15	$\xrightarrow{+1}$	16
O	15	$\xrightarrow{+1}$	16
S	19	$\xrightarrow{+1}$	20
E	5	$\xrightarrow{+1}$	6

Similarly-

	Place value		code
M	13	$\xrightarrow{+1}$	14
O	15	$\xrightarrow{+1}$	16
B	2	$\xrightarrow{+1}$	3
I	9	$\xrightarrow{+1}$	10
L	12	$\xrightarrow{+1}$	13
E	5	$\xrightarrow{+1}$	6

Now 'MOBILE' is coded as '14 16 3 10 13 6'

82. Arrange the following words in logical order :

1. Associate Professor 2. Professor
3. Assistant Professor

Select the correct answer using the codes given below.

(a) 2, 3, 1 (b) 2, 1, 3
(c) 1, 2, 3 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)

In the academic hierarchy, the logical order is typically as follows– From lower to higher.

Assistant Professor	Associate Professor	Professor
3	1	2

So the correct order will bi "2 1 3" from higher to lower

83. A person divides some items among four children. The first child gets $\frac{1}{2}$ of the total items and the second child gets $\frac{1}{4}$ of the first child. The third child gets $\frac{3}{4}$ of the first child. The fraction of the items the fourth child gets is

(a) $\frac{3}{4}$ of the total items

(b) $\frac{3}{8}$ of the total items

(c) $\frac{1}{4}$ of the total items

(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (e)

Let the total item = 1

First child $= 1 \times \frac{1}{2} = \frac{1}{2}$

Second child $= \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

Third child $= \frac{3}{4} \times \frac{1}{2} = \frac{3}{8}$

Therefore, Fourth child get Nothing

Remaining

$= 1 - \left(\frac{1}{2} + \frac{1}{8} + \frac{3}{8}\right)$

$= 1 - \left(\frac{1}{2} + \frac{1}{2}\right)$

$= 1 - 1$

84. If the area of a square is equal to the area of an equilateral triangle, the ratio of the side of the square and the side of the equilateral triangle is

(a) 1 : 1 (b) 2 : 1
(c) $\sqrt{3}:2$ (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (e)

Let the side of a square is 'a' and side of the equilateral triangle is 'b'
Now we know that-
Area of square = a^2

Area of a equilateral triangle $= \frac{\sqrt{3}}{4}b^2$

According to the question-

$$a^2 = \frac{\sqrt{3}}{4}b^2$$

$$\frac{a^2}{b^2} = \frac{\sqrt{3}}{4}$$

$$\boxed{\frac{a}{b} = \frac{\sqrt[4]{3}}{2}}$$

85. Find the wrong number in the series given below :
3, 8, 13, 21, 31, 43

(a) 21 (b) 31
(c) 8 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The Given series-

$\boxed{7}$

3 8 13 21 31 43

+4 +6 +8 +10 +12

Therefore, In the place 8, there should be 7 for the correct sequencing of the series.

86. Find the wrong number from the table given below :

2	**4**	**1**	**5**
6	**12**	**3**	**15**
18	**36**	**9**	**45**
54	**108**	**27**	**100**

(a) 9 (b) 36
(c) 100 (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The Given table-

2 ×3	4 ×3	1 ×3	5 ×3
6 ×3	12 ×3	3 ×3	15 ×3
18 ×3	36 ×3	9 ×3	45 ×3
54	108	27	100 135

In the Given table- 100 is incorrect, in the place of 100 there should be 135.

87. Let *a* = 10 cm, *b* = 3 cm and *c* = 6 cm. Which of the following is true?

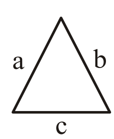
(a) *a, b* and *c* are the sides of a triangle.
(b) *a, b* and *c* form right-angled triangle.
(c) *a, b* and *c* form isosceles triangle.
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

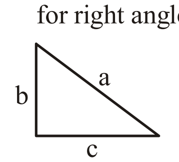
Ans. (e)

Given that–
$a = 10, b = 3, c = 6$
from the Given parameter no triangle can be formed because as per the triangle properties-

for right angled triangle



$a + b > c$
$b + c > a$
$c + a > b$



$a^2 = b^2 + c^2$

Here
$10 + 3 > 6$
but
$3 + 6 \not> 10$
Hence, No triangle can be formed

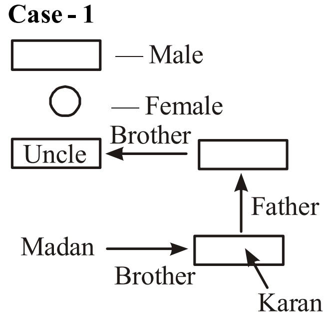
88. Madan's brother's uncle is Karan's father's brother. What can be the relationship between Madan and Karan?

(a) Madan is Karan's uncle
(b) Madan is Karan's brother
(c) Madan is Karan's son
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

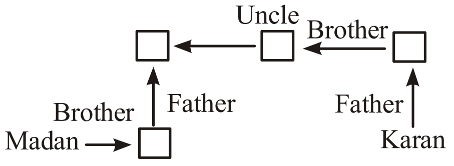
Ans. (e)

Case - 1



Madan's Gender is not defined he may be brother of Karan

Case - 2



Here, Madan & Karan both gender is not defined So we cannot define relation.

89. Find the missing letters in the series given below :

Y, T, O, ______, ______

(a) J, E (b) J, D

(c) K, F (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

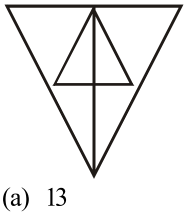
The Given Series-

–5 –5 –5 –5

Y T O J E

Hence, the missing Letter will be J & E.

90. How many triangles are there in the figure given below?



(a) 13 (b) 11

(c) 15 (d) More than one of the above

(e) None of the above

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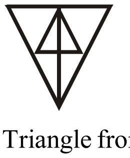
Ans. (b)



Triangle from single Image-6

Triangle from double Image-2

Triangle triple Image-2

Triangle, Comprising all part of the picture-1

Total triangle = 6 + 2 + 2 + 1 = 11

91. Asha walks 3 km southward and then turns right and walks 2 km. She again turns right and walks 3 km, and then turns towards her left and starts walking straight. In which direction is she walking now?

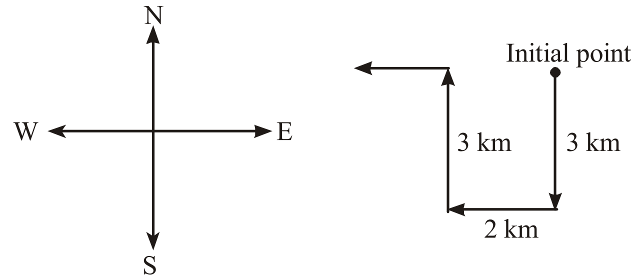
(a) North (b) West

(c) East (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (b)



Hence, finally Asha is walking towards west direction.

92. Find the wrong number in the series given below :

28, 84, 112, 196, 308, 504, 872

(a) 112 (b) 308

(c) 872 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

28 84 112 196 308 504 872 $\boxed{812}$

+28 +84 +112 +196 +308

From the above number series it is clear that in place of 872, there should be 812.

Hence, 872 is the wrong number in the series.

93. In a certain code, '256' means 'you are good'; '637' means 'we are bad' and '358' means 'good and bad'. Which of the following digits represents 'and' in that code?

(a) 2 (b) 5

(c) 8 (d) More than one of the above

(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The given code-

2 5 6 → You are good

6 3 7 → we are bad

3 5 8 → good and bad

from the above arrangment if is clear in that 'and' is coded as '8' in the question.

94. Six roads lead to a country. They may be indicated by the letters X, Y, Z and the digits 1, 2, 3. When there is a storm, Y is blocked. When there are floods, X, 1 and 2 will be affected. When road 1 is blocked, Z is also blocked. At a time, when there are floods and a storm also blows, which road can be used?

(c) Z (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

Roads	Affected or Blocked	Unblocked
X	✓	×
Y	✓	×
Z	✓	×
1	✓	×
2	✓	×
3		✓

Therefore, In the Given Situation, road 3 can be used,

95. In a certain code, DELHI is written as CCIDD. How is BOMBAY written in that code?

(a) AJMTVT (b) MJXVSU
(c) AMJXVS (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

Just Like-

$D \xrightarrow{-1} C$
$E \xrightarrow{-2} C$
$L \xrightarrow{-3} I$
$H \xrightarrow{-4} D$
$I \xrightarrow{-5} D$

Similarly

$B \xrightarrow{-1} A$
$O \xrightarrow{-2} M$
$M \xrightarrow{-3} J$
$B \xrightarrow{-4} X$
$A \xrightarrow{-5} V$
$Y \xrightarrow{-6} S$

Here, option (c) will be the correct option.

96. One of the two digits of a two-digit number is three times the other digit. If you inter- change the digits of this two-digit number and add the resulting number to the original number, you get 88. What is the original number?

(a) 62 (b) 31
(c) 64 (d) 33

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Ans. (a)

Let the Unit digit of the number is 'x', then other number will be '3x'
Now,
$3x \times 10 + x = 31x$
after interchanging, the number-
$10x + 3x = 13x$
Now
$31x + 13x = 88$
$44x = 88$
$x = 2$
Now the Original Number is- $31 \times 2 = 62$
Hence, Option (a) will be the right answer.

97. Two candles are of different lengths and thicknesses. The short and the long ones can burn respectively for 3.5 hours and 5 hours. After burning for 2 hours, the lengths of the candles become equal in length. What fraction of the long candle's height was the short candle initially?

(a) 5/7 (b) 3/5
(c) 4/5 (d) 2/7

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Ans. (a)

Let the lengths are l_1 and l_2 and thickness are x_1 and x_2
$l_1 =$ 3.5 hours and $l_2 = 5$ hours
According to the question-

$$l_1 - 2\times\frac{l_1}{3.5} = l_2 - 2\times\frac{l_2}{5}$$

$$\frac{3.5l_1 - 2l_1}{3.5} = \frac{5l_2 - 2l_2}{5}$$

$$\frac{1.5l_1}{3.5} = \frac{3l_2}{5}$$

$$\frac{l_1}{l_2} = \frac{2.1}{1.5}$$

$$\frac{l_1}{l_2} = \frac{7}{5} \text{ or } \boxed{\frac{l_2}{l_1} = \frac{5}{7}}$$

Hence, option (a) will be the right answer.

98. In a survey of 500 students of a college, it was found that 49% liked watching football, 53% liked watching hockey and 62% liked watching basketball. Also, 27% liked watching football and hockey both, 29% liked watching basketball and hockey both and 28% liked watching football and basketball both. 5% liked watching none of these games. How many students like watching all the three games?

(a) 10% (b) 15%
(c) 12% (d) 20%

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Ans. (b)

n(F) = Percentage of Student who like watching football = 49%
n(H) = Percentage of Student who like watching hockey = 53%
n(B) = Percentage of Student who like watching basketball

$n(F \cap H) = 27\%, n(B \cap H) = 29\%, n(F \cap B) = 28\%$

Since, 5% like watching none of the given games so,

$n(F \cap H \cap B) = 95\%$

Now

$n(F \cap H \cap B) = n(F) + n(H) + n(B) - n(F \cap H) - n(B \cap H)$
$- n(F \cap B) + n(F \cap H \cap B)$

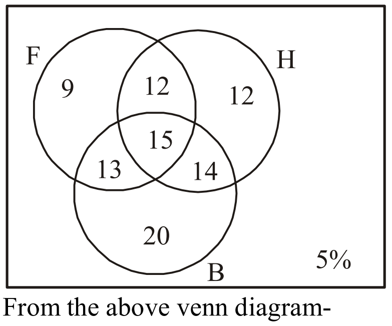
$95\% = 49\% + 53\% + 62\% - 27\% - 29\% - 28\% + n(F \cap H \cap B)$

$95\% = 164\% - 84 + n(F \cap H \cap B)$

$95\% = 80\% + n(F \cap H \cap B)$

$n(F \cap H \cap B) = 15\%$

Now-



From the above venn diagram-

Number of Students who like watching all the three games = 15%

99. Mother was asked how many gifts she had in the bag. She replied that there were all dolls but six, all cars but six, and all books but six. How many gifts had she in all?

(a) 18 (b) 27

(c) 36 (d) 9

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Ans. (d)

Let the total number of toys be x

Then Number of dolls = x – 6

Number of cars = x – 6

Number of books = x – 6

According to the question-

$(x-6)+(x-6)+(x-6) = x$

$3x - 18 = x$

$2x = 18$

$x = 9$

Hence, option (d) will be the right answer.

100. In a dairy, there are 60 cows and buffalos. The number of cows is twice that of buffalos. Buffalo X ranked seventeenth in terms of milk delivered. If there are 9 cows ahead of buffalo X, how many buffalos are after in rank in terms of milk delivered?

(c) 13 (d) 10

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Ans. (b)

Let the number of buffalo be x, then no. of cows will be 2x

Now

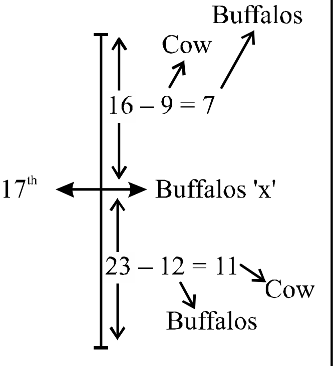
$2x + x = 60$

$3x = 60$

$x = 20$

Cows- 40

Buffalo - 20



There are total 20 buffalos and 7 of them are above buffalos 'x' then after 'x' there will be 12 buffalos

101. If you write down all the numbers from 1 to 100, then how many times do you write 3?

(a) 17 (b) 21

(c) 20 (d) 18

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Ans. (c)

According to the question-

$1 \longrightarrow 9 \longrightarrow 1$ time

$10 \longrightarrow 20 \longrightarrow 1$ time

$21 \longrightarrow 30 \longrightarrow 2$ times

$31 \longrightarrow 40 \longrightarrow 10$ times

$41 \longrightarrow 50 \longrightarrow 1$ time

$51 \longrightarrow 60 \longrightarrow 1$ time

$61 \longrightarrow 70 \longrightarrow 1$ time

$71 \longrightarrow 80 \longrightarrow 1$ time

$81 \longrightarrow 90 \longrightarrow 1$ time

$91 \longrightarrow 100 \longrightarrow 1$ time

Total = 20

Therefore, from 1 to 100, 3 comes, 20 times

102. If a cube of 25 cm side is divided into 125 smaller cubes of equal volume, then the side of smaller cube thus formed will be

(a) 3 cm (b) 5 cm

(c) 6 cm (d) 2 cm

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Ans. (b)

Let us assume that the sides of the smaller cube is x-

According to the question-

(a) 11 (b) 12

$25 \times 25 \times 25 = 125x^3$
$x^3 = 125$
$x = 5$ cm
Hence, option (b) will be the right answer.

103. A, B, C, D, E and F, not necessarily in that order, are sitting at a round table. A is between D and F, C is opposite to D; and D and E are not on neighboring chairs. Which one of the following pairs must be sitting on neighboring chairs?

(a) C and E (b) B and F
(c) A and C (d) A and B

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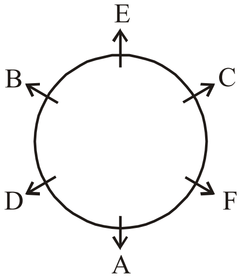
Ans. (a)

There are six person A, B, C, D, E and F sitting around a table with certain condition-
On following that condition the sitting arrangement as follows-



Here, C and E must be sitting on neighboring chairs

104. A man is facing West. He turns 45^0 in the clockwise direction and then another 180^0 in the same direction and then 270^0 in the anticlockwise direction. Which direction is he facing now?

(a) North-West
(b) West
(c) South-West
(d) South

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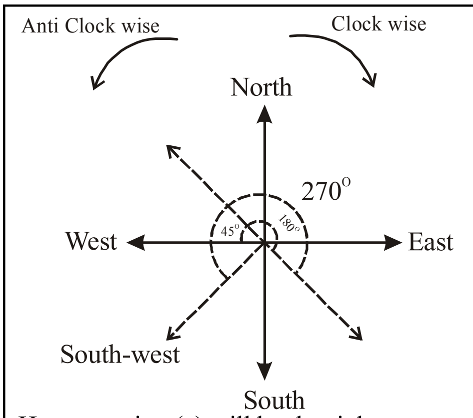
Ans. (c)



Hence, option (c) will be the right answer.

120 degrees

105. If a, b, c, d, e and *f* are six consecutive even numbers, then their average is

(a) a + 5 (b) 6(a + 5)
(c) (abcedf)/6 (d) a + 4

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Ans. (a)

Given that-
a, b, c, d, e and f are six consecutive even numbers-
$b = a + 2$
$c = a + 4$
$d = a + 6$
$e = a + 8$
$f = a + 10$
Now-

$$\text{Average} = \frac{a+b+c+d+e+f}{6}$$

$$= \frac{a+(a+2)+(a+4)+(a+6)+(a+8)+(a+10)}{6}$$

$$= \frac{6a+30}{6}$$

Average = (a + 5)
Hence, option (a) will be the right answer.

106. The question given below has a problem and two statements I and II. Decide if the information given in the statements is sufficient for answering the problem:
K, R, S and T are four players in Indian cricket team. Who is the oldest among them?
I. The total age of K and T together is more than that of S.
II. The total age of R and K together is less than that. of S.

(a) Data in statement II alone is sufficient
(b) Data in both statements together is sufficient
(c) Data in both statements together is not sufficient
(d) Data in statement I alone is sufficient

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Ans. (c)

As per the Given statement in the question-
$K + T > 5$
$R+K<5$
This implies that-
$R<T$
and $R, K<S$
Therefore either T or S is the oldest, but there is no relation between ages of T and S
Hence, data in both statement together is not sufficient.

107. An accurate clock shows 8 O'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 O'clock in the afternoon?

(c) 180 degrees (d) 150 degrees

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Ans. (c)

Given that – hour hand of the Clock moves from 8 O'clock to 2 O' Clock –
Time Gap between 8 O' Clock and 2' O' Clock = 6 hours
Now, We know that–
hour hand moves 360^0 in 12 hours
12 hours $= 360^0$
1 hours $= 30^0$
6 hours $= 180^0$
Therefore, hour hand moves 180^0 from 8 O'clock to 2 O'clock

108. If (x-1) is a factor of the equation x^3 (p)$3x^2 + 3x$ (q) 1 = 0, then signs in place of (p) and (q) are

(a) –, + (b) +, +
(c) +, – (d) –, –

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Ans. (d)

Given that $(x-1)$ is a factor of the equation
$x^3 (p)3x^2 + 3x (q)\ 1 = 0$
then $x - 1 = 0$
$x = 1$
$1(p)3 + 3(q)1 = 0$
From the option (d)-
L.H.S.
$1 - 3 + 3 - 1$
$-2 + 2 = 0$
R.H.S.
Hence, option (d) will be the right answer.

109. Mr. Prasad invested an amount of ₹13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be ₹3,508, what was the amount invested in scheme B?

(a) ₹ 6800 (b) ₹ 7100
(c) ₹ 6400 (d) ₹ 6600

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Ans. (c)

Given that-
$A + B = 13900$ ...... (1)
and
$$\frac{A \times 14\% \times 2}{100} + \frac{B \times 11\% \times 2}{100} = 3508$$
$28A + 22B = 3508 \times 100$
$14A + 11B = 175400$ .....(ii)
Multiplying by 14 in equation (i) & substract equation (ii) from it-
$14A + 14B = 13900 \times 14$

$$\begin{array}{r} 14A + 14B = 194600 \\ \underline{-\ 14A \underset{-}{+} 11B \underset{-}{=} 175400} \\ 3B = 19200 \end{array}$$
B = ₹ 6400
Hence, Option (c) will be the right answer

110. If the national day of a country was celebrated on the 4th Saturday of a month, then the date of celebration was (it is given that the first day of that month is Tuesday)

(a) 25th (b) 26th
(c) 27th (d) 24th

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Ans. (B)

Let us assume a month that have 30 day's
If the Ist day of the month is "Tuesday" then the Ist saturday of this month will be on '5'
Now
II[nd] Saturday-12
III[rd] Saturday- 19
IV[th] Saturday- 26
Hence, Option (b) will be the right answer.

111. Find the quadratic equation whose roots are the reciprocals of the roots of $2x^2+5x+3$.

(a) $3x^2 - 5x - 2 = 0$ (b) $5x2 + 3x + 2 = 0$
(c) $3x^2 + 5x + 2 = 0$ (d) None of the above

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Ans. (c)

The Given equation–
$2x^2+5x+3 = o$
We know that–
$ax^2 + bx + c = 0$ whose roots are α & β
then
$$\alpha + \beta = \frac{-b}{a}, \alpha.\beta = \frac{c}{a}$$
similarly–
$$\alpha + \beta = \frac{-5}{2}$$
$$\alpha.\beta = \frac{3}{2}$$
reciprocal of α, β, $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is
Now–
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha.\beta} = \frac{-\frac{5}{2}}{\frac{3}{2}} = \frac{-5}{3}$$

and–

$$\frac{1}{\alpha.\beta}=\frac{2}{3}$$

Now the new equation–

$$x^2-\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)x+\left(\frac{1}{\alpha}\times\frac{1}{\beta}\right)=0$$

$$x^2-\left(\frac{-5}{3}\right)x+\frac{2}{3}=0$$

$$x^2+\frac{5}{3}x+\frac{2}{3}=0$$

$3x2+5x+2=0$

Hence, Option (c) will be the right answer.

112. Suppose that in a town, 800 people are selected randomly. 280 go to work by car only, 220 go to work by bicycle only and 140 use both ways-sometimes go with a car and sometimes with a bicycle. How many people go by neither car nor bicycle?

(a) 500 (b) 160

(c) 60 (d) 140

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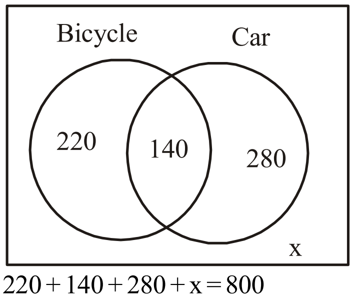
Ans. (b)

Total People = 800

Let x person does not go by neither car nor bicycle-



$220+140+280+x=800$

$640+x=800$

$x=160$

Hence, option (b) will be the right answer.

113. A worker is paid ₹ 20 for a full day work. He works 1, 1/3, 2/3, 1/8, 3/4 days in a week. What is the total amount paid for that worker?

(a) 58 (b) 57.50

(c) 56.50 (d) 57

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Ans. (b)

Given that-

Worker's full day salary = ₹20

Total days work in a week

$$=1+\frac{1}{3}+\frac{2}{3}+\frac{1}{8}+\frac{3}{4}$$

$$=1+1+\frac{1}{8}+\frac{6}{8}$$

$$=2+\frac{1}{8}+\frac{6}{8}$$

$$=\frac{23}{8}\text{ day}$$

Total amount paid $=\frac{23}{8}\times 20=\frac{115}{2}=₹57.50$

Hence, option (b) will be the right answer.

114. The odometer of a car reads 2000 km at the start of a trip and 2400 km at the end of the trip. If the trip took 8 hours, the average speed of the car will be

(a) 40 km/h (b) 75 km/h

(c) 50 km/h (d) 60 km/h

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Ans. (c)

Reading of odometer at the start of trip = 2000 km

Reading of odometer at the end of trip = 2400 km

Distance covered = 2400 - 2000 = 400 km

Time taken for the trip = 8 hours.

Now

Average speed of car $=\frac{400}{8}$ km/hr

= 50 km/hr

115. The average marks of 80 students are given to be 60. It was found later that marks of two students were 45 and 31 which were misread as 85 and 71. The corrected mean mark will be

(a) 59 (b) 53

(c) 61 (d) 49

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Ans. (a)

The average marks of 80 students = 60

Total marks of 80 students = 60 × 80 = 4800

It was found that 45 & 31 misread as 85 and 71. Now the correct total marks of 80 students

$=4800+(45+31)-(85+71)$

$=4800+76-156$

$=4720$

Now the correct average of 80 students $=\frac{4720}{80}=59$

Hence, option (a) will be the right answer.

116. If out of given six consecutive natural numbers, the sum of first three consecutive natural numbers is 27, what is the sum of rest last three consecutive natural numbers?

(c) 36 (d) 96

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Ans. (c)

Let us consider the six consecutive natural number-
$x, (x+1), (x+2), (x+3), (x+4), (x+5)$
According to the question-
$x+(x+1)+(x+2)=27$
$3x+3=27$
$x+1=9$
$x=8$
Now, the sum of last three consecutive natural number-
$=(x+3)+(x+4)+(x+5)$
$=3x+12$
$=3\times 8+12$
$=36$
Hence, option (c) will be the right answer.

117. If $3^{2x-1}+3^{2x+1}=270$ then x is equal to

(a) 5 (b) 3
(c) 2 (d) 4

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

$3^{2x-1}+3^{2x+1}=270$
$3^{2x}\times 3^{-1}+3^{2x}\times 3^{1}=270$
$$3^{2x}\left(\frac{1}{3}+3\right)=270$$
$$3^{2x}\times\frac{10}{3}=270$$
$3^{2x}=3\times 3\times 3\times 3$
$3^{2x}=3^{4}$
$2^{x}=4$
$x=2$
Hence, option (c) will be the right answer.

118. A train reaches at its destination on time if its speed is 40 km per hour from its initial station. If this train runs with a speed of 35 km per hour, it reaches its destination late by 15 minutes. Accordingly what is the distance between its initial station and destination station?

(a) 60 km (b) 70 km
(c) 80 km (d) 35 km

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Ans. (b)

Let the Distance is x
we know that-
$$\text{Speed}=\frac{\text{Distance}}{\text{Time}}$$
Now, According to the question-
$$\frac{x}{35}-\frac{x}{40}=\frac{15}{60}$$
$$x\left(\frac{1}{35}-\frac{1}{40}\right)=\frac{15}{60}$$
$$x\times\frac{5}{40\times 35}=\frac{1}{4}$$
$$x=\frac{40\times 35}{20}$$
$x=70$ km
Hence, option (b) will be the right answer.

ight answer.

119. If two numbers x and y are less than 50% and 20% respectively with any third number, the percentage x of y is equal to

(a) 62.5 (b) 65
(c) 66 (d) 62

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Ans. (a)

Let the third number is 'z'
Then $x=z\times\frac{50}{100}\Rightarrow\frac{z}{2}$
$y=z\times\frac{80}{100}\Rightarrow\frac{4z}{5}$
Now percentage x of y $=\dfrac{\frac{z}{2}}{\frac{4z}{5}}\times 100=62.5\%$
Hence, option (a) will be the right answer.

120. A shopkeeper fixes selling price of its items by 10% increase on its cost. On this fixed selling price, a discount of 10% is given. What is the percentage loss or profit on sale?

(a) 1% profit (b) 10% loss
(c) 10% profit (d) 1% loss

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Ans. (d)

Let the cost price of item = 100
then market price $=100\times\frac{110}{100}=110$
After 10% discount, the selling price $=110\times\frac{90}{100}=99$
Now
$$\text{Loss }(L\%)=\frac{CP-SP}{CP}\times 100$$
$$=\frac{100-99}{100}\times 100$$
$L\%=1\%$

121. Two numbers are such that their sum, difference and product are in the ratio 9:5:28. The difference of their square is

(a) 170 (b) 280
(c) 370 (d) 180

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Ans. (d)

Let two numbers are x and y
according to the question-
$(x+y):(x-y):xy = 9:5:28$
$x+y = 9k$ ....(i)
$x-y = 5k$ ....(ii)
from equation (i) and (ii)-
$2x = 14k$
$x = 7k$
also
$y = 2k$
Now
$xy = 28k$
$14K^2 = 28k$
$k = 2$
then the numbers are-
$x = 14, y = 4$
According to the question-
$x^2 - y^2 = 196 - 16$
$x^2 - y^2 = 180^0$

122. The cost price of an item is ₹800. At the time of its sale, a discount of 10% is given on its fixed selling price. In spite of this discount, a profit of 12.5% is earned. Accordingly, what is the fixed selling price of this item?

(a) ₹1200 (b) ₹1100
(c) ₹1000 (d) ₹1300

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (c)

Given that-
cost price of article = 800
Discount = 10%
Profit = 12.5%
We know that-

$$\frac{MP}{CP} = \frac{100 \pm \text{Profit or loss}}{100 - \text{discount}\%}$$

$$\frac{MP}{800} = \frac{100+12.5}{100-10}$$

$$\frac{MP}{800} = \frac{112.5}{90}$$

$$MP = \frac{112.5 \times 80}{9}$$

MP = ₹1000
Hence, option (c) will be the right answer.

123. If the sum of square of two numbers and difference of the squares of these numbers are 58 and 40 respectively, then the sum of these numbers is

(a) +10 (b) -10
(c) 0 (d) Both (a) and (b)

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (d)

Let the numbers are x & y
According to the question-
$x^2 + y^2 = 58$ ....(i)
$x^2 - y^2 = 40$ ....(ii)
On adding both the equation-
$2x^2 = 98$
$x^2 = 49$
$x = \pm 7$
from equation (i)-
$49 + y^2 = 58$
$y^2 = 9$
$y = \pm 3$
then sum of these numbers are +10 and – 10 which is given in the option (d).

124. How many words, no matter if they are meaningless, can be formed by the letters of the word 'DIARY"?

(a) 24 (b) 25
(c) 10 (d) More than one of the above
(e) None of the above

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Ans. (e)

No. of letters in word 'DIARY' = 5
According to questions
No. of words that can be formed by the letters of the word 'DIARY' = 5!
$= 5 \times 4 \times 3 \times 2 \times 1$
$= 120$

125. Select the missing number from the given alternatives:

44	49	37
52	?	41
58	35	53

(a) 66 (b) 56
(c) 77 (d) More than one of the above
(e) None of the above

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44	49	37
52	?	41
58	35	53

In row I $\rightarrow$ $44-37=7 \Rightarrow 7\times7=49$
In row III $\rightarrow$ $58-53=5 \Rightarrow 5\times7=35$
Similarly:
In row II $\rightarrow$ $52-41=11 \Rightarrow 11\times7=77$

126. Find the missing number in the given series following the same pattern :
15, 20, 32, 62, 118, 248, ?

(a) 428 (b) 322
(c) 368 (d) More than one of the above
(e) None of the above

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Ans. (a)

15, 20 32, 62, 118, 248, ?

15 → 20 → 32 → 62 → 118 → 248 → $\boxed{428}$

+5, +12, +30, +56, +130, +180

2^2+1, 3^2+3, 5^2+5, 7^2+7, 11^2+9, 13^2+11

Pattern $\rightarrow$ (Prime no.)^2 + (odd no.) in increasing order.
So the correct answer is option (a) 428.

127. In a code language, if FASTER is written as 2229212319 and MONK is written as 15161412, then how will GUIDE be written in the same language?

(a) 192019423 (b) 212219523
(c) 222119522 (d) More than one of the above
(e) None of the above

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Ans. (b)

As

F	A	S	T	E	R
(6)	(1)	(19)	(20)	(5)	(18)
O.P.V +1	P.V +1	O.P.V +1	P.V +1	O.P.V +1	P.V +1
21+1	1+1	8+1	20+1	22+1	18+1
22	2	9	21	23	19

and

M	O	N	K
(13)	(15)	(14)	(11)
O.P.V +1	P.V +1	O.P.V +1	P.V +1
14+1	15+1	13+1	11+1
15	16	14	12

Similarly–

G	U	I	D	E
(7)	(21)	(9)	(4)	(5)
O.P.V +1	P.V +1	O.P.V +1	P.V +1	O.P.V +1
20+1	21+1	18+1	4+1	22+1
21	22	19	5	23

Hence. GUIDE will be written as 212219523.

[Note: O.P.V. $\rightarrow$ Opposite Positional Value
P.V. $\rightarrow$ Positional Value.]

128. A rectangle has an area 30 cm2 and perimeter 26 cm. Its sides (in cm) are

(a) 10, 3 (b) 5, 6
(c) 2, 15 (d) More than one of the above
(e) None of the above

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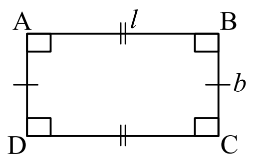
Ans. (a)

Given that,



Area of a rectangle = 30 cm^2
Perimeter of rectangle = 26 cm
Let l and b are the length and breadth of rectangle
lb = 30 cm^2 ....(i)
2 (l+b) = 26 cm
l+b = 13 cm .....(ii)
From eq^n (i) and eq^n (ii)

$\boxed{l = 10\,\text{cm}}$

and

$\boxed{b = 3\,\text{cm}}$

129. A shopkeeper offers 10% discount on an item with marked price Rs. 400. If he charges 10% GST, then the final price of the item is

(a) Rs. 380 (b) Rs. 400
(c) Rs. 396 (d) More than one of the above
(e) None of the above

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Ans. (c)

Given that
Marked price (M.P.) of an item = Rs. 400

Price of item after 10% discount = Rs. 400 $\frac{\text{Rs. } 400\times10}{100}$

= Rs. 400 – Rs. 40

= Rs. 360

GST on item $= \frac{360 \times 10}{100} = 36$

Total payable price of item = Rs. 360 + Rs. 36

= Rs. 396

130. Arrange the following words in logical and meaningful order :

1. Vice President **2. President**

3. Speaker **4. Prime Minister**

5. Members of the Parliament

(a) 5, 1, 2, 3, 4 (b) 4, 2, 1, 3, 5

(c) 2, 1, 4, 3, 5 (d) More than one of the above

(e) None of the above

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Ans. (c)

The logical and meaningful order is →

President > Vice President > Prime Minister > Speaker >
(2) (1) (4) (3)

Member of Parliament
(5)

Hence, the correct answer is option (c).

131. Reshma donates one-fourth of her property to a charity organization and divides the remaining property equally among her three children. The part of property each child gets is

(a) half (b) one-fourth

(c) two-thirds (d) More than one of the above

(e) None of the above

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Ans. (b)

Let the property of Reshma is x.

Part of property donated to charity organisation $= \frac{x}{4}$

Remaining property $= x - \frac{x}{4} = \frac{3x}{4}$

Now, Part of property one child $= \frac{\frac{3x}{4}}{3} = \frac{3x}{4 \times 3} = \frac{x}{4}$

Hence, the part of property each child gets is one fourth.

132. Which of the following statements is false?

(a) Every square is a rhombus.

(b) Every square is a rectangle.

(c) Every rhombus is a parallelogram.

(d) More than one of the above

(e) None of the above

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Ans. (e)

The correct answer is option (e).

Every square is a rectangle and also rhombus.

Every rhombus is a parallelogram. So all the statement are true.

133. In a family of some persons, B says that R is the daughter of my sister A, who is the only daughter of T. X is the child of T and Z, who is the grandmother of H. K is the mother of M, who is the only sister of H. X is unmarried. If S is the spouse of A, how is K related to S?

(a) Brother-in-law

(b) Sister-in-law

(c) Wife of brother-in-law

(d) More than one of the above

(e) None of the above

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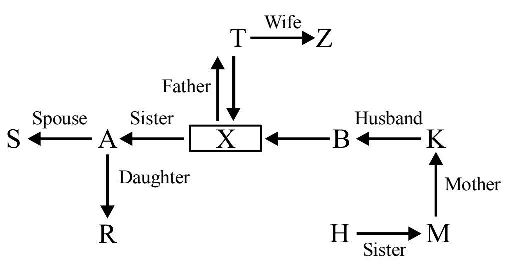
Ans. (c)

According to question



Hence, K is the wife of B, who is brother-in-law of S.

So option (c) is the correct answer i.e. K is the wife of brother-in-law of S.

134. The missing number in the sequence 2, 3, 6, 18, ?, 1944 is

(a) 154 (b) 180

(c) 108 (d) 450

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (c)

The Given series-

2 3 6 18 108 1944

2×3 6×3 18×6 108×18

Hence, option (c) will be the right answer.

135. At 4:30 AM, what is the angle formed between the hour hand and the minute hand?

(a) 60^0 (b) 45^0

(c) 30^0 (d) 90^0

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Formula for finding angle between hour hand and minute hand

$$\theta = \left[\frac{11}{2} \times \text{minute} - 30 \times \text{hour}\right]$$

At 4 : 30 am

$$\theta = \left[\frac{11}{2} \times 30 - 30 \times 4\right]$$

$= [11 \times 15 - 120]$

$= [165^0 - 120^0]$

$= 45^0$

136. If nC9 = nC8, then the value of n is

(a) 1 (b) 17

(c) 72 (d) 36

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (b)

$^nC_9 = {^nC_8}$

$$\frac{n!}{9!(n-9)!} = \frac{n!}{8!(n-8)!} \qquad \left[\because {^nC_r} = \frac{n!}{r!(n-r)!}\right]$$

$$\frac{1}{9 \times 8!(n-9)!} = \frac{1}{8! \times (n-8) \times (n-9)!}$$

$(n - 8) = 9$

$n = 17$

137. If the average of 8 observations is 16 and that of 4 observations is 64, then the average of (8 + 4) = 12 observations is

(a) 4/8 (b) 8/12

(c) 32 (d) 4

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (c)

Average of 8 observations is 16 and average of 4 observations is 64, then the average of 12 observation will be-

$$= \frac{8 \times 16 + 4 \times 64}{(8+4)}$$

$$= \frac{128 + 256}{12}$$

$$= \frac{384}{12}$$

$= 32$

Hence, option (c) will be the right answer.

138. In a school, there are 20 teachers who teach Mathematics or Physics. Of these, 12 teach Mathematics and 4 teach both Mathematics and Physics. How many teachers teach Physics?

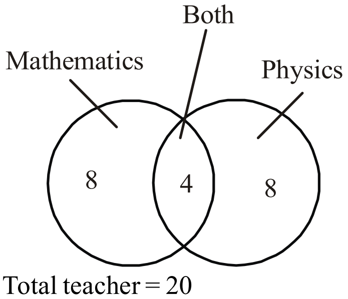
(a) 4 (b) 8

(c) 12 (d) 16

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (c)



Total teacher = 20

only maths teacher= 8

only physics teacher = 8

and teacher who teach both subject = 4

Now the physics teacher = 12

139. The 101st term of the arithmetic progression 5, 11, 17, ... is

(a) 605 (b) 505

(c) 405 (d) 305

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (a)

The Given Arithmetic progression-

5, 11, 17 ....

$a = 5$

$d = 6$

$T_n = a + (n - 1)d$

For 101st term-

$T_{101} = 5 + (101 - 1) \times 6$

$= 5 + 600$

$T_{101} = 605$

140. The two roots of the quadratic equation

$$2x^2 + \sqrt{6}x - 6 = 0 \text{ are}$$

(a) $\sqrt{6}$ and $\sqrt{6}/2$

(b) $-\sqrt{6}$ and $\sqrt{6}$

(c) $-\sqrt{6}$ and $\sqrt{6}/2$

(d) 6 and $\sqrt{6}$

(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

The given quadratic equation–

$$2x^2+\sqrt{6}x-6=0$$
$$2x^2+\left(2\sqrt{6}-\sqrt{6}\right)x-6=0$$
$$2x^2+2\sqrt{6}x-\sqrt{6}x-6=0$$
$$2x\left(x+\sqrt{6}\right)-\sqrt{6}\left(x+\sqrt{6}\right)=0$$
$$\left(2x-\sqrt{6}\right)\left(x+\sqrt{6}\right)=0$$
$$x=\frac{\sqrt{6}}{2},-\sqrt{6}$$

Hence, option (c) will be the right answer.

141. Given that
$2x + 3y = 10$
$3x – y = 4$
then x and y are respectively

(a) – 2 and 2 (b) 2 and –2
(c) 2 and –3 (d) 2 and 2
(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (d)

$2x+3y=10$ ....(i)
$3x-y=4$ ....(ii)
multiply by 3 in equation (ii)
$9x-3y=12$ ....(iii)
Adding equation (i) and (iii)
$11x=22$
$x=2$
from equation (i)-
$2\times 2+3y=10$
$3y=10-4$
$3y=6$
$y=2$
Hence, option (d) will be the right answer.

142. The Value of

$$\frac{(35+27)^2+(35-27)^2}{(35)^2+(27)^2}$$

(a) 4 (b) 8
(c) 62 (d) 2
(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (d)

$$\frac{(35+27)^2+(35-27)^2}{(35)^2+(27)^2}$$
$$=\frac{(35)^2+(27)^2+2\times35\times27+(35)^2+(27)^2-2\times35\times27}{(35)^2+(27)^2}$$
$$=\frac{2[(35)^2+(27)^2]}{[(35)^2+(27)^2]}$$
$=2$
Hence, option (d) will be the right answer.

143. The population of a village is 16500. If it increases annually at the rate of 10%, then what will be its population after two years?

(a) 19965 (b) 19925
(c) 19800 (d) 19250
(e) None of the above / More than one of the above

B.P.S.C. CDPO (English) 2022

Ans. (a)

Present population = 16500
It increase by 10% annually then after two year-
$$=16500\times\frac{110}{100}\times\frac{110}{100}$$
$=165\times 121$
$=19965$
Hence, option (a) will be the right answer.

144. Sarita purchases lemons at the rate of 2 lemons in 1 rupee and sells them at the rate of 5 lemons in 3 rupees. Her profit percentage is

(a) 12.5 (b) 18
(c) 15 (d) 20
(e) None of the above / More than one of the above

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Ans. (d)

Cost price of 2 lemons = 1 rupee
So 2 CP = 1 ... (i)
Now, selling price of 5 lemons = 3 rupees
So 5 SP = 3 ... (ii)
from equation (i) & (ii)
2 CP = 1 ... (i)
5 SP = 3 ... (ii)
multiply equation (i) by 3
6 CP = 3 ... (iii)
from (ii) & (iii)
5 SP = 6 CP ($\because$ RHS is same)
$$\frac{SP}{CP}=\frac{6}{5}$$
We know that
$$P\%=\frac{SP-CP}{CP}\times100$$
Now
$$\frac{SP-CP}{CP}\times100=\frac{6-5}{5}\times100\Rightarrow 20\%$$

145. The parallel sides of a trapezium are 12 cm and 8 cm respectively and the distance between them is 7 cm. Its area is

(a) 70 cm^2 (b) 90 cm^2
(c) 120 cm^2 (d) 140 cm^2
(e) None of the above / More than one of the above

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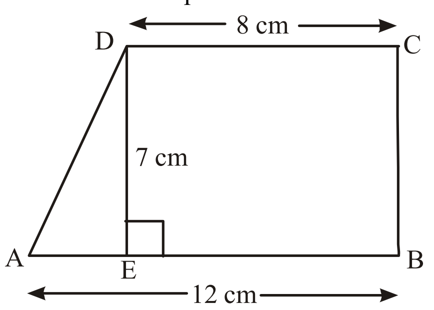
Ans. (a)

Let ABCD is a trapezium-



According to the area formula of trapezium =

Area $= \frac{1}{2} \times$ distance $\times$ (addition of parallel sides)

$= \frac{1}{2} \times 7 \times (12+8)$

$= \frac{1}{2} \times 7 \times 20$

Area = 70 cm^2

146. A and B can finish a work in 11 and 17 days respectively. They worked together and received remuneration of ₹ 5,600 after finishing the work. Their shares in the remuneration are respectively

(a) ₹ 3,600, ₹ 2,000 (b) ₹ 3,400, ₹ 2,200
(c) ₹ 3,500, ₹ 2,100 (d) ₹ 3,300, ₹ 2,300
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

A takes 11 days to finish work and B takes 17 days to complete the same work so-

Efficiency
A (11) — 17
Total work
(LCM of 11, 17)
= 187
B (17) — 11

As we know that total remuneration of 5600 will be distributed between them in the ratio of their efficiency.

Now,

(17 + 11) → 5600
28 → 5600
1 → 200
17 → 3400
11 → 2200

The share of A & B will be ₹ 3400 & ₹ 2200 respectively

147. The average weight of 60 students of a class is 50 kg. By including the class teacher, the average weight increases by 500 gm. The weight of the class teacher is

(a) 79 kg (b) 80 kg
(c) 80.5 kg (d) 81 kg
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (c)

Average weight of 60 student = 50 kg

After including teachers weight the average weight of the class increased by 500 gm

Now the weight of the Class teacher

$= 50 + 61 \times 0.5$ kg

$= 50 + 61 \times \frac{5}{10}$

$= 50 + 30.5$

weight of Class teacher = 80.5 kg

148. The missing number in the sequence 60, 180, 90, 270, ?, 405 is

(a) 675 (b) 540
(c) 360 (d) 135
(e) None of the above / More than one of the above

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Ans. (d)

$60 \xrightarrow{\times 3} 180 \xrightarrow{\div 2} 90 \xrightarrow{\times 3} 270 \xrightarrow{\div 2} \boxed{135} \xrightarrow{\times 3} 405$

Hence the missing number will be '135'

149. A clock is started at 12:00 noon. By 20 minutes past 04:00, the hour hand will turn through

(a) 125° (b) 130°
(c) 135° (d) 140°
(e) None of the above / More than one of the above

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Ans. (b)

Time between 12 : 00 noon & 20 minute past 4 : 00

= 4 hours 20 min.

$= \left(4 + \frac{1}{3}\right)$ hours $\Rightarrow \frac{13}{3}$ hours

Hour hand angle in 12 hours = 360^0

$\frac{13}{3}$ hours $= \frac{360}{12} \times \frac{13}{3} \Rightarrow 130^0$

150. Given that
113x+97y=517
97x+113y=533
then x and y are respectively
(a) 3 and 2 (b) 2 and 3
(c) 3 and 4 (d) 4 and 3
(e) None of the above / More than one of the above
B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

The Given equation-
$113x + 97y = 517$ —— (i)
$97x + 113y = 533$ —— (ii)
multiply (i) by 113 & (ii) by 97, then we get-
$12769x + 10961y = 58421$ ——(iii)
$\underline{-9409x \pm 10961y = \pm 51701}$ ——(iv)
substact (iii) & (iv)
$3360x = 6720$
$$x = \frac{6720}{3360} = 2$$
put the value of x in the equation (i)
$113 \times 2 + 97y = 517$
$226 + 97y = 517$
$97y = 517 - 226$
$97y = 291$
$y = 3$
Hence, the value of x & y will be 2 & 3 respectively.

151. The population of a town is 3528000. If it increases annually at the rate of 5%, then what will be its population after 3 years?
(a) 4084102 (b) 3984102
(c) 4048102 (d) 4148102
(e) None of the above / More than one of the above
B.P.S.C. Assistant Audit Officer 2022

Ans. (e)

Given that the population of the town
$= 3528000$
population increases annually by 5% per annum so-
$$\text{Increased population} = 3528000 \times \frac{105}{100} \times \frac{105}{100} \times \frac{105}{100}$$
$$= 3528 \times \frac{21 \times 21 \times 21}{8}$$
$= 21 \times 21 \times 21 \times 441$
$= 441 \times 441 \times 21$
$= 4084101$

152. In a school, every student plays either cricket or badminton or both. If 50.7% play cricket, 64.3% play badminton and the total number of students is 600, how many students play both cricket and badminton?
(a) 30 (b) 60
(c) 90 (d) 120
(e) None of the above / More than one of the above
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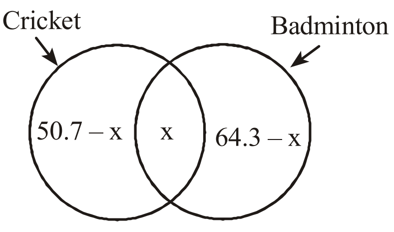
Ans. (c)

Total student = 600



let x student play both the games
So,
$50.7 - x + x + 64.3 - x = 100$
$115 - x = 100$
$x = 115 - 100$
$x = 15\%$
$$\text{Total student who play both the games} = 600 \times \frac{15}{100}$$
$= 90$ students

153. The ratio of wine and water in a mixture of 35 litres is 4 : 1. How many litres of water should be added to it so that the ratio of wine and water becomes 2 : 1?
(a) 5 litres (b) 6 liters
(c) 7 litres (d) 8 litres
(e) None of the above / More than one of the above
B.P.S.C. Assistant Audit Officer 2022

Ans. (c)

Total mixture of wine and water = 35 litres
The ratio of wine and water = 4 : 1
As we know that- water is being added to the mixture
So the quantity of wine will be same-
So-
Wine : water = 4 : 1 — Initial
+1 | addition of water
2 : 1 - Final
4 : 2
1 Unit is being added, initial quantity of water is– 1 Unit
$(4 + 1)$ Unit = 35
1 Unit = 7 litre
Hence, 7 litre water is being added to the mixture.

154. Shashi loses 25% by selling oranges at the rate of ₹ 150 per dozen. At what rate should she sell them to get a profit of 20%?
(a) ₹ 220 per dozen (b) ₹ 240 per dozen

(c) ₹ 240 per dozen (d) ₹ 250 per dozen
(e) None of the above/More than one of the above

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Ans. (c)

According to the question
75% of cost prize = 150 (since there is loss of 25%)

120% of cost price = $\left(\frac{150}{75} \times 120\right)$ Rs. (since profit is 20% than 20% is profit) = 240 Rs.

Hence to make a profit of 20%, oranges should be sold at the rate of Rs. 240 per dozen

155. The lengths of the diagonals of a rhombus are 10 cm and 24 cm. Its perimeter is

(a) 48 cm (b) 52 cm
(c) 56 cm (d) 60 cm
(e) None of the above/More than one of the above

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Ans. (b)

We know that–

$d_1^2 + d_2^2 = 4a^2$ (where a = side d_1, d_2 are diagonals of rhombus)

Side of rhombus $= \frac{1}{2} \times \sqrt{(\text{Diagonal}_1)^2 + (\text{Diagonal}_2)^2}$

$= \frac{1}{2}\sqrt{(10)^2 + (24)^2} = \frac{1}{2}\sqrt{100+576}$

$= \frac{1}{2}\sqrt{676} = \frac{1}{2} \times 26 = 13$ cm

∴ Perimeter of rhombus = 4 × side
= (4 × 13) cm = 52 cm

156. A train of length 140 m is running at a speed of 60 km/hr and a dog is running in the same direction parallel to the train at a speed of 18 km/hr. The train will cross the dog in

(a) 10 seconds (b) 11 seconds
(c) 12 seconds (d) 121 seconds
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

Relative speed = (60 – 18) km/hr
= 42 km/hr
(∴ both going in the same direction)

∴ Time taken by Train to cross dog $= \frac{\text{Length of train}}{\text{Relative Speed}}$

$= \frac{140 \text{ m}}{\left(42 \times \frac{5}{18}\right) \text{m/s}}$

ock are together then q = 0°)

$= \left(\frac{140 \times 18}{42 \times 5}\right)$

= 12 second

157. A clock is set right at 6 a.m. It gains 10 minutes in 24 hours. What will be the right time when the clock indicates 11 a.m. on the next day?

(a) 10 a.m. (b) 48 minutes past 10
(c) 50 minutes past 10 (d) 54 minutes past 10
(e) None of the above/More than one of the above]

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Ans. (b)

24 hours = (24 × 60) minutes
Actual time of (24 × 60 + 10) minutes = (24 × 60) minutes
Actual time of 1450 minutes = 1440 minutes
Total time from 6 AM to 11 AM next day
= (24 × 60) minutes + (5 × 60) minutes
= (1440 + 300) minutes = 1740 minutes
∵ Actual time of 1450 minutes = 1440 minutes

∴ Actual time of 1740 minutes = $\left(\frac{1440}{1450} \times 1740\right)$ minutes

= 1728 minutes
= (28 × 60) minutes + 48 minutes
= 28 Hrs 48 minutes
= 24 Hrs + 4 Hrs + 48 Minutes
On the next day at 11:00 am at the time when the hand of the clock shows the actual time will be = 6 AM + 4 Hrs + 48 minutes= 10:48 in the morning

Second Method

= (6 + 4) O'clock + 48 minutes
= 10:48 AM

158. At what time between 5 and 6 will the two hands of a watch coincide?

(a) 26 minutes past 5
(b) 27 minutes past 5
(c) $27\frac{3}{11}$ minutes past 5
(d) 28 minutes past 5
(e) None of the above/More than one of the above

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Ans. (c)

The angle between the hour hand and the minute hand,

$\theta = \left[30 \ \ H - 11\frac{M}{2}\right]$

H = 5 and θ = 0°

$$\therefore \theta = \left[30 \times 5 - \frac{11 \times M}{2}\right]$$

$$M = \frac{150 \times 2}{11} = \frac{300}{11}$$

$$M = 27\frac{3}{11}$$

Therefore between 5 and 6 O'clock, both the clock hands will be coincide at 5O'clock 27.31 minutes.

159. Anil and Suman together can do a work in 12 days, in which, Anil alone can do in 20 days. If Suman alone has to do this work, she will take

(a) 27 days (b) 28 days
(c) 29 days (d) 30 days
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

Let the total work = 60

Anil and Suman both can do in one day $= \frac{60}{12} = 5$ work

Work done by Anil in one day $= \frac{60}{20} = 3$

$\therefore$ Work done by Suman in one day = (5 – 3) work = 2 work

$\therefore$ Time taken by Suman to complete Total work $= \frac{60}{2}$ Days

$= 30$ Days

160. In a class of 55 students, 34 like to play Cricket and 26 like to play Badminton. Also, each student likes to play at least one of the two games. How many students like to play both Cricket and Badminton?

(a) 3 (b) 4
(c) 5 (d) 6
(e) None of the above/More than one of the above

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Ans. (c)

Number of students playing both the games
$= (34 + 26) - 55 = 60 - 55 = 5$

161. What are the two natural numbers whose product is 2400 and the sum of whose squares is 5200?

(a) 120, 20 (b) 80, 30
(c) 75, 32 (d) 60, 40
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

Let both the numbers be x and y respectively.
According to question
$x \times y = 2400$
$x^2 + y^2 = 5200$
$(x + y)^2 = x^2 + y^2 + 2xy$
$= 5200 + 2 \times 2400$
$= 5200 + 4800 = 10000$
$x + y = \sqrt{10000} = 100$ ——————(i)
and $(x - y)^2 = x^2 + y^2 - 2xy$
$= 5200 - 4800$
$= 400$
$x - y = \sqrt{400} = 20$ ——————(ii)
From the equation (i) and (ii)
$x = 60$ and $y = 40$
Hence both the numbers will be 60 and 40.

162. How many numbers between 100 and 500 are divisible by 4, 5 and 6?

(a) 5 (b) 6
(c) 7 (d) 8
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (c)

LCM of 4, 5 and 6 =

2	4, 5, 6
2	2, 5, 3
3	1, 5, 3
5	1, 5, 1
	1, 1, 1

$\therefore$ LCM = 2 × 2 × 3 × 5 = 60
Now the number of multiples of 60 which lie between 100 and 500 = 120, 180, 240, 300, 360, 420 to 480. Hence these are the 7 numbers between 100 and 500 which are divisible by 4, 5 and 6.

163. In a mixture of 70 kg, the ratio of sand and cement is 4 : 1. How much sand should be added to the mixture so that the ratio of sand and cement in it becomes 6 : 1?

(a) 24 kg (b) 28 kg
(c) 30 kg (d) 32 kg
(e) None of the above/More than one of the above

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Ans. (b)

Given that–
5 Unit $\rightarrow$ 70
$\because$ 1 unit = 14 kg.
$\therefore$ 6 unit = 14 × 6 = 84 kg.
Hence the required amount of additional sand to the mixture

3 kg.

164. How many numbers between 200 and 600 are divisible by 4, 5 and 6?

(a) 4 (b) 5

(c) 6 (d) 8

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

The number of such multiples of LCM of 4, 5 and 6, which lie between 200 and 600.

2	4,	5,	6
2	2,	5,	3
3	1,	5,	3
5	1,	5,	1
	1,	1,	1

L.C.M. $= 2 \times 2 \times 3 \times 5 = 60$

Now, such multiples of 60 which lie between 200 and 600. are 240, 300, 360, 420, 480 & 540

Thus total numbers are 6.

165. If n is any positive integer, then $(3^{4n} - 4^{3n})$ is always divisible by ______ .

(a) 7 (b) 17

(c) 112 (d) 145

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Let, n = 1, 2, 3, ...

If n = 1

Then, $3^{4n} - 4^{3n} = 3^4 - 4^3$

$\Rightarrow 81 - 64 = 17$

If n = 2

Then, $3^8 - 4^6 = 3^4 \times 3^4 - 4^3 \times 4^3$

$= 81 \times 81 - 64 \times 64$

$= (81 + 64)(81 - 64)$

$= 145 \times 17$

$\because$ If n is any positive integer then $3^{4n} - 4^{3n}$ is always divisible by 17.

166. The difference between the square of two numbers is 256000 and the sum of the numbers is 1000. The numbers are-

(a) 600, 400

(b) 640, 360

(c) 628, 372

(d) 650, 350

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

Let, two number are x and y respectively.

According to question-

$x^2 - y^2 = 256000$

$(x + y)(x - y) = 256000$ ..... (i)

and given that, $x + y = 1000$ ..... (ii)

Putting the value eq^n (ii) in eq^n (i)

$1000 \times (x - y) = 256000$

$x - y = 256$ ..... (iii)

Adding eq^n (ii) and (iii)

$2x = 1256$

$x = 628$

Putting the value of x in eq^n (ii)

$y = 1000 - 628$

$y = 372$

So, the numbers are 628 and 372.

167. The value of $\log_{\sqrt{2}}(32)$ is.

(a) $\frac{5}{2}$ (b) 5

(c) 10 (d) $\frac{1}{10}$

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

$\log_{\sqrt{2}}(32) = \log_{\sqrt{2}}(2)^5$

$= \log_{\sqrt{2}}\left(\sqrt{2}.\sqrt{2}\right)^5$

$= \log_{\sqrt{2}}\left(\sqrt{2}\right)^{10}$

$= 10\log_{\sqrt{2}}\left(\sqrt{2}\right) \quad (\because \log_x x = 1)$

Thus $\log_{\sqrt{2}}(32) = 10$

168. By selling 45 lemons for Rs. 40, a man loses 20%. How many lemon should be sell for Rs. 24 to gain 20% in the transaction?

(a) 16 (b) 18

(c) 20 (d) 22

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Selling price of 45 lemons at loss of 20%. = Rs. 40

Cost price of 45 lemons $= 40 \times \frac{100}{(100 - 20)}$

$= 40 \times \frac{100}{80} = Rs. 50$

Selling price of 45 lemons at profit of 20%

$= 50 \times \frac{120}{100} = Rs. 60$

No. of lemon for Rs. 1 at 20% profit $=\frac{45}{60}$

No. of lemon for Rs. 24 at 20% profit $=\frac{45}{60}\times 24=18$

169. A 110m long train is travelling at a speed of 58 kmph, pass a passerby walking at 4 kmph in the same direction in-

(a) 6 Seconds (b) $7\frac{1}{2}$ Seconds

(c) $7\frac{1}{3}$ Seconds (d) 8 Seconds

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (c)

Relative speed of train $=(58-4)=54$ Km/hr.

$$=\left(54\times\frac{5}{18}\right)=15 \text{ m/s}$$

Length of train =110 m.

The train will take time to cross passer by

$$=\frac{110}{15} \text{ s.}$$

$$=\frac{22}{3} \text{ s.} = 7\frac{1}{3} \text{ s.}$$

170. The perimeter of a rhombus is 52m and its shorter diagonal is 10m. The length of the longer diagonal is-

(a) 12 m (b) 18 m

(c) 10 m (d) 24 m

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

Each sides of rhombus $=\frac{52}{4}=13$ m

Let, the length of longer diagonal of rhombus is $2x$ m. Since half of both diagonal and one side form a right angled triangle, whose hypotenuse is the side of the rhombus.

So,

$$\left(\frac{10}{2}\right)^2+\left(\frac{2x}{2}\right)^2=(13)^2$$

$$(5)^2+(x)^2=169$$

$$x^2=169-25$$

$$x=\sqrt{144}=12 \text{ m}$$

So, the length of longer diagonal of rhombus

$$=(2\times 12) \text{ m}$$

$$=24 \text{ m}$$

171. In a 100m race, A runs at 8 kmph. If A gives B a start of 4m and still beats him by 15 seconds, what is the speed of B?

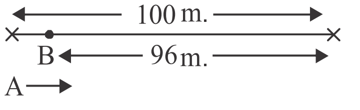
(a) 5 kmph (b) 5.76 kmph

(c) 6 kmph (d) 6.34 kmph

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)



Speed of A $=\left(8\times\frac{5}{18}\right)=\frac{20}{9}$ m/s.

Time taken by A to cover 100m $=\frac{100}{\frac{20}{9}}=45$ s.

Time taken by B to cover 96m $=(45+15)=60$ s.

Speed of B $=\frac{96\text{ m}}{60\text{ s}}=\left(\frac{96}{60}\times\frac{18}{5}\text{km/hr}\right)=5.76$ km./hr

172. A clock is set right at 8 a.m. The clock gains 10 minutes in 24 hours. What will be the true time when the clock indicates 1 p.m. on the following day?

(a) 12 noon (b) 48 minutes past 12 noon

(c) 1 p m (d) 2 pm

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

From 8 am to 8 am next day, the clock will show 8:10 am.

24 hr $=(24\times 60)$ minutes

Real time of $(24\times 60+10)$ minute

$=(24\times 60)$ minutes

Real time of 1450 minutes = 1440 minutes

Real time of $(24\times 60+5\times 60)$ minutes

$$=\left(\frac{1440}{1450}\times 1740\right) \text{ minutes}$$

[$\because$ Total time from 8 am to 1 pm the next day.

$=(24\times 60)$ min. $+(5\times 60)$ min.$=1740$]

$$=\frac{144}{145}\times 1740$$

$=144\times 12=1728$ min.

$=28$ hr. $+48$ min.

$=24$ hr. $+4$ hr. $+48$ min.

So, on next day, at 1 o'clock in the afternoon, when the clock shows the hand, actual time will be.

$=$ 8 am $+$ (4 hr $+$ 48 min)

173. The value of continued fraction.

$$2-\cfrac{1}{2-\cfrac{1}{2-\cfrac{1}{\cdots}}}\ \text{is—}$$

(a) 1 (b) – 1
(c) 2 (d) 0
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (a)

$$2-\cfrac{1}{2-\cfrac{1}{2-\cfrac{1}{\cdots}}}$$

Let,

$$x = 2-\cfrac{1}{2-\cfrac{1}{2-\cfrac{1}{\cdots}}}$$

Now, $x = 2 - \frac{1}{x}$

$x = \frac{2x-1}{x}$

$x^2 - 2x + 1 = 0$

$x^2 - x - x + 1 = 0$

$x(x-1) - 1(x-1) = 0$

$(x-1)^2 = 0$

$x = 1$

So, the value of the continued fraction is 1.

174. Arrange the following in a logical sequence:

1. Patient **2. Diagnosis**
3. Bill **4. Doctor**
5. Treatment

(a) 1, 4, 2, 3, 5 (b) 1, 4, 3, 2, 5
(c) 1, 4, 2, 5, 3 (d) 4, 1, 2, 3, 5
(e) None of the above / More than one of the above

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Ans. (c)

The logical sequence of the Given Word-

Patient → Doctor → Diagnosis → Treatment → Bill
1 4 2 5 3

Hence, The correct sequence is '14253' which is given in the option (c)

175. In a class, 72 students drink only tea, 50% of the students drink coffee, 25% of the students drink both tea and coffee and 5% of the students drink neither tea nor coffee. How many of the students drink only tea or only coffee?

(a) 113 (b) 112
(c) 114 (d) 128
(e) None of the above / More than one of the above

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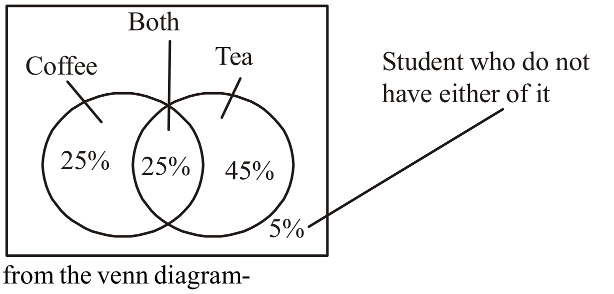
Ans. (b)

Given that-
The Student, who drink only tea = 72
Now-



from the venn diagram-
45% = 72
5% = 8
one who drink only tea-
45% = 72
one who drink only coffee-
25% = 40
Total student who drink either tea or coffee = 72 + 40 = 112

176. Arrange the words given below in a meaningful sequence:

1. Rain **2. Monsoon**
3. Rescue **4. Flood**
5. Shelter **6. Relief**

(a) 1, 2, 3, 4, 5, 6 (b) 1, 2, 4, 5, 3, 6
(c) 2, 1, 4, 3, 5, 6 (d) 4, 1, 2, 3, 5, 6
(e) None of the above / More than one of the above

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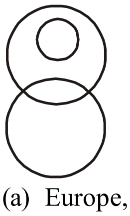
Ans. (c)

Meaningful sequence of the Given Words-

Monsoon → Rain → Flood → Rescue → Shelter → Relief
2 1 4 3 5 6

Hence, the Correct sequence is "214356" which is given in the option (c).

177. Given figure shows exactly correct relation in which of the following?



(a) Europe, Asia, Africa
(b) Odd numbers, prime numbers, odd multiples of nine
(c) Blue, red, green
(d) Players, students, actors
(e) None of the above / More than one of the above

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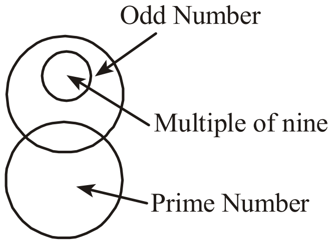
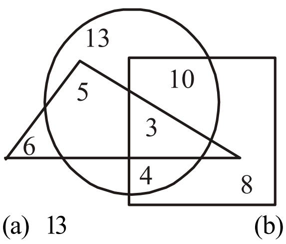
The Given venn diagram



Hence, option (b) will be the right answer of the given question.

178. In the following diagram, square represents women, triangle represents sub-inspectors of police and circle represents graduates. Which numbered area represents women graduates sub-inspector of police?



(a) 13 (b) 8
(c) 5 (d) 3
(e) None of the above / More than one of the above

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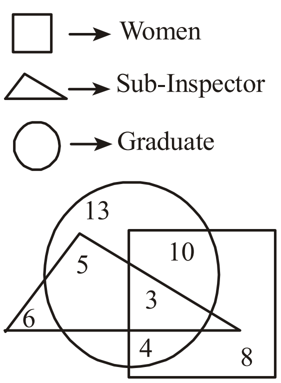
Ans. (d)



The area number, which represents women graduates sub-inspector of police = 3

179. When the reports of the annual profits of five companies L, M, N, O and P were submitted, it was found that no two companies had the same profit. M got more profit than O but less than P, whereas L got more profit than N but less than P. Which of the following companies got the highest profit?

(a) P (b) L
(c) M (d) O
(e) None of the above / More than one of the above

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Ans. (a)

Let us analyze the given information
M got more profit than O but less than P
P > M > O
L got more profit than N but less than P
P > L > N
Here, we can say that, P got the highest profit
Therefore, option (a) will be the right answer.

180. Given below are two statements: "All men are vertebrates." "Some mammals are vertebrates." Which one of the following is correct?

(a) All men are mammals.
(b) All mammals are men.
(c) Some vertebrates are mammals.
(d) All vertebrates are men.
(e) None of the above / More than one of the above

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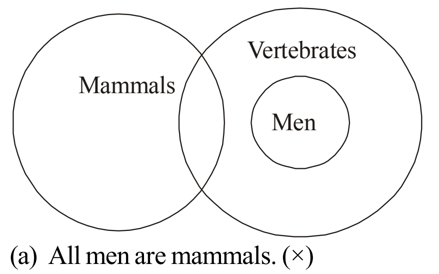
Ans. (c)

Representing the Given statement through venn diagram-
Through Option-



(a) All men are mammals. (×)
(b) All mammals are men. (×)
(c) Some vertebrates are mammals. (✓)
(d) All vertebrates are men. (×)
Hence, Option (c) will be the right answer.

181. M, N, O, P, Q, R and S are seven persons sitting in a row for a photo session. O sits to the immediate left of S and R is to the immediate right of P. N has equal number of persons on either side of him in a row. M is to the immediate left of N and there are exactly three persons between M and S. How many persons are there between R and Q?

(a) 2 (b) 3
(c) 4 (d) 1
(e) None of the above / More than one of the above

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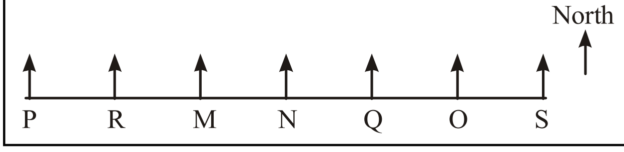
Ans. (a)

From the given instruction in the question, we will draw a appropriate sitting arrangement-



Now- we can clearly see that, there are two person (M, N) sitting between R and Q
Hence, Option (a) will be the right answer of the given question.

182. If A+B means A is the son of B; A-B means A is the daughter of B; AxB means A is the father of B; A ÷ B means A is the mother of B, then which of the following means S is the son-in-law of P?

(a) P+Q÷R×S-T (b) P×Q÷R-S+T
(c) P+Q×R-S÷T (d) P×Q-R÷S×T
(e) None of the above / More than one of the above

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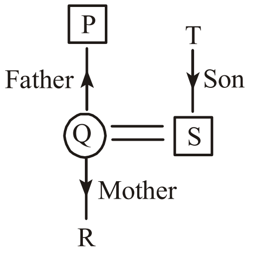
Ans. (b)

The Given code-
A + B ⇨ A is son of B
A – B ⇨ A is daughter of B
A × B ⇨ A is father of B
A ÷ B ⇨ A is mother of B
Now - We are required to show that S is the son-in-Law of P from the given option-
Now, from the option (b)-
□ ⟶ Male
○ ⟶ Female
P×Q÷R-S+T



Hence, option (b) will be the right answer-

183. The number of the digits in the number 97415628, which remain unchanged while the digits are written in descending order, is

(a) 1 (b) 2
(c) 3 (d) 4
(e) None of the above / More than one of the above

B.P.S.C. Project Manager Prelims 2021

Ans. (c)

The Number

9 7 4 1 5 6 2 8

9 8 7 6 5 4 2 1

ight answer.

Hence, there are 3 number's which position does not change when we write the digits of the number in descending order- Therefore, option (c) will be the right Answer.

184. If you write down all the numbers from 1 to 100, then how many times do you write 3?

(a) 11 (b) 18
(c) 20 (d) 21
(e) None of the above / More than one of the above

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Ans. (c)

If we write down all the number from 1 to 100 then the number of times we write 3 is 20.
3, 13, 23, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 43, 53, 63, 73, 83, 93. Hence '20' is correct answer.

185. The significant figure in the number 0.00386 is

(a) 6 (b) 5
(c) 4 (d) 3
(e) None of the above / More than one of the above

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Ans. (d)

Leading zeroes to the Left of the First non-zero digit are not significant.
All non-zero digits are significant-
According to the rule, the given number 0.00386 has three significant Numbers (3, 8, 6)
Hence, option (d) will be the right answer

186. In a row of persons, the position of Kirti from the left side of the row is 27th and position of Kirti from the right side of the row is 34th, then the total number of persons in the row is

(a) 60 (b) 61
(c) 62 (d) 59
(e) None of the above / More than one of the above

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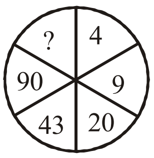
Ans. (a)

Kirti's Position from Left (L) = 27^{th}
Kirti's Position from Right (R) = 34^{th}
We know that-
Total person in Row = Left + Right – 1
T= 27 + 34 – 1
T = 61 - 1
T= 60
There are 60 person in the Row-

187. In the following figure, the missing number is



(a) 126 (b) 185

(c) 239 (d) 145

(e) None of the above / More than one of the above

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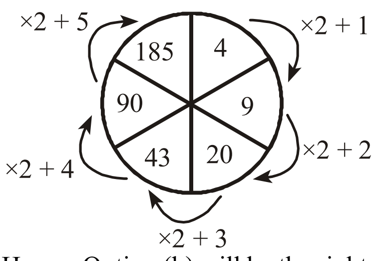
Ans. (b)



Hence, Option (b) will be the right answer.

188. If the number 12.464762 is rounded off to four places of decimal, then the correct number is

(a) 12.4648 (b) 12.4647

(c) 12.4640 (d) 12.4649

(e) None of the above / More than one of the above

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Ans. (a)

To round off the number 12.464762 to four decimal places-

Then the round off number will be 12.4648

There fore, option (a) will be the right answer.

189. If A = 26, SUN = 27, then CAT = ?

(a) 24 (b) 27

(c) 57 (d) 58

(e) None of the above / More than one of the above

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Ans. (c)

Letter Positional value Opposite

A $\longrightarrow$ 1 $\longrightarrow$ 26

Similarly

S $\rightarrow$ 19 $\rightarrow$ 8 C $\rightarrow$ 3 $\rightarrow$ 24

U $\rightarrow$ 21 $\rightarrow$ 6 A $\rightarrow$ 1 $\rightarrow$ 26

N $\rightarrow$ 14 $\rightarrow$ 13 T $\rightarrow$ 20 $\rightarrow$ 7

27 57

Now

CAT = 57

190. In the series 1, 2, 4, 7, 28, 140, 198, 205, a wrong number is

(a) 198 (b) 4

(c) 205 (d) 140

(e) None of the above / More than one of the above

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Ans. (d)

The Given series-

$\boxed{33}$

1 2 4 7 28 $\boxed{140}$ 198 205

+1 ×2 +3 ×4 +5 ×6 +7

In the Given series, 140 is wrong number, there should be 33 so we can make a meaningful number series,

Hence, Option (d) will be the right answer-

191. The next letter in the series, A, B, D, H will be

(a) L (b) N

(c) R (d) P

(e) None of the above / More than one of the above

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Ans. (d)

The Given Alphabet series-

1 2 4 8 16

A B D H $\boxed{P}$

×2 ×2 ×2 ×2

Hence, Option (d) will be the right answer-

192. The missing number in

27 : 51 :: 83 : ? is

(a) 102 (b) 117

(c) 123 (d) 138

(e) None of the above / More than one of the above

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Ans. (c)

27 : 51 :: 83 : ?

Just as- Similarly

27 $\longrightarrow$ $5^2 + 2$ 83 $\longrightarrow$ $9^2 + 2$

51 $\longrightarrow$ $7^2 + 2$ 123 $\longrightarrow$ $11^2 + 2$

Hence, Option (c) will be the right answer

193. Choose the odd one among the following.

(a) 17 (b) 27

(c) 37 (d) 47

(e) None of the above / More than one of the above

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In the given option of the question-
27 is the odd one because except 27 all are Prime Numbers, Hence, Option (b) will be the right answer.

194. In a certain code language, if the word RECTANGLE is coded as TGEVCPING, then how will be the word RHOMBUS coded in that language?

(a) TJQODWU (b) TJQNDWU
(c) TJOQDWV (d) TJQOEWU
(e) None of the above / More than one of the above

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Ans. (a)

Just as-

$R \xrightarrow{+2} T$
$E \xrightarrow{+2} G$
$C \xrightarrow{+2} E$
$T \xrightarrow{+2} V$
$A \xrightarrow{+2} C$
$N \xrightarrow{+2} P$
$G \xrightarrow{+2} I$
$L \xrightarrow{+2} N$
$E \xrightarrow{+2} G$

Similarly-

$R \xrightarrow{+2} T$
$H \xrightarrow{+2} J$
$O \xrightarrow{+2} Q$
$M \xrightarrow{+2} O$
$B \xrightarrow{+2} D$
$U \xrightarrow{+2} W$
$S \xrightarrow{+2} U$

Hence Option (a) will be the right answer.

195. In a certain code language, if CRICKET is coded as 3923564, ROCKET is coded as 913564 and KETTLE is coded as 564406, then in that language LITTLE is coded as

(a) 244060 (b) 024406
(c) 020446 (d) 200446
(e) None of the above / More than one of the above

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Ans. (b)

Here, we compare the word's code with the given unknown word "LITTLE"

C — 3	R — 9	K — 5	L — 0
R — 9	O — 1	E — 6	I — 2
$\boxed{\text{I — 2}}$	C — 3	$\boxed{\text{T — 4}}$	T — 4
C — 3	K — 5	T — 4	T — 4
K — 5	$\boxed{\text{E — 6}}$	$\boxed{\text{L — 0}}$	L — 0
E — 6	T — 4	E — 6	E — 6
T — 4			

ight answer.

Hence, Option (b) will be the right answer.

196. Choose the odd one among the following.

(a) SORE (b) SOTLU
(c) NORGAE (d) MEJNIAS
(e) None of the above / More than one of the above

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Ans. (c)

From the Given Option-

SORE ⟶ ROSE
SOTLU ⟶ LOTUS
MEJNIAS ⟶ JASMINE

But the word given in the option (c) i.e. NORGAE cannot make a meaningful word of flower
Hence, option (c) will be the right answer.

197. Complete the follwoing series :

$$\mathbf{6+\sqrt{216}, 7+\sqrt{343}, 8+\sqrt{512}, 9+\sqrt{729}, \ldots\ldots\ldots\ldots}$$

(a) $10+\sqrt{100}$ (b) $10+\sqrt{1000}$
(c) $100+\sqrt{1000}$ (d) $100+\sqrt{10}$
(e) None of the above / More than one of the above

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Ans. (b)

The given series,

$$6+\sqrt{216}, 7+\sqrt{343}, 8+\sqrt{512}, 9+\sqrt{729}, \boxed{?}$$

Now,

$$6+\sqrt{6^3}, 7+\sqrt{7^3}, 8+\sqrt{8^3}, 9+\sqrt{9^3}, 10+\sqrt{10^3}$$

$$6+\sqrt{216}, 7+\sqrt{343}, 8+\sqrt{512}, 9+\sqrt{729}, 10+\sqrt{1000}$$

Hence, Option (b) will be the right answer.

198. If 1st January, 1992 is Tuesday, then on which day of the week will be 1st January, 1993 fall?

(a) Thursday (b) Wednesday
(c) Friday (d) Saturday
(e) None of the above / More than one of the above

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Ans. (a)

1 January, 1992
Since, 1992 is perfectly divided by 4, so it is a leap year that has 366 days
Now-

$$\frac{366}{7} = \text{remainder} - 2$$

Thus, the day on 1st January 1993 is– Tuesday +2
= Thursday

199. When the clock shows 3 hours 14 minutes, what is the angle between the hands of the clock?

(a) 10^0 (b) 12^0
(c) 13^0 (d) 16^0
(e) None of the above / More than one of the above

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Ans. (c)

Clock shows 3 hours 14 minutes-
then angle between hour hand and minute hand

$$\theta = 30h - \frac{11}{2}m$$

$$\theta = 30 \times 3 - \frac{11}{2} \times 14$$

$$\theta = 90 - 11 \times 7$$

$$\theta = 90^0 - 77^0$$

$$\theta = 13^0$$

Hence, Option (c) will be the right answer

200. P, Q, R, S and T sit around a table. P sits two seats to the left of R and Q sits two seats to the right of R. If S is not sitting next to Q, who is sitting between Q and S?

(a) P (b) R
(c) T (d) Both R and P
(e) None of the above / More than one of the above

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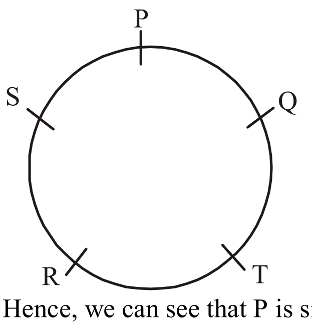
Ans. (a)

According to the question the appropriate sitting arrangement-



Hence, we can see that P is sitting between Q & S.

201. P, Q, R, S and T are five speakers who have to speak on a particular day, not necessarily in the same order. R is neither the first nor the last speaker. There are three speakers after S and three speakers ahead of T. If P speaks after Q, then who is the last speaker to speak?

(a) S (b) T
(c) P (d) R
(e) None of the above / More than one of the above

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Ans. (c)

According to the question the given arrangement are as following-

1	Q
2	S
3	R
4	T
5	P

Here, we can see that, the last speaker is 'P'
Hence, Option (c) will be the right answer.

202. Vinod walks 20 metres towards north. He then turns left and walks 40 metres. He again turns left and walks 20 metres. Further, he moves 20 metres after turning to the right. How far is he from his original position?

(a) 20 meters (b) 60 meters
(c) 30 meters (d) 50 meters
(e) None of the above / More than one of the above

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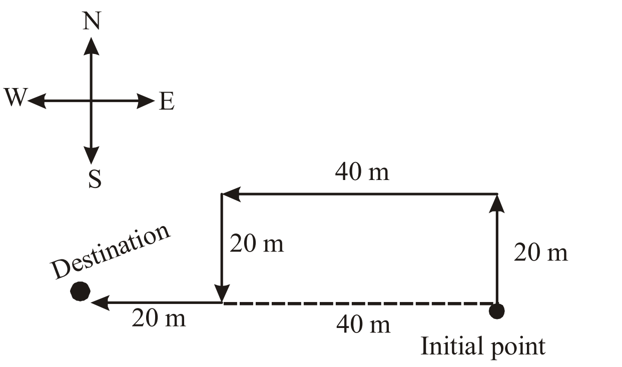
Ans. (b)



Hence, we can see that, the distance between Destination and Initial point is 60 meter, which is given in the option (b)

203. The series

$$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + ...$$

(a) geometric series
(b) harmonic series
(c) exponential series
(d) p-series
(e) None of the above / More than one of the above

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Ans. (c)

The Given Series-

$$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + ...$$

This series is an exponential series
This series is an expansion of ex

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + ...$$

Hence, option (c) will be the right answer.

204. If $\cos\theta = \sin\theta$, then the value of θ is

(a) π (b) $\frac{\pi}{12}$

(c) $\frac{\pi}{4}$ (d) $\frac{2\pi}{3}$

(e) None of the above / More than one of the above

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Ans. (c)

$\cos\theta = \sin\theta$

$\cos\theta = \cos(90^0 - \theta)$

$\theta = 90^0 - \theta$

$2\theta = 90^o \text{ or } \frac{\pi}{2}$

$\boxed{\theta = \frac{\pi}{4} \text{ or } 45^o}$

Hence, option (c) will be the right answer-

205. If $\frac{^nC_2}{^{n-1}C_2} = \frac{2}{1}, n \geq 1$ then the value of n is

(a) 2

(b) 4

(c) 1

(d) 3

(e) None of the above / More than one of the above

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Ans. (b)

$$\frac{^nC_2}{^{n-1}C_2} = \frac{2}{1}$$

$$\frac{\frac{n!}{2!(n-2)!}}{\frac{(n-1)!}{2!(n-1-2)!}} = \frac{2}{1} \qquad \left[\because {}^nC_r = \frac{n!}{r!(n-r)!}\right]$$

$$\frac{\frac{n(n-1)!}{2!(n-2)!}}{\frac{(n-1)!}{2!(n-3)!}} = \frac{2}{1}$$

$$\frac{n(n-3)!}{(n-2)!} = \frac{2}{1}$$

$$\frac{n(n-3)!}{(n-2)(n-3)!} = \frac{2}{1}$$

$n = 2\,(n-2)$

$n = 2n - 4$

$n = 4$

Hence, option (b) will be the right answer.

206. $\frac{5}{3}x^3 + \frac{5}{3}x - \frac{10}{3} = 0$, then one of the value of x is

(a) $\frac{5}{3}$ (b) $\frac{10}{3}$

(c) $\frac{3}{5}$ (d) 1

(e) None of the above / More than one of the above

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Ans. (d)

$\frac{5}{3}x^3 + \frac{5}{3}x - \frac{10}{3} = 0$

$5x^3 + 5x - 10 = 0$

$x^3 + x - 2 = 0$

Let putting the value of x = 1

$2 - 2 = 0$

$0 = 0$

Therefore-

$x^2(x-1) + x(x-1) + 2(x-1) = 0$

$(x-1)(x^2 + x + 2) = 0$

Here, one of the value of x is 1

Hence, option (d) will be the right answer.

207. If $f : N \rightarrow N$ is defined by $f(x) = 3x + 7$ and $g : N \rightarrow N$ is defined by $g(x) = 5x + 1$, then (gof) (1) is

(a) 25 (b) 51

(c) 57 (d) 26

(e) None of the above / More than one of the above

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Ans. (b)

To find (gof) (1), you need to substitute the value of f(1) into g(x)

Now,

$f(x) = 3x + 7$

$f(1) = 10$
Now, substitute this value into g(x) :
$g(x) = 5x + 1$
$g[f(1)] = g(10) = 5 \times 10 + 1$
$(gof)(1) = 51$

208. If $\log\left(\frac{a}{b}\right) + \log\left(\frac{b}{a}\right) = \log(a+b)$ then,

(a) $a + b = 0$ (b) $a + b = 1$
(c) $a = b$ (d) $ab = 0$
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

$\log\left(\frac{a}{b}\right) + \log\left(\frac{b}{a}\right) = \log(a+b)$
$\log\left(\frac{a}{b} \times \frac{b}{a}\right) = \log(a+b)$ $[\because \log m + \log n = \log mn]$
$\log 1 = \log(a + b)$
$a + b = 1$
Hence, option (b) will be the right answer.

209. Find the number which when multiplited by 15 is increased by 196.

(a) 12 (b) 14
(c) 17 (d) 19
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

Let the number is 'x'
According to the question
$x \times 15 = x + 196$
$15x - x = 196$
$14x = 196$
$x = 14$
Hence option (b) will be the right answer.

210. The solution of the equations
$2x - 3y = 23$
$4x + 3y = 43$
is

(a) $x = 11; y = -\frac{1}{3}$
(b) $x = 23; y = 43$
(c) $x = 43; y = 23$
(d) $x = 11; y = 11$
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (a)

$2x - 3y = 23$ -------(i)
$4x + 3y = 43$ -------(ii)
Adding the equation (i) and (ii)
$6x = 66$
$x = 11$
Putting the value of x in the equation (i)
$2 \times 11 - 3y = 23$
$22 - 3y = 23$
$3y = -1$
$y = -\frac{1}{3}$
Hence, option (a) will be the right answer.

211. The ratio of two number is 3 : 4 and their HCF is 4. Their LCM is

(a) 16 (b) 12
(c) 24 (d) 48
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (d)

The ratio of Two numbers are 3 : 4
Let the number-
3x and 4x
HCF (x) = 4
Now
$LCM = HCF \times 3 \times 4$
$= 4 \times 3 \times 4$
$LCM = 48$
Hence, LCM will be 48

212. The sum of the first 8 prime numbers is

(a) 75 (b) 77
(c) 60 (d) 58
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

First 8 prime number are- 2, 3, 5, 7, 11, 13, 17, 19
According to the question-
$2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 = 77$

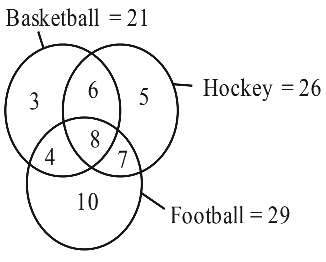
ight answer.

213. In a group of athletics teams in a school, 21 are in the basketball team, 26 in the hockey team and 29 in the football team. If 14 play hockey and basketball 12 play football and basketball, 15 play hockey and football, 8 play all the three games, then how many play football only?

(a) 10 (b) 29
(c) 21 (d) 18
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)



So, the number of players, who play football only are 10

214. Mohan can do a bit of work in 25 days which can be completed by Sohan in 20 days. Both together labour for 5 days and afterward Mohan leaves off. How long will Sohan take to complete the remaining work?

(a) 20 days (b) 11 days
(c) 14 days (d) 21 days
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans.-(b)

Let, total work = 100 (LCM of 25 and 20)

Mohan, (25) > Total Mark 100 < 4
Sohan, (20) 5

5 days work of Mohan and Sohan = (4 + 5) × 5
= 45

Remaining work = 100 – 45 = 55

Time taken by Sohan to complete remaining work $= \frac{55}{5}$

= 11 days

215. A clock is started at 12:00 noon. By 10 minutes past 5:00, the hour hand has turned through.

(a) 135^0 (b) 145^0
(c) 155^0 (d) 165^0
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

∵ Angle formed by hour hand in 1 min = $\frac{1}{2}^0$

Angle formed by hour hand in 5 hr 10 min or 310 min is

$= 310 \times \frac{1}{2}^0 = 155^0$

So, by 10 minutes past 5:00, the hour hand turned through 155^0

216. Which one of the following cannot be the square of a natural numbers?

(a) 26569 (b) 143642
(c) 30976 (d) 28561
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

The unit digit of square of any natural number can never be 2, 3, 7 and 8, So the number given in option (b) 143642 cannot be the square number.

217. Given that-

$217x + 131y = 913$

$131x + 217y = 827$

Then x and y are respectively

(a) 5 and 7 (b) 3 and 2
(c) – 5 and – 7 (d) 2 and 5
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

$217x + 131y = 913$ ..... (i)
$131x + 217y = 827$ .... (ii)
Add eq^n (ii) and eq^n (i)
$348x + 348y = 1740$
$x + y = 5$ ..... (iii)
Sub. eq^n (ii) from eq^n (i)
$86x - 86y = 86$
$x - y = 1$ ..... (iv)
Add eq^n (iii) and (iv)
$2x = 6$
$x = 3$
Put the value of x in eq^n (iv)
$3 - y = 1$
$y = 2$
So the value of x and y are 3 and 2 respectively.

218. $\frac{(598+479)^2-(598-479)^2}{598\times 479} = ?$

(a) 4 (b) 10
(c) 132 (d) 8
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

$$\frac{(598+479)^2-(598-479)^2}{598\times 479}$$

$$=\frac{4\times 598\times 479}{598\times 479} \quad \{\because (a+b)^2-(a-b)^2=4ab\}$$

$=4$

219. The population of a town is 176400. If it increases annually at the rate of 5%, then what will be its population after two years?

(a) 194481 (b) 296841
(c) 394481 (d) 396841
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

$\because$ Population of a town after two years

$$=176400\times\frac{105}{100}\times\frac{105}{100}$$

$=194481$

220. The missing number in the sequence-
4, 18, 48, 100, ?, 294, 448 is.

(a) 94 (b) 164
(c) 180 (d) 192
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

The sequence is–

4	18	48	100	$\boxed{180}$	294	448
↑	↑	↑	↑	↑	↑	↑
$(2)^3-(2)^2$	$(3)^3-(3)^2$	$(4)^3-(4)^2$	$(5)^3-(5)^2$	$(6)^3-(6)^2$	$(7)^3-(7)^2$	$(8)^3-(8)^2$

$\therefore$? = 180

221. If $^{2n}C_3 : {}^nC_2 = 44:3$, then the value of n is–

(a) 1 (b) 6
(c) 11 (d) 4
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

$$\frac{^{2n}C_3}{^nC_2}=\frac{44}{3}$$

$$\frac{\dfrac{\lfloor 2n}{\lfloor 2n-3\times\lfloor 3}}{\dfrac{\lfloor n}{\lfloor n-2\times\lfloor 2}}=\frac{44}{3}$$

$$\frac{\dfrac{2n\times(2n-1)(2n-2)}{3\times 2\times 1}}{\dfrac{n(n-1)}{2\times 1}}=\frac{44}{3}$$

$$\frac{2n\times 2\times(2n-1)(n-1)}{3n(n-1)}=\frac{44}{3}$$

$$\frac{4(2n-1)}{3}=\frac{44}{3}$$

$2n-1=11$

$2n=12$

$\therefore\ n=6$

222. If the average of m numbers is n^2 and that of n numbers is m^2, then the average of m+n numbers is—

(a) $\frac{n}{m}$ (b) $\frac{m}{n}$
(c) m n (d) m – n
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

$$\text{Average}=\frac{mn^2+nm^2}{(m+n)}$$

$$=\frac{mn(n+m)}{(m+n)}$$

$=mn$

223. The number of ways in which 12 identical pens can be distributed between two students if each student is to get at least two pens, is-

(a) 8 (b) 9
(c) 10 (d) 11
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

$\because$ Each students is to get at least two pens, if

Pen received by first student	**Pen received by sceond student**
2	10
3	9
4	8
5	7
6	6
7	5
8	4
9	3
10	2

So, total ways = 9

224. If the radii of circles A and B are in the ratio of 1.5:1, then the area of the circles A and B will be in the ratio of -

(b) 2 : 1

(c) 2.25 : 1 (d) 2.5 : 1

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Let the radius of circles A and B are $1.5x$ and x respectively.

Area of circle $= \pi \times (\text{radius})^2$

$\because$ Area of circle A : Area of circle B

$= \pi (1.5x)^2 : \pi (x)^2$

$= 2.25x^2 : x^2$

$= 2.25 : 1$

Note – If the ratio of radius or circumference of two circles are x : y then the ratio of their areas will be $x^2 : y^2$

Ratio of area of circle A and B $= (1.5)^2 : (1)^2 = 2.25 : 1 = 9 : 4$

225. One tap can fill a water tank in 3 hours and another tap can empty it in 4 hours. If the tank is one third full and both the taps are opened together, then the time taken by taps to fill the tank will be-

(a) 8 hours (b) 9 hours

(c) 10 hours (d) 11 hours

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

Let the capacity of the tank is 12 liters.

1/3 capacity of tank $= 12 \times \frac{1}{3} = 4\ l.$

Remaining capacity of tank $= 12 - 4 = 8\ l.$

Tank filled by first tap in 1 hr. $= \frac{12}{3} = 4\ l.$

and Second tap will empty in 1 hr. $= \frac{12}{4} = 3\ l.$

Both tap will fill the tank in 1 hr. $= 4 - 3 = 1\ l.$

Time taken by taps to fill remaining 8 ltr. $= \frac{8}{1} = 8$ hr.

226. The next term in the sequence 1, 3, 9, 15, 25, 35, 49, ...will be?

(a) 80 (b) 64

(c) 81 (d) 63

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

The given number series is as follows -

1	3	9	15	25	35	49	?
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
1^2	$(2)^2-1$	3^2	$(4)^2-1$	5^2	$(6)^2-1$	7^2	$(8)^2-1$

Next term $= (8)^2 - 1$

$= 64 - 1$

227. If $x^2 - y^2 = 7$ and $x - y = 1$, then the length of a diagonal of a rectangle with length and width respectively x cm and y cm will be -

(a) 5 cm (b) 6 cm

(c) 7 cm (d) 8 cm

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

Given that,

$x^2 - y^2 = 7$ and

$x - y = 1$ ....(i)

$(x - y)(x + y) = 7$

$1(x + y) = 7$

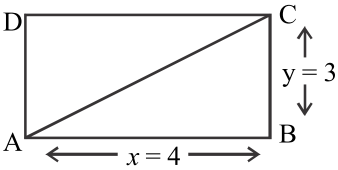
$x + y = 7$ ....(ii)

On Adding eq^n (i) and (ii)

$2x = 8$

$x = 4$

and $y = 4 - 1 = 3$ [by putting the value of x in eq^n (i)]



Length of diagonal of rectangle ABCD $= \sqrt{(AB)^2 + (BC)^2}$

$= \sqrt{(4)^2 + (3)^2}$

$= \sqrt{16 + 9} = \sqrt{25} = 5$ cm

228. In a class of 80 students, 60% students play carrom, 45% play chess and 10% student neither play carrom nor chess. What is the number of student who play chess only?

(a) 36 (b) 24

(c) 12 (d) 8

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

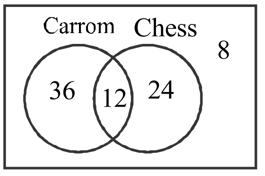
No. of students play carrom in class $= 80 \times \frac{60}{100} = 48$

and no. of students play chess in class $= 80 \times \frac{45}{100}$

$= 36$

Students who neither play carrom nor chess $= 80 \times \frac{10}{100} = 8$

No. of such students play atleast one game. $= 80 - 8 = 72$



No. of students who play both game $= (48+36) - 72$
$= 84 - 72 = 12$

No. of students who play only chess $= 36 - 12 = 24$

229. If $x^3 - y^3 = \frac{117}{8}$ and $x - y = \frac{3}{2}$, then the value of $x^2 + xy + y^2$ will be-

(a) 1 (b) $\frac{5}{8}$

(c) $\frac{39}{4}$ (d) $\frac{39}{8}$

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Given that,

$x^3 - y^3 = \frac{117}{8}$ and $x - y = \frac{3}{2}$

$x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ (**Formula**)

$\frac{117}{8} = \frac{3}{2}(x^2 + xy + y^2)$

$\therefore (x^2 + xy + y^2) = \frac{117 \times 2}{8 \times 3} \Rightarrow \frac{39}{4}$

230. If $(5)^{x^2+2x+7} = (125)^{2x+1}$ then the value of x is-

(a) 5 (b) 4

(c) 3 (d) 2

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

$(5)^{x^2+2x+7} = (125)^{2x+1}$

$(5)^{x^2+2x+7} = (5^3)^{2x+1} \Rightarrow 5^{(x^2+2x+7)} = (5)^{6x+3}$

Bases of both sides are equal, so the powers will also equal.

$x^2 + 2x + 7 = 6x + 3$

$x^2 + 2x - 6x + 7 - 3 = 0$

$x^2 - 4x + 4 = 0$

$(x-2)^2 = 0$

$x - 2 = 0$

$x = 2$

231. If $S = \sum_{n=1}^{15}\left(n + \frac{1}{3}\right)$ then the value of S is-

(a) 125 (b) $120 + \frac{1}{3}$

(c) $135 + \frac{1}{3}$ (d) 130

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

$S = \sum_{1}^{15}\left(n + \frac{1}{3}\right)$

$= \left(1 + \frac{1}{3}\right) + \left(2 + \frac{1}{3}\right) .... + \left(15 + \frac{1}{3}\right)$

$= \frac{4}{3} + \frac{7}{3} + ... + \frac{46}{3}$

$= \frac{4 + 7 + ... + 46}{3}$

$\because S_n = \frac{n}{2}\left[2a + (n-1)d\right] \ \{\text{where } a=4, d=3\}$

$= \frac{\frac{15}{2}\left[2 \times 4 + (15-1)3\right]}{3} = \frac{\frac{15}{2} \times 50}{3} = \frac{15 \times 25}{3}$

$= 5 \times 25 = 125$

232. Two trains, each of length 150 metres, are moving in opposite directions with equal speed of 90 km per hour. The time taken by the trains to cross each other will be.

(a) 3 seconds (b) 4.5 seconds

(c) 6 seconds (d) 9 seconds

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

Relative speed $= (90 + 90)$ km./hr $= 180$ km/hr

$= \left(180 \times \frac{5}{18}\right)$ m/s $= 50$ m/s

Distance = total length of both train

$= (150 + 150)$ m $= 300$ m

Time taken by the trains to cross each other

$= \frac{300}{50} = 6$ second

233. How many prime numbers are between 1 and 50?

(a) 17 (b) 15

(c) 14 (d) 16

(e) None of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

Prime numbers between 1 and 50 are

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43 and 47

There are 15 prime numbers between 1 and 50

234. A Number when divided by 342, gives a remainder 47. If the same numbers is divided by 19, what will be remainder?

(a) 0 (b) 9

(c) 18 (d) 8

(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (b)

Let the number be x

Dividen = (Divisor × Quotient) + Remainder.

$x = (342 \times \text{Quotient}) + 47$

Now dividing "dividen" by 19.

$= \left(\frac{342 \times Quotient}{19}\right) + \frac{47}{19}$

$= (18 \times \text{Quotient}) + 2\text{ Quotient} + 9\text{ Remainder}$

So, the remainder is 9.

235. A person who spends $66\frac{2}{3}\%$ of his income is able to save Rs. 1,200 per month. His monthly expense (in Rs.) is–

(a) 2,400 (b) 3,000

(c) 2,000 (d) 3,600

(e) 2,800

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (a)

Monthly saving of a person $= 100\% - 66\frac{2}{3}\%$

$= \left(100 - \frac{200}{3}\right)\%$

$= \frac{100}{3}\%$

$\because$ Monthly saving of person $= \frac{100}{3}\%$

According to questions, $\frac{100}{3}\% = \text{Rs.}1200$

$\therefore$ Monthly Expense of person $\frac{200}{3}\%$

$= \text{Rs.}1200 \times 2 = 2400\text{ Rs.}$

236. A man gains 20% by selling an article for a certain price. If he sells it at the double the price, the percentage of profit will be–

(a) 140 (b) 200

(c) 160 (d) 160

(e) 120

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (a)

Let the cost price of an article = Rs. 100

Selling price of article at 20% profit = Rs. 120

If the article is sold at double the price, the new selling price

$= 2 \times 120 \Rightarrow 240\text{ Rs.}$

Now profit % $= \left(\frac{240-100}{100} \times 100\right)\%$

$= 140\ \%$

237. 10 women can complete a work in 7 days and 10 children take 14 days to complete the work. How many days will 5 women and 10 children take to complete the work?

(a) 6 (b) 5

(c) 3 (d) 7

(e) 4

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (d)

$\because$ Work of (10×7) women = work of (14×10) children

$\therefore$ Work of 1 women = work of $\left(\frac{14 \times 10}{10 \times 7}\right)$ children

$\therefore$ Work of 1 women = work of 2 children

$\therefore$ Work of 5 women and 10 children

= Work of $(5 \times 2 + 10)$ Children

= Work of 20 Children

$M_1D_1 = M_2D_2$

$14 \times 10 = 20 \times D_2$

$D_2 = \frac{140}{20} \Rightarrow 7$ Day

238. How many numbers between 11 and 90 are divisible by 7?

(a) 10 (b) 9

(c) 13 (d) 12

(e) 11

63[rd] B.P.S.C. (Pre) Exam. 2017

Ans. (e)

Such numbers (between 11 and 90) which are divisible by 7 are as-

14, 21, 28, ... 84

The series of these numbers are arithmetic progression (AP) sequence.

$\therefore$ First term (a) = 14, d = 7

T_n (n^{th} term) = 84

$T_n = a + (n - 1)\, d$

$84 = 14 + (n - 1)\, 7$

$7\,(n - 1) = 84 - 14$

$n - 1 = \frac{70}{7} \Rightarrow 10$

$\therefore \quad n = 10 + 1 \Rightarrow 11$

239. A man can row 7.5 km per hours in still water. If in a river, running at 1.5 km per hour, it takes 50 minutes to row to a place and back then how far is the place?

(a) 3 km (b) 4 km

(c) 2 km (d) 5 km

(e) 7 km

63[rd] B.P.S.C. (Pre) Exam. 2017

Ans. (a)

Speed of boat in downstream $= \left(7\frac{1}{2} + 1\frac{1}{2}\right)$km/hr

$= 9$ km/hr

and speed of boat in upstream $= \left(7\frac{1}{2} - 1\frac{1}{2}\right)$km/hr

$= 6$ km/hr

Let, man goes x km and return x km back.

$\therefore \quad \frac{x}{9} + \frac{x}{6} = \frac{50}{60}$ hr

$\frac{4x + 6x}{36} = \frac{5}{6}$

$\frac{2x + 3x}{18} = \frac{5}{6}$

$\frac{5x}{18} = \frac{5}{6} \Rightarrow 5x = 15$

$\boxed{x = 3 \text{ km}}$

240. A sum of money invested at compound interest amounts to Rs. 4624 in 2 years and to Rs. 4913 in 3 years. The sum of money is-

(a) Rs. 4,240 (b) Rs. 4,280

(c) Rs. 4,096 (d) Rs. 4,346

(e) Rs. 4,406

63[rd] B.P.S.C. (Pre) Exam. 2017

Ans. (c)

Let, the principle is Rs. P and rate is R%.

Formula - $A = P\left(1 + \frac{x}{100}\right)^x$

According to question,

$P\left(1 + \frac{R}{100}\right)^2 = 4624$............(i)

$P\left(1 + \frac{R}{100}\right)^3 = 4913$............(ii)

On dividing eq^n (ii) by eq^n (i)

$1 + \frac{R}{100} = \frac{4913}{4624}$

$\frac{R}{100} = \frac{4913}{4624} - 1 = \frac{4913 - 4624}{4624}$

$R = \frac{289 \times 100}{4624} = \frac{25}{4}$

Putting the value of R in eq^n (i)

$P\left(1 + \frac{25}{4 \times 100}\right)^2 = 4624$

$P \times \frac{425}{400} \times \frac{425}{400} = 4624$

$P = \frac{4624 \times 400 \times 400}{425 \times 425} \Rightarrow 4096$

241. A man buys Rs. 20 shares by paying 9% dividend. Man wants to have an interest of 12% on his money. The market value of each shares is:-

(a) Rs.18 (b) Rs. 15

(c) Rs. 21 (d) Rs. 21

(e) None of the above/More than one of the above

63[rd] B.P.S.C. (Pre) Exam. 2017

Ans. (b)

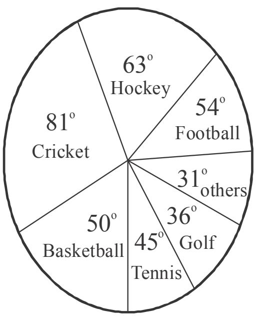
Let the market value of each share is Rs. x

$\therefore$ 12% of x = 9% of 20

$$x \times \frac{12}{100} = 20 \times \frac{9}{100}$$

$$x = \frac{20 \times 9}{12} \Rightarrow 15 \text{ Rs.}$$

242. The following pie chart shows the expenditure of the country on various sports during the particular year. Study the chart carefully and answer the question.



How much percent less is spent on football than that of cricket?

(a) $35\frac{1}{2}$ (b) 29

(c) $32\frac{1}{2}$ (d) 31

(e) $33\frac{1}{3}$

63rd B.P.S.C. (Pre) Exam. 2017

Ans. (e)

It is clear from the pie-chart that

Expenditure on football = 54°

and expenditure on cricket = 81°

$\therefore$ less, percent spent on football than that of cricket.

$$= \left(\frac{81^\circ - 54^\circ}{81^\circ} \times 100\right)\%$$

$$= \left(\frac{27^\circ}{81^\circ} \times 100\right)\%$$

$$= \frac{100}{3}\% \Rightarrow 33\frac{1}{3}\%$$

243. A invested Rs. 76,000 in a business. After some months, B joined him with Rs. 57,000. At the end of the year, the total profit was divided between them in the ratio 2:1 After how many months did B join?

(a) 6 (b) 4

(c) 3 (d) 8

(e) 5

63rd B.P.S.C. (Pre) 2017

Ans. (b)

(Let B, invest money after x months)

	A	:	B
Investment $\rightarrow$	76000	:	57000
	4	:	3
	×	:	×
Time $\rightarrow$	12	:	$(12 - x)$
Profit $\rightarrow$	2	:	1

$$\therefore \frac{4 \times 12}{3 \times (12 - x)} = \frac{2}{1}$$

$$12 - x = 8$$

$$x = 12 - 8 \Rightarrow 4 \text{ Months}$$

So, the B join the business after 4 months.

244. How many times the digit 9 occurs in the numbers between 100 to 999?

(a) 280 (b) 218

(c) 229 (d) 228

(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

Digit 9, occurs 20 times between 100 and 200.

$\therefore$ No. of times digit 9 occurs between 100 to 899.

= 20 × 8 = 160 times

digits 9 occurs between 900 to 990

= (11×9+2) Times

= 101 Times

digit 9 occurs between 991 to 999

= 9× 2 + 1

= 19 Times

So the digit 9 occurs between 100 to 999

= 160 + 101 + 19

= 280 Times

$\therefore$ So option (a) is the correct answer.

245. Find the value of expression. $\frac{4}{\sqrt[3]{9} - \sqrt[3]{3} + 1}$ **is—**

(a) $3^{1/3}+1$ (b) $3^{1/2}+1$

(c) $3^{1/3}-1$ (d) $3^{1/2}-1$

48th to 52nd B.P.S.C. (Pre) 2008

Given that, expression $\frac{4}{\sqrt[3]{9}-\sqrt[3]{3}+1}$

$$=\frac{3+1}{(3^2)^{\frac{1}{3}}-(3)^{\frac{1}{3}}+1}$$

$$=\frac{(3^{\frac{1}{3}})^3+(1^{\frac{1}{3}})^3}{(3^{\frac{1}{3}})^2-(3)^{\frac{1}{3}}\times 1^{\frac{1}{3}}+(1^{\frac{1}{3}})^2}$$

$$=\frac{(3^{\frac{1}{3}}+1^{\frac{1}{3}})\left[(3^{\frac{1}{3}})^2-(3)^{\frac{1}{3}}\times 1^{\frac{1}{3}}+(1^{\frac{1}{3}})^2\right]}{(3^{\frac{1}{3}})^2-(3)^{\frac{1}{3}}\times 1^{\frac{1}{3}}+(1^{\frac{1}{3}})^2}$$

$[\because a^3+b^3=(a+b)(a^2-ab+b^2)]$

$$=3^{\frac{1}{3}}+1$$

246. The dearness-allowance of a person with basic salary Rs. 7700 is increased to 132% from 125% and the tax deduction on both is increased from 20% to 22%. He got salary increased by.

(a) Rs. 74.00 (b) Rs. 77.00

(c) Rs.385.00 (d) Rs. 369.60

(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

Basic salary of person = Rs.7700

Initial dearness-allowance $= 7700\times\frac{125}{100}=$ Rs. 9625

Increased dearness- allowance $= 7700\times\frac{132}{100}\Rightarrow$ Rs.10164

Salary after 20% tax deduction in basic salary and dearness allowance $=(7700+9625)\frac{80}{100}$

$$=17325\times\frac{80}{100}=13860 \text{ Rs.}$$

Salary after 22% tax deduction in basic salary and increased dearness allowance.

$$=(7700+10164)\frac{78}{100}$$

$$=17864\times\frac{78}{100}=13933.92 \text{ Rs.}$$

Increased salary of person = 13933.92 – 13860 = Rs.73.92

So, the correct answer is option (e)

247. A, B, C Invested 20 lack rupees in a business in the ratio 7:2:1. The total profit is 18%. A has to pay 30% tax and B has to pay 20% tax. A's net profit is what percent of B's net profit?

(a) 118.8% (b) 180.0%

(c) 306.25% (d) 304.5%

(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

Total profit of A, B, C, $= 2000000\times\frac{18}{100}=$ Rs.360000

Proportional sum = 7 + 2 + 1 = 10

Profit of A $= 360000\times\frac{7}{10}=Rs.252000$

Profit of B $= 360000\times\frac{2}{10}=Rs.72000$

Net profit of A after paying tax

$$=252000\times\frac{70}{100}=Rs.176400$$

Net profit of B after paying tax

$$=72000\times\frac{80}{100}=Rs.57600$$

Now we can see that, the net profit of A is more than net profit of B.

Required% $=\frac{176400-57600}{57600}\times 100\%=206.25\%$

So, option (e) is the correct answer.

248. A coin is tossed above the ground with a velocity of 9.8 m/s, then it will rise to the following height :

(a) 9.8 m (b) 10 m

(c) 4.9 m (d) 49 m

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (c)

Let, the velocity of coin at the height of h meter =0

then $\because v^2=u^2-2gh$

(where u = initial velocity and v = final velocity)

$0=(9.8)^2-2(9.8)h$

$$h=\frac{(9.8)^2}{2\times(9.8)}=\frac{9.8}{2}$$

249. There are two drains, A and B in the bottom of a water tank. If only A is open, it takes 30 minutes to empty the full tank and if only B is open, it takes 20 minutes. If both drains are open for 10 minutes, then B is closed, how much time does take to empty a full tank?

(a) 18 minutes (b) 15 minutes
(c) 17 minutes (d) 20 minutes
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Part of tank, empty by drain A in 1 min $= \frac{1}{30}$

Part of tank, empty by drain B in 1 min $= \frac{1}{20}$

Part of tank, empty by both drains in 1 min

$$= \left(\frac{1}{30} + \frac{1}{20}\right)\text{Part}$$

Part of tank, empty by both drains A and B in 10 min.

$$= 10\left(\frac{1}{30} + \frac{1}{20}\right)\text{Part}$$

$$= \frac{1}{3} + \frac{1}{2} \Rightarrow \frac{5}{6} \text{ Part}$$

Remaining, $1 - \frac{5}{6} \Rightarrow \frac{1}{6}$ Part

Time taken by drain A, to empty part of tank = 30 minutes

Time taken by drain A to empty remaining part of tank

$= \frac{1}{6} \times 30 \Rightarrow 5$ min.

Total time taken to empty full tank = (10 + 5) min.
= 15 min.

So, option (b) is correct answer.

250. If $x = -\frac{2}{3}$, then $9x^2 - 3x - 11$ is equal to-

(a) – 13 (b) 13
(c) – 5 (d) – 17
(e) 17

64th B.P.S.C. (Pre) 2018

Ans. (c)

If $x = -\frac{2}{3}$

then $9x^2 - 3x - 11 = 9\left(-\frac{2}{3}\right)^2 - 3\left(-\frac{2}{3}\right) - 11$

$$= 9 \times \frac{4}{9} + 2 - 11$$
$$= 4 + 2 - 11$$
$$= 6 - 11 = -5$$

251. In an examination, each candidate takes Hindi or History or both. 66% take Hindi and 59% take History. The total number of candidates was 3000. How many candidates took both Hindi and History?

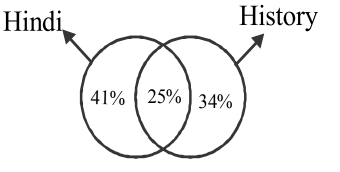
(a) 500 (b) 750
(c) 542 (d) 738
(e) 830

64th B.P.S.C. (Pre) 2018

Ans. (b)

Percentage of candidates take both Hindi and History as subject-

$= \{(66+59) - 100\}\%$
$= (125 - 100)\%$
$= 25\%$



Number of candidates taking both the subjects-

$= 3000 \times \frac{25}{100}$

$= 30 \times 25 = 750$

252. If $x = \sqrt{\frac{0.00001225}{0.00005329}}$ then, the value of x is-

(a) $\frac{35}{73}$ (b) $\frac{525}{933}$

(c) $\frac{205}{403}$ (d) $\frac{135}{233}$

(e) None of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

$x = \sqrt{\frac{0.00001225}{0.00005329}}$

$= \sqrt{\frac{1225}{5329}}$

$= \sqrt{\frac{35 \times 35}{73 \times 73}}$

$= \frac{35}{73}$

Therefore, option (a) is correct answer.

253. If $x+\frac{1}{y}=1$ and $y+\frac{1}{z}=1$, then the value of $z+\frac{1}{x}=1$ is-

(a) $x-y$ (b) 1

(c) Uncountable (d) 2

(e) None of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

Given that

$$x+\frac{1}{y}=1$$

$$x=1-\frac{1}{y}=\frac{y-1}{y}$$

$$\therefore \frac{1}{x}=\frac{y}{y-1} \quad ....(i)$$

and $y+\frac{1}{z}=1$

$$\frac{1}{z}=1-y$$

$$\therefore z=\frac{1}{1-y} \quad ....(ii)$$

Adding eq^n (i) and (ii)

$$z+\frac{1}{x}=\frac{y}{y-1}+\frac{1}{1-y}$$

$$=\frac{y}{y-1}-\frac{1}{y-1}$$

$$=\frac{1}{y-1}(y-1)=1$$

254. The average age of 4 sisters is 7 years. If we add the age of mother, the average increases by 6 years. Find the age of mother?

(a) 46 years (b) 39 years

(c) 37 years (d) 47 years

(e) 57 years

64th B.P.S.C. (Pre) 2018

Ans. (c)

Average age of 4 sisters = 7 years

Total age of 4 sisters $=7\times4=28$ years

Average age of all members including mother = 7 + 6 = 13 years

Total age of all members (Mother + 4 sisters)= 13 × 5 = 65 years

Mother age = (65 – 28) years

= 37 years

255. If $3^{x+8}=27^{2x+1}$ then the value of x is-

(a) 9 (b) 1

(c) –1 (d) 10

(e) –10

64th B.P.S.C. (Pre) 2018

Ans. (b)

$$3^{x+8}=27^{2x+1}$$

$$3^{x+8}=(3^3)^{2x+1}$$

$$3^{x+8}=3^{6x+3}$$

Since, the bases of both the sides are equal, hence the power will also be equal.

$$x+8=6x+3$$

$$6x-x=8-3$$

$$5x=5$$

$$x=1$$

256. If $S=\sum_{n=1}^{10}\left(2n+\frac{1}{2}\right)$, then S is-

(a) $55\frac{1}{2}$ (b) 56

(c) 111 (d) 115

(e) $110\frac{1}{2}$

64th B.P.S.C. (Pre) 2018

Ans. (d)

$$S=\sum_{n=1}^{10}\left(2n+\frac{1}{2}\right)$$

n = 1, 2, 3, ..., 10

$$S_1=2\times1+\frac{1}{2}=\frac{5}{2}$$

$$S_2=2\times2+\frac{1}{2}=\frac{9}{2}$$

$$S_3=2\times3+\frac{1}{2}=\frac{13}{2}$$

$$S_{10}=2\times10+\frac{1}{2}=\frac{41}{2}$$

$$\therefore S=\frac{5}{2}+\frac{9}{2}+\frac{13}{2}+........+\frac{41}{2}$$

$$=\frac{1}{2}[5+9+13+........,+41]$$

$$=\frac{1}{2}\left[\frac{10}{2}\{2\times5+(10-1)4\}\right]\therefore S_n=\frac{n}{2}[2a+(n-1)d]$$
$$=\frac{1}{2}\times5\times[10+9\times4]$$
$$=\frac{1}{2}\times5\times46=5\times23\Rightarrow115$$

257. The value of x is a consecutive whole number and the first four value of the expression x^3+3y-3 are 7, 20, 45, 88, then the fifth value will be-

(a) 137 (b) 155
(c) 158 (d) 143
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Given that, the expression is x^3+3y-3
where the value of x is a whole number
or $x=\{0,1,2,3,4.........\}$
putting $x=0$ in expression
$(0)^3+3y-3=7$
$$y=\frac{10}{3}$$
putting, $x=1$ in expression
$(1)^3+3y-3=20$
$$y=\frac{22}{3}$$
putting, $x=2$ in expression
$(2)^3+3y-3=45$
$$y=\frac{40}{3}$$
On putting, $x=3$ in expression
$(3)^3+3y-3=88$
$$y=\frac{64}{3}$$
So, the next value of x will be-
$$\frac{10}{3}+4=\frac{22}{3},\ \frac{22}{3}+6=\frac{40}{3}$$
$$\frac{40}{3}+8=\frac{64}{3},\ \frac{64}{3}+10=\frac{94}{3}$$
then the fifth value of expression will be-
$$(4)^3+3\times\frac{94}{3}-3$$
$=64+94-3$
$=64+91$
$=155$
So, option (b) is correct answer-

258. Find the value of $x-[y-\{z-(x-\overline{y-z})\}]$

(a) $x+y+z$ (b) $x-y-z$
(c) 1 (d) 0

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

$$x-[y-\{z-(x-\overline{y-z})\}]$$
$$=x-[y-\{z-(x-y+z)\}]$$
$$=x-[y-\{z-x+y-z\}]$$
$$=x-[y-z+x-y+z]$$
$$=x-y+z-x+y-z$$
$$=0$$

259. The divisor of the expression $x(x^2-1)(3x+2)$ for every integer x is-

(a) 13 (b) 15
(c) 24 (d) 25

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (c)

Given that, the expression is $F(x)=x(x^2-1)(3x+2)$
For each integer-
On putting, $x=1,2,3,4,5.......$
$x=1, F(1)=0=24\times0$
$x=2, F(2)=2(2^2-1).(3\times2+2)=2\times3\times8=24\times2$
$x=3, F(3)=3(3^2-1).(3\times3+2)=3\times11\times8=24\times11$
$x=4, F(4)=4(4^2-1).(3\times4+2)=4\times15\times14=24\times35$
$x=5, F(5)=5(5^2-1).(3\times5+2)=5\times24\times17=24\times85$
Thus, the divisor of the given expression $x.(x^2-1).(3x+2)$ for each integer is $x=24$

260. The inverse of matrix $\begin{bmatrix}5&2\\7&3\end{bmatrix}$ is-

(a) $\begin{bmatrix}3&2\\-7&5\end{bmatrix}$

(b) $\begin{bmatrix}3&2\\7&5\end{bmatrix}$

(c) $\begin{bmatrix}3&-2\\-7&5\end{bmatrix}$

(d) $\begin{bmatrix}-3&2\\7&-5\end{bmatrix}$

53rd to 55th B.P.S.C. (Pre) 2011

Let, $A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}$

$\therefore$ $\text{Adj } A = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$

and $|A| = 3 \times 5 - 7 \times 2$
$= 15 - 14$
$= 1$

$\therefore \quad A^{-1} = \dfrac{\text{Adj}A}{|A|} = \dfrac{1}{1}\begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$

$= \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$

261. The solution of the differential equation

$\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = e^{5x}$ **is-**

(a) $y = C_1e^{x} + C_2e^{2x} + \dfrac{1}{12}e^{5x}$

(b) $y = C_1e^{-x} + C_2e^{2x} + \dfrac{1}{12}e^{5x}$

(c) $y = C_1e^{x} + C_2e^{-2x} + \dfrac{1}{12}e^{5x}$

(d) $y = C_1e^{x} + C_2e^{2x} + \dfrac{1}{5}e^{5x}$

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

$\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = e^{5x}$

Let, $\dfrac{d^2y}{dx^2} = D^2$

$D^2 - 3D + 2 = 0$
$D^2 - 2D - D + 2 = 0$
$D(D-2) - 1(D-2) = 0$
$(D-2)(D-1) = 0$
$D = 2, 1$

then $y = C_1e^{x} + C_2e^{2x}$

Specific solution

$\text{P.I.} = \dfrac{1}{f(D)}e^{5x}$

$= \dfrac{1}{D^2 - 3D + 2}e^{5x}$

$= e^{5x}.\dfrac{1}{25 - 15 + 2}$

$= \dfrac{1}{12}e^{5x}$

then, General solution.

$y = C_1e^{x} + C_2e^{2x} + \dfrac{1}{12}e^{5x}$

where C_1 & C_2 are constant.

262. If function $f : I^{+} \to R,\ f(x) = \log x$

then the value of $f(x) + f(y)$ is-

(a) $f(xy)$ (b) $f(x+y)$

(c) $f\left(\dfrac{x}{y}\right)$ (d) $f\left(\dfrac{y}{x}\right)$

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

$\because$ $f(x) = \log x$ .... (i)

Substituting y in place of x

$\therefore$ $f(y) = \log y$ .... (ii)

Substituting $x = v$

$f(v) = \log v$ ....(iii)

$\therefore$ $f(x) + f(y) = \log x + \log y$

$= \log xy$ ($\because$ $\log m + \log n = \log mn$)

Let, $x\,y = v$

$\therefore$ $f(x) + f(y) = \log v$

$\therefore$ $f(x) + f(y) = f(v)$ (from eq^n (iii))

$\therefore$ $f(x) + f(y) = f(xy)$ ($\because$ $v = xy$)

263. If triangle PQR has $\angle P = 120^o$ and PQ = PR, then $\angle Q$ and $\angle R$ will be respectively:

(a) $60^o, 30^o$ (b) $30^o, 40^o$

(c) $30^o, 30^o$ (d) $20^o, 40^o$

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

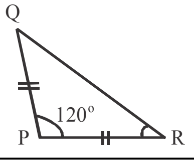
$\because$ In ΔPQR $\angle P = 120^o$

and PQ = PR

$\therefore$ If two sides of a triangle are equal, then the angles opposites to it will also be.

$\therefore$ $\angle Q = \angle R$ ....(i)

$\because$ $\angle P + \angle Q + \angle R = 180^o$



$\therefore \quad 120 + \angle Q + \angle Q = 180^o$

$(\because$ From eq^n (i) $\angle Q = \angle R)$

$$2\angle Q = 180^o - 120^o$$
$$2\angle Q = 60^o$$
$$\angle Q = 30^o$$
$$\therefore \quad \angle R = 30^o$$

264. The coordinates of the mid-point joining the point P (4, 6) and Q (–4, 8) will be–

(a) (2, 7) (b) (7, 2)

(c) (7, 0) (d) (0, 7)

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The coordinates of the mid-point joining the point

P (x_1, y_1) and Q (x_2, y_2)

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$\therefore$ The coordinates of the mid-point joining the point (4, 6) and (–4, 8)

$$= \left(\frac{4-4}{2}, \frac{6+8}{2}\right)$$
$$= (0, \quad 7)$$

265. The triangle formed by the points (2, 7), (4, -1) and (-2, 6) will be -

(a) Equilateral (b) Right angle

(c) Isosceles (d) None of these

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (b)

The three points of the triangle respectively.

$$A = (2, 7)\ B = (4, -1), C = (-2, 6)$$

Length of first side, AB

$$= \sqrt{(4-2)^2 - (-1-7)^2}$$
$$= \sqrt{2^2 + (-8)^2} = \sqrt{68}$$

Length of second side, BC

$$= \sqrt{(-2-4)^2 + (+6+1)^2}$$
$$= \sqrt{(-6)^2 + 7^2} = \sqrt{85}$$

Length of the third side, CA

$$= \sqrt{(-2-2)^2 + (6-7)^2}$$
$$= \sqrt{(-4)^2 + (-1)^2} = \sqrt{17}$$

According to Pythagoras theorem–

$$BC^2 = AB^2 + CA^2$$
$$(\sqrt{85})^2 = (\sqrt{68})^2 + (\sqrt{17})^2$$
$$85 = 85$$

i.e. ΔABC is a right angled triangle.

266. The length of the minute hand of a clock is 12 cm, then the area covered by minute in one minute will be–

(a) 22.12 sq. cm

(b) 23.10 sq. cm

(c) 24.12 sq. cm

(d) None of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The angle subtended by the min. hand of the clock in One minutes $= 6^o$

$\therefore$ Area covered by min. hand in one minute $= \dfrac{\pi r^2 \theta}{360^o}$

$$= \frac{22}{7} \times \frac{12 \times 12 \times 6^o}{360^o} \text{ cm}^2$$
$$= \frac{22}{7 \times 60} \times 144 \text{ cm}^2$$
$$= \frac{11 \times 72}{7 \times 15} \Rightarrow 7.54 \text{ cm}^2$$

267. A player completes a run of 2200 meters by completing 7 rounds of circular path, then the radius of the circular path is- $\left(\pi = \frac{22}{7}\right)$

(a) 30 meters

(b) 40 meters

(c) 50 meters

(d) 60 meters

53rd to 55th B.P.S.C. (Pre) 2011

Let, the radius of circular path = r

∴ Distance covered by player in one round $= 2\pi r$

∴ According to question

$$7 \times 2\pi r = 2200$$

$$r = \frac{2200 \times 7}{14 \times 22}$$

$$= \frac{100}{2} \Rightarrow 50 \text{ meters.}$$

268. The variance of 7, 7, 7, 7 , 7 will be-

(a) 7 (b) 0

(c) 2 (d) 7.5

56th to 59th B.P.S.C. (Pre) 2015

Ans. (b)

The variance of observations is represented by σ^2

$$\sigma^2 = \frac{(x-\bar{x})^2}{\text{no. of observation}}$$

Now,

$$\bar{x} = \frac{7+7+7+7+7}{5} = \frac{35}{5} = 7$$

then,

$(x-\bar{x})^2 = 0$ (because all observaitons are same)

So Variance, $(\sigma)^2 = \frac{0}{5} = 0$

269. The median of 2, 4, 6, 8, 10, 12, 14, 16 is-

(a) 8 (b) 9

(c) 10 (d) 11

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Given that, the series is 2, 4, 6, 8, 10, 12, 14, 16

(Total 8 term i.e. the no- of terms is even)

$$\therefore \quad \text{Median} = \frac{\frac{n}{2} th\, term + \left(\frac{n}{2}+1\right) th\, term}{2}$$

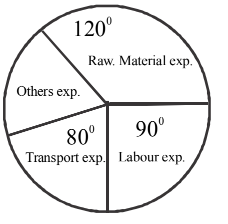
$$= \frac{\frac{8}{2} th\, term + \left(\frac{8}{2}+1\right) th\, term}{2}$$

$$= \frac{4\, th\, term + 5\, th\, term}{2}$$

$$= \frac{8+10}{2} \Rightarrow 9$$

answer

270. In the following pie-chart, the various expenditures of an industry are represented. Total expenditure on raw material is Rs. 30 lakh, if labour expenditure is increased by 5%. How much expenditure on other expenses is reduced to maintain the profit?



(a) 5.9% (b) 12.86%

(c) 6.43% (d) 6.21%

(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

According to question,

Total expenditure on raw material = 30 lakh

$120^0 = 30$ lakh

$$1^0 = \frac{30}{120} \text{ lakh}$$

$$= \frac{1}{4} \text{ lakh}$$

Exp. on labour, $90^0 = \frac{90}{4}$ lakh

Increased expenditure on labour $= \frac{90}{4} \times \frac{5}{100}$

$$= \frac{9}{8} \text{ lakh}$$

In order to keep the profit constant, the expenditure must be constant.

Area of other exp. $= 360^0 - (120^0 + 80^0 + 90^0)$

$= 70^0$

$$= \frac{70}{4} \text{ lakh} \left(\because 1^0 = \frac{1}{4} \text{Lakh}\right)$$

deduction in other expenses $= \frac{9}{8}$ lakh

$\frac{70}{4}$ lakh x % of $= \frac{9}{8}$ lakh

$$x = \frac{9}{8} \times \frac{4}{70} \times 100\%$$

$= 6.43\%$

271. Find the value of –

$$\frac{\mathbf{0.05\times0.05\times0.05-0.04\times0.04\times0.04}}{\mathbf{0.05\times0.05+0.05\times0.04+0.04\times0.04}}$$

(a) 1.01 (b) 0.1
(c) 0 (d) 0.01

40th B.P.S.C. (Pre) 1995

Ans. (d)

$$\frac{(0.05)^3-(0.04)^3}{(0.05)^2+(0.05)(0.04)+(0.04)^2}$$
$=0.05-0.04$
$=0.01$
Note $[a^3-b^3=(a-b)(a^2+ab+b^2)]$

272. If $x^2+\frac{1}{x^2}=3^4$, then $x+\frac{1}{x}$ is equal to—

(a) $\sqrt{43}$ (b) $\sqrt{37}$
(c) $\sqrt{46}$ (d) $\sqrt{83}$

44th B.P.S.C. (Pre) 2000-01

Ans. (d)

$$x^2+\frac{1}{x^2}=3^4$$
$$x^2+\frac{1}{x^2}+2=3^4+2$$
$$\left(x+\frac{1}{x}\right)^2=83$$
$$\therefore \left(x+\frac{1}{x}\right)=\sqrt{83}$$

273. Which of the following statements is not correct.

(a) π is an irrational number
(b) e is a rational number
(c) $\sqrt{2}$ is an even number
(d) ∞ is a real number

43rd B.P.S.C. (Pre) 1999

Ans. (c)

$\sqrt{2}$ is not an even number.

274. Choose the false statements—

(a) Mathematical work written by Laghu Bhaskarya Bhaskar
(b) Is also enumerated in the vachali inscription
(c) Decimal Method discovered by the Europeans
(d) Mathematical Tilak is a work of eleventh century

40th B.P.S.C. (Pre) 1995

Ans. (c)

Decimal Method discovered by the Indian.

275. Which of the following statements is not true?

(a) The square of every negative number is positive.
(b) A negative number does not have a square root.
(c) $-\infty$ is a negative number.
(d) $-\pi$ is an algebraic number.

44th B.P.S.C. (Pre) 2000-01

Ans. (c)

(i) The square of every negative number is positive, this statement is true.
(ii) A negative number root does not have a square root, its square root is imaginary.
(iv) π is an algebraic number. In fact, π its the ratio of circumference and diameter, It is not possible for the ratio to be negative.

276. Choose the difference of 5 miles and 8 km in yards–

(a) 51.1 (b) 51
(c) 52.2 (d) 52

42nd B.P.S.C. (Pre) 1997-98

Ans. (a)

Scale -
1 miles = 1760 yards
1 km = 1093.61 yards
According to question,
5 miles = (5 × 1760) yards
= 8800 yards
and 8 km. = (8 × 1093.61) yards
= 8748.88 yards
$\therefore$ Required difference = (8800 - 8748.88) yards
= 51.12 yards
= 51.1 yards

277. The difference of two positive numbers is 72 and when one is divided by the other, the dividend is 4. Choose the numbers–

(a) 40, 112 (b) 24, 96
(c) 32, 128 (d) 30, 102

42nd B.P.S.C. (Pre) 1997-98

Ans. (b)

Let, the two number be a and b respectively,
According to question,
a-b = 72 .... (i)
$$\frac{a}{b}=4$$

on,

a = 4 b .... (ii)

Substituting the value of eq^n (i) in equation (ii)

$4b - b = 72$

$b = \frac{72}{3} = 24$

$\therefore a = 24 \times 4 = 96$

So, the number will be 24 and 96.

278. How may Prime (twin prime) numbers are between 70 and 100 ?

(a) 1 (b) 0
(c) 2 (d) 3

42nd B.P.S.C. (Pre) 1997-98

Ans. (a)

Twin Prime number are such prime numbers in which there is a difference of two e.g. (3, 5) and (5, 3)

According to question–

There are 6 prime number between 70 and 100

71, 73, 79, 83, 89, 97

Hence the number of twin prime between 70 and 100 is 1 (71 & 73).

279. Find the lowest number, which should be multiplied into 56,700, so that it becomes a perfect square-

(a) 3 (b) 6
(c) 7 (d) 9

39th B.P.S.C. (Pre) 1994

Ans. (c)

$56,700 = 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 5 \times 5 \times 7$

It is clear that, on multiplying by 7, number 56700 will become a perfect square

280. The sum of three prime number is 100. One of the number is 36 more than the other numbers. The highest prime number is–

(a) 31 (b) 67
(c) 73 (d) 79

41st B.P.S.C. (Pre) 1996

Ans. (b)

Let the first prime number x is

$\therefore$ Second prime number $= x + 36$

Third number $= 100 - (x + x + 36)$

$= 100 - (2x + 36)$

$= 64 - 2x$

x must be such a prime number that the result obtained by putting it in $(64-2x)$ should also prime number.

$\therefore$ On putting $x = 31$

Third Number = 2 (2 is a prime number)

So the highest prime number is = 31+36 = 67

281. If $7^{13}+1$ is divisible by 6 then the remainder is–

(a) 0 (b) 1
(c) 2 (d) 3

41st B.P.S.C. (Pre) 1996

Ans. (c)

On dividing $7^{13}+1$, by 6

Obtained remainder, $\frac{7^{13}+1}{6} = \frac{(6+1)^{13}+1}{6}$

$= \frac{(+1)^{13}+1}{6} = \frac{1+1}{6}$

$= \frac{2}{6}$

Thus, remainder = 2

282. In a class test, x and y obtained the following marks:

Subject \ Student	x	y	Total Mark
English	84	92	100
Hindi	80	79	100
Mathematics	90	88	100
History	69	60	100

Who got more marks and how many.

(a) x got 1% more marks
(b) x got 2% more marks
(c) y got 1% more marks
(d) y got 2% more marks

40th B.P.S.C. (Pre) 1995

Ans. (a)

According to question total marks obtained by x

$= 84+80+90+69 = 323$

marks percentage of $x = \frac{323 \times 100}{400} = 80.75\%$

60 =319

marks percentage of $y = \frac{319 \times 100}{400} = 79.75\%$

Percentage difference of x and $y = 80.75 - 79.75 = 1\%$

Hence x got 1% more marks.

283. If $33\frac{1}{3}\%$ of $x = 11$, the value of x will be?

(a) 11 (b) 25
(c) 33 (d) 44

39th B.P.S.C. (Pre) 1994

Ans. (c)

$$33\frac{1}{3}\% \text{ of } x = 11$$

$$x \times \frac{100}{3}\% = 11$$

$$x \times \frac{100}{300} = 11$$

$$x = \frac{11 \times 300}{100} = 33$$

284. A man suffers a loss of 20% by selling a cycle for Rs. 450. By selling the cycle at 20% profits, it selling price will be–

(a) Rs. 652.50 (b) Rs. 765
(c) Rs. 562.50 (d) Rs. 675

43rd B.P.S.C. (Pre) 1999

Ans. (d)

Selling price = Rs.450, the loss % = 20%

$$\text{Cost price} = \frac{100 \times \text{Selling price}}{100\text{-Loss\%}}$$

$$\text{Cost price of cycle} = \frac{100 \times 450}{80}$$

$$= \text{Rs. } 562.50$$

Selling price of cycle of 20% profit =

$$\frac{\text{Cost price} \times (100 + \text{profit \%})}{100}$$

$$= \frac{562.5 \times 120}{100}$$

∴ The selling price of cycle at 20% profit = Rs. 675

285. Choose the compound interest on Rs. 400 for 2 years at 5% per annum compounded annually.

(a) Rs. 414 (b) Rs. 441
(c) Rs. 410 (d) Rs. 401

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Compound Interest = Amount - Principal

$$\text{Amount} = P\left(1 + \frac{R}{100}\right)^{\text{Time}}$$

$$= 4000\left(1 + \frac{5}{100}\right)^2$$

$$= 4000\left(\frac{21}{20}\right)^2 = \text{Rs. } 4410$$

Compound Interest = (4410 – 4000) Rs. = 410.

286. The market price of 1 share of Rs. 10 face value of a company is Rs. 25. The company declared a dividend of 10%. The dividend of 1500 shares of the company will be–

(a) 3750 (b) 15000
(c) 7500 (d) 1500

47th B.P.S.C. (Pre) 2005

Ans. (d)

Market price = 10
No. of shares = 1500
Market price of 1500 share = 15000
dividend = 10%

$$\text{Now dividend} = 15000 \times \frac{10}{100} = 1500$$

287. The simple interest on a sum of money for 2 years is Rs. 160, while the compound interest is Rs. 164. Find the rate of interest?

(a) 4% (b) 5%
(c) 6% (d) 10%

41st B.P.S.C. (Pre) 1996

Ans. (b)

Simple Interest for 2 years = Rs.160
Compound Interest for 2 years = Rs.164
Difference = Rs. 4
i.e. interest for one year's interest = Rs.4

$$\text{Simple Interest of 1 year} = \frac{160}{2} = \text{Rs. } 80$$

$$\text{Rate of interest} = \frac{(\text{Compound Interest - Simple Interest}) \times 100}{\text{Simple Interest} \times \text{Time}}$$

$$= \frac{\times \ 4 \quad 100}{80 \times 1}$$

∴ Rate of interest = 5%

288. **A person buys 11 books for Rs. 10 and sells 10 books for Rs. 11. What percent profit will he make?**

(a) 12% (b) 12.5%
(c) 21.5% (d) 21%

42nd B.P.S.C. (Pre) 1997-98

Ans. (d)

Cost price of 1 book = $\frac{10}{11}$ Rs.

Selling price of 1 book = $\frac{11}{10}$ Rs.

Profit = Selling Price - Cost price

$$= \frac{11}{10} - \frac{10}{11}$$

$$= \frac{21}{110}$$

$$\text{Profit \%} = \frac{\text{Profit} \times 100\%}{\text{Cost Price}}$$

$$= \frac{\frac{21}{110} \times 100}{\frac{10}{11}}$$

$$= \frac{21}{110} \times \frac{11}{10} \times 100$$

= 21%

289. **In an examination, 20% students failed in English, 35% failed in Hindi. 27% of the students failed in both. If the number of passed student is 288, then the number of students who appeared in the examination is–**

(a) 600 (b) 500
(c) 400 (d) 300

39th B.P.S.C. (Pre) 1994

Ans. (c)

Failed students in English = n(E) = 20%

Failed students in Hindi = n(H) = 35%

Students failed in both subjects = $n(E \cap H)$ = 27%

Total failed students = $n(E \cup H)$

$n(E \cup H) = n(E) + n(H) - n(E \cap H)$

= (20 + 35 − 27)%

$n(E \cup H)$ = 28%

∴ Total percentage of failed students = 28%

Total percentage of passed students. = (100-28)%

= 72%

According to question,

72% = 288

$$100\% = \frac{288 \times 100}{72}$$

= 400

290. **Consider the following statements–**

Assertion (A) : To increase a number by 40%, it is sufficient to multiply that by 1.4

Reason (R) : To reduce a number by 40%, it is sufficient to divide that number by 1.4,

(a) A is true but R is false
(b) R is true but A is false
(c) Both A and R are true
(d) Both A and R are false

39th B.P.S.C. (Pre) 1994

Ans. (a)

Let the number is 1, if it is increased by 40% then its total value

= 1+ 40% of 1

= 1.4

But if it is decreased by 40%. then the number 1-40% of 1

= 0.6

Therefore, on reduction by 40% the number has to be multiplied by 0.6, dividing by 1.4 cannot yield 0.6 per unit.

291. **If the population of a city increases at the rate of 4% per annum and there is an additional increase at the rate of 1% per annum due to refugees, then the increase population will after two years.**

(a) 10% (b) 10.25%
(c) 10.50% (d) 10.75%

40th B.P.S.C. (Pre) 1995

Ans. (b)

Let, population = 100

Increased population after 1 years

= 100 + 4 + 1

= 105

Population after 2 years = 105 + $\frac{5}{100}$ of 105

= 110.25

Increase in population = 10.25

Percentage increased in population after 2 years = 10.25%

292. **80% of the voter participated in an election. There were only two candidates Ram and Shyam. Ram got 42% of total votes and defeated Shyam by 400 votes. Thus how**

get?–

(a) 4200 votes
(b) 3800 votes
(c) 3,360 votes
(d) 3000 votes

41st B.P.S.C. (Pre) 1996

Ans. (a)

$\because$ Ram got 42% votes
$\therefore$ Shyam got $(80\text{-}42)\% = 38\%$ votes
Thus, the difference of their votes
$(42\text{-}38)\% = 400$ (According to question)
$4\% = 400$
$1\% = 100$
Hence, Ram got votes $= 42 \times 100 = 4200$

293. If $x\%$ of y is equal to $y\%$ of z , then z will be equal to-

(a) x
(b) y
(c) $x/100$
(d) $y/100$

40th B.P.S.C. (Pre) 1995

Ans. (a)

x % of y = y % of z

$$y \times \frac{x}{100} = z \times \frac{y}{100}$$

$z = x$

294. If a : b :: c: d then–

(a) (a + b) : b : (c + d)
(b) (a - b) : b :: (c - d):d
(c) (a + b + c) : d
(d) (a - c) : b :(d - a)

40th B.P.S.C. (Pre) 1995

Ans. (b)

If a : b :: c : d
then, (a + b) : b :: (c + d) : d
or (a – b) : b :: (c – d) : d
or a × d = b × c
Hence, option (b) is the correct answer

295. While walking at $\frac{3}{4}$ of his normal rate, a man is delayed by 1.5 hr. His normal time is–

(a) 1hr
(b) $4\frac{1}{2}$ hr
(c) $3\frac{1}{2}$ hr
(d) $6\frac{1}{2}$ hr

41st B.P.S.C. (Pre) 1996

Ans. (b)

Let normal speed $= x$ km/hr and
Time $= y$ hr

distance $= x\,y$ $\quad (\therefore \text{distance} = \text{Speed} \times \text{Time})$

Time taken with $\frac{3}{4}x$

$$\frac{xy}{\frac{3}{4}x} = y + \frac{3}{2} \qquad \left(\text{Time} = \frac{\text{distance}}{\text{speed}}\right)$$

$$\frac{4y}{3} = \frac{2y+3}{2}$$

$8y = 6y + 9$

$$y = \frac{9}{2}\text{ hr} = 4\frac{1}{2}\text{hr.}$$

$$= 4\frac{1}{2}\text{h}$$

296. The distance between two stations A and B is 70 km. A train starts from A to B with a speed of 80 kmph and at the same time another train starts from B to A with a speed of 60 kmph, so in how much time they meet each other later?

(a) 60 minutes
(b) 45 minutes
(c) 30 minutes
(d) 15 minutes

41st B.P.S.C. (Pre) 1996

Ans. (c)

Given that,

$(A \to B)$ Speed=80 km/h

$(B \to A)$=60 km/h

($\therefore$ Both trains are running in opposite direction)
Relative speed = 140 km/hr
$\therefore$ The optimum time taken to meet the two trains is the relative running distance between the two stations. to relative speed of both trains.

$$= \frac{70}{140}$$

$$= \frac{1}{2}\text{ hr} = 30\text{ minutes}$$

297. Two trains from Delhi to Amritsar run at 10 A.M. and 11 A.M. on the same day at 60 km/hr and 75 km/hr. At what distance from Delhi will the trains meet?

(a) 150 km
(b) 200 km
(c) 250 km
(d) 300 km

39th B.P.S.C. (Pre) 1994

The first train left 1 hr ago. So it covered a distance of 60 km.

$\therefore$ Relative speed of both trains $= (75-60)$ km/hr

$= 15$ km/hr

($\therefore$ Both trains are running in same direction)

$\therefore$ Required Time $= \dfrac{\text{distance}}{\text{speed}}$

$= \dfrac{60}{15}$

$= 4$ hr

distance covered by another train in 4 hour

$= (4 \times 75) = 300$ km

298. While climbing a tower of 180 steps, a man rests for 2 minutes after climbing every 30 steps how long did he rest till he reached the top of the tower?

(a) 30 minutes (b) 12 minutes

(c) 10 minutes (d) 9 minutes

40th B.P.S.C. (Pre) 1995

Ans. (c)

Person took 2-2 minutes rest on 30th, 60th, 90th, 120th and 150th stairs. so he rested for a total of 10 minutes.

299. If 10 workers dig a well in 4 days then how many workers will be required to dig the same well in half a day?

(a) 5 (b) 40

(c) 60 (d) 80

40th B.P.S.C. (Pre) 1995

Ans. (d)

$\because$ 10 workers dig a well in 4 days

$\therefore (4 \times 10)$ workers can dig a well in 1/a day

$\therefore (4 \times 10 \times 2)$ workers can dig a well in 1/2 day

$= 80$ workers

Second Method-

$M_1D_1 = M_2D_2$

$10 \times 4 = m_2 \times \dfrac{1}{2}$

$M_2 = (10 \times 4 \times 2)$ workers

$= 80$ workers

300. A cyclist covers a distance of 15 km, 20 km and 25 km at the speed of 15 km/hr, 25 km/hr and 15 km/hr respectively what will be his average speed?

(a) 30 km/hr (b) 20 km/hr

(c) 22.5 km/hr (d) 21.4 km/hr

47th B.P.S.C. (Pre) 2005

Ans. (*)

$$\text{Average speed} = \frac{\text{Total distance covered}}{\text{total time taken}}$$

$$= \frac{15+20+25}{\frac{15}{15}+\frac{20}{25}+\frac{25}{15}} \qquad \left(\text{speed} = \frac{\text{distance}}{\text{time}}\right)$$

$$= \frac{60}{1+\frac{4}{5}+\frac{5}{3}}$$

$$= \frac{60}{\frac{15+12+25}{15}}$$

$$= \frac{60 \times 15}{52} = \frac{225}{13}$$

$= 17.3$ km/hr

301. The price of an article increased by 5% from 1995 to 1996, 8% from 1996 to 1997 and 77% from 1997 to 1998, then the average increase in price from 1995 to 1998 is-

(a) 24% (b) 32%

(c) 26% (d) 30%

45th B.P.S.C. (Pre) 2002

Ans. (d)

$$\text{Average increase\%} = \left(\frac{5+8+77}{3}\right)\% = 30\%$$

302. Find the average of first ten prime numbers−

(a) 12.7 (b) 13

(c) 12.9 (d) 12.8

Ans. (c) **42th B.P.S.C. (Pre) 1997-98**

The first ten prime numbers are—2, 3, 5, 7, 11, 13, 17, 19, 23, 29

$$\text{Average} = \frac{2+3+5+7+11+13+17+19+23+29}{10}$$

$$= \frac{129}{10} = 12.9$$

303. The average temperature for Monday, Tuesday and Wednesday was 42^{0}C. The average temp. for Tuesday, Wednesday and Thursday was 43^{0}C. If the average temp. on Thursday is 44^{0}C, then the temp on Monday was-

(a) 41^{0}C (b) 42^{0}C

(c) 43^{0}C (d) 44^{0}C

39th B.P.S.C. (Pre) 1994

Ans. (a)

Let, the temp. on Monday, Tuesday, Wednesday, Thursday are a, b, c and d respectively.

According to question,

$$\frac{a+b+c}{3} = 42^0C \Rightarrow a+b+c = 126^0C$$

$$\frac{b+c+d}{3} = 43^0C \Rightarrow b+c+d = 129^0C$$

$d = 44^0C$ (According to question)

$b+c = (129^0 - 44^0)C$

$= 85^0C$

$a = (a+b+c) - (b+c)$

$= 126^0 - 85^0 = 41^0C$

304. If the average annual income of 10 teachers is Rs. 25,000 and the annual income of two teachers is Rs. 20,000. then the annual income of the remaining teachers will be-

(a) 26,250 Rs. (b) 25,500 Rs.

(c) 23,200 Rs. (d) 22,000 Rs.

40th B.P.S.C. (Pre) 1995

Ans. (a)

Total annual income of 10 teachers $= 25000 \times 10$ Rs.

Total annual income of 2 teachers $= 20000 \times 2$ Rs.

Total annual income of remaining 8 teachers

$= (250000 - 40000)$ Rs.

Average income of 8 teachers $= \dfrac{250000 - 40000}{8}$

= Rs. 26250

305. The sum of eleven consecutive numbers is 220, then the middle number is-

(a) 25 (b) 20

(c) 15 (d) 10

41st B.P.S.C. (Pre) 1996

Ans. (b)

Let middle number $= x$

So the eleven consecutive numbers are—

$(x-5), (x-4), (x-3), (x-2), (x-1), x, (x+1), (x+2), (x+3), (x+4), (x+5)$,

According to question,

$(x-5)+(x-4)+(x-3)+(x-2)+(x-1)+x+(x+1)+(x+2)+(x+3)+(x+4)+(x+5) = 220$

$11x = 220$

$x = 20$

306. A man's age at the time of his death was twenty-ninth of the year of the year he was born. If the age at the time of death was in complete years, then what was his age in the year 1900?

(a) 74 years (b) 45 years

(c) 44 years (d) 54 years

40th B.P.S.C. (Pre) 1995

Ans. (c)

Let the age of man is 1900 = x year.

It is clear that his birth year was $(1900 - x)$

According to q^n, $(1900 - x,)$ is divisible by 29.

$\dfrac{1900 - x}{29}$ is an integer.

$1900 - x = 1856$ (1856 is also divisible by 29)

$\therefore x = 44$ years

307. Which of the following statements is false?

(a) Two acute angles in any right triangle are complementary angle.

(b) The diagonal of a rhombus bisect each other at right angles.

(c) If the angle of a quadrilateral bisect each other, then the shape of a Parallelogram is formed.

(d) Every triangle have one acute angle.

44th B.P.S.C. (Pre) 2000-01

Ans. (d)

Each triangle must have almost one acute angle. This statement almost is wrong, because all angle of a triangle can be acute angle.

308. The perimeter of a square is (4a+8) unit, the area of square is-

(a) $(a^2 + 4a + 4)$ unit^2

(b) $(b - a - 8)$ unit^2

(c) $(a + 2 + 4)$ unit^2

(d) $(4a^2 + a - 4)$ unit^2

40th B.P.S.C. (Pre) 1995

Perimeter of square $= 4a + 8$

Side of square $= \dfrac{4a+8}{4} = (a+2)$ Unit

here, side of square $= \dfrac{\text{Perimeter}}{4}$

Area of square, $(\text{side})^2 = (a+2)^2$ Unit^2

$= (a^2 + 4a + 4)$ Unit^2

309. If the sides of a triangle are 11m, 60m and 61m, then choose the height of triangle relative to longer side.

(a) 10 m (b) 9 m

(c) 10.8 m (d) 9.8 m

42nd B.P.S.C. (Pre) 1997-98

Ans. (c)

It is clear that given triangle is right angled triangle Δ because $(11)^2 + (60)^2 = (61)^2$

Area of triangle $= \dfrac{1}{2} \times 60 \times 11 = 330$ sq. m.

$(\because$ Area of right angle triangle $= \dfrac{1}{2} \times \text{base} \times \text{height})$

Let height on hypotenuse $= h$

Now, let the hypotenuse as base and perpendicular as base then,

Area of triangle $=$ area of Δ ABC

$330 = \dfrac{1}{2} \times h \times 61$

$h = 10.81$ m

So, height $= 10.81$ m

310. The diameter of a solid copper sphere is 6cm. The sphere is melted and shaped into a new wire whose diameter is 0.2 cm. Then the length of wire is-

(a) 36 m (b) 360 m

(c) 24 m (d) 360 cm

42nd B.P.S.C. (Pre) 1997-98

Ans. (a)

Radius of sphere $= \dfrac{6}{2} = 3$cm

Volume of sphere $= \dfrac{4}{3}\pi r^3$

$= \dfrac{4}{3}\pi (3)^3$

A solid is melted and shaped into a wire it is clear that the new shape is cylindrical.

Let the length is l and given that diameter $= 0.2$cm

$\therefore$ radius of wire $= \dfrac{0.2}{2} = 0.1$cm

Volume of sphere $=$ Volume of cylinder

$\dfrac{4}{3}\pi (3)^3 = \pi (0.1)^2 l$

$l = \dfrac{4}{3} \times \dfrac{27}{(0.1)^2}$

$l = 3600$cm, $l = 36$m

311. Three cubes of sides 3cm, 4cm and 5cm are made of a metal, they are melted to form a new cube, the side of a new cube is-

(a) 6 cm (b) $5\frac{1}{2}$ cm

(c) $4\frac{1}{2}$ cm (d) 4 cm

41st B.P.S.C. (Pre) 1996

Ans. (a)

$(\text{Side of new cube})^3 =$ Total volume of all three cubes.

$= 3^3 + 4^3 + 5^3$

$= 216$

side of new cube $= (216)^{1/3}$

$= 6$ cm

312. The sides of a wooden block is 5 cm × 10 cm × 20 cm How many such complete blocks would be required to made solid density of wood of least measure.

(a) 6 (b) 8

(c) 12 (d) 16

40th B.P.S.C. (Pre) 1995

Ans. (b)

Given that, volume of wooden box $= 5 \times 10 \times 20 = 1000$ cm^2

HCF of 5, 10, 20 $= 5$ cm

Volume of cube of side 5 cm $= 5 \times 5 \times 5$ cm^3

Total number of blocks $= \dfrac{1000}{5 \times 5 \times 5}$

$= 8$

313. A engine whose wheel circumference is $7\frac{1}{2}$ m. takes 9 second to revolve 7 round, then the speed of that vehicle in km/hr will be?

(b) 21

(c) 30 (d) 35

40th B.P.S.C. (Pre) 1995

Ans. (b)

∵ A wheel covers a distance equal to its circumference in one revolution.

∴ distance covered by the wheel in one round = $7\frac{1}{2}$ m

According to question,

Takes 9 second to revolve 7 rounds

∵ distance covered in 9 second= $7\times\frac{15}{2}$ m/s

∴ distance covered in 1 second = $\frac{7\times15}{2\times9}$ m/s

∴ speed = $\frac{7\times15}{18}$ m/s

∴ speed of train = $\frac{7\times15}{18}\times\frac{18}{5}$ km/hr

= 21 km/hr

314. Select of value of such that $2x + ay = 1$ and $3x - 5y = 7$ has no solution?

(a) $\frac{3}{10}$ (b) $-\frac{10}{3}$

(c) $\frac{10}{3}$ (d) $-\frac{3}{10}$

43rd B.P.S.C. (Pre) 1999

Ans. (b)

Given that,

$2x + ay = 1$

and $3x - 5y = 7$

Condition for the equation which has no solution

$ax + by + c = 0$ .... (i)

$dx + ey + f = 0$ .... (ii)

than $\frac{a}{d} = \frac{b}{e} + \frac{c}{f}$

According to question, $2x + ay = 1$ and

$3x - 5y = 7$

Now, $\frac{2}{3} = \frac{a}{-5} + \frac{1}{7}$

$\boxed{a = \frac{-10}{3}}$

315. ${}^nc_1 + {}^nc_2$ .... $+ {}^nc_n = 255$, then the value of n will be.-

(a) 6 (b) 8

(c) 4 (d) 10

47th B.P.S.C. (Pre) 2005

Ans. (b)

We know that— ${}^nc_0 + {}^nc_1 + {}^nc_2$............$+ {}^nc_n = 2^n$

${}^nc_1 + {}^nc_2 + {}^nc_3$............ ${}^nc_n = 2^n - {}^nc_0$

Given that— ${}^nc_1 + {}^nc_2$............$+ {}^nc_n = 255$

$2^n - {}^nc_0 = 255$

$2^n - 1 = 255$

$2^n = 256 = 2^8$

Since, both sides have the same base so the powers will also same

$\boxed{n = 8}$

316. Series $\sum\frac{n.2^n}{n^n}$ is-

(a) Convergent

(b) Divergent

(c) Convergent and divergent

(d) None of these

46th B.P.S.C. (Pre) 2004

Ans. (a)

$\sum\frac{n.2^n}{n^n}$

putting n = 1, 2, 3, ...

$\sum\frac{n.2^n}{n^n} = 2, 2, \frac{24}{27}, \frac{1}{4}$,........ (which are in decreasing order)

Hence the series is convergent.

317. If the roots of equation $x^2 + 11x + 50 = 0$ are α, β is, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = ?$

(a) 0.21 (b) 0.42

(c) 0.84 (d) 1.0

46th B.P.S.C. (Pre) 2004

Ans. (b)

Equation $x^2 + 11x + 50 = 0$

$\alpha + \beta = \frac{-11}{1}$ (Sum of roots $= \frac{-b}{a}$)

$\alpha . \beta = \frac{50}{1}$ (products of root $= \frac{c}{a}$)

$$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{\alpha^2+\beta^2}{\alpha\beta}$$

$$=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$$

Putting the value of $\alpha+\beta$ and $\alpha\,.\,\beta$

$$=\frac{(-11)^2-2(50)}{50}$$

$$=\frac{21}{50}=0.42$$

318. The value of k, so that point (1, 2, 3,), (k, 0, 4) and (-2, 4, 2) lie in a line–

(a) 4 (b) 3

(c) 1 (d) 2

47th B.P.S.C. (Pre) 2005

Ans. (a)

The given points will be linear if

$$\begin{vmatrix} 1 & 2 & 3 \\ k & 0 & 4 \\ -2 & 4 & 2 \end{vmatrix}=0$$

$$\Rightarrow 1\begin{vmatrix} 0 & 4 \\ 4 & 2 \end{vmatrix}-2\begin{vmatrix} k & 4 \\ -2 & 2 \end{vmatrix}+3\begin{vmatrix} k & 0 \\ -2 & 4 \end{vmatrix}=0$$

$\Rightarrow$ 1 (0 - 16) - 2 (2k + 8) + 3 (4k) = 0

$\Rightarrow$ -16 - 4k - 16 + 12k = 0

$\Rightarrow$ 8k = 32

$\Rightarrow$ k = 4

319. x = {a, b}, y = {2, 3} and z = {3, 4}, then choose $x \times (y \cap z)$

(a) {(a, 3), (b, 3)}

(b) {(3, a), (3, b)}

(c) {(a, 2), (a, 4), (b, 2) (b, 4)}

(d) {a, b, 2, 3, 4}

43rd B.P.S.C. (Pre) 1999

Ans. (a)

Given that x = {a,b}, y = {(2, 3)} & z = {3, 4}

$y \cap z = \{3\}$

$x \times (y \cap z) = (a, b) \times \{3\}$

$x \times (y \cap z) = (a, 3), (b, 3)$

(b) 5

320. If f: {x, y, z} $\to$ {a, b), is defined by f = {(x - a), (y, b)} then f–

(a) is a unitary function

(b) is not a function

(c) is an interpolative function

(d) is an empty set

43rd B.P.S.C. (Pre) 1999

Ans. (c)

f: {x, y, z} $\to$ {a, b), f = {(x - a), (y, b)}, then f is an interpolative function

321. Set (x : $x = x + 1$) is

(a) Set of real number

(b) Set of complex number

(c) Set of prime number

(d) Empty set

44th B.P.S.C. (Pre) 2000-01

Ans. (a)

For any value of x, the value of (x : $x = x+1$) is real.

322. Choose co-suffix for (1, 6), (2, 9), (3, 6), (4,7), (5, 8), (6, 5), (7, 12), (8, 3), (9, 17), (10, 1)-

(a) 0.4 (b) 0.6

(c) 4 (d) 6

44th B.P.S.C. (Pre) 2000-01

Ans. (a)

$$\bar{x}=\frac{1+2+3+4+5+6+7+8+9+10}{10}$$

$= 5.5$

$$\bar{y}=\frac{6+9+6+7+8+5+12+3+17+1}{10}$$

$= 7.4$

n = 10

$\Sigma\ (x-\bar{x})(y-\bar{y}) = 4.0$

$$(\sigma)^2=\frac{1}{n}\Sigma\ (x-\bar{x})(y-\bar{y})=\frac{1}{10}\times 4$$

$= 0.4$

323. If A and B are two sets such that $A \cup B$ has 18, A has 8 and B has 15 members then choose the number of members in $A \cap B$–

(c) 23 (d) 26

42th B.P.S.C. (Pre) 1997-98

Ans. (b)

We knot that–

$A \cup B = A + B - (A \cap B)$

$A \cap B = A + B - (A \cup B)$

$= 8 + 15 - 18$ (According to question)

$= 5$

324. If $(x - a)$ is factor of $x^3 - a^2x + x + 2$ then find the value of a.

(a) 1 (b) 2

(c) -2 (d) -1

42th B.P.S.C. (Pre) 1997-98

Ans. (c)

$(x - a)$ is a factor of given equation

$\therefore x - a = 0$

$x = a$

Substituting the value in the given equation—

$a^3 - a^3 + a + 2 = 0$

$\therefore a = -2$

325. Let matrix A= $\begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix}$, B= $\begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix}$ If (A+B) (A-B) = $A^2 - B^2$ then

(a) $\alpha = \beta = -1$ (b) $\alpha = \beta = 0$

(c) $\alpha = \beta = 1$ (d) $\alpha = 1, \beta = -1$

46th B.P.S.C. (Pre) 2004

Ans. (b)

$$A^2 = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -3 \\ 6 & 2 \end{vmatrix}$$

$= -2 + 18$

$= 16$

$$B^2 = \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix} \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1+\alpha\beta & 2\alpha \\ 2\beta & \alpha\beta+1 \end{vmatrix}$$

cle by 1 unit $= r + 1$

$= (1 + \alpha\beta)^2 - 4\alpha\beta$

$= (1 - \alpha\beta)^2$

$$A + B = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & \alpha - 1 \\ 2+\beta & 3 \end{vmatrix}$$

$$A - B = \begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} - \begin{vmatrix} 1 & \alpha \\ \beta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -\alpha - 1 \\ 2-\beta & 1 \end{vmatrix}$$

According to question, substituting the value in equation

$(A + B)(A - B) = A^2 - B^2$ — $\alpha = \beta = 0$

326. Let $f(x) = ax^7 + bx^3 - cx - 5$ where a, b, c are constant, if $f(-7) = 7$, then $f(7)$ is equal to-

(a) -17 (b) -7

(c) 14 (d) 17

45th B.P.S.C. (Pre) 2002

Ans. (a)

$f(x) = ax^7 + bx^3 - cx - 5$

$f(-7) = a(-7)^7 + b(-7)^3 - c(-7) - 5$

$+7 = -a(7)^7 - b(7)^3 + 7c - 5$ $(\because f(-7) = +7)$

$7 = -\{a(7)^7 + b(7)^3 - 7c\} - 5$

$12 = -\{a(7)^7 + b(7)^3 - 7c - 5 + 5\}$

$12 = -\{f(7) + 5\}$

$12 = -f(7) - 5$

$\therefore f(7) = -17$

327. If the radius of a circle is increased by 1 unit then its area becomes double. Find the radius of circle?

(a) $2\sqrt{2}$ (b) $\sqrt{2}(\sqrt{2} - 1)$

(c) $\sqrt{2} - 1$ (d) $1 + \sqrt{2}$

46th B.P.S.C. (Pre) 2004

Ans. (d)

Let radius $= r$

Area of circle $= \pi r^2$

New area of circle = $\pi\ (r+1)^2$

According to question.

$2\pi\ r^2 = \pi\ (r+1)^2$

$2r^2 = (r+1)^2$

$r^2 - 2r - 1 = 0$

$r = \dfrac{2 \pm 2\sqrt{2}}{2}$

$r = \dfrac{2 + 2\sqrt{2}}{2}$ (r cannot be negative)

$= 1 + \sqrt{2}$

328. A sector of a circle of radius 7 cm subtends an angle of 120^0. Choose the area of the sector–

(a) 51.31cm^2 (b) 53.13 cm^2

(c) 53.31cm^2 (d) 53.03 cm^2

40th B.P.S.C. (Pre) 1995

Ans. (a)

Area of sector $= \dfrac{\text{Angle}}{360^0}\pi r^2$

$= \dfrac{120^0}{360^0} \times \pi\ (7)^2$

$= \dfrac{1}{3} \times \dfrac{22}{7} \times 7 \times 7 = 51.31\text{cm}^2$

329. If the radius of a circle is doubled, then choose the percentage of its increased area.

(a) 400 (b) 100

(c) 300 (d) 200

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Half of diameter means radius

Let the radius of circle is r.

$\therefore$ Area of circle = $\pi\ r^2$

New radius after doubling the radius = $2r$

Area of new circle = $\pi\ (2.r)^2$

$= 4\pi\ r^2$

Percentage of increased area.

$= \dfrac{4\pi r^2 - \pi r^2}{\pi r^2} \times 100$

= 300 percentage

330. Two pillars are standing on flat ground, whose height are 6m and 11m. If the distance between their lower ends on the ground is 12 m. then. What will be the distance between their ends?

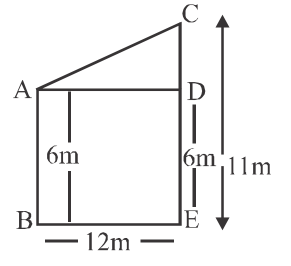
(a) 16 m (b) 15 m

(c) 13 m (d) 14 m

45th B.P.S.C. (Pre) 2001-02

Ans. (c)

AB and AC are pillars, whose heights are 6m and 11m.



CD = CE - ED

= 11 - 6 (ED = AB = 6m)

= 5m

In Δ ACD

AD = 12m ($\therefore$ AD = BE)

CD = 5m

$AC^2 = AD^2 + CD^2$

$AC = \sqrt{12^2 + 5^2}$

$\boxed{AC = 13\text{m}}$

331. The maximum length of pillar that can be placed in a room of length 12m, breadth 9 cm and height 8m is?

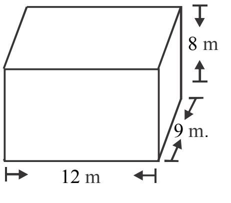
(a) 864 m (b) 10 m

(c) 17 m (d) 43 m

(e) None of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)



Maximum length of a room that can be placed in room = diagonal of room $= \sqrt{(l)^2 + (b)^2 + (h)^2}$

$$= \sqrt{(12)^2 + (9)^2 + (8)^2}$$

$$= \sqrt{144 + 81 + 64}$$

$$= \sqrt{289} = 17 \text{ m}$$

332. A train, whose length is 150m, travelling towards North at a speed of 144 km/hr can cross 250 m long bridge in-

(a) In 20 seconds
(b) In 100 seconds
(c) In 45 seconds
(d) In 10 seconds
(e) In 28 seconds

64th B.P.S.C. (Pre) 2018

Ans. (d)

Speed of train = 144 km/hr

$$= \left(144 \times \frac{5}{18}\right) \text{ m/s}$$

= 40 m/s

distance covered by train to pass bridge = Length of train + length of bridge

= (150+ 250) m = 400 m

Time taken by train to pass bridge $= \frac{400}{40}$ seconds

= 10 seconds

333. Driver of a car drives at an average speed of 40 km/hr between a hill station at a distance of 200 km from the plane. While travelling back he covers the same distance at an average speed of 20 km/hr. What will be the average speed of train in total 400 km?

(a) 25 km/hr (b) 30 km/hr
(c) 24.33 km/hr (d) 26.67 km/hr

45th B.P.S.C. (Pre) 2001-02

Ans. (d)

Total time to cover 200 km with speed of 40 km/hr $= \frac{200}{40}$

= 5 hr

Time taken by train at a speed of 20 km/h

$$= \frac{200}{20} = 10 \text{ hr}$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$= \frac{200+200}{10+5}$$

= 26.67 km/hr

334. If $y = |x|, x \neq 0$ then choose the value of $\frac{dy}{dx}$

(a) $\frac{|x|}{x}$ (b) 1
(c) $\frac{x}{|x|}$ (d) -1

44th B.P.S.C. (Pre) 2000-01

Ans. (*)

(i) If $x > 0$ then $|x| = x$

$y = x$

$$\frac{dy}{dx} = 1$$

(ii) If $x < 0$ then $|x| = -x$

$y = -x$

$$\frac{dy}{dx} = -1$$

So, $$\frac{dy}{dx} = \pm x$$

$\therefore$ None of the options is correct

335. If the function $f(x)$ is expressed as $f(x) = \frac{x - |x|}{x}$ then–

(a) Function is continuous everywhere
(b) Function is not continuous
(c) Function is continuous where $x = 0$
(d) Function is continuous for all x except zero

45th B.P.S.C. (Pre) 2002

Ans. (d)

Given $f(x) = \frac{x - |x|}{x}$

$$f(x) = \begin{cases} \frac{x+x}{x} & x < 0 \\ \frac{x-x}{} & x > 0 \end{cases}$$

$$f(x)=\begin{cases}\frac{2x}{x} & x<0\\ 0 & x>0\end{cases}$$

$$f^1(x)=\begin{cases}0 & x<0\\ 0 & x>0\end{cases}$$

Now, $\lim_{x\to of}=\lim_{x-10}=0$

We got constant value so the function is continuous for all values of x except $x=0$

336. Series $\sum\frac{x^2-2}{x^2+1}x^2, x<0$

(a) Convergent for $x\geq 1$
(b) Convergent for $x = 1$
(c) Divergent for $x\geq 1$
(d) Divergent for x ?1

40th B.P.S.C. (Pre) 1995

Ans. (c)

From CR. Test

$x^2\neq 0$

$$lim_{x\to 0}\left(\frac{n^2-1}{n^2+1}x\right)^{\frac{1}{n}}$$

While $x>0$

$x\geq 1$

So, it is divergent for $x\geq 1$

337. If function $f(x) = (x+1)$ and $g(x) = x^2 - 3$ then $\frac{gf\{(-1)\}}{f\{g(-1)\}}$ is equal to–

(a) 0
(b) 1
(c) 2
(d) 3

46th B.P.S.C. (Pre) 2004

Ans. (d)

$f(-1) = -1+1$
$= 0$
$g(-1) = 1^2 - 3$
$= -2$

$$\frac{gf\{(-1)\}}{f\{g(-1)\}}=\frac{g\ (0)}{f\ (-2)}$$

$$=\frac{0^2-3}{-2+1}$$

$$=\frac{-3}{-1}=3$$

338. $f(x)=\frac{1}{1-x}$, the value of $f[f(f(x))]$ will be–

(a) $\frac{1}{(1-x)^2}$
(b) x^2
(c) $(1-x)$
(d) x

47th B.P.S.C. (Pre) 2005

Ans. (d)

$f(x)=\frac{1}{1-x}$ (Given)

$$f(f(x))=\frac{1}{1-\frac{1}{(1-x)}}$$

$$=\frac{1-x}{1-x-1}$$

$$=\frac{1-x}{-x}$$

$$=\frac{x-1}{x}$$

Again $f[f(f(x))]=\frac{\frac{1}{(x-1)}-1}{\frac{1}{(1-x)}}$

$$=\frac{1-1+x}{(1-x)}\times\frac{(1-x)}{1}=x$$

339. Curve $f(n)=x^3, x=0$ to–

(a) 26
(b) 25
(c) 2
(d) 0

47th B.P.S.C. (Pre) 2005

Ans. (d)

$y=x^3$

$\frac{dy}{dx}=3x^2$ ....(i)

For the maximum and minimum value, $\frac{dy}{dx}=0$

$\frac{d^2y}{dx_2}=6x$

$\frac{d^3y}{dx_3}=6\neq 0$

Hence at: $x=0$, the function is neither maximum nor minimum. This the point of change.

340. The demand function of a good is $x = \frac{1}{3}$ (24 - 2p), where x represents the number of units demanded and p represents per unit root. The value of p which will maximum the income?

(a) 4 (b) 8
(c) 6 (d) 5

47[th] B.P.S.C. (Pre) 2005

Ans. (c)

$x = \frac{1}{3}(2x - 2p)$ ....(i)

$x = 8 - \frac{2}{3}p$

$p = 12 - \frac{3}{2}x$

Revenue function R(p) = px = $p\left(\frac{24-2p}{3}\right)$ ....[by (i)]

$= 8p - \frac{2}{3}p^2$

For maximum income–

$\frac{dR(p)}{dp} = \frac{d}{dp}\left(8p - \frac{2}{3}p^2\right) = 0$

$8 - \frac{4}{3}p = 0$

$8 = \frac{4}{3}p$

$\boxed{p = 6}$

341. The maximum value of $z = 50x + 15y$ under the condition $5x + y \le 100$, and $x + y \le 60$ $x \ge 0$, $y \ge$ will be.

(a) 1500 (b) 1250
(c) 900 (d) 1000

47[th] B.P.S.C. (Pre) 2005

Ans. (b)

$5x + y \le 100$ ....(i)

$x + y \le 60$ ....(ii)

Substracting eq[n] (ii) from (i)

$4x \le 40$

$x \le 10$

$y \le 50$

Given that $x \ge 0, y \ge 0$

$\begin{pmatrix} 10 \ge x \ge 0 \\ 50 \ge y \ge 0 \end{pmatrix}$

$z = 50x + 15y$

For the maximum value of z, the value of x and y should be maximum

$\therefore x = 10$

$y = 50$

$z = 50 \times 10 + 15 \times 50$

$z = 1250$

342. No. of solutions of equation $|x|^2 - 3|x| + 2 = 0$ is-

(a) 2 (b) 3
(c) 4 (d) 5

46[th] B.P.S.C. (Pre) 2004

Ans. (c)

If $x > 0$ then $|x| = x$

$|x|^2 - 3|x| + 2 = 0$

$x^2 - 3x + 2 = 0$

$x(x-2) - 1(x-2) = 0$

$(x-2)(x-1) = 0$

$x = 1$ and 2

If $x < 1$ then $|x| = -x$

$|x|^2 - 3|x| + 2$

$(-x)^2 - 3(-x) + 2 = 0$

$x^2 + 3x + 2 = 0$

$x^2 + 2x + x + 2 = 0$

$x(x+2) + 1(x+2) = 0$

$(x+2)(x+1) = 0$

$\therefore x = -1$ and -2

It is clear that, x has four values are 1, -1, –2 and 2

343. $\lim_{x \to 0} \frac{xe^x - \log(1+x)}{x^2}$ is equal to–

(a) 1/2 (b) 3/2
(c) 0 (d) 5/2

45[th] B.P.S.C. (Pre) 2002

Ans. (b)

$\lim_{x \to 0} \frac{xe^x - \log(1+x)}{x^2}\left[\frac{0}{0}\right]$

$$\frac{\frac{d}{dx}\left[xe^x - \log(1+x)\right]}{\frac{d}{dx}(x)^2}$$

$$lim_{x \to 0} \frac{xe^x + e^x - \frac{1}{1+x}}{2x}\left[\frac{0}{0}\right]$$

Again by L- hospital rule

$$lim_{x \to 0} \frac{\frac{\mathrm{d}}{\mathrm{dx}}\left(xe^x + e^x - \frac{1}{(1+x)}\right)}{\frac{d}{dx}(2x)}$$

$$lim_{x \to 0} \frac{xe^x + e^x + e^x + \frac{1}{(1+x)^2}}{2}$$

$$= \frac{0+1+1+1}{2} = \frac{3}{2}$$

344. $f(x) = \frac{x-5}{x+5}$ **,** $x \pm 5$**, then the obtained of** $f^{-1}(x)$ **is–**

(a) R (b) R - {1}

(c) $(-\infty, 1)$ (d) $(1, \infty)$

46th B.P.S.C. (Pre) 2004

Ans. (b)

$$f(x) = y = \frac{x-5}{x+5}$$

Swap x and y–

$$x = \frac{y-5}{y+5}$$

$$x(y+5) = y-5$$

$$xy + 5x = y-5$$

$$xy - y = -5x - 5$$

$$y(x-1) = -5x-5$$

$$y = \frac{5(x+1)}{1-x}$$

so $f^{-1}(x) = \frac{5(x+1)}{(1-x)}$

Clearly $x \neq 1$

So, obtained = $R = \{x \neq 1\}$

345. The value of $lim_{x \to 0} \frac{1}{1-e^{-1/x}}$ **is–**

(a) 1 (b) 0

(c) $-\infty$ (d) ∞

46th B.P.S.C. (Pre) 2004

Ans. (a)

$$lim_{x \to 0} \frac{1}{1-e^{-1/x}}$$

$$\Rightarrow lim_{x \to 0} \frac{1}{1 - \frac{1}{e^{1/x}}}$$

$$= 1$$

346. $x = $ **a Cos**θ **+ b Sin**θ

y = a Sinθ **- b Cos**θ

$x^2 + y^2$ **is equal to-**

(a) $a^2 - b^2$ (b) $b^2 - a^2$

(c) $a^2 + b^2$ (d) $a^2 + 2ab$

46th B.P.S.C. (Pre) 2004

Ans. (c)

$x = a\,\mathrm{Cos}\theta + b\,\mathrm{Sin}\theta$

$x^2 = (a\,\mathrm{Cos}\theta + b\,\mathrm{Sin}\theta)^2$

$y = (a\,\mathrm{Sin}\theta - b\,\mathrm{Cos}\theta)$

$y^2 = (a\,\mathrm{Sin}\theta - b\,\mathrm{Cos}\theta)^2$

$x^2 + y^2 = (a\,\mathrm{Cos}\theta + b\,\mathrm{Sin}\theta)^2 + (a\,\mathrm{Sin}\theta - b\,\mathrm{Cos}\theta)^2$

$= a^2(\mathrm{Sin}^2\theta + \mathrm{Cos}^2\theta) + b^2(\sin^2\theta + \mathrm{Cos}^2\theta)$

$+ 2ab\,\mathrm{Cos}\theta\,\mathrm{Sin}\theta - 2ab\,\mathrm{Sin}\theta.\mathrm{Cos}\theta$

$x^2 + y^2 = a^2 + b^2$

347. Choose the value of $13^0 12'$ from the following–

(a) $\frac{11}{150}\pi$ (b) $\frac{13}{150}\pi$

(c) $\frac{6}{125}\pi$ (d) $\frac{11}{25}\pi$

44th B.P.S.C. (Pre) 2000-01

Ans. (a)

$$13^0 12' = 13^0 \left(\frac{12}{60}\right)^0$$

$$= 13^0 + \frac{1^o}{5}$$

$$= \frac{66^0}{5}$$

$$\therefore 180^0 = \pi$$

$$\frac{66^0}{5} = \frac{\frac{66}{5}\times\pi}{180}$$

$$= \frac{11}{150}\pi$$

348. If a = 13, b = 14, c =15 then find the value of tan $\frac{c}{2}$ –

(a) 1/2 (b) 2/3
(c) 4/7 (d) 1/7

44th B.P.S.C. (Pre) 2000-01

Ans. (b)

$$s = \frac{a+b+c}{2}$$

$$= \frac{13+14+15}{2}$$

s = 21

$$\tan\frac{c}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$= \sqrt{\frac{(21-13)(21-14)}{21(21-15)}} = \frac{2}{3}$$

349. Equation $x = 5 + 3\,Cos\,\alpha$ and y = 7 + $3 Sin\alpha$ is combinally a–

(a) Parabola (b) Circle
(c) Hyperbola (d) Line pair

44th B.P.S.C. (Pre) 2000-01

Ans. (b)

$x = 5 + 3Cos\alpha$ $\quad$ $y = 7 + 3Sin\alpha$

$Cos\alpha = \frac{x-5}{3}$ $\quad$ $Sin\alpha = \frac{y-7}{3}$

On squaring both sides-

$$Cos^2\alpha = \left(\frac{x-5}{3}\right)^2, \quad Sin^2\alpha = \left(\frac{y-7}{3}\right)^2$$

$$\Rightarrow \because Sin^2\alpha + Cos^2\alpha = 1$$

$$\Rightarrow \therefore \left(\frac{x-5}{3}\right)^2 + \left(\frac{y-7}{3}\right)^2 = 1$$

$\Rightarrow (x-5)^2 + (y-7)^2 = 3^2$

$\Rightarrow$ It is an equation of circle whose radius is 3 and centre is (5,7).

Note—$x^2 + y^2 = a^2$

is an equation of circle

350. $x^2y^2 = a^2(x^2 + y^2)$ intangible parallel to the coordinates for the curve-

(a) $x = \pm a$, y = 0
(b) $x = \pm a$, y = $\pm a$
(c) $x = a$, y = 0
(d) $x = -a$, y = - a

45th B.P.S.C. (Pre) 2002

Ans. (b)

$x^2y^2 = a^2(x^2 + y^2)$

$$y = \pm\frac{ax}{\sqrt{x^2 - a^2}}$$

y is imaginary when $-a < x < a$ and x is imaginary when $-a < y < a$. Hence, any part of curve is not between $x = \pm a$ and $y = \pm a$.

Hence $y = \pm a$ is tangent parallel to the axis.

and $x = \pm a$ ....

351. The radius of the concentric circle touching the line x - y = 0 and the circle $x^2 + y^2 - 6x - 2y - 3 = 0$ is.

(a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$

(c) 1 (d) 2

46th B.P.S.C. (Pre) 2004

Ans. (a)

The radius of the concentric circle touching the straight line $x - y = 0$, centre (3, 1) of circle $x^2 + y^2 - 6x - 2y - 3 = 0$ equal to perpendicular dropped from the center.

Radius of concentric circle $= \frac{3-1}{\sqrt{1^2+1^2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

352. The value of the subtle angle between the normal's of the planes $2x - y + z = 6$ and $x + y + 2z = 7$ will be-

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{}$

4

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

47th B.P.S.C. (Pre) 2005

Ans. (c)

$2x - y + z = 6$ here $a_1 = 2$, $b_1 = -1$, $c_1 = 1$
$x + y + 2z = 7$ here $a_2 = 1$, $b_2 = 1$, $c_2 = 2$

$$\text{Cos}\theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$= \frac{2-1+2}{\sqrt{6}\ \sqrt{6}}$$

$$= \frac{3}{6}$$

$$Cos\theta = \frac{1}{2} = \text{Cos}60^0 = \text{Cos} = \frac{\pi}{3}$$

$$\therefore\ \theta = \frac{\pi}{3}$$

353. If in a code, DECIDE is written as 453945 then what will be written in place of ABIDE?

(a) 94521 (b) 49521

(c) 12945 (d) 49251

40th B.P.S.C. (Pre) 1995

Ans. (c)

Clearly, It is a alphabetical number of english language.

D	E	C	I	D	E
↓	↓	↓	↓	↓	↓
4	**5**	**3**	**9**	**4**	**5**

So, A B I D E ⟹ 1,2,9,4,5

354. In a code HORSE is written as DRONG then, in the some code, MONKEY is written as–

(a) XDJMNL
(b) YEKNOM
(c) ESRIHD
(d) GNQRDM

39th B.P.S.C. (Pre) 1994

Ans. (*)

There is an error in this question due to which no correct answer can drawn If the question had HOPSE instead of HORSE, than the correct answer would be option (a).

355. Which of the following numbers is in the blank space 2, 4, 3, 9, 4, 16, 5, 25, 6 ....

(a) 36 (b) 39

(c) 81 (d) 243

39th B.P.S.C. (Pre) 1994

Ans. (a)

The given series is increasing as follows–

+5 +7 +9 +11
4 3 9 4 16 5 25 6 $\boxed{36}$
+1 +1 +1

356. *x, y,* z, u are in increasing order and v, *y, w* are in decreasing order. Which of the following has neither increasing nor decreasing order.

(a) *x, y,* z are in increasing order (as per question)
(b) w, *y,* z are in increasing order (as per question)
(c) *x,* v, w are not in any order.
(d) v, *y,* w are in decreasing order (as per question)

39th B.P.S.C. (Pre) 1994

Ans. (c)

Option (c) is correct.

357. My clock is 3 minutes behind on Monday at 2 PM and 5 minutes ahead on Wednesday at 2PM. When did it show the right time?

(a) Tuesday at 6 am
(b) Tuesday at 8 am
(c) Wednesday at 4 am
(d) Wednesday at 8 am

39th B.P.S.C. (Pre) 1994

Ans. (b)

∵ According to question, clock gained 8 min time interval in 48 hours

∴ The time interval per minutes is achieved in 6 hours. The clock was 3 minutes behind, hence just after completing 3 minutes time duration, clock will show right time.

Time taken by the clock in 3-minutes interval $= 3 \times 6 = 18$

General Science

I. PHYSICS

Measurement/Unit

1. In the following which is fundamental physical quantity?

(a) Force (b) Velocity
(c) Electric current (d) Work
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

In physics, there are seven fundamental physical quantities (which are measured in base or fundamental units) : Length, Mass, Time, Electric current, Temperature, Luminous intensity and Amount of a substance.

2. The unit of electric power is :

(a) Ampere (b) Volt
(c) Coulomb (d) Watt
(e) None of the above / More than one of the above

65th B.P.S.C. (Pre) 2019
64th B.P.S.C. (Pre) 2018

Ans. (d)

Electric power is the rate per unit time, at which electrical energy is transferred by an electric circuit. The SI unit of electric power is Watt. Ampere is the unit of electric current, Volt is the unit of electrical potential and Coulomb is the unit of electric charge in the SI system.

3. At the time of a short circuit, the current in the circuit

(a) varies continuously (b) reduces substantially
(c) does not change (d) increases heavily

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (d)

The potential difference/voltage in the domestic circuit is provided by live and neutral wire. When due to any reason, live wire and neutral wire get connected to each other i.e. short circuit, a huge amount of current starts flowing in the circuit because the path of current has shortened and a low resistance path for current has been made. That's why at the time of short circuit the current in the circuit increases heavily. To stop the overloading of current in the circuit, electric fuse is used.

4. Frequency is measured in :

(a) hertz (b) metre/second
(c) radian (d) watt
(e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

The Hertz (symbol : Hz) is the derived unit of frequency in the International System of Units (SI system) and is defined as one cycle per second.

5. What is measured in hertz?

(a) Frequency (b) Energy
(c) Heat (d) Quality
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (a)

See the explanation of the above question.

6. Bolometer is used to measure

(a) frequency (b) heat radiations
(c) velocity (d) wavelength
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (b)

A bolometer is used to detect Infrared rays. A bolometer is a device used for detecting and measuring the heat and radiation of microwave energy.

7. The hearing frequency range of cat is

(a) 20 Hz to 20 kHz
(b) 20 Hz to 50 kHz
(c) 45 Hz to 64 kHz
(d) 10 Hz to 100 kHz
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (*)

The hearing range of the cat for sounds of 70 dB SPL extends from 48 Hz to 85 kHz, giving it one of the broadest hearing ranges among mammals. Analysis suggests that cats evolved extended high-frequency hearing without sacrificing of low-frequency hearing.

8. 'Ohm-meter' is unit of :

(a) Resistance (b) Conductance

(c) Resistivity (d) Charge

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

Electrical resistivity is an intrinsic property that quantifies how strongly a given material opposes the flow of electric current. A low resistivity indicates a material that readily allows the movement of electric charge. The SI unit of electrical resistivity is ohm-meter (Ωm). It is commonly represented by the-

Greek letter ρ (rho) defined as $\left[\rho = \frac{RA}{l}\right]$

Here,

R = electrical resistance of the material

l = Length, A = Cross section area, ρ = resistivity.

9. 'Light-year' is a unit of :

(a) Time (b) Distance

(c) Speed (d) Force

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

66th B.P.S.C. (Pre) 2020

Ans. (b)

Light-year is a unit of astronomical distance equivalent to the distance that light travels in one year, which is 9.46×10^{15} metres.

10. Angstrom is a unit of

(a) Wavelength (b) Energy

(c) Frequency (d) Velocity

(e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

Angstrom (Å) is the unit of length used mainly for measuring the wavelengths of electromagnetic waves which is equal to 10^{-10} meter or 0.1 nanometer. It is named after the 19th century Swedish physicist Anders Jonas Angstrom.

11. 1 zepto is equal to

(a) 10^{-25} (b) 10^{25}

(c) 10^{-21} (d) 10^{-22}

(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

In metric system, 1 zeptometer $=10^{-21}$ meter.

12. 1 femtometer is equal to how many meters?

(a) 10^{-12} m (b) 10^{-14} m

(c) 10^{-15} m (d) 10^{-24} m

(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (c)

1 femtometer is equal to 10^{-15} meters. The symbol of femtometer is Fm.

13. One micron is equal to –

(a) 1/10 mm (b) 1/100 mm

(c) 1/1000 mm (d) 1/10,000 mm

39th B.P.S.C. (Pre) 1994

Ans. (c)

$$1 \text{ micron} = 10^{-6} \text{ m} = 10^{-6} \times 10^{3} \text{ mm} = \frac{1}{10^3} \text{mm} = \frac{1}{1000} \text{mm}$$

14. A micron is related to centimetre as

(a) 1 micron = 10^{-4} cm (b) 1 micron = 10^{-8} cm

(c) 1 micron = 10^{-6} cm (d) 1 micron = 10^{-5} cm

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Ans. (a)

1 micron (μm) = 10^{-6} m

$\because$ 1 m = 100 cm

$\therefore$ 1 μm = $10^{-6} \times 10^{2}$ cm

1 μm = 10^{-4} cm

15. A distance of 1 km means –

(a) 100 m (b) 1000 cm

(c) 1000 m (d) 100 cm

45th B.P.S.C. (Pre) 2001

Ans. (c)

The SI unit for distance is meter. 1 km is equal to 1000m. 1 m is equal to 100 cm.

16. One picogram is equal to –

(a) 10^{-6} gram (b) 10^{-9} gram

(c) 10^{-12} gram (d) 10^{-15} gram

42nd B.P.S.C. (Pre) 1997

Ans. (c)

The smaller units of measuring mass are Milligram, Microgram, Picogram and Femtogram etc.

1 Picogram = 10^{-12} gram 1 Milligram = 10^{-3} gram

1 Microgram = 10^{-6} gram 1 Nanogram = 10^{-9} gram

1 Femtogram = 10^{-15} gram